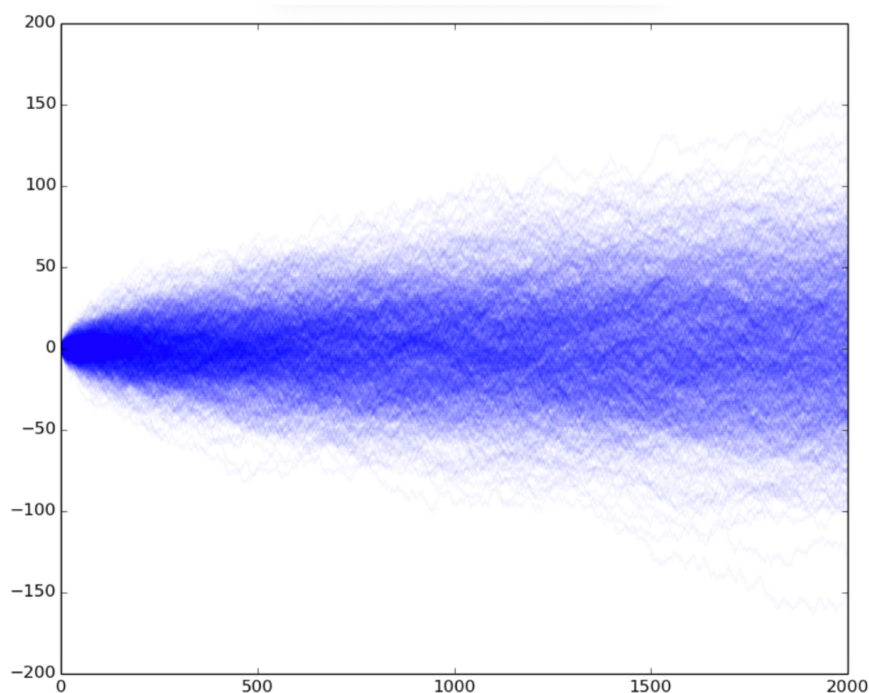




Probability Theory

With Analytic Foundations for Statistics and Learning Theory

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Abstract

These notes are a set of personal notes that I wrote as I studied probability with a measure-theoretic approach. The current draft focuses on three pillars: (1) measure theory; (2) Fourier analysis tools that reappear frequently in probability; (3) core probability. This edition is incomplete. In future revisions, I would like to add material on stochastic processes, applications to statistics and machine learning (including information-theoretic and statistical-physics viewpoints), as well as selected advanced topics such as large deviation theory. I also intend to include a dedicated section on random vectors and significantly increase the number of examples and illustrations. Readers who find the early measure-theoretic sections too abstract may safely skip ahead and return to them later when needed.

Contents

1	Measure Theory	4
1.1	Measurable Spaces	4
1.2	Measures	5
1.3	Measurable Functions	8
1.4	Integration	12
1.4.1	Lebesgue integration	13
1.4.2	Construction of the Lebesgue Integral	14
1.4.3	Integrable Functions	18
1.5	Product Measures and Fubini–Tonelli	24
1.6	Convergence Theorems	27
1.7	L^p Spaces and Inequalities	31
1.8	Densities and Measure Decompositions	39
1.9	Change of Variables	40
2	Harmonic Analysis	43
2.1	Hilbert Spaces	43
2.2	Fourier Series	49
2.3	Fourier Coefficients	54
2.4	Convolutions and Kernels	54
2.4.1	Convolutions	55
2.4.2	Gaussian Kernels and Approximate Identities	57
2.5	Completeness of the Trigonometric System	61
2.6	Fourier Transform	64
2.6.1	Fourier transform on L^1	65
2.6.2	Fourier Transform on L^2	68
2.7	Discrete Fourier Transform	72
2.7.1	Discrete Time Fourier Transform	72
2.7.2	Discrete Fourier Transform	73
3	Probability Foundations	76
3.1	A Toy Example	76
3.2	Probability Spaces, Events, and Probability Measures	77
3.3	Examples of Probability Measures	79
3.4	Independence and Conditional Probability	80
3.5	Random Variables	82
3.6	Law of a Random Variable	84
3.7	Cumulative Distribution Function	87
3.8	Independent Random Variables	90
3.9	Transformations of Random Variables	94
4	Expectation and Moments	97
4.1	Expectation	97
4.2	Moments and L^p -integrability	101
4.3	Classical Inequalities	102
4.4	Variance and Covariance	104
5	Generating functions	109
5.1	Moment Generating Function	109
5.2	Probability Generating Function	112
5.3	Characteristic function	114
6	Convergence of Random Variables and Limit Theorems	120
6.1	Almost Sure Convergence	120
6.2	Convergence in Probability	121
6.3	Convergence in L^p	122
6.4	Convergence in Distribution	123
6.5	Hierarchy of Convergence	127
6.6	Further Convergence Results	130
6.7	Uniform Integrability	132

6.8	Laws of Large Numbers	135
6.9	Central Limit Theorem	138
7	Conditioning	140
7.1	Conditional expectation	140
7.2	Conditioning with respect to a random variable	142
7.3	Conditional probability	143
7.4	Conditional variance	146
8	Classical Distributions	148
8.1	Discrete Distributions	148
8.1.1	Discrete Uniform Distribution	148
8.1.2	Bernoulli distribution	149
8.1.3	Binomial distribution	149
8.1.4	Geometric distribution	153
8.1.5	Negative binomial distribution	154
8.1.6	Poisson distribution	157
8.2	Continuous Distributions	159
8.2.1	Uniform distribution	159
8.2.2	Exponential distribution	161
8.2.3	Normal distribution	163
8.2.4	Lognormal distribution	168
8.2.5	Gamma distribution	169
8.2.6	Beta distribution	174
8.2.7	Chi-square distribution	178
8.2.8	Cauchy distribution	178
8.2.9	Pareto distribution	182
A	Complex Analysis	183

1 Measure Theory

Given a set X , measure theory studies *measures*, functions that assign a real number to subsets of X , representing their "size". The main references for this section are [1], [2], [3], [4], [5].

1.1 Measurable Spaces

Definition 1.1 (σ -algebra). If X is a nonempty set, a σ -algebra $\mathcal{F} \subseteq 2^X$ is a collection of subsets of X satisfying the following properties:

- i) *Non-empty:* $X \in \mathcal{F}$;
- ii) *Complement closure:* If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- iii) *Stability under countable unions:* If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{F}$.

The elements of $\mathcal{F}$ are called measurable sets. The pair $(X, \mathcal{F})$ is a measurable space.

Remark 1.2. We define σ -algebras because they are the algebras relevant for probability theory. However, relaxing the third requirement to only *finite unions* (i.e., if $A, B \in \mathcal{F}$, then $A \cup B \in \mathcal{F}$), we obtain an even weaker structure called an *algebra*, denoted $\mathcal{A}$.

Proposition 1.3. A σ -algebra contains $\emptyset$ and is closed under finite and countable unions and intersections.

Proof.

Since $X \in \mathcal{F}$, then $X^c = \emptyset \in \mathcal{F}$; in particular, finite unions of elements of $\mathcal{F}$ can be seen as countable unions of elements of $\mathcal{F}$, with all but finitely many terms equal to the empty set. Now let $(A_n)_{n \in \mathbb{N}}$ be a countable family of σ -measurable sets. By property ii) we have $A_n^c \in \mathcal{F}$ for every $n \in \mathbb{N}$; moreover, by property iii) $\bigcup_{n \in \mathbb{N}} A_n^c \in \mathcal{F}$. Finally, $\bigcap_{n \in \mathbb{N}} A_n = (\bigcup_{n \in \mathbb{N}} A_n^c)^c$, which belongs to $\mathcal{F}$ by property ii). $\square$

Example 1.4. The collections $\{\emptyset, X\}$ and 2^X are σ -algebras on X , called respectively the *trivial* and the *discrete* σ -algebra. If $X = \{1, 2, 3\}$, then

$$\mathcal{F} = \{\emptyset, \{1\}, \{2, 3\}, \{1, 2, 3\}\}$$

is a σ -algebra on X , as you can verify.

Definition 1.5 (Generated σ -algebra). For any collection $\mathcal{I} \subset 2^X$, the σ -algebra generated by $\mathcal{I}$ is the intersection of every σ -algebra containing $\mathcal{I}$, that is,

$$\mathcal{F}(\mathcal{I}) := \bigcap_{\substack{\mathcal{A} \text{ } \sigma\text{-algebra} \\ \mathcal{I} \subseteq \mathcal{A}}} \mathcal{A}.$$

Proposition 1.6. $\mathcal{F}(\mathcal{I})$ is the smallest σ -algebra containing $\mathcal{I}$. In particular, if $\mathcal{I}$ is a σ -algebra, then $\mathcal{F}(\mathcal{I}) = \mathcal{I}$.

Proof.

We prove that a countable intersection of σ -algebras is a σ -algebra. Let $(\mathcal{A}_n)_{n \in \mathbb{N}}$ be a family of σ -algebras. Since $X \in \mathcal{A}_n$ for every n , we have $X \in \bigcap_{n \in \mathbb{N}} \mathcal{A}_n$. Next, if $A \in \bigcap_{n \in \mathbb{N}} \mathcal{A}_n$, then $A^c \in \mathcal{A}_n$ for every n , hence $A^c \in \bigcap_{n \in \mathbb{N}} \mathcal{A}_n$. Finally, if $A_1, A_2, \dots \in \bigcap_{n \in \mathbb{N}} \mathcal{A}_n$, then $A_i \in \mathcal{A}_n$ for every $i \in \mathbb{N}$ and every $n \in \mathbb{N}$, thus $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{A}_n$ for every $n \in \mathbb{N}$, and therefore $\bigcup_{i \in \mathbb{N}} A_i \in \bigcap_{n \in \mathbb{N}} \mathcal{A}_n$.

Now, note that the intersection defining $\mathcal{F}(\mathcal{I})$ is nonempty, since there exists at least one σ -algebra containing $\mathcal{I}$ (namely 2^X). Consequently, $\mathcal{F}(\mathcal{I})$ is a σ -algebra. Finally, if $\mathcal{A}$ is a σ -algebra containing $\mathcal{I}$, then $\mathcal{A} \supseteq \mathcal{F}(\mathcal{I})$, so $\mathcal{F}(\mathcal{I})$ is contained in every such $\mathcal{A}$. $\square$

Example 1.7. Let $X = \{1, 2, 3, 4\}$ and $\mathcal{I} = \{\{1, 2\}\}$. Then $\mathcal{F}(\mathcal{I}) = \{\emptyset, \{1, 2\}, \{3, 4\}, X\}$, as you shall verify.

Example 1.8. Let $X = \mathbb{R}$, and consider

$$\mathcal{I} = \{A, B\} = \{(-\infty, 0], (-\infty, 1]\}.$$

Then $A^c = (0, +\infty)$, $B^c = (1, +\infty)$, $A^c \cap B = (0, 1]$ and $A \cup B^c = (-\infty, 0] \cup (1, +\infty)$ all belong to $\mathcal{F}(\mathcal{I})$, so that

$$\mathcal{F}(\mathcal{I}) = \{\emptyset, (-\infty, 0], (0, 1], (1, +\infty), (-\infty, 1], (0, +\infty), (-\infty, 0] \cup (1, +\infty), \mathbb{R}\}.$$

Remark 1.9 (Topology). We recall some topological notions that will be used later. A *topological space* (X, τ) consists of a set X together with a collection τ of subsets of X , called *topology*, that contains $\emptyset$ and X , and is closed under arbitrary unions and finite intersections. We then call *open sets* the sets in τ and *closed sets* the complements of the open sets. The arbitrary intersection of closed sets is closed, and the finite union of closed sets is closed. Of course, a set may be both open and closed, or neither open nor closed. Consider these two examples:

- Take the collections

$$\tau_1 = \{\emptyset, \{a, b, c\}, \{a\}, \{b\}\}, \quad \tau_2 = \{\emptyset, \{a, b, c\}, \{a\}, \{b, c\}\}, \quad \text{and} \quad \tau_3 = \{\emptyset, \{a, b, c\}, \{a\}, \{b, c\}, \{a, c\}\}.$$

Then τ_1 is not a topology, while τ_2 and τ_3 are topologies, as you can check.

- The canonical topology $\tau_{\mathbb{R}}$ of the real line $\mathbb{R}$ defines *open sets* those that are arbitrary unions of open balls $B_r(x) = \{y \in \mathbb{R} : |x - y| < r\}$.

Let (X, τ) be a topological space. A *neighbourhood* of $x \in X$ is an open set (arbitrarily small) containing x . A point $x \in X$ belongs to the *closure* of a set S if every neighbourhood of x intersects S . We call *support* of a function $f : X \rightarrow \mathbb{R}$ the set

$$\text{supp } f := \overline{\{x \in X : f(x) \neq 0\}},$$

that is, the set of points $x \in X$ that have a neighbourhood where f is *not* identically zero. For instance, consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}.$$

This function is C^∞ and positive on $(0, +\infty)$. Its support, however, is $[0, +\infty)$, as one can verify.

1.2 Measures

Definition 1.10 (Measure). A *measure* on a measurable space $(X, \mathcal{F})$ is a function $\mu : \mathcal{F} \rightarrow [0, +\infty]$ that is *σ -additive* (i.e., countably additive): if $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu(A_n).$$

Remark 1.11. The definition of a measure essentially clarifies the need of σ -algebras, as defined in the previous section. Since we want to take countable unions (to express σ -additivity) and complements (to express differences), a σ -algebra is the minimal natural domain on which a measure μ can be meaningfully defined.

Proposition 1.12 (Properties of measures). Let μ be a measure on $(X, \mathcal{F})$. Then:

- i) $\mu(\emptyset) = 0$;
- ii) *Monotonicity:* if $A \subseteq B$, then $\mu(A) \leq \mu(B)$. If moreover $\mu(A) < +\infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$.
- iii) *Monotone Continuity:*

i) (Continuity from below) For every increasing sequence $E_1 \subseteq E_2 \subseteq \dots$ (i.e., $E_n \uparrow E$),

$$\lim_{n \rightarrow \infty} \mu(E_n) = \mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \mu(E).$$

ii) (Continuity from above) For every decreasing sequence $E_1 \supseteq E_2 \supseteq \dots$ (i.e., $E_n \downarrow E$) with $\mu(E_1) < \infty$,

$$\lim_{n \rightarrow \infty} \mu(E_n) = \mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \mu(E).$$

- iv) *Subadditivity:* for every sequence $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$, $\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} \mu(A_n)$.

Proof.

The first property follows from σ -additivity applied on countably many empty sets:

$$\mu(\emptyset) = \mu\left(\bigcup_{n \in \mathbb{N}} \emptyset\right) = \sum_{n=1}^{\infty} \mu(\emptyset) \implies \mu(\emptyset) = 0.$$

If $A \subseteq B$, then

$$\mu(B) = \mu(A \cup (B \setminus A)) \stackrel{1}{=} \mu(A) + \mu(B \setminus A) \stackrel{2}{\geq} \mu(A),$$

where 1 uses σ -additivity (because A and $B \setminus A$ are disjoint) and 2 uses the nonnegativity of μ . Moreover, if $\mu(A) < \infty$, the above identity implies

$$\mu(B \setminus A) = \mu(B) - \mu(A).$$

Next, we prove monotone continuity properties.

- **Continuity from below.** We define the *disjoint increments*

$$F_1 := E_1, \quad F_n := E_n \setminus E_{n-1} \quad (n \geq 2).$$

Then $(F_n)_{n \in \mathbb{N}}$ are disjoint and, for every n , $E_n = \bigcup_{k=1}^n F_k$ and $E = \bigcup_{k=1}^{\infty} F_k$. Thus

$$\mu(E) = \mu\left(\bigcup_{k=1}^{\infty} F_k\right) = \sum_{k=1}^{\infty} \mu(F_k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mu(F_k) = \lim_{n \rightarrow \infty} \mu(E_n).$$

- **Continuity from above.** We define the *sequence of complements*

$$F_n := E_1 \setminus E_n, \quad (n \geq 1).$$

Then (F_n) is increasing and $\bigcup_{n=1}^{\infty} F_n = E_1 \setminus E$. By continuity from below,

$$\mu(E_1) - \mu(E) = \mu(E_1 \setminus E) = \mu\left(\bigcup_{n=1}^{\infty} F_n\right) = \lim_{n \rightarrow \infty} \mu(F_n) = \lim_{n \rightarrow \infty} \mu(E_1) - \mu(E_n).$$

Hence,

$$\lim_{n \rightarrow \infty} \mu(E_n) = \mu(E).$$

Finally, we show that μ is subadditive. We define the *partial unions*

$$B_n := \bigcup_{k=1}^n A_k.$$

Then $(B_n)_{n \in \mathbb{N}}$ is increasing and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$. Thus

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \lim_{n \rightarrow \infty} \mu(B_n) = \lim_{n \rightarrow \infty} \mu\left(\bigcup_{k=1}^n A_k\right) \leq \lim_{n \rightarrow \infty} \sum_{k=1}^n \mu(A_k) = \sum_{n=1}^{\infty} \mu(A_n).$$

□

Definition 1.13 (Measure space). A *measure space* $(X, \mathcal{F}, \mu)$ is a σ -measurable space $(X, \mathcal{F})$ endowed with a measure μ on X .

Broadly speaking, a measure provides a notion of "size" for measurable sets. A priori, there are many ways to define a measure. When X is finite or countable, the situation is particularly simple: one typically works with the full σ -algebra $\mathcal{F} = 2^X$, and many natural measures can be described explicitly:

Example 1.14 (Discrete measures on countable sets). If X is countable, we can define a measure $\mu : 2^X \rightarrow [0, +\infty]$ by assigning a nonnegative weight $w_x \geq 0$ to each point $x \in X$. For every $A \subseteq X$, set

$$\mu(A) := \sum_{x \in A} w_x.$$

It is straightforward to verify that μ is a measure. If $w_x \equiv 1$ for all $x \in X$, one obtains the *counting measure*,

i.e. $\mu(A) = |A|$ for every $A \subseteq X$.

Now let $X = \mathbb{N}$ and let μ be the counting measure. If we define $E_n = \{n, n+1, \dots\}$, then $(E_n)_{n \in \mathbb{N}}$ is a decreasing sequence with $\bigcap_{n \in \mathbb{N}} E_n = \emptyset$. Hence

$$\mu\left(\bigcap_{n \in \mathbb{N}} E_n\right) = \mu(\emptyset) = 0 \neq +\infty = \lim_{n \rightarrow \infty} \mu(E_n),$$

which seems to contradict continuity from above. This shows that the assumption $\mu(E_1) < +\infty$ is necessary.

Another important special case is obtained by concentrating all the mass at a single point. Before that, let us define, once and for all, a useful function called *indicator* of a set.

Definition 1.15 (Indicator Function). Let X be a set and $A \subseteq X$. We define the *indicator function* $\mathbf{1}_A : X \rightarrow [0, 1]$ as

$$\mathbf{1}_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}.$$

In other words, the indicator function records whether the point x belongs to the set A .

Remark 1.16. In some contexts, the notation χ_A is used, and the function is called the *characteristic function*. In probability theory, however, this terminology may be ambiguous, since the term *characteristic function* is commonly used to denote the Fourier transform of a probability measure.

Proposition 1.17 (Properties of the Indicator Function). Let $A, B \subseteq X$.

- i) $\mathbf{1}_{A \cap B} = \mathbf{1}_A \cdot \mathbf{1}_B$;
- ii) If $A \cap B = \emptyset$, then $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B$;
- iii) If $A = \bigcup_{i=1}^n B_i$ and $B_1, \dots, B_n$ are disjoint, then $\mathbf{1}_A = \sum_{i=1}^n \mathbf{1}_{B_i}$.

The first two properties are immediate from the definition. The third property is an inductive consequence of the second.

Example 1.18 (Dirac Measure). Fix $x \in X$ and define the *Dirac measure at x* by

$$\delta_x(A) := \mathbf{1}_A(x), \quad A \subseteq X.$$

Equivalently, $\delta_x(A) = 1$ if $x \in A$, and $\delta_x(A) = 0$ otherwise. In this notation, any discrete probability measure on X can be written as $\sum_{n \geq 1} p_n \delta_{x_n}$, where $p_n \geq 0$ and $\sum_{n \geq 1} p_n = 1$.

On uncountable sets, however, one cannot expect to define a sufficiently “natural” measure on *all* subsets while retaining reasonable structural properties.

Example 1.19 (Why not all sets are measurable). Let $X = \mathbb{R}$. One may try to formalize the notion of “length” by requiring:

- i) $\lambda([a, b]) = b - a$ for all $a \leq b$, and singletons do not contribute individually to the measure;
- ii) translation invariance: for every set $E \subseteq \mathbb{R}$ and every $t \in \mathbb{R}$, if $E + t := \{e + t : e \in E\}$, then $\lambda(E + t) = \lambda(E)$.

Let us define

$$x \oplus y = \begin{cases} x + y & x + y \leq 1 \\ x + y - 1 & x + y > 1 \end{cases}.$$

We say that $x \sim y$ if there exists a rational number $r \in \mathbb{Q} \cap (0, 1]$ such that $y = x \oplus r$. Let us consider the quotient space $(0, 1] / \sim$. Using the axiom of choice, we can take one representative element from each equivalence class. Let V be the set of such chosen elements. For $r \in \mathbb{R}$, define $V \oplus r = \{v \oplus r : v \in V\}$. Then if $r_1, r_2 \in \mathbb{Q} \cap (0, 1]$ and $r_1 \neq r_2$, the sets $(V \oplus r_1)$ and $(V \oplus r_2)$ are disjoint; on the other hand, $\bigcup_{r \in \mathbb{Q} \cap (0, 1]} (V \oplus r) = (0, 1]$. Hence,

$$1 = \lambda((0, 1]) = \lambda\left(\bigcup_{r \in \mathbb{Q} \cap (0, 1]} (V \oplus r)\right) = \sum_{r \in \mathbb{Q} \cap (0, 1]} \lambda(V \oplus r).$$

By translation invariance, $\lambda(V \oplus r) = \lambda(V)$ for any $r \in \mathbb{R}$. Therefore

$$1 = \lambda((0, 1]) = \sum_{r \in \mathbb{Q} \cap (0, 1]} \lambda(V \oplus r) = \sum_{r \in \mathbb{Q} \cap (0, 1]} \lambda(V).$$

Since the rational numbers in $(0, 1]$ are countable, if $\lambda(V) > 0$ then $\lambda((0, 1]) = \infty$. If $\lambda(V) = 0$, then $\lambda((0, 1]) = 0$. This contradicts the assumption that $\lambda((0, 1]) = 1$. Hence, we have reached a contradiction.

We conclude that there is *no* countably additive measure defined on *all* subsets of $\mathbb{R}$ satisfying the latter requirements. A related phenomenon occurs in $\mathbb{R}^3$: the Banach-Tarski paradox shows that a solid ball can be decomposed into finitely many non-measurable pieces and reassembled into two balls of the same size. The subtlety is that the pieces produced by the construction are non-measurable.

To “recover” a useful theory of length on $\mathbb{R}$, the standard approach is to *exclude pathological sets*: by restricting the domain of μ to a suitable σ -algebra $\mathcal{F} \subsetneq 2^{\mathbb{R}}$, one can construct measures with the desired invariance and regularity properties, at the price of declaring some subsets non-measurable.

More generally, if (X, τ) is a topological space (see Remark 1.9), there is a canonical σ -algebra associated with the topology, called the *Borel σ -algebra*. It is designed to be the smallest σ -algebra compatible with the topology, and it is therefore standard for defining measures on topological spaces.

Definition 1.20 (Borel σ -algebra). In a topological space (X, τ) , the Borel σ -algebra $\mathcal{B}(X) = \mathcal{F}(\tau)$ is the σ -algebra generated by τ , i.e., it is the smallest σ -algebra such that every open set (and hence every closed set) is measurable.

Example 1.21 (Borel σ -algebra on $\mathbb{R}$). The open sets of $\mathbb{R}$ are arbitrary unions of open intervals (a, b) . Hence $\mathcal{B}(\mathbb{R})$ is the σ -algebra generated by the open intervals (a, b) , $a < b$. Together, $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra containing arbitrary unions of open intervals. Equivalently, $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra that makes every open set (and therefore every closed set) measurable.

Proposition 1.22. Let $\mathcal{I} = \{(-\infty, a) : a \in \mathbb{R}\}$. Then $\mathcal{F}(\mathcal{I}) = \mathcal{B}(\mathbb{R})$.

Proof.

Since $\mathcal{I} \subseteq \tau_{\mathbb{R}}$, we have $\mathcal{F}(\mathcal{I}) \subseteq \mathcal{B}(\mathbb{R})$. We now prove $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{F}(\mathcal{I})$. It suffices to prove that, for every $a, b \in \mathbb{R}$ with $a < b$, $(a, b) \in \mathcal{F}(\mathcal{I})$. Indeed, a well-known property of $\tau_{\mathbb{R}}$ is that every open set is a countable union of open intervals (a, b) . Therefore open sets would be in $\mathcal{F}(\mathcal{I})$, and hence $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{F}(\mathcal{I})$. In order to prove that every open interval (a, b) is in $\mathcal{F}(\mathcal{I})$, we observe that

$$(a, b) = (-\infty, b) \cap (a, +\infty) = (-\infty, b) \cap \left[\bigcup_{n=1}^{\infty} [a + 1/n, \infty) \right] = \underbrace{(-\infty, b)}_{\in \mathcal{I} \subseteq \mathcal{F}(\mathcal{I})} \cap \left[\bigcup_{n=1}^{\infty} \underbrace{(-\infty, a + 1/n)^c}_{\in \mathcal{F}(\mathcal{I})} \right].$$

□

Finally, we introduce the Lebesgue measure on $\mathbb{R}$ (or, more generally, on $\mathbb{R}^n$), commonly used to formalize Riemann integration and to provide a notion of length and volume. Its construction is nontrivial and requires tools that we do not develop in these notes (see, for instance, [3]). Instead, it is convenient to characterize it by listing its basic properties. We denote Lebesgue measure on $\mathbb{R}^n$ by λ^n . It is a measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, normalized by

$$\lambda^n([0, 1]^n) = 1.$$

Moreover, $\lambda^n(\{a\}) = 0$ for every $a \in \mathbb{R}^n$, and λ^n is translation invariant and homogeneous:

$$\lambda^n(E + s) = \lambda^n(E), \quad \lambda^n(\alpha E) = \alpha^n \lambda^n(E),$$

for every $E \in \mathcal{B}(\mathbb{R}^n)$, $s \in \mathbb{R}^n$, and $\alpha \geq 0$, where $E + s := \{x + s : x \in E\}$ and $\alpha E := \{\alpha x : x \in E\}$. In particular, λ^n extends the familiar geometric notions of length and volume. These facts will be formalized more systematically after we introduce the Lebesgue integral.

Remark 1.23 (Completion of Lebesgue measure). In many applications, including probability, it is convenient to work not only with the measure space $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \lambda^n)$, but also with the *completion* $(\mathbb{R}^n, \overline{\mathcal{B}}(\mathbb{R}^n), \overline{\lambda}^n)$. Concretely, $\overline{\mathcal{B}}(\mathbb{R}^n)$ is obtained by adjoining all subsets of λ^n -null Borel sets, i.e. sets $B \subseteq \mathbb{R}^n$ such that $\lambda^n(B) = 0$. This convention is often taken as a default assumption in analysis and probability, and we will adopt it tacitly when needed.

1.3 Measurable Functions

As we will see later, measurable sets are those to which a measure can be assigned, while measurable functions are those for which an integral can be defined.

Definition 1.24 (Measurable function). Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. A function $f : X \rightarrow Y$ is *measurable* if

$$f^{-1}(B) \in \mathcal{A} \quad \text{for every } B \in \mathcal{B}.$$

Remark 1.25. Observe that the definition of measurability is structurally similar to that of continuity. Indeed, if (X, τ_X) and (Y, τ_Y) are topological spaces, a function $f : X \rightarrow Y$ is *continuous* if $f^{-1}(S) \in \tau_X$ for every $S \in \tau_Y$.

Proposition 1.26 (Link between measurable functions and measurable sets). *In a measurable space $(X, \mathcal{A})$, a set $A \subseteq X$ is measurable if and only if the indicator function $\mathbf{1}_A : (X, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable.*

Proof.

Fix $A \subseteq X$ and $B \in \mathcal{B}(\mathbb{R})$. Since $\mathbf{1}_A$ only takes values in $\{0, 1\}$, its preimage depends only on whether 0 and/or 1 belong to B :

$$\mathbf{1}_A^{-1}(B) = \begin{cases} \emptyset & \text{if } 0 \notin B \text{ and } 1 \notin B \\ A & \text{if } 0 \notin B \text{ and } 1 \in B \\ A^c & \text{if } 0 \in B \text{ and } 1 \notin B \\ X & \text{if } 0 \in B \text{ and } 1 \in B \end{cases}.$$

($\Rightarrow$) Assume $A \in \mathcal{A}$. Then $A^c \in \mathcal{A}$, and also $\emptyset \in \mathcal{A}$ and $X \in \mathcal{A}$. Since $\mathbf{1}_A^{-1}(B) \in \{\emptyset, A, A^c, X\} \subseteq \mathcal{A}$, we have $\mathbf{1}_A^{-1}(B) \in \mathcal{A}$.

($\Leftarrow$) Assume $\mathbf{1}_A^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{B}(\mathbb{R})$. Since $\{1\} \in \mathcal{B}(\mathbb{R})$, we get

$$A = \mathbf{1}_A^{-1}(\{1\}) \in \mathcal{A}.$$

This proves the equivalence. $\square$

Next, we study basic stability properties of measurable functions. In particular, finding a non-measurable function is essentially as hard as finding a non-measurable set.

Lemma 1.27 (Composition of measurable functions). *If $(X, \mathcal{A})$, $(Y, \mathcal{B})$ and $(Z, \mathcal{C})$ are measurable spaces and $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are measurable, then $g \circ f : X \rightarrow Z$ is measurable.*

Proof.

It relies on the identity $(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C))$ from set theory and the definition of measurability. $\square$

In practice, to check that a function is measurable, one does not need to look at the inverse image of all the sets in the σ -algebra of the target space: some "good candidates" are enough, as the next proposition shows:

Proposition 1.28. *Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces, and let $\mathcal{C}$ be a family of subsets of Y such that $\mathcal{B} \subseteq \sigma(\mathcal{C})$. If $f : X \rightarrow Y$ is such that $f^{-1}(C) \in \mathcal{A}$ for all $C \in \mathcal{C}$, then $f : (X, \mathcal{A}) \rightarrow (Y, \mathcal{B})$ is measurable.*

Proof.

Let $\overline{\mathcal{B}} = \{B \subseteq Y \mid f^{-1}(B) \in \mathcal{A}\}$. The theorem then reads: if $\mathcal{C} \subseteq \overline{\mathcal{B}}$ and $\mathcal{B} \subseteq \sigma(\mathcal{C})$, then $\mathcal{B} \subseteq \overline{\mathcal{B}}$. We claim that $\overline{\mathcal{B}}$ is a σ -algebra:

- Clearly $f^{-1}(Y) = X \in \mathcal{A}$, hence $Y \in \overline{\mathcal{B}}$;
- If $B \in \overline{\mathcal{B}}$, then $f^{-1}(B) \in \mathcal{A}$. Consequently, also $(f^{-1}(B))^c = f^{-1}(B^c) \in \mathcal{A}$, hence $B^c \in \overline{\mathcal{B}}$;
- If $(B_n)_{n \in \mathbb{N}} \subseteq \overline{\mathcal{B}}$, then $(f^{-1}(B_n))_{n \in \mathbb{N}} \subseteq \mathcal{A}$. Consequently,

$$\bigcup_{n \in \mathbb{N}} f^{-1}(B_n) = f^{-1}\left(\bigcup_{n \in \mathbb{N}} B_n\right) \in \mathcal{A},$$

hence $\bigcup_{n \in \mathbb{N}} B_n \in \overline{\mathcal{B}}$.

Since $\overline{\mathcal{B}}$ is a σ -algebra and contains $\mathcal{C}$, we have $\sigma(\mathcal{C}) \subseteq \overline{\mathcal{B}}$, and we conclude by the assumption $\mathcal{B} \subseteq \sigma(\mathcal{C})$ that $\mathcal{B} \subseteq \overline{\mathcal{B}}$. $\square$

Example 1.29. If $f : (X, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then to check that f is measurable it is enough (for instance) to verify that $f^{-1}([a, +\infty)) = \{x \in X \mid f(x) > a\} \in \mathcal{A}$ for all $a \in \mathbb{R}$. It is clear that also $f^{-1}([a, +\infty))$ or $f^{-1}((-\infty, a))$ or $f^{-1}((-\infty, a])$ are valid candidates. Indeed, we recall the elementary identities, valid for every $a \in \mathbb{R}$,

$$[a, +\infty) = \bigcap_{n \geq 1} (a - 1/n, +\infty), \quad (-\infty, a) = \bigcup_{n \geq 1} (-\infty, a - 1/n], \quad (-\infty, a] = \bigcap_{n \geq 1} (-\infty, a + 1/n).$$

Applying f^{-1} and using that preimages commute with unions and intersections, we obtain

$$\begin{aligned} f^{-1}([a, +\infty)) &= \bigcap_{n \geq 1} f^{-1}((a - 1/n, +\infty)) \\ f^{-1}((-\infty, a)) &= \bigcup_{n \geq 1} f^{-1}((-\infty, a - 1/n]) \\ f^{-1}((-\infty, a]) &= \bigcap_{n \geq 1} f^{-1}((-\infty, a + 1/n)). \end{aligned}$$

Therefore, if one knows that the preimages of the sets $(a, +\infty)$ are in $\mathcal{A}$, then the right-hand sides are countable unions or intersections of sets in $\mathcal{A}$, hence belong to $\mathcal{A}$ as well.

Proposition 1.30 (Continuous functions are measurable). *A continuous function $f : (X, \tau_X) \rightarrow (Y, \tau_Y)$ between topological spaces (X, τ_X) and (Y, τ_Y) is measurable as a function $f : (X, \mathcal{B}(X)) \rightarrow (Y, \mathcal{B}(Y))$.*

Proof.

It suffices to apply Proposition 1.28 by choosing $\mathcal{C} := \tau_Y$. Indeed, $\mathcal{B}(Y) \subseteq \sigma(\tau_Y)$, and $f^{-1}(C) \in \tau_X \subseteq \mathcal{B}(X)$ for every $C \in \tau_Y$. $\square$

Proposition 1.31. *Let $f, g : (X, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable. Then the functions*

$$f + g, \quad fg, \quad \max(f, g), \quad \min(f, g), \quad f_+ = \max(f, 0), \quad f_- = \max(-f, 0), \quad |f|$$

are measurable.

Proof.

Let us start by proving that the sum of measurable functions is measurable. We'd like to write the sum as a composition of measurable functions, so as to apply Lemma 1.27. We thus write $f + g = \varphi \circ \psi$, where

$$\begin{aligned} \psi : (X, \mathcal{A}) &\rightarrow (\mathbb{R}^2, \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})) & \varphi : (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2)) &\rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})) \\ x &\mapsto (f(x), g(x)) & (a, b) &\mapsto a + b, \end{aligned}$$

and $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \sigma(\{A \times B : A, B \in \mathcal{B}(\mathbb{R})\})$. If we can prove that $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^2)$, as well as φ and ψ measurable, then Lemma 1.27 implies that $f + g$ is measurable.

Step 1: We prove that $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^2)$. Pick $A, B \in \mathcal{B}(\mathbb{R})$. Then

$$A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B),$$

where $\pi_1, \pi_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ are the coordinate projection maps, i.e., $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$.

Since A and B are Borel sets in $\mathcal{B}(\mathbb{R})$ and π_1, π_2 are continuous, we have $\pi_1^{-1}(A) \in \mathcal{B}(\mathbb{R}^2)$ and $\pi_2^{-1}(B) \in \mathcal{B}(\mathbb{R}^2)$. In particular, $A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B) \in \mathcal{B}(\mathbb{R}^2)$, for any $A, B \in \mathcal{B}(\mathbb{R})$. This implies $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \sigma(\{A \times B : A, B \in \mathcal{B}(\mathbb{R})\}) \subseteq \mathcal{B}(\mathbb{R}^2)$.

We now prove the converse inclusion. We recall the standard result from topology that every open set $O \in \tau_{\mathbb{R}^2}$ can be written as a countable union of rectangles $(a, b) \times (c, d)$ with rational endpoints, i.e. $a, b, c, d \in \mathbb{Q}$, $a < b$, $c < d$. For each such rectangle $R = (a, b) \times (c, d)$, we have $(a, b) \in \mathcal{B}(\mathbb{R})$ and $(c, d) \in \mathcal{B}(\mathbb{R})$, hence $R \in \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. Therefore, every open set is a countable union of elements of $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$, and thus $\tau_{\mathbb{R}^2} \subseteq \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. Taking σ -closures gives

$$\mathcal{B}(\mathbb{R}^2) = \sigma(\tau_{\mathbb{R}^2}) \subseteq \sigma(\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}).$$

Step 2: We show that φ is measurable. It suffices to prove that $\varphi^{-1}((a, +\infty)) = \{(x, y) \in \mathbb{R}^2 : x + y >$

$a\} \in \mathcal{B}(\mathbb{R}^2)$ for every $a \in \mathbb{R}$, by Example 1.29. But $\{(x, y) \in \mathbb{R}^2 : x + y > a\}$ is an open set of $\mathbb{R}^2$ for every $a \in \mathbb{R}$, hence it belongs to $\tau_{\mathbb{R}^2}$ and, in particular, to $\sigma(\tau_{\mathbb{R}^2}) = \mathcal{B}(\mathbb{R}^2)$.

Step 3: We show that ψ is measurable. Applying Proposition 1.28, we do not need to verify the property for every element of $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$: it suffices to select a family $\mathcal{C}$ of subsets of $\mathbb{R}^2$ such that $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) \subseteq \sigma(\mathcal{C})$. By definition, $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \sigma(\{A \times B : A, B \in \mathcal{B}(\mathbb{R})\})$, so we can verify measurability on the family $\mathcal{C} = \{A \times B : A, B \in \mathcal{B}(\mathbb{R})\}$. If $A \times B \in \mathcal{C}$, then

$$\psi^{-1}(A \times B) = \{x \in X : (f(x), g(x)) \in A \times B\} = \{x \in X : f(x) \in A, g(x) \in B\} = f^{-1}(A) \cap g^{-1}(B).$$

Since f and g are measurable, $f^{-1}(A) \in \mathcal{A}$ and $g^{-1}(B) \in \mathcal{A}$, hence $f^{-1}(A) \cap g^{-1}(B) \in \mathcal{A}$.

Combining the previous three steps and Lemma 1.27, we conclude that measurability is stable under sums.

Step 4: We observe that $fg = \varphi_1 \circ \psi$, $\max(f, g) = \varphi_2 \circ \psi$ and $\min(f, g) = \varphi_3 \circ \psi$ with ψ defined as before, $\varphi_1(a, b) = ab$, $\varphi_2(a, b) = \max(a, b)$ and $\varphi_3(a, b) = \min(a, b)$. Therefore, the only thing that changes in the previous reasoning is that φ_1 , φ_2 and φ_3 should be measurable. The proof is identical to Step 2: observe that

$$\{(x, y) \in \mathbb{R}^2 : xy > a\}, \quad \{(x, y) \in \mathbb{R}^2 : \min(x, y) > a\}, \quad \text{and} \quad \{(x, y) \in \mathbb{R}^2 : \max(x, y) > a\}$$

are open sets of $\mathbb{R}^2$ for every $a \in \mathbb{R}$. Therefore, they belong to $\tau_{\mathbb{R}^2}$ and, in particular, to $\sigma(\tau_{\mathbb{R}^2}) = \mathcal{B}(\mathbb{R}^2)$.

Step 5: The functions f_+ and f_- are measurable because $\max(f, g)$ is measurable for every measurable f, g , and the constant $g \equiv 0$ is clearly measurable. The function $|f|$ is measurable because $|f| = \max(f, -f)$, and both f and $-f$ are measurable. $\square$

Remark 1.32. Step 2 and Step 4 are perhaps redundant if one concludes directly from Proposition 1.30.

Remark 1.33. With a similar procedure, one shows that if f, g are measurable and $g(x) \neq 0$ for all $x \in X$, then f/g is measurable.

To conclude this section, we prove an important stability property: the class of measurable functions is closed under pointwise limits. From now on, we work on the extended real line $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$. The space $\overline{\mathbb{R}}$ carries a natural topology obtained by adjoining to the usual topology on $\mathbb{R}$ the sets $[-\infty, a)$ and $(b, +\infty]$, for $a, b \in \mathbb{R}$. Consequently, the Borel σ -algebra $\mathcal{B}(\overline{\mathbb{R}})$ is generated either by the family $\{[-\infty, a) : a \in \mathbb{R}\}$ or, equivalently, by $\{(a, +\infty] : a \in \mathbb{R}\}$.

Lemma 1.34. *Let $(X, \mathcal{F})$ be a measurable space and $(f_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions $f_n : (X, \mathcal{F}) \rightarrow (\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$. Then the functions*

$$\sup_{n \in \mathbb{N}} f_n, \quad \inf_{n \in \mathbb{N}} f_n, \quad \limsup_{n \rightarrow \infty} f_n, \quad \liminf_{n \rightarrow \infty} f_n$$

are measurable.

Proof.

We use the criterion of Example 1.29: it suffices to show that the preimages of intervals of the form $(a, +\infty]$ are measurable. Let $f := \sup_{n \in \mathbb{N}} f_n$. For $a \in \mathbb{R}$,

$$\{x \in X : f(x) > a\} = \{x \in X : \sup_{n \in \mathbb{N}} f_n(x) > a\} = \bigcup_{n \in \mathbb{N}} \{x \in X : f_n(x) > a\}.$$

Each set $\{x \in X : f_n(x) > a\} = f_n^{-1}((a, +\infty])$ belongs to $\mathcal{F}$ because f_n is measurable; hence the union belongs to $\mathcal{F}$, and f is measurable. The proof for $\inf_{n \in \mathbb{N}} f_n$ is analogous, using the sets $\{x \in X : f_n(x) < a\} = f_n^{-1}([-\infty, a))$.

As for $\limsup$ and $\liminf$, we recall the identities

$$\limsup_{n \rightarrow \infty} f_n = \inf_{n \in \mathbb{N}} \sup_{k \geq n} f_k, \quad \liminf_{n \rightarrow \infty} f_n = \sup_{n \in \mathbb{N}} \inf_{k \geq n} f_k.$$

For each n , the family $(f_k)_{k \geq n}$ is countable and measurable, hence $\sup_{k \geq n} f_k$ and $\inf_{k \geq n} f_k$ are measurable by the previous point. Applying the same reasoning to the sequences $n \mapsto \sup_{k \geq n} f_k$ and $n \mapsto \inf_{k \geq n} f_k$ we conclude the proof. $\square$

Theorem 1.35 (Pointwise Limit of Measurable Functions). Let $(X, \mathcal{F})$ be a measurable space and $(f_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions $f_n : (X, \mathcal{F}) \rightarrow (\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$. If $f_n(x) \rightarrow f(x)$ for every $x \in X$, where $f : (X, \mathcal{F}) \rightarrow (\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$, then f is measurable.

Proof.

If $f_n \rightarrow f$ pointwise, then for every $x \in X$ one has

$$f(x) = \liminf_{n \rightarrow \infty} f_n(x) = \limsup_{n \rightarrow \infty} f_n(x),$$

and both are measurable by the previous lemma. We conclude that f is measurable. $\square$

1.4 Integration

The main goal of measure theory is to develop a theory for integration that is more robust and general than *Riemann integration*. Although Riemann integration has the apparent advantage of a more naive definition, it has a major drawback in terms of generalizability. In particular, it is *very* difficult to decide whether an arbitrary function is Riemann integrable. In a sense, the complications of the classical notion of integrability can be summarized in one central issue: the class of Riemann-integrable functions is not stable (i.e., not closed) under pointwise limits.

Example 1.36 (Non-Closure of the Space of Riemann-integrable Functions). For each $n \geq 1$, let

$$f_n(x) = \begin{cases} 1 & \text{if } x = p/q, \text{ where } p, q \in \mathbb{N}, 1 \leq p \leq q \leq n, \gcd(p, q) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

For every n , f_n differs from 0 at at most n^2 points. Therefore, it has only finitely many discontinuities and is Riemann integrable, with $\int_0^1 f_n(x) dx = 0$. However, for every $x \in \mathbb{R}$,

$$f(x) := \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \cap [0, 1], \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$f|_{[0,1]} = \mathbf{1}_{\mathbb{Q} \cap [0,1]},$$

which is discontinuous everywhere and *not* Riemann integrable. By contrast, f is Lebesgue integrable, with $\int_{\mathbb{R}} f(x) d\lambda(x) = 0$, as expected.

Naive Approach. The basic idea is that, while in Riemann integration we subdivide the domain of a function, in Lebesgue integration we subdivide its range. Following [3], we can sketch the shift in perspective between Riemann and Lebesgue integration as follows:

$$\text{Riemann integration:} \quad \int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{i}{n}\right) \cdot \frac{1}{n},$$

$$\text{Lebesgue integration:} \quad \int_{[0,1]} f d\mu = \lim_{n \rightarrow \infty} \sum_{k=-\infty}^{+\infty} \frac{k}{n} \cdot \mu\left(\left\{x \in X : \frac{k}{n} \leq f(x) < \frac{k+1}{n}\right\}\right).$$

In the Riemann approach, one finds the area under the graph of f by splitting the domain $[0, 1]$ into n subintervals of length $1/n$ and summing rectangles with base $1/n$ and height given by the value of f at each subinterval. In the Lebesgue approach, one splits the range $(-\infty, +\infty)$ into the intervals $[\frac{k}{n}, \frac{k+1}{n})$, $k \in \mathbb{Z}$, and sums rectangles with height k/n and base given by the size of the level set $\{x \in X : \frac{k}{n} \leq f(x) < \frac{k+1}{n}\}$. We formalize this approach in four steps:

- We define the Lebesgue integral for a well-behaved class of functions called *simple functions*.
- We define the Lebesgue integral for nonnegative measurable functions by approximation from below with simple functions.
- We define the Lebesgue integral for *integrable* (signed) functions, i.e. the largest class of functions for which the integral is finite.
- We explain why measurability and integrability are understood up to sets of measure zero (i.e. almost everywhere).

1.4.1 Lebesgue integration

Definition 1.37 (Simple function). Let $(X, \mathcal{F}, \mu)$ be a measure space. A function $\varphi : X \rightarrow \mathbb{R}$ is *simple* if it takes only finitely many values $\alpha_1 < \dots < \alpha_n$. Every simple function admits a unique canonical representation as

$$\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i},$$

provided we define $A_i := \varphi^{-1}(\{\alpha_i\})$, which are pairwise disjoint and nonempty.

Remark 1.38. Without assumptions on A_i , any simple function admits infinitely many representations. For instance, $2\mathbf{1}_{\mathbb{Q}}(x) + \mathbf{1}_{\mathbb{R} \setminus \mathbb{Q}}(x) = \mathbf{1}_{\mathbb{R}}(x) + \mathbf{1}_{\mathbb{Q}}(x)$. The first representation is canonical, while the second is not.

Proposition 1.39. A simple function $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$ is measurable if and only if A_i is measurable for every $i = 1, \dots, n$.

Proof.

By Proposition 1.26, A_i is measurable if and only if $\mathbf{1}_{A_i}$ is measurable. If A_i is measurable for every $i = 1, \dots, n$, φ is a finite sum of measurable functions, hence measurable. Conversely, if φ is measurable, $A_i = \varphi^{-1}(\{\alpha_i\})$ is measurable. $\square$

Definition 1.40 (Null set). Given a measure space $(X, \mathcal{F}, \mu)$, a set $\mathcal{N} \in \mathcal{F}$ is called *null* if $\mu(\mathcal{N}) = 0$.¹

Definition 1.41 (Almost everywhere). Let $(X, \mathcal{F}, \mu)$ be a measure space and let $P(x)$ be a property depending on $x \in X$. We say that P holds μ -almost everywhere (abbreviated μ -a.e.) if there exists a null set $N \in \mathcal{F}$ such that $P(x)$ holds for each $x \in N^c$.

Example 1.42 (Almost everywhere equality). If f and g are measurable functions on $(X, \mathcal{F}, \mu)$, we say that $f = g$ μ -a.e., if $\mu(\{x \in X : f(x) \neq g(x)\}) = 0$.

Definition 1.43 (Lebesgue integral). Let $(X, \mathcal{F}, \mu)$ be a measure space, $E \in \mathcal{F}$ and φ, f, Φ and ζ measurable functions.

- If $\varphi : X \rightarrow [0, +\infty]$ is simple and nonnegative with a canonical representation $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$, then the *Lebesgue integral* of φ on E is given by

$$\int_E \varphi d\mu := \sum_{i=1}^n \alpha_i \mu(E \cap A_i).$$

- If $f : X \rightarrow [0, +\infty]$ is nonnegative, then the *Lebesgue integral* of f on E is given by

$$\int_E f d\mu := \sup_{\varphi \in \mathcal{S}_f} \int_E \varphi d\mu, \quad \text{where } \mathcal{S}_f = \{\varphi : \varphi \text{ is simple measurable and } 0 \leq \varphi \leq f\}, \quad (1.1)$$

- If $\Phi : X \rightarrow \overline{\mathbb{R}}$ satisfies $\int_E |\Phi| d\mu < +\infty$, then the *Lebesgue integral* of Φ on E is given by

$$\int_E \Phi d\mu := \int_E \Phi^+ d\mu - \int_E \Phi^- d\mu,$$

where $\Phi^+ = \max(\Phi, 0)$ and $\Phi^- = \max(-\Phi, 0)$. Measurable functions satisfying $\int_E |\Phi| d\mu < +\infty$ are called *integrable* (actually, μ -integrable) on E . The class of μ -integrable functions is labeled as $\mathcal{L}^1(X, \mu)$, $\mathcal{L}^1(X)$, $\mathcal{L}^1(\mu)$ or simply $\mathcal{L}^1$.

- If $\zeta : \hat{E} \rightarrow \overline{\mathbb{R}}$ is defined on a measurable $\hat{E} \subsetneq E$, and $\mu(E \setminus \hat{E}) = 0$ (that is, up to a null set), we say that ζ is *measurable on E* if $\zeta^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\overline{\mathbb{R}})$. If ζ is measurable on E and integrable on $\hat{E}$, then the *Lebesgue integral* of ζ on E is given by

$$\int_E \zeta d\mu := \int_{\hat{E}} \zeta d\mu.$$

¹Some authors define a *null set* as any set $N \subseteq X$ contained in a measurable set of μ -measure zero. This extends the terminology to possibly non-measurable sets. This subtlety is analogous to the common ambiguity in the notion of a neighbourhood in $\mathbb{R}$: one may define it either as an open interval $(a, b) \ni x$, or as an open set containing some interval $(a, b) \ni x$. In practice, the two conventions are equivalent.

Remark 1.44. The only functions that can't be Lebesgue integrated over E are non-measurable functions and non-integrable functions that do not satisfy $\int_E |\Phi| d\mu < \infty$. The latter requirement rules out indeterminate expressions of the form $+\infty - \infty$. For instance, the function $f(x) = x$ is not Lebesgue integrable on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, since both $\int_{\mathbb{R}} f^+ d\lambda$ and $\int_{\mathbb{R}} f^- d\lambda$ are infinite, which would lead to the undefined difference $+\infty - \infty$. Note that the Lebesgue integral does allow diverging integrals, as long as they do not involve indeterminate forms.

Defining properties of the Lebesgue integral. We proceed by highlighting the architectural properties of the theory of integration. We start by stating the three fundamental theorems in their full generality to provide a clear roadmap of the results we aim to establish. All other results are either corollaries of these theorems or intermediate results used to derive them. Immediately following this overview, we build the theory constructively, starting from simple functions and extending to the general case.

The first property we state is monotonicity, which governs the *order* structure of the integral.

Theorem 1.45 (Monotonicity of the Lebesgue integral). *Let $(X, \mathcal{F}, \mu)$ be a measure space, let $E \in \mathcal{F}$, and let $\Phi, \Psi : X \rightarrow \mathbb{R}$ be integrable on E . If $\Phi \leq \Psi$ μ -a.e. on E , then*

$$\int_E \Phi d\mu \leq \int_E \Psi d\mu.$$

The second property is linearity, which gives the *algebraic* structure of the integral.

Theorem 1.46 (Linearity of the Lebesgue integral). *Let $(X, \mathcal{F}, \mu)$ be a measure space, let $E \in \mathcal{F}$, and let $\Phi, \Psi : X \rightarrow \mathbb{R}$ be integrable on E . If α, β are real-valued constants, then $\alpha\Phi + \beta\Psi$ is integrable on E and*

$$\int_E (\alpha\Phi + \beta\Psi) d\mu = \alpha \int_E \Phi d\mu + \beta \int_E \Psi d\mu.$$

The third property is considered the deepest result of the theory of integration. It is known as the Monotone Convergence Theorem (MCT), or *Beppo Levi's Theorem*. It gives the topological property of the Lebesgue integral, i.e., how to deal with limits of integrals.

Theorem 1.47 (Monotone Convergence). *Let $(X, \mathcal{F}, \mu)$ be a measure space, $(f_n)_{n \in \mathbb{N}}$ be an increasing sequence of nonnegative measurable functions from $(X, \mathcal{F}, \mu)$ to $[0, +\infty]$, and assume that it converges pointwise to a function f , i.e.,*

$$\lim_{n \rightarrow \infty} f_n(x) = f(x), \quad \forall x \in X.$$

Then f is measurable and

$$\int_X f d\mu = \int_X \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int_X f_n d\mu.$$

1.4.2 Construction of the Lebesgue Integral

Next, we start by proving a list of intermediate results for simple functions:

Proposition 1.48. *Let $(X, \mathcal{F}, \mu)$ be a measure space, $E \in \mathcal{F}$ a measurable set, φ, ψ simple nonnegative measurable functions from $(X, \mathcal{F}, \mu)$ to $[0, +\infty]$ and $\alpha, \beta \in [0, +\infty)$. Then:*

- i) If φ has two different representations $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i} = \sum_{j=1}^k \beta_j \mathbf{1}_{B_j}$, then $\sum_{i=1}^n \alpha_i \mu(E \cap A_i) = \sum_{j=1}^k \beta_j \mu(E \cap B_j)$, that is, the integral of a simple function does not depend on the representation chosen.*
- ii) $\int_E \varphi d\mu = \int_X \varphi \mathbf{1}_E d\mu$;*
- iii) If $\varphi \leq \psi$, then $\int_E \varphi d\mu \leq \int_E \psi d\mu$;*
- iv) $\int_E (\alpha\varphi + \beta\psi) d\mu = \alpha \int_E \varphi d\mu + \beta \int_E \psi d\mu$.*

Proof.

(i). Take two different representations of φ , namely $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i} = \sum_{j=1}^k \beta_j \mathbf{1}_{B_j}$. For simplicity, here we assume $\{A_1, \dots, A_n\}$ and $\{B_1, \dots, B_k\}$ are measurable partitions of X . If not, the proof is the

same but notationally involved. In this case,

$$\begin{aligned}
 \sum_{i=1}^n \alpha_i \mu(E \cap A_i) &= \sum_{i=1}^n \alpha_i \mu \left(\bigcup_{j=1}^k E \cap A_i \cap B_j \right) \stackrel{1}{=} \sum_{i=1}^n \alpha_i \left(\sum_{j=1}^k \mu(E \cap A_i \cap B_j) \right) \\
 &= \sum_{i=1}^n \sum_{j=1}^k \alpha_i \mu(E \cap A_i \cap B_j) \stackrel{2}{=} \sum_{i=1}^n \sum_{j=1}^k \beta_j \mu(E \cap A_i \cap B_j) \\
 &= \sum_{j=1}^k \beta_j \left(\sum_{i=1}^n \mu(E \cap A_i \cap B_j) \right) \stackrel{1}{=} \sum_{j=1}^k \beta_j \mu \left(\bigcup_{i=1}^n E \cap A_i \cap B_j \right) = \sum_{j=1}^k \beta_j \mu(E \cap B_j),
 \end{aligned}$$

where 1 uses the fact that B_j 's (resp. A_i 's) are disjoint and the additivity of μ , and 2 uses the fact that, for every $x \in E$, $\sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}(x) = \varphi(x) = \sum_{j=1}^k \beta_j \mathbf{1}_{B_j}(x)$, and, consequently, $\alpha_i = \beta_j$ on $A_i \cap B_j$ (whenever non-empty).

(ii). Writing $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$, we deduce

$$\int_X \varphi \mathbf{1}_E d\mu = \int_X \left(\sum_{i=1}^n \alpha_i \mathbf{1}_{E \cap A_i} \right) d\mu = \sum_{i=1}^n \alpha_i \mu(E \cap A_i) = \int_E \varphi d\mu.$$

(iii). Take canonical representations $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$ and $\psi = \sum_{j=1}^k \beta_j \mathbf{1}_{B_j}$. Partition X into sets $(A_i \cap B_j)_{i,j}$. For any $x \in A_i \cap B_j$, we have $\varphi(x) = \alpha_i$ and $\psi(x) = \beta_j$, thus $\alpha_i \leq \beta_j$ whenever $A_i \cap B_j \neq \emptyset$. Hence:

$$\int_E \varphi d\mu = \sum_{i,j} \alpha_i \mu(E \cap A_i \cap B_j) \leq \sum_{i,j} \beta_j \mu(E \cap A_i \cap B_j) = \int_E \psi d\mu.$$

(iv). Take canonical representations $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$ and $\psi = \sum_{j=1}^k \beta_j \mathbf{1}_{B_j}$. The sets $\{A_i \cap B_j\}_{i,j}$ partition X , and on each set $A_i \cap B_j$ the function $\alpha\varphi + \beta\psi$ takes the value $\alpha\alpha_i + \beta\beta_j$. Therefore,

$$\begin{aligned}
 \int_E (\alpha\varphi + \beta\psi) d\mu &= \int_E \left(\sum_{i=1}^n \sum_{j=1}^k (\alpha\alpha_i + \beta\beta_j) \mathbf{1}_{E \cap A_i \cap B_j} \right) d\mu = \sum_{i=1}^n \sum_{j=1}^k (\alpha\alpha_i + \beta\beta_j) \mu(E \cap A_i \cap B_j) \\
 &= \sum_{i=1}^n \alpha\alpha_i \sum_{j=1}^k \mu(E \cap A_i \cap B_j) + \sum_{j=1}^k \beta\beta_j \sum_{i=1}^n \mu(E \cap A_i \cap B_j) \\
 &= \alpha \sum_{i=1}^n \alpha_i \mu(E \cap A_i) + \beta \sum_{j=1}^k \beta_j \mu(E \cap B_j) = \alpha \int_E \varphi d\mu + \beta \int_E \psi d\mu.
 \end{aligned}$$

□

The next goal is to prove monotonicity, linearity, and the MCT for nonnegative measurable functions. We start with monotonicity:

Lemma 1.49. *Let $(X, \mathcal{F}, \mu)$ be a measure space, let $E \in \mathcal{F}$ be a measurable set, and let f, g be nonnegative measurable functions from $(X, \mathcal{F}, \mu)$ to $[0, +\infty]$. If $f \leq g$ on E , then*

$$\int_E f d\mu \leq \int_E g d\mu.$$

Proof.

The proof follows from the fact that if $A \subseteq B$, then $\sup_{x \in A} F(x) \leq \sup_{x \in B} F(x)$ for some function F . Since $f \leq g$, we deduce $\mathcal{S}_f \subseteq \mathcal{S}_g$. This implies

$$\int_E f d\mu = \sup_{\varphi \in \mathcal{S}_f} \int_E \varphi d\mu \leq \sup_{\varphi \in \mathcal{S}_g} \int_E \varphi d\mu = \int_E g d\mu.$$

□

Monotonicity is then used to establish the MCT. The following proof of the MCT is inspired by [2]:

Theorem 1.50 (Monotone Convergence). *Let $(X, \mathcal{F}, \mu)$ be a measure space. Let $(f_n)_{n \in \mathbb{N}}$ be an increasing sequence of nonnegative measurable functions $X \rightarrow [0, +\infty]$. Let f be its pointwise limit, i.e.,*

$$f(x) = \lim_{n \rightarrow \infty} f_n(x), \quad \forall x \in X.$$

Then f is measurable and

$$\int_X f \, d\mu = \int_X \lim_{n \rightarrow \infty} f_n \, d\mu = \lim_{n \rightarrow \infty} \int_X f_n \, d\mu.$$

Proof.

Let us observe that, for every $n \in \mathbb{N}$, it holds (by Lemma 1.49)

$$f_n \leq f_{n+1} \leq f \implies \int_X f_n \, d\mu \leq \int_X f_{n+1} \, d\mu \leq \int_X f \, d\mu.$$

As a consequence, the sequence $(\int_X f_n \, d\mu)_{n \in \mathbb{N}}$ is increasing and bounded from above by $\int_X f \, d\mu$: if we can show that also

$$\lim_{n \rightarrow \infty} \int_X f_n \, d\mu \geq \int_X f \, d\mu, \quad (1.2)$$

we would conclude that $\int_X f \, d\mu = \lim_{n \rightarrow \infty} \int_X f_n \, d\mu$, which is the desired result.

Lemma 1.51. *Let $\varphi = \sum_{i=1}^m \alpha_i \mathbf{1}_{A_i}$ be a simple, measurable, nonnegative function written in canonical representation, such that $\varphi \leq f$, i.e., $\varphi \in \mathcal{S}_f$. Take $c \in (0, 1)$ and define*

$$B_i^n = \{x \in A_i : f_n(x) \geq c\varphi(x)\}.$$

Then for every $i \in \{1, \dots, m\}$, $B_i^1 \subseteq B_i^2 \subseteq \dots \subseteq A_i$ and $\bigcup_{n \in \mathbb{N}} B_i^n = A_i$. As a consequence, for every $i \in \{1, \dots, m\}$, $\mu(B_i^n) \rightarrow \mu(A_i)$ as $n \rightarrow +\infty$.

Proof. Fix $i \in \{1, \dots, m\}$. The proof follows from a clever combination of the following observations, each true independently:

- i) $f(x) \geq f_{n+1}(x) \geq f_n(x) \geq 0$ for every $x \in A_i$. This holds by hypothesis;
 - ii) $B_i^n \subseteq A_i$ for every $n \in \mathbb{N}$. This holds by definition of B_i^n ;
 - iii) Let $x \in A_i$. Then $x \in B_i^n$ if and only if $f_n(x) \geq c\varphi(x)$. This holds by definition of B_i^n .
 - iv) $f(x) \geq c\varphi(x) \geq 0$, because $\varphi \in \mathcal{S}_f$.
 - v) If $f(x) = 0$, then $f(x) = f_n(x) = c\varphi(x) = 0$ for every $n \in \mathbb{N}$, because of property (i) and (iv); if $f(x) \neq 0$, then $f(x) > c\varphi(x) \geq 0$, because $\varphi \in \mathcal{S}_f$ and $0 < c < 1$.
- If $x \in B_i^n$ for some $n \in \mathbb{N}$, then

$$f_{n+1}(x) \stackrel{(i)}{\geq} f_n(x) \stackrel{(iii)}{\geq} c\varphi(x) \stackrel{(iii)}{\implies} x \in B_i^{n+1}.$$

Hence $B_i^1 \subseteq B_i^2 \subseteq \dots \stackrel{(ii)}{\subseteq} A_i$. In particular, $\bigcup_{n \in \mathbb{N}} B_i^n \subseteq A_i$.

- Let us prove $\bigcup_{n \in \mathbb{N}} B_i^n = A_i$. We already proved one direction. For the other direction, assume $x \in A_i$. If $f(x) = 0$, then $f_n(x) \geq c\varphi(x)$ for every $n \in \mathbb{N}$ by property (v), hence $x \in \bigcup_{n \in \mathbb{N}} B_i^n$. If $f(x) \neq 0$, then $f(x) > c\varphi(x)$ by property (v). However, since $f_n(x) \uparrow f(x)$, there exists n^* sufficiently large such that $f_{n^*}(x) > c\varphi(x)$ as well. Hence $x \in B_i^{n^*} \subseteq \bigcup_{n \in \mathbb{N}} B_i^n$.
- B_i^n is measurable: since $\varphi = \alpha_i$ on A_i , we have $B_i^n = A_i \cap f_n^{-1}([c\alpha_i, +\infty])$.
- By continuity from below in Proposition 1.12, it follows that $\mu(B_i^n) \rightarrow \mu(A_i)$ as $n \rightarrow +\infty$ for every $i \in \{1, \dots, m\}$.

□

Observe that, by construction, for every $n \in \mathbb{N}$, it holds

$$f_n \geq \sum_{i=1}^m c\alpha_i \mathbf{1}_{B_i^n}.$$

This is because if $x \in B_{i_0}^n$ for some $i_0 \in \{1, \dots, m\}$, then $f_n(x) \geq c\varphi(x) = c\alpha_{i_0}$ and, since $B_i^n \subseteq A_i$ and the sets $(A_i)_{i=1}^m$ are disjoint, we have $\sum_{i=1}^m c\alpha_i \mathbf{1}_{B_i^n}(x) = c\alpha_{i_0}$. If instead $x \notin B_i^n$ for every $i \in \{1, \dots, m\}$, then $f_n(x) \geq 0 = \sum_{i=1}^m c\alpha_i \mathbf{1}_{B_i^n}(x)$. Thus

$$\int_X f_n d\mu \geq \int_X \left[\sum_{i=1}^m c\alpha_i \mathbf{1}_{B_i^n} \right] d\mu = \sum_{i=1}^m c\alpha_i \mu(B_i^n).$$

Thanks to the previous lemma, $\mu(B_i^n) \rightarrow \mu(A_i)$ as $n \rightarrow \infty$. Hence

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu \geq \lim_{n \rightarrow \infty} \sum_{i=1}^m c\alpha_i \mu(B_i^n) = \sum_{i=1}^m c\alpha_i \mu(A_i) = c \int_X \varphi d\mu.$$

Observe that this inequality holds for every $\varphi \in \mathcal{S}_f$. Taking the supremum over $\varphi \in \mathcal{S}_f$, we deduce

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu \geq c \int_X f d\mu = c \int_X \lim_{n \rightarrow \infty} f_n d\mu.$$

Since $c \in (0, 1)$ is arbitrary, we let $c \rightarrow 1$ to obtain (1.2), concluding the proof. □

Now that we have proved the MCT, we can establish the last fundamental property of the Lebesgue integral, namely linearity. We first need a lemma bridging these two concepts.

Lemma 1.52 (Simple approximation). *Let $(X, \mathcal{F}, \mu)$ be a measure space and let $f : X \rightarrow [0, +\infty]$ be a nonnegative measurable function. Then there exists a sequence $(\varphi_n)_{n \in \mathbb{N}}$ of simple, nonnegative, measurable functions such that:*

- i) *The sequence is increasing: $0 \leq \varphi_1(x) \leq \varphi_2(x) \leq \dots \leq f(x)$ for every $x \in X$;*
- ii) *The sequence converges pointwise to f : $\lim_{n \rightarrow \infty} \varphi_n(x) = f(x)$ for every $x \in X$.*

In this case, we say that (φ_n) increases to f from below and we write $\varphi_n \uparrow f$. Moreover,

$$\int_E f d\mu = \sup_{\varphi \in \mathcal{S}_f} \int_E \varphi d\mu = \lim_{n \rightarrow \infty} \int_E \varphi_n d\mu.$$

Proof.

The first part of the proof is constructive. For instance, one may define

$$\varphi_n(x) = \min \left(n, \frac{\lfloor 2^n f(x) \rfloor}{2^n} \right) = \begin{cases} \frac{k}{2^n} & \text{if } \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \text{ for } k = 0, \dots, n2^n - 1, \\ n & \text{if } f(x) \geq n. \end{cases}$$

By construction, $0 \leq \varphi_n \leq f$, and $\varphi_n(x) \rightarrow f(x)$ pointwise as $n \rightarrow \infty$. We now show that (φ_n) is increasing. Fix $x \in X$.

- If $f(x) \geq n+1$, then $\varphi_{n+1}(x) = n+1 > n = \varphi_n(x)$.
- If $n \leq f(x) < n+1$, then $\varphi_n(x) = n$ and $2^{n+1}f(x) \geq n2^{n+1}$, hence $\lfloor 2^{n+1}f(x) \rfloor \geq n2^{n+1}$. Consequently,

$$\varphi_{n+1}(x) = \frac{\lfloor 2^{n+1}f(x) \rfloor}{2^{n+1}} \geq \frac{n2^{n+1}}{2^{n+1}} = n = \varphi_n(x).$$

- If $f(x) < n$, we use the elementary inequality $\lfloor 2y \rfloor \geq 2\lfloor y \rfloor$:

$$\varphi_{n+1}(x) = \frac{\lfloor 2 \cdot 2^n f(x) \rfloor}{2^{n+1}} \geq \frac{2\lfloor 2^n f(x) \rfloor}{2 \cdot 2^n} = \varphi_n(x).$$

The second part of the proof is an application of Theorem 1.50 to the sequence (φ_n) . □

Remark 1.53. This result demonstrates that the Lebesgue integral for nonnegative measurable functions can be equivalently defined as the limit of the integrals of an increasing sequence of simple functions.

Theorem 1.54 (Linearity of the Lebesgue Integral). *Let $(X, \mathcal{F}, \mu)$ be a measure space, let $E \in \mathcal{F}$ be a measurable set, let f, g be nonnegative measurable functions from $(X, \mathcal{F}, \mu)$ to $[0, +\infty]$, and let $\alpha, \beta \in [0, +\infty)$ be constants. Then*

$$\int_E (\alpha f + \beta g) d\mu = \alpha \int_E f d\mu + \beta \int_E g d\mu.$$

Proof.

We prove homogeneity and additivity separately. While homogeneity follows directly from the definition, additivity requires the Monotone Convergence Theorem (via Lemma 1.52).

Homogeneity. If $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$ is a simple, nonnegative, measurable function given in its canonical representation, then (by Proposition 1.48)

$$\int_E \alpha \varphi d\mu = \sum_{i=1}^n (\alpha \alpha_i) \mu(E \cap A_i) = \alpha \sum_{i=1}^n \alpha_i \mu(E \cap A_i) = \alpha \int_E \varphi d\mu.$$

Now consider a general function f . Observe that $\mathcal{S}_{\alpha f} = \{\alpha \varphi \mid \varphi \in \mathcal{S}_f\}$, because $\varphi \leq f \iff \alpha \varphi \leq \alpha f$ (assuming $\alpha > 0$; the case $\alpha = 0$ is trivial). Hence,

$$\int_E \alpha f d\mu = \sup_{\zeta \in \mathcal{S}_{\alpha f}} \int_E \zeta d\mu = \sup_{\varphi \in \mathcal{S}_f} \int_E \alpha \varphi d\mu = \alpha \sup_{\varphi \in \mathcal{S}_f} \int_E \varphi d\mu = \alpha \int_E f d\mu.$$

Additivity. We want to prove

$$\int_X (f + g) d\mu = \int_X f d\mu + \int_X g d\mu. \quad (1.3)$$

By Lemma 1.52, there exist increasing sequences of nonnegative simple measurable functions $(\varphi_n)_{n \in \mathbb{N}}$ and $(\psi_n)_{n \in \mathbb{N}}$ converging pointwise to f and g , respectively. Since $\varphi_n \leq f$ and $\psi_n \leq g$ for every $n \in \mathbb{N}$, we have $0 \leq \varphi_n + \psi_n \leq f + g$ and

$$\lim_{n \rightarrow \infty} (\varphi_n(x) + \psi_n(x)) = f(x) + g(x), \quad \forall x \in X.$$

Since φ_n and ψ_n are simple functions, $(\varphi_n + \psi_n)_{n \in \mathbb{N}}$ is an increasing sequence of nonnegative simple functions converging to $f + g$. By Lemma 1.52, we have:

$$\begin{aligned} \int_X f d\mu &= \lim_{n \rightarrow \infty} \int_X \varphi_n d\mu, & \int_X g d\mu &= \lim_{n \rightarrow \infty} \int_X \psi_n d\mu, \\ \int_X (f + g) d\mu &= \lim_{n \rightarrow \infty} \int_X (\varphi_n + \psi_n) d\mu. \end{aligned}$$

By linearity of the integral for simple functions (Proposition 1.48), we know that $\int_X (\varphi_n + \psi_n) d\mu = \int_X \varphi_n d\mu + \int_X \psi_n d\mu$. Therefore,

$$\int_X (f + g) d\mu = \lim_{n \rightarrow \infty} \left(\int_X \varphi_n d\mu + \int_X \psi_n d\mu \right) = \lim_{n \rightarrow \infty} \int_X \varphi_n d\mu + \lim_{n \rightarrow \infty} \int_X \psi_n d\mu,$$

which proves (1.3). Finally, applying additivity to $f \mathbf{1}_E$ and $g \mathbf{1}_E$, we deduce $\int_E (f + g) d\mu = \int_E f d\mu + \int_E g d\mu$. $\square$

1.4.3 Integrable Functions

Now that we've proven the deepest theorems of the theory, we only need to generalize them to μ -a.e. integrable functions and recover further results.

Theorem 1.55 (Exchanging integrals and series). *Let $(X, \mathcal{F}, \mu)$ be a measure space and let $(f_n)_{n \in \mathbb{N}}$ be a sequence of nonnegative measurable functions $X \rightarrow [0, +\infty]$. Then*

$$\int_X \sum_{n=1}^{\infty} f_n d\mu = \sum_{n=1}^{\infty} \int_X f_n d\mu.$$

Observe that this property can be interpreted as σ -additivity for the Lebesgue integral.

Proof.

For every $x \in X$, let $S_n(x) := \sum_{i=1}^n f_i(x)$ and $f(x) := \sum_{n=1}^{\infty} f_n(x)$. Since each f_n is nonnegative, the sequence (S_n) is increasing, measurable, and converges *pointwise* to f . By the Monotone Convergence Theorem,

$$\int_X \sum_{n=1}^{\infty} f_n \, d\mu = \int_X \lim_{n \rightarrow \infty} S_n \, d\mu = \lim_{n \rightarrow \infty} \int_X S_n \, d\mu.$$

Finally, for every $n \in \mathbb{N}$ we have $S_n = \sum_{i=1}^n f_i$, hence, by finite additivity (Proposition 1.48),

$$\int_X S_n \, d\mu = \sum_{i=1}^n \int_X f_i \, d\mu.$$

Taking the limit as $n \rightarrow \infty$ yields

$$\int_X \sum_{n=1}^{\infty} f_n \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu,$$

as claimed. $\square$

Corollary 1.56 (Exchanging infinite series). *If $a : \mathbb{N} \times \mathbb{N} \rightarrow [0, +\infty]$ is a nonnegative function, then*

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a(i, j) = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a(i, j).$$

Proof.

Let $(f_j)_{j \in \mathbb{N}}$ be the sequence of nonnegative functions defined by $f_j(i) := a(i, j)$ for every $(i, j) \in \mathbb{N} \times \mathbb{N}$. Let us work in the measure space on $\mathbb{N}$ with counting measure μ . Then f_j are measurable functions, and

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a(i, j) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} f_j(i) = \int_{\mathbb{N}} \sum_{j=1}^{\infty} f_j \, d\mu = \sum_{j=1}^{\infty} \int_{\mathbb{N}} f_j \, d\mu = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} f_j(i) = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a(i, j).$$

$\square$

Proposition 1.57. *Let $(X, \mathcal{F}, \mu)$ be a measure space, let $E \in \mathcal{F}$ be a measurable set, and let Φ, Ψ be integrable functions from $(X, \mathcal{F}, \mu)$ to $\mathbb{R}$.*

- i) *If $\Phi \leq \Psi$, then $\int_E \Phi \, d\mu \leq \int_E \Psi \, d\mu$.*
- ii) *$|\int_E \Phi \, d\mu| \leq \int_E |\Phi| \, d\mu$.*
- iii) *$\int_E \Phi \, d\mu = \int_X \Phi \mathbf{1}_E \, d\mu$.*
- iv) *If $\alpha, \beta \in \mathbb{R}$ are constants, then $\int_E (\alpha\Phi + \beta\Psi) \, d\mu = \alpha \int_E \Phi \, d\mu + \beta \int_E \Psi \, d\mu$.*

Proof.

(i). If $\Phi \leq \Psi$, then clearly $\Phi^+ \leq \Psi^+$ and $\Psi^- \leq \Phi^-$. Therefore, by Lemma 1.49 (monotonicity for nonnegative functions) applied to the pairs (Φ^+, Ψ^+) and (Ψ^-, Φ^-) , we have

$$\int_E \Phi^+ \, d\mu \leq \int_E \Psi^+ \, d\mu \quad \text{and} \quad \int_E \Psi^- \, d\mu \leq \int_E \Phi^- \, d\mu.$$

Subtracting the inequalities, we deduce

$$\int_E \Phi \, d\mu = \int_E \Phi^+ \, d\mu - \int_E \Phi^- \, d\mu \leq \int_E \Psi^+ \, d\mu - \int_E \Psi^- \, d\mu = \int_E \Psi \, d\mu.$$

(ii). Suppose that $\int_E \Phi \, d\mu \geq 0$. Since $\Phi \leq |\Phi|$, by point (i) we have

$$\left| \int_E \Phi \, d\mu \right| = \int_E \Phi \, d\mu \leq \int_E |\Phi| \, d\mu.$$

Similarly, if $\int_E \Phi \, d\mu < 0$, since $-\Phi \leq |\Phi|$, by point (i) we have

$$\left| \int_E \Phi \, d\mu \right| = - \int_E \Phi \, d\mu = \int_E (-\Phi) \, d\mu \leq \int_E |\Phi| \, d\mu.$$

(iii). We observe that we proved this property only for simple functions in Proposition 1.48. We now prove that $\int_E f \, d\mu = \int_X f \mathbf{1}_E \, d\mu$ for a general nonnegative measurable f . Observe that $\varphi \in \mathcal{S}_f \implies \varphi \mathbf{1}_E \in \mathcal{S}_{f \mathbf{1}_E}$, since $\varphi \leq f \implies \varphi \mathbf{1}_E \leq f \mathbf{1}_E$. Thus

$$\int_E f \, d\mu = \sup_{\varphi \in \mathcal{S}_f} \int_E \varphi \, d\mu = \sup_{\varphi \in \mathcal{S}_f} \int_X \varphi \mathbf{1}_E \, d\mu \leq \sup_{\psi \in \mathcal{S}_{f \mathbf{1}_E}} \int_X \psi \, d\mu = \int_X f \mathbf{1}_E \, d\mu. \quad (1.4)$$

On the other hand, observe that

$$\int_X f \mathbf{1}_E \, d\mu = \sup_{\varphi \in \mathcal{S}_{f \mathbf{1}_E}} \int_X \varphi \, d\mu \stackrel{(1)}{=} \sup_{\varphi \in \mathcal{S}_{f \mathbf{1}_E}} \int_E \varphi \, d\mu = \int_E f \mathbf{1}_E \, d\mu \stackrel{2}{\leq} \int_E f \, d\mu, \quad (1.5)$$

where (1) holds because if $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i} \in \mathcal{S}_{f \mathbf{1}_E}$, then $\varphi(x) = 0$ for every $x \notin E$, so $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i \cap E}$. Hence,

$$\int_X \varphi \, d\mu = \sum_{i=1}^n \alpha_i \mu(A_i) = \sum_{i=1}^n \alpha_i \mu(A_i \cap E) = \int_E \varphi \, d\mu.$$

The inequality (2) follows from the monotonicity of the Lebesgue integral (since $f \mathbf{1}_E \leq f$). Combining (1.4) and (1.5) we deduce the claimed result.

Finally, for a general integrable Φ , since $(\Phi \mathbf{1}_E)^+ = \Phi^+ \mathbf{1}_E$ and $(\Phi \mathbf{1}_E)^- = \Phi^- \mathbf{1}_E$, it holds

$$\int_X \Phi \mathbf{1}_E \, d\mu = \int_X \Phi^+ \mathbf{1}_E \, d\mu - \int_X \Phi^- \mathbf{1}_E \, d\mu = \int_E \Phi^+ \, d\mu - \int_E \Phi^- \, d\mu = \int_E \Phi \, d\mu.$$

(iv). Let us start from homogeneity: $\int_E c\Phi \, d\mu = c \int_E \Phi \, d\mu$ for every $c \in \mathbb{R}$. First, observe that

$$\int_E |c\Phi| \, d\mu \stackrel{(1)}{\leq} \int_E |c| |\Phi| \, d\mu \stackrel{(2)}{=} |c| \int_E |\Phi| \, d\mu < \infty,$$

where (1) is point (i) and (2) is the homogeneity of nonnegative measurable functions. Therefore, the expression $\int_E c\Phi \, d\mu$ is well-defined.

- If $c \geq 0$, then $(c\Phi)^+ = c\Phi^+$ and $(c\Phi)^- = c\Phi^-$. Therefore,

$$\int_E c\Phi \, d\mu = \int_E c\Phi^+ \, d\mu - \int_E c\Phi^- \, d\mu = c \left(\int_E \Phi^+ \, d\mu - \int_E \Phi^- \, d\mu \right) = c \int_E \Phi \, d\mu.$$

- If $c \leq 0$, $(c\Phi)^+ = -c\Phi^-$ and $(c\Phi)^- = -c\Phi^+$. Therefore,

$$\begin{aligned} \int_E c\Phi \, d\mu &= \int_E (c\Phi)^+ \, d\mu - \int_E (c\Phi)^- \, d\mu = \int_E -c\Phi^- \, d\mu - \int_E -c\Phi^+ \, d\mu \\ &= -c \left(\int_E \Phi^- \, d\mu - \int_E \Phi^+ \, d\mu \right) = -c \left(- \int_E \Phi \, d\mu \right) = c \int_E \Phi \, d\mu. \end{aligned}$$

Finally, we show that $\int_X (\Phi + \Psi) \, d\mu = \int_X \Phi \, d\mu + \int_X \Psi \, d\mu$. First, observe that the left hand-side is well-defined:

$$\int_X |\Phi + \Psi| \, d\mu \leq \int_X (|\Phi| + |\Psi|) \, d\mu = \int_X |\Phi| \, d\mu + \int_X |\Psi| \, d\mu < \infty.$$

Unfortunately, $(\Phi + \Psi)^+$ can't be decomposed as $\Phi^+ + \Psi^+$ or similar. What can be done is to observe the following possible decompositions:

$$\Phi + \Psi = (\Phi + \Psi)^+ - (\Phi + \Psi)^- = (\Phi^+ - \Phi^-) + (\Psi^+ - \Psi^-).$$

Rearranging the terms, we deduce

$$(\Phi + \Psi)^+ + \Phi^- + \Psi^- = (\Phi + \Psi)^- + \Phi^+ + \Psi^+.$$

Now we can integrate both sides and apply additivity for the integral of nonnegative measurable functions:

$$\begin{aligned} \int_X [(\Phi + \Psi)^+ + \Phi^- + \Psi^-] d\mu &= \int_X [(\Phi + \Psi)^- + \Phi^+ + \Psi^+] d\mu \\ \int_X (\Phi + \Psi)^+ d\mu + \int_X \Phi^- d\mu + \int_X \Psi^- d\mu &= \int_X (\Phi + \Psi)^- d\mu + \int_X \Phi^+ d\mu + \int_X \Psi^+ d\mu \\ \underbrace{\int_X (\Phi + \Psi)^+ d\mu - \int_X (\Phi + \Psi)^- d\mu}_{\int (\Phi + \Psi) d\mu} &= \underbrace{\int_X \Phi^+ d\mu - \int_X \Phi^- d\mu}_{\int \Phi d\mu} + \underbrace{\int_X \Psi^+ d\mu - \int_X \Psi^- d\mu}_{\int \Psi d\mu} \\ \int_X (\Phi + \Psi) d\mu &= \int_X \Phi d\mu + \int_X \Psi d\mu \end{aligned}$$

Of course, applying the additivity to $\Phi \mathbf{1}_E$ and $\Psi \mathbf{1}_E$, we deduce $\int_E (\Phi + \Psi) d\mu = \int_E \Phi d\mu + \int_E \Psi d\mu$. $\square$

Proposition 1.58 (Invariance modulo null sets). *Let $(X, \mathcal{F}, \mu)$ be a measure space, let $A, B \in \mathcal{F}$ be measurable sets, and let Φ, Ψ be integrable functions from $(X, \mathcal{F}, \mu)$ to $\mathbb{R}$. Then:*

- i) *If $A \cap B = \emptyset$, then $\int_{A \cup B} \Phi d\mu = \int_A \Phi d\mu + \int_B \Phi d\mu$.*
- ii) *If $A \subseteq B$ and $\Phi \geq 0$, then $\int_A \Phi d\mu \leq \int_B \Phi d\mu$*
- iii) *If $\mu(A) = 0$, then $\int_A \Phi d\mu = 0$;*
- iv) *If $\Phi = \Psi$ μ -a.e., then $\int_A \Phi d\mu = \int_A \Psi d\mu$.*
- v) *If $\Phi \leq \Psi$ μ -a.e., then $\int_A \Phi d\mu \leq \int_A \Psi d\mu$.*
- vi) *If $\Phi \geq 0$ μ -a.e. on A , then $\int_A \Phi d\mu \geq 0$.*

Proof.

(i). Since A and B are disjoint, $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B$. By the linearity of the Lebesgue integral (Proposition 1.57),

$$\int_{A \cup B} \Phi d\mu = \int_X \Phi \mathbf{1}_{A \cup B} d\mu = \int_X \Phi (\mathbf{1}_A + \mathbf{1}_B) d\mu = \int_X \Phi \mathbf{1}_A d\mu + \int_X \Phi \mathbf{1}_B d\mu = \int_A \Phi d\mu + \int_B \Phi d\mu.$$

(ii). If $A \subseteq B$ and $\Phi \geq 0$, then $0 \leq \Phi \mathbf{1}_A \leq \Phi \mathbf{1}_B$ and hence

$$0 \leq \int_A \Phi d\mu = \int_X \Phi \mathbf{1}_A d\mu \leq \int_X \Phi \mathbf{1}_B d\mu = \int_B \Phi d\mu.$$

(iii). If $\mu(A) = 0$, let $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$ be a simple nonnegative measurable function written in canonical representation. We note that $\mu(A \cap A_i) \leq \mu(A) = 0$ for every $i = 1, \dots, n$. Therefore,

$$\int_A \varphi d\mu = \sum_{i=1}^n \alpha_i \mu(A \cap A_i) = \sum_{i=1}^n \alpha_i \cdot 0 = 0.$$

Hence,

$$\int_A \Phi d\mu = \int_A \Phi^+ d\mu - \int_A \Phi^- d\mu = \sup_{\varphi \in \mathcal{S}_{\Phi^+}} \int_A \varphi d\mu - \sup_{\varphi \in \mathcal{S}_{\Phi^-}} \int_A \varphi d\mu = \sup_{\varphi \in \mathcal{S}_{\Phi^+}} 0 - \sup_{\varphi \in \mathcal{S}_{\Phi^-}} 0 = 0.$$

(iv). Let $\mathcal{N} := \{x \in A : \Phi(x) \neq \Psi(x)\}$. By hypothesis $\mu(\mathcal{N}) = 0$. If $\mathcal{N} = \emptyset$, then $\Phi \equiv \Psi$ on A , hence $\int_A \Phi d\mu = \int_A \Psi d\mu$. Otherwise, we can decompose, thanks to point (i), as

$$\int_A \Phi d\mu = \int_{\mathcal{N}} \Phi d\mu + \int_{A \setminus \mathcal{N}} \Phi d\mu \stackrel{(\dagger)}{=} 0 + \int_{A \setminus \mathcal{N}} \Phi d\mu = \int_{A \setminus \mathcal{N}} \Psi d\mu \stackrel{(\dagger)}{=} \int_{\mathcal{N}} \Psi d\mu + \int_{A \setminus \mathcal{N}} \Psi d\mu = \int_A \Psi d\mu,$$

where the equalities marked with $(\dagger)$ hold by point (iii).

(v). Let $\mathcal{N} := \{x \in A : \Phi(x) > \Psi(x)\}$. By hypothesis $\mu(\mathcal{N}) = 0$. We proceed as in point (iv), splitting the integral on $\mathcal{N}$ and $A \setminus \mathcal{N}$. The integrals over $\mathcal{N}$ vanish; on $A \setminus \mathcal{N}$ we have $\Phi \leq \Psi$, so the result follows from monotonicity.

(vi). Apply (v) to Φ and 0. □

Corollary 1.59. Let $(X, \mathcal{F}, \mu)$ be a measure space and $f : (X, \mathcal{F}) \rightarrow [0, +\infty]$ a nonnegative measurable function. Then

$$\int_X f d\mu = 0 \iff [f = 0 \text{ } \mu\text{-a.e.}].$$

Proof.

If $f = 0$ μ -a.e., then $\int_X f d\mu = 0$ by Proposition 1.58. Conversely, let $\mathcal{N} = \{x \in X : f(x) > 0\}$. We want to show that $\mu(\mathcal{N}) = 0$. This set can be written as

$$\mathcal{N} = \bigcup_{n=1}^{\infty} \mathcal{N}_n = \bigcup_{n=1}^{\infty} \left\{x \in X : f(x) \geq \frac{1}{n}\right\} \implies \mu(\mathcal{N}) = \mu\left(\bigcup_{n=1}^{\infty} \mathcal{N}_n\right) \leq \sum_{n=1}^{\infty} \mu(\mathcal{N}_n),$$

by Proposition 1.12. It remains to prove that $\mu(\mathcal{N}_n) = 0$ for every $n \in \mathbb{N}$. Fix $n \in \mathbb{N}$. For every $x \in \mathcal{N}_n$, it holds $f(x) \geq 1/n$, hence

$$0 = \int_X f d\mu \geq \int_{\mathcal{N}_n} f d\mu \geq \int_{\mathcal{N}_n} \frac{1}{n} d\mu = \frac{1}{n} \mu(\mathcal{N}_n) \implies \mu(\mathcal{N}_n) = 0,$$

concluding the proof. □

Corollary 1.60. Let $(X, \mathcal{F}, \mu)$ be a measure space and $\Phi : (X, \mathcal{F}) \rightarrow \overline{\mathbb{R}}$ an integrable function. Then

$$\int_E \Phi d\mu = 0 \quad \forall E \in \mathcal{F} \implies [\Phi = 0 \text{ } \mu\text{-a.e.}].$$

Proof.

Write $\Phi = \Phi^+ - \Phi^-$. Let $A := \{x \in X : \Phi^+(x) > 0\}$ and $B := \{x \in X : \Phi^-(x) > 0\}$. Then

$$0 = \int_A \Phi d\mu = \int_A \Phi^+ d\mu \quad \text{and} \quad 0 = \int_B \Phi d\mu = \int_B \Phi^- d\mu.$$

Since $\Phi^\pm \geq 0$, by the previous corollary it holds $\Phi^+ = 0$ μ -a.e. and $\Phi^- = 0$ μ -a.e.; therefore, $\Phi = \Phi^+ - \Phi^- = 0$ μ -a.e. □

Corollary 1.61. Let $(X, \mathcal{F}, \mu)$ be a measure space and let $f : (X, \mathcal{F}) \rightarrow [0, +\infty]$ be measurable. If $\int_X f d\mu < \infty$, then $f < \infty$ μ -a.e.

Proof.

Let $\mathcal{N} = \{x \in X : f(x) = \infty\}$. We want to show that $\mu(\mathcal{N}) = 0$. If $\mu(\mathcal{N}) = \varepsilon > 0$, then $\int_X f d\mu \geq \int_{\mathcal{N}} f d\mu = \infty \cdot \varepsilon = \infty$, which contradicts the assumption that $\int_X f d\mu \in \mathbb{R}$. □

Corollary 1.62. Let $(X, \mathcal{F}, \mu)$ be a measure space and f, g be nonnegative measurable functions from $(X, \mathcal{F}, \mu)$ to $[0, +\infty]$. Moreover, assume $f \leq g$ and $\int_X g d\mu < +\infty$. Then

$$\int_X f d\mu = \int_X g d\mu \iff [f = g \text{ } \mu\text{-a.e.}].$$

Proof.

If $f = g$ μ -a.e., then $\int_X f d\mu = \int_X g d\mu$ by Proposition 1.58. Conversely, let $h := g - f$. Then $0 \leq h \leq g$, so that $\int_X h d\mu \leq \int_X g d\mu < \infty$ and h is nonnegative and measurable. Applying Corollary 1.59 to h we deduce

$$\int_X f d\mu = \int_X g d\mu \iff \int_X (g - f) d\mu = 0 \iff \int_X h d\mu = 0 \iff [h = 0 \text{ } \mu\text{-a.e.}] \iff [f = g \text{ } \mu\text{-a.e.}].$$

□

Corollary 1.63 (Countable Additivity of the Integral). *Let $(X, \mathcal{F}, \mu)$ be a measure space and let $\Phi : (X, \mathcal{F}) \rightarrow \overline{\mathbb{R}}$ be an integrable function. Let $(A_n)_{n \in \mathbb{N}}$ be a sequence of disjoint measurable sets in $\mathcal{F}$, and let $A = \bigcup_{n=1}^{\infty} A_n$. Then:*

$$\int_A \Phi d\mu = \sum_{n=1}^{\infty} \int_{A_n} \Phi d\mu.$$

Proof.

We write

$$\begin{aligned} \int_A \Phi d\mu &= \int_A \Phi^+ d\mu - \int_A \Phi^- d\mu = \int_A \sum_{n=1}^{\infty} \Phi^+ \mathbf{1}_{A_n} d\mu - \int_A \sum_{n=1}^{\infty} \Phi^- \mathbf{1}_{A_n} d\mu \\ &= \sum_{n=1}^{\infty} \int_{A_n} \Phi^+ d\mu - \sum_{n=1}^{\infty} \int_{A_n} \Phi^- d\mu = \sum_{n=1}^{\infty} \left[\int_{A_n} \Phi^+ d\mu - \int_{A_n} \Phi^- d\mu \right] \\ &= \sum_{n=1}^{\infty} \left[\int_{A_n} \Phi^+ d\mu - \int_{A_n} \Phi^- d\mu \right] = \sum_{n=1}^{\infty} \int_{A_n} \Phi d\mu. \end{aligned}$$

The exchange of series and integrals follows from Theorem 1.55 applied to the nonnegative functions $\Phi^+ \mathbf{1}_{A_n}$ and $\Phi^- \mathbf{1}_{A_n}$. The remaining steps use linearity and the identity $\int_A \Psi \mathbf{1}_{A_n} d\mu = \int_{A_n} \Psi d\mu$. □

Example 1.64. We are now able to compute the integral

$$\int_{\mathbb{R}} \mathbf{1}_{\mathbb{Q} \cap [0,1]} d\mu,$$

which was discussed in Example 1.36. There are two ways to proceed.

The first is to observe that $f_n \uparrow f$ pointwise, where $(f_n)_{n \in \mathbb{N}}$ and f are defined in Example 1.36. Since $(f_n)_{n \in \mathbb{N}}$ is a sequence of nonnegative measurable functions, we can apply the Monotone Convergence Theorem (Theorem 1.50):

$$\int_{\mathbb{R}} \mathbf{1}_{\mathbb{Q} \cap [0,1]} d\mu = \int_{\mathbb{R}} f d\mu = \int_{\mathbb{R}} \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n d\mu = \lim_{n \rightarrow \infty} 0 = 0.$$

Alternatively, reorder the rational numbers so that the sequence $(q_n)_{n \in \mathbb{N}}$ corresponds to the set of all rational numbers (which is countable). We define the sequence of disjoint sets $(\{q_n\})_{n \in \mathbb{N}}$, and we apply countable additivity of the measure to deduce

$$\mu(\mathbb{Q}) = \mu\left(\bigcup_{n=1}^{\infty} \{q_n\}\right) = \sum_{n=1}^{\infty} \mu(\{q_n\}) = \sum_{n=1}^{\infty} 0 = 0.$$

Then

$$\int_{\mathbb{R}} \mathbf{1}_{\mathbb{Q} \cap [0,1]} d\mu = \int_{\mathbb{R} \setminus \mathbb{Q}} \mathbf{1}_{\mathbb{Q} \cap [0,1]} d\mu + \int_{\mathbb{Q}} \mathbf{1}_{\mathbb{Q} \cap [0,1]} d\mu = 0 + 0 = 0,$$

where the first integral is zero because the integrand vanishes on $\mathbb{R} \setminus \mathbb{Q}$, while the second integral is zero because the integration domain has measure zero. Finally, one can also see that $\int_{\mathbb{Q}} 1 d\mu = 0$ directly using Corollary 1.63.

Remark 1.65. It is worth mentioning that the Monotone Convergence Theorem (MCT) provides a unified justification for several classical facts regarding Riemann integration. In particular, it shows how Lebesgue integration and Riemann integration coincide for Riemann integrable functions, and clarifies the relationship between the Lebesgue integral and the definitions of improper Riemann integrals.

For instance, consider a nonnegative continuous function $f : [0, +\infty) \rightarrow [0, +\infty)$. In the Riemann theory, the integral over the unbounded domain is *defined* as a limit:

$$\int_0^{+\infty} f(x) \, dx := \lim_{n \rightarrow \infty} \int_0^n f(x) \, dx.$$

From the perspective of Lebesgue theory, this equality is not a definition but a consequence. We can define the sequence of truncated functions $f_n := f \cdot \mathbf{1}_{[0, n]}$. This sequence is nonnegative, measurable, and increases pointwise to f . Therefore, by the MCT:

$$\int_{[0, +\infty)} f \, d\mu = \lim_{n \rightarrow \infty} \int_{[0, +\infty)} f_n \, d\mu = \lim_{n \rightarrow \infty} \int_0^n f(x) \, dx.$$

1.5 Product Measures and Fubini–Tonelli

In this section we develop tools for computing integrals over product spaces.

Definition 1.66 (Product σ -algebra). Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. A *measurable rectangle* is a set of the form $A \times B$, with $A \in \mathcal{A}$ and $B \in \mathcal{B}$. The *product σ -algebra* on $X \times Y$ is the σ -algebra generated by *measurable rectangles*, i.e.

$$\mathcal{A} \otimes \mathcal{B} := \mathcal{F}(\{A \times B : A \in \mathcal{A}, B \in \mathcal{B}\}).$$

The pair $(X \times Y, \mathcal{A} \otimes \mathcal{B})$ is called the *product measurable space*.

Definition 1.67 (Sections). Let $E \subseteq X \times Y$. For $x \in X$ and $y \in Y$ we define the sections

$$E_x := \{y \in Y : (x, y) \in E\} \subseteq Y \quad E^y := \{x \in X : (x, y) \in E\} \subseteq X.$$

If $f : X \times Y \rightarrow \overline{\mathbb{R}}$, we define the sections $f_x : Y \rightarrow \overline{\mathbb{R}}$ and $f^y : X \rightarrow \overline{\mathbb{R}}$ as

$$f_x : y \mapsto f(x, y) \quad f^y : x \mapsto f(x, y).$$

Lemma 1.68 (Measurability of sections). If $E \in \mathcal{A} \otimes \mathcal{B}$, then $E_x \in \mathcal{B}$ for every $x \in X$ and $E^y \in \mathcal{A}$ for every $y \in Y$. Equivalently, if f is $(\mathcal{A} \otimes \mathcal{B})$ -measurable, then for every fixed x the map $y \mapsto f(x, y)$ is $\mathcal{B}$ -measurable, and for every fixed y the map $x \mapsto f(x, y)$ is $\mathcal{A}$ -measurable.

Proof.

We prove the claims only for $E_x \in \mathcal{B}$ and $y \mapsto f(x, y)$ $\mathcal{B}$ -measurable, because the other results are symmetric. Let

$$\mathcal{C} := \{E \subseteq X \times Y : E_x \in \mathcal{B} \text{ for every } x \in X\}.$$

We claim that $\mathcal{C}$ is a σ -algebra on $X \times Y$:

- $(X \times Y)_x = Y \in \mathcal{B}$, hence $X \times Y \in \mathcal{C}$;
- If $E \in \mathcal{C}$, then $(E^c)_x = (E_x)^c \in \mathcal{B}$, so $E^c \in \mathcal{C}$;
- If $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{C}$, then $(\bigcup_{n \in \mathbb{N}} E_n)_x = \bigcup_{n \in \mathbb{N}} (E_n)_x \in \mathcal{B}$, so $\bigcup_{n \in \mathbb{N}} E_n \in \mathcal{C}$.

Since $\mathcal{C}$ is a σ -algebra containing $A \times B$ for every $A \in \mathcal{A}$, $B \in \mathcal{B}$, then $E_x \in \mathcal{B}$ for every $x \in X$.

As for functions, note that if $x \in X$ and $D \in \mathcal{B}(\mathbb{R})$, then

$$(f_x)^{-1}(D) = \{y \in Y : f(x, y) \in D\} = \{y \in Y : (x, y) \in f^{-1}(D)\} = [f^{-1}(D)]_x.$$

Since f is measurable, $E := f^{-1}(D) \in \mathcal{A} \otimes \mathcal{B}$. By the first part of the proof, $E_x \in \mathcal{B}$. Thus $(f_x)^{-1}(D) \in \mathcal{B}$, proving that f_x is measurable. $\square$

σ -finiteness. Before continuing with the construction, we pause to define an important class of measure spaces. A measure space $(X, \mathcal{F}, \mu)$ is said to be σ -finite if there exists a countable family of measurable sets $\{E_n\}_{n \in \mathbb{N}}$ such that $\mu(E_n) < \infty$ for every $n \in \mathbb{N}$ and $X = \bigcup_{n \in \mathbb{N}} E_n$. For example, $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ is σ -finite, since $\mathbb{R} = \bigcup_{n=1}^{\infty} [-n, n]$. Another instance is $(\mathbb{N}, 2^{\mathbb{N}}, \#)$, where $\#$ is the counting measure: indeed, $\mathbb{N} = \bigcup_{n=1}^{\infty} \{n\}$ and $\#(\{n\}) = 1 < \infty$. In contrast, $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \#)$ is not σ -finite, since any infinite set has infinite measure, and $\mathbb{R}$ cannot be covered by countably many finite sets. It is clear that σ -finiteness is a mild assumption.² However,

²In fact, on a σ -compact topological space (X, τ_X) , every measure μ that is finite on compact sets is σ -finite. In turn, every "reasonable" space we work with (such as manifolds) is assumed to be locally compact Hausdorff and second-countable, which implies σ -compactness. A counterexample is $\mathbb{R}$ endowed with the discrete topology, which is not σ -compact.

this assumption is necessary in at least four important cases, each of them foundational for probability theory: (i) the uniqueness of the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$; (ii) the uniqueness of the product measure on $(X \times Y, \mathcal{A} \otimes \mathcal{B})$; (iii) the Radon-Nikodym Theorem; (iv) the Kolmogorov extension theorem. Here, we introduce it as a prerequisite for the applicability of Fubini's Theorem.

Product Measure. We aim to define a measure for the product measurable space $(X \times Y, \mathcal{A} \otimes \mathcal{B})$. Such a measure, denoted by $\mu \otimes \nu$, should satisfy the natural property that for every measurable rectangle $A \times B$ (with $A \in \mathcal{A}$ and $B \in \mathcal{B}$):

$$(\mu \otimes \nu)(A \times B) = \mu(A) \cdot \nu(B). \quad (1.6)$$

This is the measure-theoretic formalization of what we do in physics when we compute the momentum as mass times velocity. Note that (1.6) can't be a definition for $\mu \otimes \nu$. We don't even know whether such a measure exists, and if it does, whether it is a valid measure for the whole σ -algebra. The *existence* and *uniqueness* of such a measure is a deep theorem (see [3] for a proof). Its proof relies on the same tools used in the construction of the Lebesgue measure, namely the *Carathéodory Extension Theorem* and the *Monotone Class Lemma*. Assuming that the measure spaces are σ -finite, the uniqueness is guaranteed. The strategy is to first prove uniqueness of such measure, then defining the measure via sections:

$$(\mu \otimes \nu)(E) = \int_X \nu(E_x) d\mu(x) = \int_Y \mu(E^y) d\nu(y), \quad \forall E \in \mathcal{A} \otimes \mathcal{B}. \quad (1.7)$$

This formula is well-defined thanks to Lemma 1.68 (and the *monotone class lemma*), and it holds for rectangles. The hard part, which we skip, is to verify that the set function defined by (1.7) is indeed a measure on $\mathcal{A} \otimes \mathcal{B}$ and that it satisfies the defining property for rectangles:

$$(\mu \otimes \nu)(A \times B) = \int_A \nu(B) d\mu = \mu(A) \nu(B).$$

At this point, under the assumption of σ -finiteness, this constructed measure is the *unique* product measure. Finally, by induction, one can define a unique measure μ on any finite product space $(\prod_{i=1}^n X_i, \bigotimes_{i=1}^n \mathcal{A}_i)$.

Remark 1.69 (Infinite Products and Kolmogorov Extension). The construction above handles finite products. As for *infinite* product spaces, the construction resembles that of product topologies on infinite cartesian products. Let $X = \prod_{j \in J} X_j$, where J may be finite, countable or uncountable. Let us define a *cylinder* as

$$C = \{\omega \in X : \omega_{j_1} \in A_{j_1}, \dots, \omega_{j_n} \in A_{j_n}\},$$

where $\{j_1, \dots, j_n\} \subseteq J$ is a finite subset of indices and $A_{j_k} \in \mathcal{F}_{j_k}$. Then the product σ -algebra $\bigotimes_{j \in J} \mathcal{A}_j$ is the one generated by *cylinders*.

- If J is countable, and μ_j is a measure on X_j for every $j \in J$, then it is possible to define a product probability measure $\text{Pr} = \bigotimes_{j=1}^{\infty}$.
- If J is uncountable, we cannot simply multiply measures. The *Kolmogorov Extension Theorem* guarantees that if these finite distributions are consistent (measurable and compatible under projection), there exists a unique measure on the infinite product space that extends them.

Fubini-Tonelli. (1.7) and (1.6) are suggesting us that, maybe,

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y). \quad (1.8)$$

Actually, the well-posedness of (1.7) is equivalent to (1.8) for indicator functions. By linearity, this extends immediately to simple functions (finite linear combinations of indicators). For a general nonnegative measurable function f , we choose a sequence of simple functions such that $0 \leq f_n \uparrow f$ pointwise. First, by the Monotone Convergence Theorem (MCT) on the product space $X \times Y$, we have:

$$\lim_{n \rightarrow \infty} \int_{X \times Y} f_n d(\mu \otimes \nu) = \int_{X \times Y} f d(\mu \otimes \nu).$$

Next, consider the inner integral. For each fixed $x \in X$, the sequence $y \mapsto f_n(x, y)$ increases to $y \mapsto f(x, y)$. By MCT on (Y, ν) :

$$\varphi_n(x) := \int_Y f_n(x, y) d\nu(y) \uparrow \int_Y f(x, y) d\nu(y) =: \varphi(x).$$

Since the equality holds for simple functions, we know that $\int_{X \times Y} f_n d(\mu \otimes \nu) = \int_X \varphi_n(x) d\mu(x)$. Applying MCT again, this time on (X, μ) to the sequence φ_n , we deduce:

$$\lim_{n \rightarrow \infty} \int_X \varphi_n(x) d\mu(x) = \int_X \varphi(x) d\mu(x) = \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x).$$

Comparing the limits proves the first equality in (1.8). The second follows by symmetry.

Finally, we extend this result to general integrable functions using the decomposition into positive and negative parts. The culmination of this construction is the *Fubini-Tonelli Theorem*.

- **Tonelli's Theorem** dictates how to compute integrals for nonnegative measurable functions (where the value $+\infty$ is allowed).
- **Fubini's Theorem** dictates how to compute integrals for integrable functions (i.e., functions Φ such that $\int_{X \times Y} |\Phi| d(\mu \otimes \nu) < \infty$).

The theorem is structurally analogous to Definition 1.43:

Theorem 1.70 (Fubini-Tonelli). *Let $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$ be two σ -finite measure spaces, f a measurable function and Φ a $(\mu \otimes \nu)$ -a.e. measurable function.*

- **Tonelli.** *If $f : X \times Y \rightarrow [0, +\infty]$ is nonnegative, let*

$$\varphi(x) := \int_Y f_x(y) d\nu, \quad \psi(y) := \int_X f^y(x) d\mu.$$

Then $\varphi : X \rightarrow [0, +\infty]$, $\psi : Y \rightarrow [0, +\infty]$ are measurable, and

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_X \varphi d\mu = \int_Y \psi d\nu.$$

- **Fubini.** *If $\Phi : X \times Y \rightarrow \overline{\mathbb{R}}$ satisfies $\int_{X \times Y} |\Phi| d(\mu \otimes \nu) < +\infty$ (integrability), let*

$$\varphi(x) := \int_Y \Phi_x(y) d\nu, \quad \psi(y) := \int_X \Phi^y(x) d\mu.$$

Then:

- *For μ -a.e. $x \in X$, the section Φ_x is measurable and integrable over Y ; similarly, for ν -a.e. $y \in Y$, the section Φ^y is measurable and integrable over X .*
- *The functions φ and ψ are well-defined almost everywhere, measurable, and integrable with respect to μ and ν respectively.*
- *It holds*

$$\int_{X \times Y} \Phi d(\mu \otimes \nu) = \int_X \varphi d\mu = \int_Y \psi d\nu.$$

Remark 1.71 (Importance of σ -finiteness). Observe the technical assumption of σ -finiteness for the measure spaces $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$. Without this assumption, the theorem may fail. [3] gives the following counterexample: take $X = Y = [0, 1]$ with the Borel σ -algebra. Let $\mu = \lambda$ be the Lebesgue measure and $\nu = \#$ be the counting measure. Let $f(x, y) = \mathbf{1}_\Delta(x, y)$ be the indicator function of the diagonal $\Delta = \{(x, y) \in X \times Y \mid x = y\}$. We see that:

$$\int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_X 1 d\mu(x) = 1,$$

since for fixed x , the set $\{y : y = x\}$ is a point, which has counting measure 1. On the other hand:

$$\int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y) = \int_Y 0 d\nu(y) = 0,$$

since for fixed y , the set $\{x : x = y\}$ is a point, which has Lebesgue measure 0. Thus, the iterated integrals differ.

Example 1.72. Let $D = \{(x, y) \in \mathbb{R}^2 : x \leq 2, -x/2 + 1 \leq y \leq e^x\}$ and let us compute $\int_D xy dx dy$. After drawing the domain, we realize x ranges from 0 to 2. Hence

$$\begin{aligned} \int_D xy dx dy &= \int_0^2 \int_{-x/2+1}^{e^x} xy dy dx = \int_0^2 x \left[\frac{y^2}{2} \right]_{-x/2+1}^{e^x} dx = \frac{1}{2} \int_0^2 x \left(e^{2x} - \left(1 - \frac{x}{2}\right)^2 \right) dx \\ &= \frac{1}{2} \int_0^2 \left(x e^{2x} - x + x^2 - \frac{x^3}{4} \right) dx = \frac{3}{8} e^4 - \frac{1}{24}. \end{aligned}$$

Example 1.73 (Counterexample to Fubini's Theorem). Let us try to compute

$$\int_{(0,1)^2} f(x, y) dx dy = \int_{(0,1)^2} \frac{x^2 - y^2}{(x^2 + y^2)^2} dx dy.$$

To compute it, we shall observe the (very nontrivial) fact that

$$\frac{x^2 - y^2}{(x^2 + y^2)^2} = \frac{\partial}{\partial x} \left(\frac{-x}{x^2 + y^2} \right) = \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2} \right).$$

Now, applying Fubini's Theorem:

$$\begin{aligned} \int_0^1 \int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2} dx dy &= \int_0^1 \left[\frac{-x}{x^2 + y^2} \right]_0^1 dy = \int_0^1 \frac{-1}{1 + y^2} - 0 dy = -[\arctan y]_0^1 = -\frac{\pi}{4} \\ \int_0^1 \int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2} dy dx &= \int_0^1 \left[\frac{y}{x^2 + y^2} \right]_0^1 dx = \int_0^1 \frac{1}{x^2 + 1} - 0 dx = [\arctan x]_0^1 = \frac{\pi}{4}. \end{aligned}$$

As we can see, Fubini's theorem fails. To understand why, let us check the integrability of f :

$$\int_0^1 \int_0^1 \left| \frac{x^2 - y^2}{(x^2 + y^2)^2} \right| dx dy \stackrel{?}{<} +\infty$$

By symmetry, $x^2 - y^2 \geq 0$ as long as $x \geq y$. Therefore, the integral is symmetric with respect to the line $x = y$. In particular,

$$\begin{aligned} \int_0^1 \int_0^1 \left| \frac{x^2 - y^2}{(x^2 + y^2)^2} \right| dx dy &= 2 \int_0^1 \int_0^x \frac{x^2 - y^2}{(x^2 + y^2)^2} dy dx = 2 \int_0^1 \left[\frac{y}{x^2 + y^2} \right]_0^x dx \\ &= 2 \int_0^1 \frac{x}{2x^2} dx = \int_0^1 \frac{1}{x} dx = +\infty. \end{aligned}$$

We conclude that Fubini's theorem fails because the function is measurable but not integrable.

1.6 Convergence Theorems

In this section, we analyze theorems that allow us to perform advanced manipulations with series and integrals, specifically justifying the interchange of the following operators:

$$\sum_{n \in \mathbb{N}} \int_X \lim_{n \rightarrow \infty}$$

We begin by recalling the fundamental result, often considered the cornerstone of the theory:

Theorem 1.74 (Monotone Convergence). *Let $(X, \mathcal{F}, \mu)$ be a measure space. Let $(f_n)_{n \in \mathbb{N}}$ be an increasing sequence of nonnegative measurable functions from $(X, \mathcal{F}, \mu)$ to $[0, +\infty]$, and assume that it converges pointwise to a function f , i.e.,*

$$\lim_{n \rightarrow \infty} f_n(x) = f(x), \quad \forall x \in X.$$

Then f is measurable and

$$\int_X f d\mu = \int_X \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int_X f_n d\mu.$$

As immediate corollaries, we have already established the following results:

Theorem 1.75 (Exchanging integral and series). *Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of nonnegative measurable functions from $(X, \mathcal{F}, \mu)$ to $[0, +\infty]$. Then*

$$\int_X \sum_{n=1}^{\infty} f_n d\mu = \sum_{n=1}^{\infty} \int_X f_n d\mu.$$

Remark 1.76. This property can be interpreted in two insightful ways:

- i) As the σ -additivity of the Lebesgue integral with respect to the sum of functions.
- ii) As an application of **Tonelli's Theorem** to the product space $X \times \mathbb{N}$, equipped with the product measure $\mu \otimes \#$ (where $\#$ is the counting measure on $\mathbb{N}$).

Corollary 1.77 (Exchanging Infinite Series). *If $a : \mathbb{N} \times \mathbb{N} \rightarrow [0, +\infty]$ is a nonnegative function, then*

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a(i, j) = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a(i, j).$$

A variation of the Monotone Convergence Theorem is the following lemma, which drops the monotonicity assumption and sacrifices the equality for an inequality.

Lemma 1.78 (Fatou). *Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of nonnegative measurable functions from $(X, \mathcal{F}, \mu)$ to $[0, +\infty]$. Then*

$$\int_X \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n \, d\mu.$$

Proof.

It is a one line proof:

$$\int_X \liminf_{n \rightarrow \infty} f_n \, d\mu \stackrel{1}{=} \int_X \lim_{n \rightarrow \infty} \inf_{m \geq n} f_m \, d\mu \stackrel{2}{=} \lim_{n \rightarrow \infty} \int_X \inf_{m \geq n} f_m \, d\mu \stackrel{3}{\leq} \lim_{n \rightarrow \infty} \inf_{m \geq n} \int_X f_m \, d\mu \stackrel{4}{=} \liminf_{n \rightarrow \infty} \int_X f_n \, d\mu.$$

Steps 1 and 4 are the definition of $\liminf$, step 2 is Theorem 1.50 applied to the nonnegative increasing function $n \mapsto \inf_{m \geq n} f_m$. As for step 3, observe that for every $k \geq n$, we have $\inf_{m \geq n} f_m \leq f_k$. By monotonicity of the integral,

$$\int_X \inf_{m \geq n} f_m \, d\mu \leq \int_X f_k \, d\mu, \quad \forall k \geq n.$$

Taking the infimum over $k \geq n$ on the right-hand side, we obtain

$$\int_X \inf_{m \geq n} f_m \, d\mu \leq \inf_{k \geq n} \int_X f_k \, d\mu,$$

which justifies the inequality before taking the limit. $\square$

Remark 1.79. In functional analysis, Fatou's Lemma is regarded as *lower semicontinuity* of the L^1 -norm with respect to pointwise (or weak) convergence. That is, if $f_n \rightarrow f$ a.e., then the "energy" of the limit is at most the limit of the energies:

$$\|f\|_{L^1} \leq \liminf_{n \rightarrow \infty} \|f_n\|_{L^1}.$$

Theorem 1.80 (Dominated Convergence). *Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions from $(X, \mathcal{F}, \mu)$ to $\mathbb{R}$ and assume that it converges μ -almost everywhere to a function $f : X \rightarrow \mathbb{R}$, i.e.,*

$$\lim_{n \rightarrow \infty} f_n(x) = f(x), \quad \mu\text{-a.e. } x \in X$$

Assume that there exists an integrable function $g \in \mathcal{L}^1(X, \mathcal{F}, \mu)$ that, for every $n \in \mathbb{N}$,

$$|f_n(x)| \leq g(x) \quad \mu\text{-a.e. } x \in X.$$

Then f is integrable and

$$\lim_{n \rightarrow \infty} \int_X |f - f_n| \, d\mu = 0.$$

In particular,

$$\lim_{n \rightarrow \infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Proof.

We first show that we may assume, without loss of generality, that the hypotheses hold pointwise on X . Let N_1 be a null set such that $f_n(x) \rightarrow f(x)$ for every $x \in N_1^c$, and let N_2 be a null set such that $|f_n(x)| \leq g(x)$ for every $x \in N_2^c$ and every $n \in \mathbb{N}$. Set $S := N_1 \cup N_2$, which is still a null set, by subadditivity of the measure. Let us replace the functions f_n with $\hat{f}_n = f_n \mathbf{1}_{S^c}$ and f with $\hat{f} = f \mathbf{1}_{S^c}$. These replacements change the functions only on the null set S . Therefore, $\int_X |\hat{f}_n - \hat{f}| d\mu = \int_X |f_n - f| d\mu$, because the integral differs on a set of measure 0. Note, however, that the corresponding conditions $\lim_{n \rightarrow \infty} \hat{f}_n(x) = \hat{f}(x)$ and $|\hat{f}_n(x)| \leq g(x)$ now hold pointwise.

Therefore, hereafter we simply assume that $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ holds for every $x \in X$ and $|f_n(x)| \leq g(x)$ holds for every $x \in X$ and $n \in \mathbb{N}$. Since $|f_n| \leq g$ and $f_n \rightarrow f$ pointwise, we also have $|f| \leq g$ pointwise, hence f is integrable and

$$\int_X |f| d\mu \leq \int_X g d\mu < \infty.$$

By the triangle inequality, $|f_n(x) - f(x)| \leq |f_n(x)| + |f(x)| \leq 2g(x)$ for every $x \in X$. Therefore, it is convenient to define the sequence of nonnegative functions $(h_n)_{n \in \mathbb{N}}$ by

$$h_n(x) := 2g(x) - |f_n(x) - f(x)|.$$

Moreover, since $f_n \rightarrow f$, then $h_n \rightarrow 2g$. Then, we have

$$\begin{aligned} \int_X 2g d\mu &= \int_X \lim_{n \rightarrow \infty} h_n d\mu = \int_X \liminf_{n \rightarrow \infty} h_n d\mu \\ &\leq \liminf_{n \rightarrow \infty} \int_X h_n d\mu = \liminf_{n \rightarrow \infty} \int_X (2g - |f_n - f|) d\mu = \int_X 2g d\mu - \limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu, \end{aligned}$$

where the inequality comes from Lemma 1.78. Since $g \in \mathcal{L}^1$, the term $\int_X 2g d\mu$ is finite. Subtracting it from both sides, we obtain:

$$0 \leq -\limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu \implies \limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu \leq 0.$$

Since the integral of the nonnegative function $|f_n - f|$ is always nonnegative, the limit exists and must be zero:

$$\lim_{n \rightarrow \infty} \int_X |f_n - f| d\mu = 0.$$

Finally, by the triangle inequality for integrals:

$$\left| \int_X f_n d\mu - \int_X f d\mu \right| = \left| \int_X (f_n - f) d\mu \right| \leq \int_X |f_n - f| d\mu \rightarrow 0.$$

□

Example 1.81. Let us compute

$$\mathcal{I} := \int_0^1 \frac{-\log(1-x)}{x} dx.$$

Observe that the integrand function has a removable singularity at $x = 0$, where $\lim_{x \rightarrow 0} \frac{-\log(1-x)}{x} = 1$, and diverges to $+\infty$ at $x = 1$. Let us rewrite it as an infinite series by expanding $-\log(1-x)$:

$$\mathcal{I} = \int_0^1 \frac{-\log(1-x)}{x} dx = \int_0^1 \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^n}{n} dx = \int_0^1 \sum_{n=1}^{\infty} \frac{x^{n-1}}{n} dx.$$

Since the function $f_n(x) = x^{n-1}/n$ is measurable and nonnegative for every $x \in (0, 1)$ and every $n \in \mathbb{N}$, we can apply Theorem 1.55:

$$\mathcal{I} = \int_0^1 \sum_{n=1}^{\infty} \frac{x^{n-1}}{n} dx = \sum_{n=1}^{\infty} \frac{1}{n} \int_0^1 x^{n-1} dx = \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

The last series is known to be $\pi^2/6$ (Basel problem). We will prove this result in Example 2.25.

Example 1.82. Let us compute

$$\mathcal{I} := \int_0^1 \frac{e^x - 1}{x} dx.$$

We rewrite it as an infinite series by expanding $e^x - 1$, and then we exchange series and integral because the terms are nonnegative for every $n \in \mathbb{N}$ and every $x \in (0, 1)$:

$$\mathcal{I} = \int_0^1 \frac{e^x - 1}{x} dx = \int_0^1 \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!} dx = \sum_{n=1}^{\infty} \int_0^1 \frac{x^{n-1}}{n!} dx = \sum_{n=1}^{\infty} \frac{1}{n \cdot n!} \approx 1.317$$

Example 1.83 (Exponential Spike). Consider the measure space $((0, +\infty), \mathcal{B}((0, +\infty)), \lambda)$ and define, for every $n \in \mathbb{N}$,

$$f_n(x) := ne^{-nx}, \quad x > 0.$$

Each f_n is measurable and nonnegative, and yet neither MCT nor DCT can be applied. MCT cannot be applied because the sequence $(f_n)_{n \in \mathbb{N}}$ is not increasing. Indeed, for any fixed $x > 0$, we have

$$f_n(x) > f_{n+1}(x) \iff ne^{-nx} > (n+1)e^{-n(x+1/n)} \iff e^x > \frac{n+1}{n}.$$

Since $\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1 < e^x$, the sequence $(f_n(x))_{n \in \mathbb{N}}$ is eventually decreasing towards 0 for every $x > 0$. To see that DCT does not apply, suppose there exists a function g integrable over $(0, \infty)$ such that $f_n \leq g$ λ -a.e. for all $n \in \mathbb{N}$. Fix $\delta \in (0, 1)$ and take $x \in (0, \delta)$. Choose $n := \lceil 1/x \rceil$, so that $1/x \leq n < 1/x + 1$, which implies $1 \leq nx < 1 + x < 2$. Then

$$g(x) \geq f_n(x) = ne^{-nx} \geq \frac{1}{x} e^{-2}.$$

Hence, $g(x) \geq e^{-2}/x$ λ -a.e. in $(0, \delta)$. But, no matter how small δ is,

$$\int_{(0, \delta)} |g| d\lambda \geq \int_0^\delta \frac{e^{-2}}{x} dx = +\infty,$$

meaning that g is not integrable.

Hence, the only achievable result is Fatou's Lemma. Let us compute explicitly both sides:

- *Integral of the Limit.* Fix $x \in (0, \infty)$. Then $f_n(x) = ne^{-nx} \rightarrow 0$ as $n \rightarrow \infty$. Therefore,

$$\int_0^\infty \liminf_{n \rightarrow \infty} f_n d\lambda = \int_0^\infty 0 d\lambda = 0.$$

- *Limit of the Integral.* Fix $n \in \mathbb{N}$. Then

$$\liminf_{n \rightarrow \infty} \int_0^\infty f_n(x) dx = \liminf_{n \rightarrow \infty} \int_0^\infty ne^{-nx} dx = \liminf_{n \rightarrow \infty} [-e^{-nx}]_0^\infty = \liminf_{n \rightarrow \infty} 1 = 1.$$

This example admits a profound physical interpretation. If we regard $f_n(x)$ as a mass density at time step n , the equality $\int_0^\infty f_n(x) dx = 1$ implies that the total mass of the system is conserved at any finite stage. However, the vanishing integral in the limit ($\int_0^\infty \lim_{n \rightarrow \infty} f_n dx = 0$) indicates that the mass is lost at infinity because it concentrates in an arbitrarily small neighborhood of the origin $x = 0$. To recover the mass lost, one must extend the space of functions to the space of *Distributions* $\mathcal{D}'$ (or Radon measures), generalized functions that capture exactly this concentration of unit mass at a single point, a behavior that the classical Lebesgue integral "cannot see" since it assigns measure zero to singletons.

Example 1.84. Let us compute the limit

$$\lim_{n \rightarrow \infty} \int_0^\infty \underbrace{\frac{n \sin(x/n)}{x(1+x^2)}}_{=: f_n(x)} dx.$$

Since

$$|f_n(x)| = \frac{n |\sin(x/n)|}{|x|(1+x^2)} \leq \frac{n|x/n|}{|x|(1+x^2)} = \frac{1}{1+x^2} =: g(x)$$

and

$$\int_0^\infty g(x)dx = \int_0^\infty \frac{1}{1+x^2}dx = [\arctan(x)]_0^\infty = \frac{\pi}{2} < \infty,$$

then the sequence is uniformly dominated by g . Applying the Dominated Convergence theorem,

$$\lim_{n \rightarrow \infty} \int_0^\infty \frac{n \sin(x/n)}{x(1+x^2)}dx = \int_0^\infty \lim_{n \rightarrow \infty} \frac{n \sin(x/n)}{x(1+x^2)}dx = \int_0^\infty \lim_{n \rightarrow \infty} \frac{n(x/n)}{x(1+x^2)}dx = \int_0^\infty \frac{1}{1+x^2}dx = \frac{\pi}{2}.$$

Example 1.85. Let us compute the limit

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{1}{1+x^n}dx.$$

Since

$$\left| \frac{1}{1+x^n} \right| \leq 1 =: g(x) \quad \text{and} \quad \int_0^1 g(x)dx = \int_0^1 1dx = 1,$$

by Dominated Convergence Theorem

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{1}{1+x^n}dx = \int_0^1 \lim_{n \rightarrow \infty} \frac{1}{1+x^n}dx = \int_0^1 1dx = 1.$$

We summarize the previous results here:

- **Swapping two integrals** ($\int_X \int_Y$): The main tool is the **Fubini-Tonelli Theorem**.
- **Swapping two series** ($\sum_i \sum_j$): This is a discrete case of Fubini-Tonelli (using the counting measure).
- **Exchanging integral and series** ($\int \sum$): This can be viewed either as a case of Fubini-Tonelli (product of X and $\mathbb{N}$) or as a limit process.
 - *Nonnegative terms*: Always possible by **MCT** (or Tonelli).
 - *General terms (Fubini)*: Possible if $\int \sum |f_n| < \infty$. Note that if the series of integrals of absolute values converges, then the integral of the series converges and they are equal.
 - *General terms (DCT)*: Possible if partial sums are dominated by an integrable function.
- **Exchanging integral and limit** ($\int \lim$):
 - *MCT*: Use if $0 \leq f_n \uparrow f$.
 - *DCT*: Use if $f_n \rightarrow f$ a.e. and $|f_n| \leq g \in L^1$.
 - *Fatou*: If neither applies (and $f_n \geq 0$), we only get an inequality: $\int \liminf \leq \liminf \int$.

1.7 L^p Spaces and Inequalities

In this section we study a special class of integrable functions that will be useful when defining the moments of a random variable. We follow [1], [3] and [5] as the main references.

Definition 1.86 (p -integrability). Let $(X, \mathcal{F}, \mu)$ be a measure space and $f : X \rightarrow \mathbb{R}$ a measurable function. For any $p \in [1, \infty)$, let

$$\|f\|_p := \left(\int_X |f|^p d\mu \right)^{1/p}.$$

We say that f is p -integrable if $\|f\|_p < \infty$, and we denote by $\mathcal{L}^p(X)$ the space of p -integrable functions.

Definition 1.87 (Essential supremum). Let $(X, \mathcal{F}, \mu)$ be a measure space and $f : X \rightarrow \mathbb{R}$ a measurable function. The *essential supremum* of f is defined as

$$\operatorname{ess\,sup}_{x \in X} f := \inf \left\{ a \in \mathbb{R} : \mu(f^{-1}((a, \infty))) = 0 \right\}$$

if the set is nonempty, and $\operatorname{ess\,sup} f = +\infty$ otherwise. Similarly, one defines

$$\operatorname{ess\,inf}_{x \in X} f := -\operatorname{ess\,sup}_{x \in X} (-f).$$

Definition 1.88 (Essential boundedness). Let $(X, \mathcal{F}, \mu)$ be a measure space and $f : X \rightarrow \mathbb{R}$ a measurable function. We say that f is *essentially bounded* if $\operatorname{ess\,sup} |f| < +\infty$. In this case, we set

$$\|f\|_\infty := \operatorname{ess\,sup} |f|.$$

We denote by $\mathcal{L}^\infty(X)$ the space of essentially bounded functions on X .

Constructing the vector space $L^p(X)$. Due to the invariance modulo null sets of the Lebesgue integral (Proposition 1.58), two functions that differ only on a null set are *essentially* the same. Motivated by this, we consider the following equivalence relation on $\mathcal{L}^p(X)$: $\mathcal{L}^p(X)$:

$$f \sim g \iff f \stackrel{\text{a.e.}}{=} g.$$

We denote by $L^p(X)$ the quotient space $\mathcal{L}^p(X)/\sim$. It is easy to see that $\sim$ is an equivalence relation and $L^p(X)$, endowed with the operations

$$[f] + [g] := [f + g] \quad \text{and} \quad \lambda[f] := [\lambda f], \quad \forall f, g \in \mathcal{L}^p(X), \lambda \in \mathbb{R},$$

is a vector space. To see why, we prove the following trivial bound:

Lemma 1.89. *Let $p \in [1, \infty)$ and $a, b \in \mathbb{R}$. Then*

$$|a + b|^p \leq 2^{p-1}(|a|^p + |b|^p). \quad (1.9)$$

Proof. By Jensen's inequality applied to the function x^p for $x \in \mathbb{R}_+$:

$$\left(\frac{|a| + |b|}{2} \right)^p \leq \frac{|a|^p + |b|^p}{2} \implies |a + b|^p \leq (|a| + |b|)^p \leq 2^{p-1}(|a|^p + |b|^p).$$

□

Thanks to (1.9), we can bound the integral of the sum:

$$\int_X |f + g|^p d\mu \leq 2^{p-1} \left(\int_X |f|^p d\mu + \int_X |g|^p d\mu \right) < \infty. \quad (1.10)$$

This confirms that $f + g \in L^p(X)$ whenever $f, g \in L^p(X)$. To simplify notation, we hereafter write $f \in L^p(X)$ in place of $[f] \in L^p(X)$. For every $p \in [1, \infty]$, we refer to $L^p(X)$ as the L^p -space on X .

Remark 1.90. The elements in L^p are equivalence classes, not functions; as a consequence, it is meaningless to evaluate point-values of elements in L^p , since different functions in the same equivalence class may yield different values at that point.

Theorem 1.91 (Jensen's inequality). *Let $(X, \mathcal{F}, \mu)$ be a measure space with $\mu(X) = 1$, let $f \in L^1(X)$, and let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. Then*

$$\varphi \left(\int_X f d\mu \right) \leq \int_X (\varphi \circ f) d\mu.$$

Proof.

We start with the following lemma:

Lemma 1.92. *If $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function and $x_0 < x_1$ are two given points, then, for every $x_t \in (x_0, x_1)$ it holds*

$$\frac{\varphi(x_t) - \varphi(x_0)}{x_t - x_0} \leq \frac{\varphi(x_1) - \varphi(x_0)}{x_1 - x_0} \leq \frac{\varphi(x_1) - \varphi(x_t)}{x_1 - x_t}.$$

Consequently, for every $x_0 \in \mathbb{R}$ there exist $a, b \in \mathbb{R}$ such that $\varphi(x_0) = ax_0 + b$ and $\varphi(x) \geq ax + b$ for every $x \in \mathbb{R}$.

Proof. The proof involves mere computations. Parametrize x_t as $x_t = (1 - t)x_0 + tx_1$. Observe that

$$\begin{aligned} \frac{\varphi(x_t) - \varphi(x_0)}{[(1 - t)x_0 + tx_1] - x_0} &\leq \frac{\varphi(x_1) - \varphi(x_0)}{x_1 - x_0} \\ \frac{\varphi(x_t) - \varphi(x_0)}{t(x_1 - x_0)} &\leq \frac{\varphi(x_1) - \varphi(x_0)}{x_1 - x_0} \\ \varphi(x_t) - \varphi(x_0) &\leq t\varphi(x_1) - t\varphi(x_0) \\ \varphi(x_t) &\leq (1 - t)\varphi(x_0) + t\varphi(x_1) \\ \varphi((1 - t)x_0 + tx_1) &\leq (1 - t)\varphi(x_0) + t\varphi(x_1), \end{aligned}$$

which holds by definition of convexity of φ . Similarly,

$$\begin{aligned}\frac{\varphi(x_1) - \varphi(x_0)}{x_1 - x_0} &\leq \frac{\varphi(x_1) - \varphi(x_t)}{[x_1 - (1-t)x_0] - tx_1} \\ \frac{\varphi(x_1) - \varphi(x_0)}{\cancel{x_1} - \cancel{x_0}} &\leq \frac{\varphi(x_1) - \varphi(x_t)}{(1-t)\cancel{x_1} - \cancel{x_0}} \\ (1-t)[\varphi(x_1) - \varphi(x_0)] &\leq \varphi(x_1) - \varphi(x_t) \\ \varphi(x_t) &\leq (1-t)\varphi(x_0) + t\varphi(x_1).\end{aligned}$$

The inequalities above imply that the difference quotient $x \mapsto \frac{\varphi(x) - \varphi(x_0)}{x - x_0}$ is increasing; thus the left and right derivatives exist and satisfy $\varphi'_-(x_0) \leq \varphi'_+(x_0)$. Choosing any a in this interval and $b = \varphi(x_0) - ax_0$ yields the result. $\square$

Let $x_0 = \int_X f \, d\mu$. A direct application of this lemma gives

$$\varphi\left(\int_X f \, d\mu\right) = \varphi(x_0) = ax_0 + b = a \int_X f \, d\mu + b \stackrel{(\dagger)}{=} \int_X (af + b) \, d\mu \leq \int_X (\varphi \circ f) \, d\mu,$$

where $(\dagger)$ is valid since $b = b\mu(X) = \int_X b \, d\mu$. $\square$

Theorem 1.93 (Hölder's inequality). *Let $p, q \in (1, \infty)$ satisfy*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then, for any $f \in L^p(X)$ and $g \in L^q(X)$, we have that $fg \in L^1(X)$ and

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

Moreover, if $f \in L^\infty(X)$ and $g \in L^1(X)$, then $fg \in L^1(X)$ and

$$\|fg\|_1 \leq \|f\|_\infty \|g\|_1.$$

Proof.

The conclusion is trivial if $\|f\|_p = 0$ or $\|g\|_q = 0$. Assume next that $\|f\|_p > 0$ and $\|g\|_q > 0$. By convexity of the exponential function $\exp(x)$, for every $x_0, x_1 \in \mathbb{R}$,

$$te^{x_0} + (1-t)e^{x_1} \geq e^{tx_0 + (1-t)x_1}.$$

Applying this inequality on the pair $(p \log \frac{|f(x)|}{\|f\|_p}, q \log \frac{|g(x)|}{\|g\|_q})$, with weights $1/p$ and $1/q$, for every $x \in X$ such that $f(x) \neq 0$ and $g(x) \neq 0$:

$$\frac{|f(x)|}{\|f\|_p} \frac{|g(x)|}{\|g\|_q} = \exp\left(\frac{1}{p} \log \frac{|f(x)|}{\|f\|_p} + \frac{1}{q} \log \frac{|g(x)|}{\|g\|_q}\right) \leq \frac{e^{p \log \frac{|f(x)|}{\|f\|_p}}}{p} + \frac{e^{q \log \frac{|g(x)|}{\|g\|_q}}}{q} = \frac{|f(x)|^p}{p \cdot \|f\|_p^p} + \frac{|g(x)|^q}{q \cdot \|g\|_q^q}.$$

(The inequality is trivial when $f(x) = 0$ or $g(x) = 0$.)

Integrating over X with respect to μ yields

$$\frac{\int_X |fg| \, d\mu}{\|f\|_p \|g\|_q} \leq \frac{1}{p} \frac{\int_X |f|^p \, d\mu}{\|f\|_p^p} + \frac{1}{q} \frac{\int_X |g|^q \, d\mu}{\|g\|_q^q} = \frac{1}{p} + \frac{1}{q} = 1,$$

hence

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

As a byproduct, $fg \in L^1(X)$.

Let $f \in L^\infty(X)$ and $g \in L^1(X)$. By definition of the essential supremum, we have $|f(x)| \leq \|f\|_\infty$ for μ -almost every $x \in X$. Multiplying by $|g(x)|$, we obtain:

$$|f(x)g(x)| \leq \|f\|_\infty |g(x)| \quad \mu\text{-a.e.}$$

Integrating both sides over X :

$$\|fg\|_1 = \int_X |f(x)g(x)| \, d\mu \leq \int_X \|f\|_\infty |g(x)| \, d\mu = \|f\|_\infty \int_X |g(x)| \, d\mu = \|f\|_\infty \|g\|_1.$$

This confirms $fg \in L^1(X)$ and proves the inequality. $\square$

Theorem 1.94 (Minkowski's inequality). *Let $p \in [1, \infty]$ and let $(X, \mathcal{F}, \mu)$ be a measure space. If $f, g \in L^p(X)$, then $f + g \in L^p(X)$ and we have the triangle inequality*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p,$$

with equality for $p \in (1, \infty)$ if and only if f and g are positive scalar multiples of each other.

Proof.

For $p = 1$, the inequality is immediate from $|f + g| \leq |f| + |g|$ by integrating. For $p = \infty$, it follows from $|f(x) + g(x)| \leq |f(x)| + |g(x)|$ and taking essential suprema.

Let $p \in (1, \infty)$. Thanks to (1.10), we can legitimately talk about $\|f + g\|_p$. If it is zero, then Minkowski's inequality holds. We now assume that $\|f + g\|_p \in (0, \infty)$.

Using the pointwise inequality $|f + g| \leq |f| + |g|$ and then Hölder's inequality, we find that

$$\begin{aligned} \|f + g\|_p^p &= \int_X |f + g|^p \, d\mu = \int_X |f + g| \cdot |f + g|^{p-1} \, d\mu \\ &\leq \int_X (|f| + |g|) |f + g|^{p-1} \, d\mu \\ &= \int_X |f| |f + g|^{p-1} \, d\mu + \int_X |g| |f + g|^{p-1} \, d\mu \\ &\leq \|f\|_p \| |f + g|^{p-1} \|_q + \|g\|_p \| |f + g|^{p-1} \|_q, \end{aligned}$$

where $q = \frac{p}{p-1}$ is the conjugate exponent of p . Hence

$$\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p) \| |f + g|^{p-1} \|_q.$$

Finally,

$$\| |f + g|^{p-1} \|_q = \left(\int_X |f + g|^{(p-1)q} \, d\mu \right)^{1/q} = \left(\int_X |f + g|^p \, d\mu \right)^{(p-1)/p} = \|f + g\|_p^{p-1}.$$

Therefore,

$$\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p) \|f + g\|_p^{p-1}.$$

Since $\|f + g\|_p > 0$, dividing both sides by $\|f + g\|_p^{p-1}$ yields

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p,$$

which is Minkowski's inequality. $\square$

Proposition 1.95 (L^p norm). *For every $p \in [1, \infty]$, $\|\cdot\|_p$ is a norm. Hence, $L^p(X)$ is a normed vector space.*

Proof.

Assume first $p \in [1, \infty)$. By definition, for every $f \in L^p(X)$,

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p} \geq 0,$$

with equality if and only if $f \equiv 0$ μ -a.e. (by Corollary 1.59) which, in L^p , implies $f = 0$. Moreover, for

every $\alpha \in \mathbb{R}$,

$$\|\alpha f\|_p = \left(\int_X |\alpha f|^p d\mu \right)^{1/p} = \left(|\alpha|^p \int_X |f|^p d\mu \right)^{1/p} = |\alpha| \left(\int_X |f|^p d\mu \right)^{1/p} = |\alpha| \|f\|_p.$$

Finally, triangle inequality corresponds to Theorem 1.94.

Assume now $p = \infty$. By definition, for every $f \in L^\infty(X)$,

$$\|f\|_\infty = \operatorname{ess\,sup}_{x \in X} |f| \geq 0,$$

with equality if and only if $f \equiv 0$ μ -a.e. which, in L^∞ , implies $f = 0$. Moreover, for every $\alpha \in \mathbb{R}$,

$$\|\alpha f\|_\infty = \operatorname{ess\,sup}_{x \in X} |\alpha f| = |\alpha| \operatorname{ess\,sup}_{x \in X} |f| = |\alpha| \|f\|_\infty.$$

Finally, triangle inequality follows from the definition of essential supremum: if $f, g \in L^\infty(X)$, then $|f(x) + g(x)| \leq |f(x)| + |g(x)|$ for every $x \in X$ hence

$$\operatorname{ess\,sup}_{x \in X} |f + g| \leq \operatorname{ess\,sup}_{x \in X} |f| + \operatorname{ess\,sup}_{x \in X} |g| \iff \|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty.$$

□

The following Lemma will be used for proving Theorem 2.45.

Lemma 1.96. *Let $(f_n)_{n \in \mathbb{N}} \subseteq L^1(X)$ be a sequence of integrable functions on a measure space $(X, \mathcal{F}, \mu)$. If $f_n \rightarrow f$ in $L^1(X)$ for some $f \in L^1(X)$, then there exists a subsequence $(f_{n_k})_{k \in \mathbb{N}}$ such that*

$$f_{n_k}(x) \rightarrow f(x) \quad \mu\text{-a.e.}$$

Proof.

Since $f_n \rightarrow f$ in $L^1(X)$, we can extract a subsequence $(f_{n_k})_{k \in \mathbb{N}}$ such that

$$\|f_{n_k} - f\|_{L^1} < \frac{1}{4^k}, \quad \forall k \in \mathbb{N}.$$

Define, for every $k \in \mathbb{N}$, the measurable set

$$A_k := \left\{ x \in X : |f_{n_k}(x) - f(x)| > \frac{1}{2^k} \right\}.$$

Clearly, for every $k \in \mathbb{N}$, it holds that

$$\mu(A_k) \leq 2^k \int_X |f_{n_k} - f| d\mu = 2^k \|f_{n_k} - f\|_{L^1} \leq 2^k \cdot 2^{-2k} = \frac{1}{2^k},$$

which implies

$$\sum_{k=1}^{\infty} \mu(A_k) \leq 1 < \infty.$$

Now, consider the set of points where infinitely many A_k occur:

$$A := \limsup_{k \rightarrow \infty} A_k := \bigcap_{N=1}^{\infty} \bigcup_{k=N}^{\infty} A_k.$$

By continuity from above and countable subadditivity,

$$\mu(A) = \lim_{N \rightarrow \infty} \mu \left(\bigcup_{k=N}^{\infty} A_k \right) \leq \lim_{N \rightarrow \infty} \sum_{k=N}^{\infty} \mu(A_k) = 0.$$

Therefore, $\mu(A) = 0$. Finally, fix $x \in X \setminus A$. By definition of $\limsup$, there exists an integer $M(x)$ such that $x \notin A_k$ for all $k \geq M(x)$, that is,

$$|f_{n_k}(x) - f(x)| \leq \frac{1}{2^k}, \quad \forall k \geq M(x).$$

Since $1/2^k \rightarrow 0$, we conclude that $f_{n_k}(x) \rightarrow f(x)$ for every $x \in X \setminus A$, and hence for μ -a.e. $x \in X$. □

The next theorem shows that the L^p spaces are Banach, i.e., every *Cauchy sequence* converges. A sequence $(a_n)_{n \in \mathbb{N}}$ is called a Cauchy sequence if, for every $\varepsilon > 0$, there exists an index $n_0 \in \mathbb{N}$ such that $\|a_n - a_m\| < \varepsilon$ for every $n, m \geq n_0$. The proof is largely taken from [6].

Theorem 1.97 (L^p spaces are Banach). *For every $p \in [1, \infty]$, the space $L^p(X, \mu)$ is complete.*

Proof.

Let $(f_n)_{n \in \mathbb{N}} \subseteq L^p(X, \mu)$ be a Cauchy sequence. It suffices to show that $(f_n)_{n \in \mathbb{N}}$ admits a convergent subsequence. Indeed, assume that a subsequence $(f_{n_i})_{i \in \mathbb{N}}$ converges to some f in L^p . Fix $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$\|f_n - f_m\|_{L^p} < \varepsilon/2 \quad \forall n, m \geq N,$$

and choose $i_0 \in \mathbb{N}$ such that $n_{i_0} \geq N$ and

$$\|f_{n_{i_0}} - f\|_{L^p} < \varepsilon/2.$$

Then, for every $n \geq N$,

$$\|f_n - f\|_{L^p} \leq \|f_n - f_{n_{i_0}}\|_{L^p} + \|f_{n_{i_0}} - f\|_{L^p} < \varepsilon.$$

Hence $f_n \rightarrow f$ in L^p .

Thanks to this property, we extract an increasing subsequence $(f_{n_i})_{i \in \mathbb{N}}$ such that

$$\|f_{n_{i+1}} - f_{n_i}\|_{L^p} \leq 2^{-i}, \quad \forall i \in \mathbb{N},$$

and we show that it converges to some $f \in L^p$. This would conclude the proof. Define

$$S_k := \sum_{i=1}^k |f_{n_{i+1}} - f_{n_i}|, \quad S := \sum_{i=1}^{\infty} |f_{n_{i+1}} - f_{n_i}| = \lim_{k \rightarrow \infty} S_k.$$

Case 1: $p < \infty$. By Minkowski's inequality (1.94), for every $k \in \mathbb{N}$,

$$\|S_k\|_{L^p} = \left\| \sum_{i=1}^k |f_{n_{i+1}} - f_{n_i}| \right\|_{L^p} \leq \sum_{i=1}^k \|f_{n_{i+1}} - f_{n_i}\|_{L^p} \leq \sum_{i=1}^k 2^{-i} \leq 1.$$

Moreover, $(S_k)_{k \in \mathbb{N}}$ is pointwise nondecreasing and $S_k \rightarrow S$ pointwise. Hence, by the Monotone Convergence Theorem 1.50,

$$\|S\|_{L^p}^p = \int_X S^p d\mu = \lim_{k \rightarrow \infty} \int_X S_k^p d\mu = \lim_{k \rightarrow \infty} \|S_k\|_{L^p}^p \leq 1,$$

so $S \in L^p(X, \mu)$. In particular, $S(x) < \infty$ for μ -a.e. x (Corollary 1.61).

Define, for every x such that $S(x) < \infty$,

$$f(x) := f_{n_1}(x) + \sum_{i=1}^{\infty} (f_{n_{i+1}}(x) - f_{n_i}(x)),$$

and set $f(x) := 0$ otherwise (on a null set). Since

$$\sum_{i=1}^{\infty} |f_{n_{i+1}}(x) - f_{n_i}(x)| \leq S(x) < \infty,$$

the series defining $f(x)$ converges absolutely for μ -a.e. x . Moreover, for every $m \in \mathbb{N}$ and μ -a.e. x ,

$$f(x) - f_{n_m}(x) = \sum_{i=m}^{\infty} (f_{n_{i+1}}(x) - f_{n_i}(x)),$$

hence

$$|f(x) - f_{n_m}(x)| \leq \sum_{i=m}^{\infty} |f_{n_{i+1}}(x) - f_{n_i}(x)|.$$

Taking L^p -norms and using Minkowski's inequality again yields

$$\|f - f_{n_m}\|_{L^p} \leq \sum_{i=m}^{\infty} \|f_{n_{i+1}} - f_{n_i}\|_{L^p} \leq \sum_{i=m}^{\infty} 2^{-i} \xrightarrow{m \rightarrow \infty} 0.$$

Therefore $f_{n_m} \rightarrow f$ in L^p .

Case 2. $p = \infty$ For every $k \in \mathbb{N}$ set

$$S_k(x) := \sum_{i=1}^k |f_{n_{i+1}}(x) - f_{n_i}(x)|, \quad S(x) := \sum_{i=1}^{\infty} |f_{n_{i+1}}(x) - f_{n_i}(x)|.$$

Then, for every $k \in \mathbb{N}$ and every $x \in X$,

$$S_k(x) \leq \sum_{i=1}^k \|f_{n_{i+1}} - f_{n_i}\|_{\infty} \leq \sum_{i=1}^k 2^{-i} \leq 1,$$

hence $S(x) \leq 1$ for all x , i.e. $S \in L^{\infty}(X, \mu)$. Therefore the series

$$\sum_{i=1}^{\infty} (f_{n_{i+1}}(x) - f_{n_i}(x))$$

converges absolutely for every x , and we may define

$$f(x) := f_{n_1}(x) + \sum_{i=1}^{\infty} (f_{n_{i+1}}(x) - f_{n_i}(x)).$$

For every $m \in \mathbb{N}$ and every $x \in X$,

$$|f(x) - f_{n_m}(x)| \leq \sum_{i=m}^{\infty} |f_{n_{i+1}}(x) - f_{n_i}(x)| \leq \sum_{i=m}^{\infty} \|f_{n_{i+1}} - f_{n_i}\|_{\infty} \leq \sum_{i=m}^{\infty} 2^{-i},$$

so taking the supremum over x yields

$$\|f - f_{n_m}\|_{\infty} \leq \sum_{i=m}^{\infty} 2^{-i} \xrightarrow{m \rightarrow \infty} 0.$$

Thus $f_{n_m} \rightarrow f$ in L^{∞} . □

How smooth are L^p functions? The following approximation theorem addresses this question: we can approximate (in L^p sense) any L^p -integrable function on an open subset $\Omega \subseteq \mathbb{R}^n$ by a sequence of continuous functions with compact support.

Theorem 1.98 (Density of C_c in L^p). *Let X be a locally compact Hausdorff space, and let μ be a regular Borel measure on X such that $\mu(K) < \infty$ for every compact set K . For every $f \in L^p(X, \mu)$, there exists a sequence $(f_n)_{n \in \mathbb{N}} \subseteq C_c(X, \mathbb{R})$ such that $f_n \rightarrow f$ in $L^p(X, \mu)$, namely*

$$\|f_n - f\|_{L^p} \xrightarrow{n \rightarrow \infty} 0,$$

which is equivalent to

$$\lim_{n \rightarrow \infty} \int_X |f_n - f|^p d\mu = 0.$$

Remark 1.99. If X is a topological space, the *support* of a function $f : X \rightarrow \mathbb{R}$ is defined as $\text{supp } f := \overline{\{x \in X : f(x) \neq 0\}}$, that is, the set of points $x \in X$ that have a neighbourhood where f is *not* identically zero. We say that f has *compact support* if $\text{supp } f$ is compact. If $X = \mathbb{R}^n$, the *Heine-Borel theorem* states that $\text{supp } f$ is compact if and only if it is closed and bounded. Intuitively, this means that a function with compact support on $\mathbb{R}^n$ "lives" in a finite region of space, vanishing identically outside of a sufficiently large ball.

Theorem 1.98 can be obtained from a deeper approximation result, known as *Lusin's theorem*. We reproduce the statement from [3].

Theorem 1.100 (Lusin). Let X be locally compact Hausdorff space, μ be a regular Borel measure on X and $\mu(K) < \infty$ for every compact set K . For every measurable function f with finite measure support ($\mu(\text{supp } f) < \infty$) and every $\varepsilon > 0$, there exists a continuous function with compact support converging to f in measure, i.e., $\mu(\{x : f(x) \neq g(x)\}) < \varepsilon$.

The next theorem has important applications in probability theory: roughly speaking, if $f \in L^q$, then $f \in L^p$ for every $1 \leq p \leq q$, with an explicit constant (depending only on p , q , and $\mu(X)$) controlling the norms. Mathematically, one can write

$$L^q \hookrightarrow L^p, \quad 1 \leq p \leq q$$

Theorem 1.101 (Lyapunov's inequality). Let $(X, \mathcal{F}, \mu)$ be a measure space with $\mu(X) < \infty$, and let $1 \leq p \leq q \leq \infty$. Then $L^q(X) \subseteq L^p(X)$ and, for every $f \in L^q(X)$,

$$\|f\|_p \leq \mu(X)^{\frac{1}{p} - \frac{1}{q}} \|f\|_q,$$

with the convention $1/\infty := 0$.

Proof.

If $p = q$ the claim is trivial. Assume $p < q$. If $q = \infty$, the result is immediate since $|f| \leq \|f\|_\infty$ a.e. implies $\int |f|^p \leq \|f\|_\infty^p \mu(X)$. Assume $q < \infty$. We apply Hölder's inequality on the pair of functions $(f, g) := (|f|^p, 1)$ with pair of conjugate exponents $(\frac{q}{p}, \frac{q}{q-p})$:

$$\int_X |f|^p \cdot 1 \, d\mu \leq \left(\int_X (|f|^p)^{\frac{q}{p}} \, d\mu \right)^{\frac{p}{q}} \left(\int_X 1^{\frac{q}{q-p}} \, d\mu \right)^{1-\frac{p}{q}} = \|f\|_q^p \cdot \mu(X)^{1-\frac{p}{q}}.$$

Taking the p -th root on both sides yields $\|f\|_p \leq \|f\|_q \cdot \mu(X)^{\frac{1}{p} - \frac{1}{q}}$. In particular, $L^q(X) \subseteq L^p(X)$. $\square$

Example 1.102 (A counterexample on $\mathbb{R}$). Observe that on $\mathbb{R}$ the assumption $\lambda(\mathbb{R}) < \infty$ fails. Consequently, the theorem does not hold. For instance, take $p = 2$ and $q = 3$, and define the real-valued function

$$f(x) := |x|^{-2/5} \mathbf{1}_{x \geq 1}.$$

Then

$$\|f\|_2^2 = \int_1^\infty x^{-4/5} \, dx = \infty,$$

while

$$\|f\|_3^3 = \int_1^\infty x^{-6/5} \, dx = \lim_{t \rightarrow +\infty} [-5x^{-1/5}]_1^t = 5.$$

Hence, $f \in L^3$ but $f \notin L^2$, contradicting $L^3 \subseteq L^2$.

Lemma 1.103 (Limit of L^p norms). Let $f \in L^1 \cap L^\infty$. Then $f \in L^p(X)$ for every $p \in [1, \infty]$ and

$$\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty.$$

Proof.

For $p \geq 1$,

$$\|f\|_p^p = \int_X |f|^p \, d\mu = \int_X |f|^{p-1} |f| \, d\mu \leq \|f\|_\infty^{p-1} \int_X |f| \, d\mu = \|f\|_\infty^{p-1} \|f\|_1,$$

Taking the p -th root:

$$\|f\|_p \leq (\|f\|_\infty^{p-1} \|f\|_1)^{1/p} = \|f\|_\infty^{1-1/p} \|f\|_1^{1/p}, \quad (1.11)$$

As $p \rightarrow \infty$, the exponents tend to 1 and 0, respectively. Even if we do not know whether $\lim_{p \rightarrow \infty} \|f\|_p$ exists, (1.11) lets us bound the lim sup, which always exists:

$$\limsup_{p \rightarrow \infty} \|f\|_p \leq \|f\|_\infty.$$

As for the converse inequality, if $\|f\|_\infty = 0$, the result is trivial. Assume $\|f\|_\infty > 0$. Fix any constant $0 < a < \|f\|_\infty$ and consider the set $E_a := \{x \in X : |f(x)| > a\}$. Of course, $\mu(E_a) < \infty$: indeed,

$$\mu(E_a) = \int_{E_a} d\mu \leq \int_{E_a} \frac{|f|}{a} \, d\mu = \frac{1}{a} \int_{E_a} |f| \, d\mu \leq \frac{1}{a} \int_X |f| \, d\mu < \infty.$$

Now, estimate the L^p norm from below: since $|f|^p \geq a^p$ on E_a , then

$$\|f\|_p^p = \int_X |f|^p d\mu \geq \int_{E_a} |f|^p d\mu \geq \int_{E_a} a^p d\mu = a^p \mu(E_a) \implies \|f\|_p \geq a \sqrt[p]{\mu(E_a)}$$

Taking the limit as $p \rightarrow \infty$:

$$\liminf_{p \rightarrow \infty} \|f\|_p \geq a \cdot \lim_{p \rightarrow \infty} \mu(E_a)^{1/p} = a \cdot 1 = a,$$

where we used the fact that $\mu(E_a)$ is a finite, positive number. Since this inequality holds for any $a < \|f\|_\infty$, taking the supremum over such a yields:

$$\liminf_{p \rightarrow \infty} \|f\|_p \geq \|f\|_\infty.$$

This concludes the proof. □

1.8 Densities and Measure Decompositions

In probability theory, it is pedagogically effective to distinguish between "discrete" measurable functions and "continuous" measurable functions. *Discrete* measurable functions concentrate the mass on a countable set of points, while *continuous* measurable functions spread the mass smoothly over intervals, without concentration on single points. Consider, for instance, these different behaviours:

- Let $(\mathbb{N}, 2^{\mathbb{N}}, \#)$ be the measurable set of the natural numbers endowed with the counting measure, and let $f : \mathbb{N} \rightarrow [0, \infty)$ be defined as $f(n) = \frac{2^n}{n!}$. Then, for every $S \in 2^{\mathbb{N}}$,

$$\int_S f d\# = \sum_{n \in S} \frac{2^n}{n!}.$$

Here, the measure of a single point is positive ($\#(\{n\}) = 1$) and the integral is built from these atoms.

- Let $f : (\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda) \rightarrow [0, \infty)$ be defined as $f(x) = xe^{-x} \mathbf{1}_{(0, +\infty)}$. Then, for every interval $(a, b) \subseteq (0, +\infty)$,

$$\int_{(a,b)} f d\lambda = \int_a^b xe^{-x} d\lambda.$$

While this distinction is natural, it is mathematically mature to ask whether it is exhaustive. Is every measure (or measurable function) strictly either discrete or continuous? Or perhaps a simple sum of the two? The answer, unfortunately, is negative. There exist pathological objects (such as the Cantor distribution) that are "continuous" in the sense that single points have zero mass, yet are "singular" because they live on sets of zero Lebesgue measure. This complexity forces us to look more closely at what "discrete" and "continuous" mean in measure theory, and at what may sit between them.

In what follows, we assume that $(X, \mathcal{F})$ is a measurable space with canonical measure ν , and we aim to classify a measure μ based on ν . We follow [2], [3] and [5]. Proofs for this section will be provided in future editions.

Definition 1.104 (Absolute Continuity). Let μ, ν be measures on $(X, \mathcal{F})$. We say that μ is *absolutely continuous* with respect to ν , and we write $\mu \ll \nu$, if

$$\nu(A) = 0 \implies \mu(A) = 0 \quad \forall A \in \mathcal{F}.$$

Definition 1.105 (Singularity). Let μ, ν be measures on $(X, \mathcal{F})$. We say that μ is *singular* with respect to ν , and we write $\mu \perp \nu$, if there exists $N \in \mathcal{F}$ such that

$$\mu(N) = 0 \quad \text{and} \quad \nu(N^c) = 0.$$

Equivalently, μ is concentrated on a ν -null set.

Lemma 1.106. Let μ, ν be measures on $(X, \mathcal{F})$. If $\mu \perp \nu$ and $\mu \ll \nu$, then $\mu = 0$.

Proof.

Let N be such that $\mu(N) = 0$ and $\nu(N^c) = 0$. Since $\mu \ll \nu$, $\nu(N^c) = 0$ implies $\mu(N^c) = 0$. Hence $\mu(X) = \mu(N) + \mu(N^c) = 0$, so $\mu = 0$. $\square$

Theorem 1.107. Let μ, ν be σ -finite measures on $(X, \mathcal{F})$. Then there exists a measure μ_s and a nonnegative measurable function $f : (X, \mathcal{F}) \rightarrow \mathbb{R}$ such that

$$\mu_s \perp \nu \quad \text{and} \quad \mu = f \cdot \nu + \mu_s.$$

Moreover, in this decomposition the measure μ_s is unique and the function f is unique ν -a.e.

Proof. See [3]. $\square$

Remark 1.108. This decomposition resembles an orthogonal decomposition in an absolutely continuous part $f \cdot \nu$ and a singular part μ_s . Observe that the decomposition depends on the chosen reference measure ν . Hence, the same measure μ may be absolutely continuous with respect to some ν_1 and singular with respect to another ν_2 .

Theorem 1.109 (Radon-Nikodym). Let μ, ν be σ -finite measures on $(X, \mathcal{F})$. If $\mu \ll \nu$, there exists a measurable function $f : (X, \mathcal{F}) \rightarrow [0, +\infty]$, unique ν -a.e., such that

$$\mu(A) = \int_A f d\nu \quad \forall A \in \mathcal{F}.$$

We write $f = \frac{d\mu}{d\nu}$ and call it the density of μ with respect to ν . Moreover, $\frac{d\mu}{d\nu} \in \mathcal{L}^1(X, \nu)$ if and only if μ is finite. In that case, $\mu(X) = \int_X f d\nu$.

Proof. See [3]. $\square$

Corollary 1.110 (Lebesgue Decomposition). Let μ, ν be σ -finite measures on $(X, \mathcal{F})$. Then

$$\mu = \mu_{ac} + \mu_s, \quad \mu_{ac} \ll \nu, \quad \mu_s \perp \nu.$$

This decomposition is unique, thus

$$\mu(A) = \int_A f d\nu + \mu_s(A).$$

Proof. See [3]. $\square$

Theorem 1.111 (Change of measure). Let μ, ν be σ -finite measures on the measurable space $(X, \mathcal{F})$, and assume $\nu \ll \mu$. Denoting by $f = \frac{d\nu}{d\mu}$ the Radon-Nikodym derivative of ν with respect to μ ,

$$\int_X g d\nu = \int_X g \cdot f d\mu,$$

for every $g : X \rightarrow \overline{\mathbb{R}}$ for which the integral is well-defined (g nonnegative and measurable or g ν -integrable).

Proof. See [3]. $\square$

1.9 Change of Variables

In this section we develop results for computing integrals in several dimensions.

Definition 1.112 (Pushforward measure). Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces, and let $T : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ be measurable. Given a measure μ on $(X, \mathcal{F})$, the *pushforward* of μ through T is the measure $T_{\#}\mu$ on $(Y, \mathcal{G})$ defined by

$$(T_{\#}\mu)(B) := \mu(T^{-1}(B)), \quad \forall B \in \mathcal{G}.$$

Proposition 1.113 (Properties of the pushforward). Let $(X, \mathcal{F})$, $(Y, \mathcal{G})$, $(Z, \mathcal{H})$ be measurable spaces, and let $T : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $S : (Y, \mathcal{G}) \rightarrow (Z, \mathcal{H})$ be measurable maps. If μ is a measure on $(X, \mathcal{F})$, then:

- i) $T_{\#}\mu$ is a measure on $(Y, \mathcal{G})$;
- ii) $(id_X)_{\#}\mu = \mu$;
- iii) $(S \circ T)_{\#}\mu = S_{\#}(T_{\#}\mu)$.

Proof.

(i). Since $\mu(B) \geq 0$ for every $B \in \mathcal{F}$, then $(T_{\#}\mu)(B) = \mu(T^{-1}(B)) \geq 0$; let us verify countable additivity. Take a sequence $(B_n)_{n \in \mathbb{N}} \subseteq \mathcal{G}$ of pairwise disjoint measurable sets. Then $(T^{-1}(B_n))_{n \in \mathbb{N}}$ are also pairwise disjoint and measurable. Hence

$$(T_{\#}\mu)\left(\bigcup_{n \in \mathbb{N}} B_n\right) = \mu\left(T^{-1}\left(\bigcup_{n \in \mathbb{N}} B_n\right)\right) = \mu\left(\bigcup_{n \in \mathbb{N}} T^{-1}(B_n)\right) = \sum_{n=1}^{\infty} \mu(T^{-1}(B_n)) = \sum_{n=1}^{\infty} (T_{\#}\mu)(B_n).$$

(ii). For every $B \in \mathcal{F}$, $(\text{id}_X)_{\#}\mu(B) = \mu(B)$.

(iii). For any $C \in \mathcal{H}$,

$$((S \circ T)_{\#}\mu)(C) = \mu((S \circ T)^{-1}(C)) = \mu(T^{-1}(S^{-1}(C))) = (T_{\#}\mu)(S^{-1}(C)) = (S_{\#}(T_{\#}\mu))(C).$$

□

Theorem 1.114 (Change of variables). Let $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G})$ be measurable spaces, and let $f : X \rightarrow Y$ be measurable. Then for every $\mathcal{G}$ -measurable function $g : Y \rightarrow [0, \infty]$ one has

$$\int_Y g d(f_{\#}\mu) = \int_X (g \circ f) d\mu.$$

Equivalently,

$$\int_Y g(y) d(\mu(f^{-1}))(y) = \int_X g(f(x)) d\mu(x).$$

Remark 1.115. The condition $g : Y \rightarrow [0, +\infty]$ is imposed only to avoid integrability issues. One may equivalently assume $g : Y \rightarrow \mathbb{R}$ and $g \in L^1(Y)$.

Proof.

Let $\nu := \mu(f^{-1}) = f_{\#}\mu$. We first prove $\int_Y \varphi d(f_{\#}\mu) = \int_X (\varphi \circ f) d\mu$ for a simple function $\varphi = \sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$ written in canonical form. By linearity of the Lebesgue integral:

$$\begin{aligned} \int_Y \varphi d(f_{\#}\mu) &= \int_Y \left(\sum_{i=1}^n \alpha_i \mathbf{1}_{A_i} \right) d(f_{\#}\mu) = \sum_{i=1}^n \alpha_i \int_Y \mathbf{1}_{A_i} d(f_{\#}\mu) = \sum_{i=1}^n \alpha_i (f_{\#}\mu)(A_i) \\ &= \sum_{i=1}^n \alpha_i \cdot \mu(f^{-1}(A_i)) = \sum_{i=1}^n \alpha_i \cdot \int_X \mathbf{1}_{f^{-1}(A_i)} d\mu = \int_X \left(\sum_{i=1}^n \alpha_i \mathbf{1}_{f^{-1}(A_i)} \right) d\mu \\ &\stackrel{\dagger}{=} \int_X (\varphi \circ f) d\mu, \end{aligned}$$

where $\dagger$ holds since $x \in f^{-1}(A_i)$ if and only if $f(x) \in A_i$. Now let $g : Y \rightarrow [0, \infty]$ be measurable. Choose a sequence of nonnegative simple functions $(g_n)_{n \in \mathbb{N}}$ such that $g_n \uparrow g$ pointwise on Y (Lemma 1.52). Then also $g_n \circ f \uparrow g \circ f$ pointwise on X . By the Monotone Convergence Theorem (Theorem 1.50) applied twice,

$$\int_Y g d(f_{\#}\mu) = \lim_{n \rightarrow \infty} \int_Y g_n d(f_{\#}\mu) = \lim_{n \rightarrow \infty} \int_X (g_n \circ f) d\mu = \int_X \lim_{n \rightarrow \infty} (g_n \circ f) d\mu = \int_X (g \circ f) d\mu.$$

□

Example 1.116 (Jacobian determinant as a density). Let $T : U \subseteq \mathbb{R}^d \rightarrow V \subseteq \mathbb{R}^d$ be a C^1 diffeomorphism between open sets, and let λ denote the Lebesgue measure. We compare the abstract pushforward formula (Theorem 1.114) with the classical change of variables for a measurable function $g : V \rightarrow [0, +\infty]$:

$$\int_V g(y) d(T_{\#}\lambda)(y) = \int_U g(T(x)) d\lambda(x); \quad (1.12)$$

$$\int_V g(y) dy = \int_U g(T(x)) |\det DT(x)| dx. \quad (1.13)$$

Using the linear algebra identity $|\det DT(x)| = |\det DT^{-1}(T(x))|^{-1}$ in (1.13), we obtain:

$$\int_V g(y) dy = \int_U \frac{g(T(x))}{|\det DT^{-1}(T(x))|} d\lambda(x).$$

By applying (1.12) to the function $y \mapsto \frac{g(y)}{|\det DT^{-1}(y)|}$, the RHS becomes:

$$\int_V g(y) \, dy = \int_V \frac{g(y)}{|\det DT^{-1}(y)|} \, d(T_{\#}\lambda)(y).$$

Since this holds for every g , we conclude that $dy = |\det DT^{-1}|^{-1} d(T_{\#}\lambda)$, or equivalently:

$$\frac{d(T_{\#}\lambda)}{d\lambda} = |\det DT^{-1}|.$$

The previous example is formalized in the following Proposition, due to [5]:

Proposition 1.117. *Let U, V be two open subsets of $\mathbb{R}^d$ and let $T : U \rightarrow V$ be one to one, continuously differentiable with $\det J_{T^{-1}} \neq 0$ on V , where $J_{T^{-1}}$ is the Jacobian matrix of T^{-1} ; then*

$$\frac{d(\lambda T^{-1})}{d\lambda} = |\det J_{T^{-1}}|.$$

2 Harmonic Analysis

In this section we review the key results of Fourier analysis. We work over the complex field and assume familiarity with basic complex-valued integration (we provide some basics of complex analysis in the Appendix A). We mainly follow [7], [8] and [9].

Definition 2.1 (Complex integration). Let $(X, \mathcal{F}, \mu)$ be a measure space and let $f : X \rightarrow \mathbb{C}$ be a complex-valued function. We can uniquely decompose f into its real and imaginary parts as $f(x) = u(x) + iv(x)$, where $u, v : X \rightarrow \mathbb{R}$ are real-valued functions.

We say that f is integrable if both u and v are μ -measurable and integrable. In this case, the Lebesgue integral of f is defined linearly as:

$$\int_X f d\mu = \int_X u d\lambda + i \int_X v d\mu.$$

Setting. We work on the space $L^2([-\pi, \pi])$ (abbreviated as L^2), namely the space of (equivalence classes of) measurable functions $f : [-\pi, \pi] \rightarrow \mathbb{C}$ such that

$$\int_{-\pi}^{\pi} |f(x)|^2 d\lambda(x) = \int_{[-\pi, \pi]} (u^2(x) + v^2(x)) d\lambda(x) < \infty,$$

where λ denotes Lebesgue measure and $f(x) = u(x) + iv(x)$. We endow $L^2([-\pi, \pi])$ with the inner product $\langle \cdot, \cdot \rangle : L^2 \times L^2 \rightarrow \mathbb{C}$ defined by

$$\langle f, g \rangle := \int_{-\pi}^{\pi} f(x) \overline{g(x)} d\lambda(x), \quad \forall f, g \in L^2([-\pi, \pi]). \quad (2.1)$$

Lemma 2.2. The map $\langle \cdot, \cdot \rangle : L^2 \times L^2 \rightarrow \mathbb{C}$ defined by (2.1) is an inner product.

Proof.

Let $\alpha, \beta \in \mathbb{C}$ be constants and $f, g, h \in L^2([-\pi, \pi])$. Then, by linearity of the Lebesgue integral (these properties hold also in $\mathbb{C}$),

$$\begin{aligned} \langle f, g \rangle &= \int_{[-\pi, \pi]} f \bar{g} d\lambda = \int_{[-\pi, \pi]} \overline{\bar{f} g} d\lambda = \overline{\int_{[-\pi, \pi]} \bar{f} g d\lambda} = \overline{\langle g, f \rangle} \\ \langle \alpha f + \beta g, h \rangle &= \int_{[-\pi, \pi]} (\alpha f + \beta g) \bar{h} d\lambda = \alpha \int_{[-\pi, \pi]} f \bar{h} d\lambda + \beta \int_{[-\pi, \pi]} g \bar{h} d\lambda = \alpha \langle f, h \rangle + \beta \langle g, h \rangle \\ \langle f, f \rangle &= \int_{[-\pi, \pi]} |f|^2 d\lambda \geq 0. \end{aligned}$$

Moreover, the last inequality is tight if and only if $f = 0$ λ -a.e. (by 1.59), which implies $f = 0$ in $L^2([-\pi, \pi])$. $\square$

Since $\langle \cdot, \cdot \rangle$ is an inner product, it induces the norm

$$\|f\|_2 := \sqrt{\langle f, f \rangle} = \left(\int_{-\pi}^{\pi} |f(x)|^2 d\lambda(x) \right)^{1/2} \in [0, \infty), \quad \forall f \in L^2([-\pi, \pi]).$$

Recall that a sequence $(a_n)_{n \in \mathbb{N}} \subseteq H$ in a normed space $(H, \|\cdot\|)$ is *Cauchy* if for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that $\|a_n - a_m\| < \varepsilon$ for all $n, m \geq n_0$. A normed space is *complete* if every Cauchy sequence converges. By Theorem 1.97, $(L^2([-\pi, \pi]), \|\cdot\|_2)$ is complete. Hence, it is Hilbert:

Definition 2.3 (Hilbert space). A *Hilbert space* is an inner product space $(H, \langle \cdot, \cdot \rangle)$ that is complete with respect to the norm induced by the inner product.

2.1 Hilbert Spaces

In what follows we work in a generic Hilbert space $(H, \langle \cdot, \cdot \rangle)$.

Lemma 2.4 (Parallelogram identity). Let $(H, \|\cdot\|)$ be a Hilbert space. If $f, g \in H$, then

$$\|f + g\|^2 + \|f - g\|^2 = 2\|f\|^2 + 2\|g\|^2 \quad (2.2)$$

Proof.

$$\|f + g\|^2 + \|f - g\|^2 = \|f\|^2 + \|g\|^2 + 2\operatorname{Re}(\langle f, g \rangle) + \|f\|^2 + \|g\|^2 - 2\operatorname{Re}(\langle f, g \rangle) = 2\|f\|^2 + 2\|g\|^2.$$

□

Definition 2.5 (Orthogonality). Let $(H, \|\cdot\|)$ be a Hilbert space. We say that $f \in H$ and $g \in H$ are orthogonal, written $f \perp g$, if

$$\langle f, g \rangle = 0.$$

Lemma 2.6 (Pythagoras). If $f_1, \dots, f_n$ are pairwise orthogonal in H , then

$$\left\| \sum_{k=1}^n f_k \right\|^2 = \sum_{k=1}^n \|f_k\|^2.$$

Proof.

If $f_1 \perp f_2$, then

$$\|f_1 + f_2\|^2 = \langle f_1 + f_2, f_1 + f_2 \rangle = \langle f_1, f_1 \rangle + \langle f_1, f_2 \rangle + \langle f_2, f_1 \rangle + \langle f_2, f_2 \rangle = \|f_1\|^2 + \|f_2\|^2.$$

The general case follows by induction. □

Definition 2.7 (Convexity). Let $(H, \|\cdot\|)$ be a Hilbert space. We say that a subset $K \subseteq H$ is *convex* if, for any $f, g \in K$ and any $\lambda \in [0, 1]$, $\lambda f + (1 - \lambda)g \in K$.

The following proofs and results are adapted from [9].

Theorem 2.8 (Orthogonal Projection). Let $K \subseteq H$ be a nonempty closed convex set. Then, for any $f \in H$ there exists a unique element $F = p_K(f) \in K$ called the orthogonal projection of f onto K , such that

$$\|f - F\| = \inf_{g \in K} \|f - g\|.$$

Proof.

Let $d := \inf_{g \in K} \|f - g\|$.

Existence. Let $(g_n)_{n \in \mathbb{N}} \in K$ be a sequence such that

$$\lim_{n \rightarrow \infty} \|f - g_n\| = \inf_{g \in K} \|f - g\|.$$

Then $(g_n)_{n \in \mathbb{N}}$ is Cauchy. For, let $m, n \in \mathbb{N}$. Let us apply (2.2) to the vectors $f - g_n$ and $f - g_m$ and rearranging the terms:

$$\begin{aligned} \|(f - g_n) + (f - g_m)\|^2 + \|(f - g_n) - (f - g_m)\|^2 &= 2\|f - g_n\|^2 + 2\|f - g_m\|^2 \\ 4\left\|f - \frac{g_n + g_m}{2}\right\|^2 + \|g_n - g_m\|^2 &= 2\|f - g_n\|^2 + 2\|f - g_m\|^2 \\ \|g_n - g_m\|^2 &= 2\|f - g_n\|^2 + 2\|f - g_m\|^2 - 4\left\|f - \frac{g_n + g_m}{2}\right\|^2. \end{aligned}$$

This is because $\|f - g_n\|^2$ and $\|f - g_m\|^2$ converge to d^2 , by definition of the sequence $(g_n)_{n \in \mathbb{N}}$. Moreover, since $(g_n + g_m)/2 \in K$ by convexity, then $\left\|f - \frac{g_n + g_m}{2}\right\|^2 \geq \inf_{g \in K} \|f - g\| = d^2$. As a consequence,

$$0 \leq \|g_n - g_m\|^2 \leq 2\|f - g_n\|^2 + 2\|f - g_m\|^2 - 4d^2 \rightarrow 2d^2 + 2d^2 - 4d^2 = 0,$$

concluding that $(g_n)_{n \in \mathbb{N}}$ is Cauchy. By completeness of H , it converges to a unique $F \in H$. Moreover, since K is closed, then $F \in K$. Then

$$\|f - F\| = \lim_{n \rightarrow \infty} \|f - g_n\| = \inf_{g \in K} \|f - g\| = d.$$

Uniqueness. Suppose there exist $F_1, F_2 \in K$ such that

$$\|f - F_1\| = \|f - F_2\| = d.$$

We repeat exactly the same steps as before, applying (2.2) to the vectors $f - F_1$ and $f - F_2$ and reshuffling:

$$\begin{aligned}\|F_1 - F_2\|^2 &= 2\|f - F_1\|^2 + 2\|f - F_2\|^2 - 4\left\|f - \frac{F_1 + F_2}{2}\right\|^2 \\ \|F_1 - F_2\|^2 &= 2d^2 + 2d^2 - 4\left\|f - \frac{F_1 + F_2}{2}\right\|^2.\end{aligned}$$

Again, since $(F_1 + F_2)/2 \in K$ by convexity, then $\left\|f - \frac{F_1 + F_2}{2}\right\|^2 \geq d^2$, hence

$$\|F_1 - F_2\|^2 = 4d^2 - 4\left\|f - \frac{F_1 + F_2}{2}\right\|^2 \leq 4d^2 - 4d^2 = 0 \implies F_1 = F_2.$$

□

The following theorem tells us that, given a closed subspace $M \subseteq H$, every vector f can be decomposed into an orthogonal projection $p_M(f)$ and an orthogonal component $p_{M^\perp}(f) := f - p_M(f)$ such that $p_{M^\perp}(f) \perp m$ for every $m \in M$.

Corollary 2.9 (Orthogonal Decomposition). *Let $M \subseteq H$ be a nonempty closed subspace. For every $f \in H$ there exists a unique $p_M(f) \in M$ such that*

$$\|f - p_M(f)\| = \inf_{g \in M} \|f - g\|.$$

Moreover, $f - p_M(f) \perp M$, i.e. $\langle f - p_M(f), m \rangle = 0$ for all $m \in M$.

Proof.

Existence and uniqueness follow from Theorem 2.8 since M is closed and convex. Since M is a subspace, $p_M(f) + tm \in M$ for all $t \in \mathbb{R}$. Consider the quadratic

$$\phi(t) := \|f - (p_M(f) + tm)\|^2 = \|f - p_M(f) - tm\|^2 = \|f - p_M(f)\|^2 - 2t \cdot \operatorname{Re}(\langle f - p_M(f), m \rangle) + t^2 \|m\|^2.$$

If $m = 0$, then clearly $\langle f - p_M(f), m \rangle = 0$, there is nothing to prove. Otherwise, ϕ is a quadratic polynomial whose unique minimizer is

$$-\frac{2\operatorname{Re}(\langle f - p_M(f), m \rangle)}{2\|m\|^2} = \frac{\operatorname{Re}(\langle f - p_M(f), m \rangle)}{\|m\|^2}.$$

On the other hand, $t = 0$ must be a minimizer because $p_M(f)$ minimizes $\|f - g\|$ over $g \in M$. Hence

$$\frac{\operatorname{Re}(\langle f - p_M(f), m \rangle)}{\|m\|^2} = 0 \implies \operatorname{Re}(\langle f - p_M(f), m \rangle) = 0 \quad \forall m \in M.$$

To conclude, apply the same argument with $im \in M$ (since M is a complex subspace). This gives $0 = \operatorname{Re}(\langle f - p_M(f), im \rangle) = \operatorname{Im}(\langle f - p_M(f), m \rangle)$, concluding the proof. □

Definition 2.10 (Orthonormal System). Let $(H, \|\cdot\|)$ be a Hilbert space. We say that a *countable* subset $\{e_n\}_{n \in \mathbb{N}} \subseteq H$ is an *orthonormal family* if, for every $i, j \in \mathbb{N}$,

$$\langle e_i, e_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}.$$

Definition 2.11 (Span). Let $A \subseteq H$ be a subset of a Hilbert space $(H, \|\cdot\|)$. The *span* of A is the *linear subspace generated by A* , that is,

$$\operatorname{span}(A) = \left\{ \sum_{i=1}^n \mu_i f_i \mid n \geq 1, \mu_i \in \mathbb{C}, f_i \in A \right\}.$$

Lemma 2.12 (Orthogonal Projection Formula). *Let $\{e_1, \dots, e_n\}$ be an orthonormal family, and let $M_n = \operatorname{span}\{e_1, \dots, e_n\}$. For every $f \in H$,*

$$p_{M_n}(f) = \sum_{i=1}^n \langle f, e_i \rangle e_i.$$

Proof.

Observe that, for every $j \in \{1, \dots, n\}$,

$$\left\langle f - \sum_{i=1}^n \langle f, e_i \rangle e_i, e_j \right\rangle = \langle f, e_j \rangle - \sum_{i=1}^n \langle f, e_i \rangle \langle e_i, e_j \rangle = \langle f, e_j \rangle - \langle f, e_j \rangle = 0,$$

so $f - \sum_{i=1}^n \langle f, e_i \rangle e_i \perp M_n$. By Corollary 2.9, this implies $\sum_{i=1}^n \langle f, e_i \rangle e_i = p_{M_n}(f)$. $\square$

Lemma 2.13 (Best quadratic approximation). *Let $\{e_1, \dots, e_n\}$ be orthonormal and let $M_n = \text{span}\{e_1, \dots, e_n\}$. For every $f \in H$ and any scalars $\mu_1, \dots, \mu_n$,*

$$\left\| f - \sum_{i=1}^n \mu_i e_i \right\|^2 = \left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 + \sum_{i=1}^n |\mu_i - \langle f, e_i \rangle|^2.$$

Proof.

By Lemma 2.12, $f - \sum_{i=1}^n \langle f, e_i \rangle e_i \perp M_n$, so that

$$f - \sum_{i=1}^n \langle f, e_i \rangle e_i \perp \sum_{i=1}^n (\mu_i - \langle f, e_i \rangle) e_i.$$

By Lemma 2.6 applied twice,

$$\begin{aligned} \left\| f - \sum_{i=1}^n \mu_i e_i \right\|^2 &= \left\| \left[f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right] - \left[\sum_{i=1}^n (\mu_i - \langle f, e_i \rangle) e_i \right] \right\|^2 \\ &= \left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 + \left\| \sum_{i=1}^n (\mu_i - \langle f, e_i \rangle) e_i \right\|^2 \\ &= \left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 + \sum_{i=1}^n |\mu_i - \langle f, e_i \rangle|^2. \end{aligned}$$

$\square$

Theorem 2.14 (Bessel's inequality). *Let $(H, \|\cdot\|)$ be a Hilbert space and let $\{e_n\}_{n \in \mathbb{N}}$ be an orthonormal system.*

- For every $n \in \mathbb{N}$ and every $f \in H$,

$$\left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 = \|f\|^2 - \sum_{i=1}^n |\langle f, e_i \rangle|^2.$$

- For every $f \in H$,

$$\sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2 \leq \|f\|^2. \quad (\text{Bessel's Inequality})$$

In particular, the series in the left-hand side converges.

Proof.

It suffices to choose $(\mu_i)_{i \leq n} = \{0\}^n$ in Lemma 2.13 to deduce the identity:

$$\begin{aligned} \left\| f - \sum_{i=1}^n 0 \cdot e_i \right\|^2 &= \left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 + \sum_{i=1}^n |0 - \langle f, e_i \rangle|^2 \\ \left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 &= \|f\|^2 - \sum_{i=1}^n |\langle f, e_i \rangle|^2. \end{aligned}$$

Passing to the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 = \|f\|^2 - \sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2.$$

$$\sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2 = \|f\|^2 - \lim_{n \rightarrow \infty} \left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 \leq \|f\|^2,$$

by nonnegativity of the norm. $\square$

Proposition 2.15 (Fourier series). *Let $(H, \|\cdot\|)$ be a Hilbert space and let $\{e_n\}_{n \in \mathbb{N}}$ be an orthonormal system. For every sequence $(\mu_n)_{n \in \mathbb{N}} \subseteq \mathbb{C}$,*

$$\sum_{i=1}^{\infty} \mu_i e_i \in H \iff \sum_{i=1}^{\infty} |\mu_i|^2 < \infty.$$

As a consequence, combining Bessel's inequality we deduce that

$$\sum_{i=1}^{\infty} \langle f, e_i \rangle e_i$$

converges in H (not necessarily to f) and it's called Fourier series of f relative to the family $\{e_n\}_{n \in \mathbb{N}}$.

Proof.

Let $s_n := \sum_{i=1}^n \mu_i e_i$ for every $n \in \mathbb{N}$. Then, for every $n, p \in \mathbb{N}$,

$$\|s_{n+p} - s_n\|^2 = \left\| \sum_{i=n+1}^{n+p} \mu_i e_i \right\|^2 = \sum_{i=n+1}^{n+p} |\mu_i|^2 = \left| \sum_{i=1}^{n+p} |\mu_i|^2 - \sum_{i=1}^n |\mu_i|^2 \right|. \quad (2.3)$$

Observe that $\sum_{i=1}^{\infty} \mu_i e_i \in H$ is equivalent to stating that $(s_n)_{n \in \mathbb{N}}$ converges in H . In a Hilbert space, $(s_n)_{n \in \mathbb{N}}$ converges if and only if it is Cauchy. In turn, this corresponds to:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} : \forall n \geq N, \forall p \in \mathbb{N}, \quad \|s_{n+p} - s_n\| < \varepsilon.$$

Finally, thanks to Equation (2.3), we can again rewrite this last condition as

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} : \forall n \geq N, \forall p \in \mathbb{N}, \quad \left| \sum_{i=1}^{n+p} |\mu_i|^2 - \sum_{i=1}^n |\mu_i|^2 \right| < \varepsilon^2. \quad (2.4)$$

Hence, $\sum_{i=1}^{\infty} \mu_i e_i \in H$ if and only if (2.4) holds. On the other hand, (2.4) means that, by definition, $\left(\sum_{i=1}^n |\mu_i|^2 \right)_{n \in \mathbb{N}}$ is Cauchy in $\mathbb{R}$. Since $(\mathbb{R}, |\cdot|)$ is complete, it follows that $\left(\sum_{i=1}^n |\mu_i|^2 \right)_{n \in \mathbb{N}}$ is convergent, or in other words,

$$\sum_{i=1}^{\infty} |\mu_i|^2 < \infty.$$

This chain of equivalences shows the claim. $\square$

Remark 2.16. Let H be a normed vector space. A set $M \subseteq H$ is dense in H if $\overline{M} = H$. Equivalently, for every $f \in H$ and every $\varepsilon > 0$, there exists $g \in M$ such that $\|f - g\| < \varepsilon$.

Definition 2.17 (Complete orthonormal system). Let $(H, \|\cdot\|)$ be a Hilbert space. An orthonormal family $\{e_n\}_{n \in \mathbb{N}} \subseteq H$ is *complete* if $\text{span}(\{e_n\}_{n \in \mathbb{N}})$ is dense in H . A complete orthonormal family is an *orthonormal basis*, or *Hilbert basis*.

Theorem 2.18 (Hilbert Basis). *Let $(H, \|\cdot\|)$ be a Hilbert space, and let $\{e_n\}_{n \in \mathbb{N}} \subseteq H$ be an orthonormal family. Then the following are equivalent:*

- i) $\{e_n\}_{n \in \mathbb{N}}$ is a Hilbert basis;
- ii) $\text{span}(\{e_n\}_{n \in \mathbb{N}})$ is dense in H ;

iii) For every $f \in H$ and $\varepsilon > 0$, there exist coefficients $\mu_1, \dots, \mu_n$ such that

$$\left\| f - \sum_{i=1}^n \mu_i e_i \right\| < \varepsilon.$$

iv) For every $f \in H$, it holds

$$f = \sum_{i=1}^{\infty} \langle f, e_i \rangle e_i; \quad (2.5)$$

v) Every $f \in H$ satisfies Parseval's identity

$$\|f\|^2 = \sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2; \quad (2.6)$$

vi) The only vector orthogonal to each vector of the family is the null vector:

$$\langle f, e_i \rangle = 0 \quad \forall i \in \mathbb{N} \implies f = 0.$$

Proof.

(i $\Leftrightarrow$ ii $\Leftrightarrow$ iii). Follows by definition.

(ii $\Rightarrow$ iv). Let $M_n = \text{span}\{e_1, \dots, e_n\}$. Since $\text{span}(\{e_n\}_{n \in \mathbb{N}}) = \bigcup_{n \in \mathbb{N}} M_n$ is dense in H , we have $d(f, M_n) \rightarrow 0$ as $n \rightarrow \infty$ for every $f \in H$. By Lemma 2.12,

$$\left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2 = \|f - p_{M_n}(f)\|^2 = d^2(f, M_n) \xrightarrow{n \rightarrow \infty} 0.$$

This yields (2.5).

(iv $\Rightarrow$ v). By Theorem 2.14,

$$\|f\|^2 = \sum_{i=1}^n |\langle f, e_i \rangle|^2 + \left\| f - \sum_{i=1}^n \langle f, e_i \rangle e_i \right\|^2.$$

Letting $n \rightarrow \infty$, we deduce (2.6).

(v $\Rightarrow$ vi). If $\langle f, e_i \rangle = 0$ for every $i \in \mathbb{N}$, then $\|f\|^2 = \sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2 = 0$, hence $f = 0$.

(vi $\Rightarrow$ ii). Let $M := \overline{\text{span}(\{e_n\}_{n \in \mathbb{N}})}$. Suppose, by contradiction, that $M \neq H$. Then there exists $f \in H \setminus M$. Since M is a nonempty closed (convex) subspace of H , by Theorem 2.8 there exists a unique orthogonal projection $p_M(f) \in M$ such that

$$f - p_M(f) \perp m \quad \forall m \in M.$$

In particular, since $e_i \in M$ for every $i \in \mathbb{N}$, it holds

$$\langle f - p_M(f), e_i \rangle = 0, \quad \forall i \in \mathbb{N}.$$

By assumption (vi), this implies $f = p_M(f)$. However, $p_M(f) \in M$ and $f \notin M$, contradiction. $\square$

Let us give some intuition for what is going on. Theorem 2.18 says that a Hilbert basis $\mathcal{F}$ satisfies the following properties:

- i) Every vector lies in the closure of the linear span $\text{span}(\mathcal{F})$, meaning that there exists a suitable choice of coefficients such that a linear combination of elements of $\mathcal{F}$ converges to any vector in H in the $\|\cdot\|$ norm.
- ii) Moreover, there is a closed-form choice of these coefficients, given by (2.5). This formula can be interpreted as expressing each $f \in H$ in an orthonormal coordinate system: the vectors $e_1, e_2, \dots \in \mathcal{F}$ play the role of coordinate axes, and the scalars $\langle f, e_i \rangle$ are the corresponding coordinates.
- iii) Since the system is orthonormal, the Pythagorean theorem applies: the norm of f is the square root of the sum of the squares of its coordinates (Equation (2.6)). Moreover, this infinite sum always converges by Theorem 2.14.

- iv) The squared norm $\|f\|^2$ of a vector in a Hilbert space is often interpreted as its *energy*. In this light, Parseval's identity (2.6) expresses a principle of energy conservation: the total energy of a vector is exactly equal to the sum of the energies carried by its individual orthogonal components.
- v) Finally, a Hilbert basis is maximal: if we try to enlarge it by adding a vector orthogonal to every element of $\mathcal{F}$, that vector must be 0. Hence it cannot be added while preserving orthonormality.
- vi) As long as an orthonormal family is not complete (equivalently, not maximal), some components of a vector are not captured by the coefficients $\{\langle f, e \rangle\}_{e \in \mathcal{F}}$, and one can still add orthogonal directions. This yields Bessel's inequality 2.14. When the orthonormal family is complete, no energy is left outside the expansion, and Bessel's inequality becomes Parseval's identity (2.6).

Remark 2.19 (Existence of Hilbert bases). The existence of a Hilbert basis is strictly tied to the dimension and the topological properties of the space H .

- If H is **finite-dimensional**, finding a Hilbert basis is straightforward: one can simply take any basis of the space and apply the Gram-Schmidt orthogonalization process.
- If H is infinite-dimensional but **separable** (meaning it contains a countable dense subset), we can still apply the Gram-Schmidt process to a suitably chosen countable sequence to explicitly construct a *countable* Hilbert basis.
- In an arbitrary Hilbert space, establishing the existence of a complete orthonormal system requires **Zorn's Lemma**. By considering the partially ordered set of all orthonormal families in H ordered by inclusion, Zorn's Lemma guarantees the existence of a maximal orthonormal family, which is exactly a Hilbert basis.

This reliance on the Axiom of Choice (which is equivalent to Zorn's Lemma) should come as no surprise. Indeed, the Axiom of Choice is equally required to prove the existence of a *Hamel basis* for general vector spaces.

2.2 Fourier Series

The summary of the previous section is that $(L^2([-\pi, \pi], \lambda, \mathbb{C}), \|\cdot\|)$ is a Hilbert space. In what follows, we work in a similar space: the space of square-integrable functions $f: \mathbb{T} \rightarrow \mathbb{C}$ modulo null sets. Here, $\mathbb{T} \cong \mathbb{R}/2\pi\mathbb{Z}$ is the space $[-\pi, \pi]$ where we identify the endpoints as the same point (hence, it is topologically equivalent to a circle, where the points identified by the angles $-\pi$ and π correspond to the same point). In other words, the space $L^2(\mathbb{T})$ is the space of square-integrable functions $f: [-\pi, \pi] \rightarrow \mathbb{C}$ such that $f(-\pi) = f(\pi)$ or, equivalently, the space of 2π -periodic functions $f: \mathbb{R} \rightarrow \mathbb{C}$. Later, we will extend this framework to the space of functions $f: \mathbb{R}^n \rightarrow \mathbb{C}$ that are 2π -periodic in each variable, with domain $\mathbb{T}^n$. For the one-dimensional case $n = 1$, a Hilbert basis for $L^2(\mathbb{T})$ is given by the set:

$$\left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\} = \left\{ \frac{1}{\sqrt{2\pi}}, \frac{e^{iz}}{\sqrt{2\pi}}, \frac{e^{-iz}}{\sqrt{2\pi}}, \frac{e^{2iz}}{\sqrt{2\pi}}, \frac{e^{-2iz}}{\sqrt{2\pi}}, \dots \right\}.$$

Proposition 2.20. *The set $\mathcal{F} = \left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\}$ is an orthonormal system.*

Proof.

For every $m, n \in \mathbb{Z}$, it holds

$$\begin{aligned} \left\langle \frac{1}{\sqrt{2\pi}} e^{imz}, \frac{1}{\sqrt{2\pi}} e^{inz} \right\rangle &= \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} e^{imz} \cdot \overline{\frac{1}{\sqrt{2\pi}} e^{inz}} dz \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(m-n)z} dz = \begin{cases} 1 & \text{if } m = n \\ \frac{1}{2\pi i(m-n)} [e^{i(m-n)z}]_{-\pi}^{\pi} = 0 & \text{if } m \neq n \end{cases}. \end{aligned}$$

□

Assuming further that the family $\mathcal{F}$ is complete (whose proof we postpone to Section 2.5 for brevity), Theorem 2.18 ensures that every function $f \in L^2([-\pi, \pi], \mathbb{C})$ admits the expansion:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad \text{where} \quad c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx, \quad (\text{Complex Fourier Series})$$

where the series converges in the L^2 norm

Although the orthogonal decomposition of a function via the trigonometric family $\mathcal{F} = \left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\}$ is elegant, it is often more convenient to analyze the same family by making the connection with trigonometric functions more explicit. In particular, thanks to the identities

$$e^{inx} = \cos(nx) + i \sin(nx), \quad \cos(nx) = \frac{e^{inx} + e^{-inx}}{2}, \quad \sin(nx) = \frac{e^{inx} - e^{-inx}}{2i},$$

we can rewrite ([Complex Fourier Series](#)) as

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} c_n e^{inx} = c_0 + \sum_{n=1}^{\infty} (c_n e^{inx} + c_{-n} e^{-inx}) \\ &= c_0 + \sum_{n=1}^{\infty} c_n [\cos(nx) + i \sin(nx)] + c_{-n} [\cos(nx) - i \sin(nx)] \\ &= c_0 + \sum_{n=1}^{\infty} (c_n + c_{-n}) \cos(nx) + i(c_n - c_{-n}) \sin(nx). \end{aligned}$$

Setting

$$c_0 := \frac{a_0}{2}, \quad c_n + c_{-n} = a_n, \quad i(c_n - c_{-n}) = b_n, \quad \forall n = 1, 2, \dots,$$

we deduce the following alternative form for the Fourier series:

$$\boxed{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]}. \quad (2.7)$$

Naturally, the span of $\mathcal{F}_{\mathbb{C}} = \left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\}$ coincides with the span of

$$\mathcal{F}_{\mathbb{R}} = \{1, \cos(nx), \sin(mx) : n, m \in \mathbb{N}^+\},$$

since the previous steps define a bijection between the representations ([Complex Fourier Series](#)) and (2.7). With the appropriate normalization, we can identify an alternative trigonometric system that constitutes a Hilbert basis:

$$\mathcal{F}_{trig} = \left\{ \frac{1}{\sqrt{2\pi}}, \frac{\cos(nx)}{\sqrt{\pi}}, \frac{\sin(nx)}{\sqrt{\pi}} : n \in \mathbb{N}^+ \right\}.$$

Proposition 2.21. *The set $\mathcal{F}_{\mathbb{R}} = \left\{ \frac{1}{\sqrt{2\pi}}, \frac{\cos(nx)}{\sqrt{\pi}}, \frac{\sin(mx)}{\sqrt{\pi}} : n, m \in \mathbb{N}^+ \right\}$ is an orthonormal system.*

Proof.

Recall that, for every $x, y \in \mathbb{R}$,

$$\begin{aligned} \cos(x+y) &= \cos x \cos y - \sin x \sin y \\ \cos(x-y) &= \cos x \cos y + \sin x \sin y \\ \sin x \sin y &= \frac{1}{2} [\cos(x-y) - \cos(x+y)] \\ \cos x \cos y &= \frac{1}{2} [\cos(x+y) + \cos(x-y)] \\ \sin(x+y) &= \sin x \cos y + \sin y \cos x \\ \sin(x-y) &= \sin x \cos y - \sin y \cos x \\ \sin x \cos y &= \frac{1}{2} [\sin(x+y) + \sin(x-y)] \end{aligned}$$

Hence, for every $m, n \in \mathbb{N}^+$, it holds

$$\begin{aligned}
 \left\langle \frac{\sin(nx)}{\sqrt{\pi}}, \frac{\sin(mx)}{\sqrt{\pi}} \right\rangle &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (\cos((n-m)x) - \cos((n+m)x)) dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos((n-m)x) dx - \frac{1}{2\pi} \left[\frac{\sin((n+m)x)}{n+m} \right]_{-\pi}^{\pi} \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos((n-m)x) dx \\
 &= \begin{cases} \frac{1}{2\pi} \left[\frac{\sin((n-m)x)}{n-m} \right]_{-\pi}^{\pi} = 0 & \text{if } m \neq n, \\ \frac{1}{2\pi} \int_{-\pi}^{\pi} dx = 1 & \text{if } m = n. \end{cases}
 \end{aligned}$$

Similarly, for every $m, n \in \mathbb{N}^+$, it holds

$$\begin{aligned}
 \left\langle \frac{\cos(nx)}{\sqrt{\pi}}, \frac{\cos(mx)}{\sqrt{\pi}} \right\rangle &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (\cos((n-m)x) + \cos((n+m)x)) dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos((n-m)x) dx + \frac{1}{2\pi} \left[\frac{\sin((n+m)x)}{n+m} \right]_{-\pi}^{\pi} \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos((n-m)x) dx \\
 &= \begin{cases} \frac{1}{2\pi} \left[\frac{\sin((n-m)x)}{n-m} \right]_{-\pi}^{\pi} = 0 & \text{if } m \neq n, \\ \frac{1}{2\pi} \int_{-\pi}^{\pi} dx = 1 & \text{if } m = n. \end{cases}
 \end{aligned}$$

Likewise, for every $m, n \in \mathbb{N}^+$, it holds

$$\begin{aligned}
 \left\langle \frac{\sin(nx)}{\sqrt{\pi}}, \frac{\cos(mx)}{\sqrt{\pi}} \right\rangle &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(nx) \cos(mx) dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (\sin((n-m)x) + \sin((n+m)x)) dx \\
 &= \frac{1}{2\pi} \left[-\frac{\cos((n-m)x)}{n-m} \right]_{-\pi}^{\pi} + \frac{1}{2\pi} \left[-\frac{\cos((n+m)x)}{n+m} \right]_{-\pi}^{\pi} \\
 &= 0,
 \end{aligned}$$

where in the case $n = m$ the first term is $\int_{-\pi}^{\pi} \sin(0) dx = 0$.

Finally, for every $n \in \mathbb{N}^+$, we have

$$\begin{aligned}
 \left\langle \frac{\sin(nx)}{\sqrt{\pi}}, \frac{1}{\sqrt{2\pi}} \right\rangle &= \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} \sin(nx) dx = -\frac{1}{\sqrt{2\pi}} [\cos(nx)]_{-\pi}^{\pi} = 0, \\
 \left\langle \frac{\cos(nx)}{\sqrt{\pi}}, \frac{1}{\sqrt{2\pi}} \right\rangle &= \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} \cos(nx) dx = \frac{1}{\sqrt{2\pi}} [\sin(nx)]_{-\pi}^{\pi} = 0, \\
 \left\langle \frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{2\pi}} \right\rangle &= \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 dx = 1.
 \end{aligned}$$

□

This justifies Equation 2.7, which can be restated by looking at Theorem 2.18: every function $f \in L^2([-\pi, \pi], \mathbb{C})$ can be written (in L^2 sense) as

$$\boxed{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]}, \quad (\text{Real Fourier Series})$$

where the coefficients are given by

$$\begin{aligned} \frac{a_0}{2} &= \frac{1}{\sqrt{2\pi}} \left\langle f, \frac{1}{\sqrt{2\pi}} \right\rangle = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{+\pi} f(t) \frac{1}{\sqrt{2\pi}} dt = \boxed{\frac{1}{2\pi} \int_{-\pi}^{+\pi} f(t) dt} \\ a_n &= \frac{1}{\sqrt{\pi}} \left\langle f, \frac{\cos(nt)}{\sqrt{\pi}} \right\rangle = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{+\pi} f(t) \frac{\cos(nt)}{\sqrt{\pi}} dt = \boxed{\frac{1}{\pi} \int_{-\pi}^{+\pi} f(t) \cos(nt) dt} \quad n = 1, 2, \dots \\ b_n &= \frac{1}{\sqrt{\pi}} \left\langle f, \frac{\sin(nt)}{\sqrt{\pi}} \right\rangle = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{+\pi} f(t) \frac{\sin(nt)}{\sqrt{\pi}} dt = \boxed{\frac{1}{\pi} \int_{-\pi}^{+\pi} f(t) \sin(nt) dt} \quad n = 1, 2, \dots \end{aligned}$$

This series converges in $L^2([-\pi, \pi])$.

Remark 2.22 (Fourier Series in PDEs). In engineering applications, Fourier expansions are particularly useful for solving linear PDEs. The key point is that, after decomposing an initial condition into orthogonal coordinates (called *modes* or *frequencies*), the evolution in space and time can often be tracked mode by mode: each frequency evolves in a controlled way, and the PDE reduces to a family of scalar ODEs for the Fourier coefficients.³ While our abstract theory is developed on $L^2(\mathbb{T})$, concrete problems often involve intervals of the form $(-\ell, \ell)$ or $(0, \ell)$. The periodic framework extends to $(-\ell, \ell)$ via the rescaling $\tilde{f}(x) := f(\frac{\ell}{\pi}x)$ for $x \in [-\pi, \pi]$. On the half-interval $(0, \ell)$ one often uses *Fourier sine* or *Fourier cosine* series. The reason is that these are just restrictions of the full Fourier series on $(-\ell, \ell)$ after a canonical extension:

- **Odd extension:** $f_{\text{odd}}(x)$ is defined as $f(x)$ on $(0, \ell)$ and $-f(-x)$ on $(-\ell, 0)$. Its Fourier expansion consists only of sine terms (because $f_{\text{odd}}(x)$ is odd, hence the even terms in the Fourier expansion are zero), which form a complete system $\{\sin(\frac{n\pi x}{\ell})\}_{n \geq 1}$ in $L^2(0, \ell)$.
- **Even extension:** $f_{\text{even}}(x)$ is defined as $f(x)$ on $(0, \ell)$ and $f(-x)$ on $(-\ell, 0)$. Its expansion consists only of cosine terms (because $f_{\text{even}}(x)$ is even, hence the odd terms in the Fourier expansion are zero), forming the complete system $\{1, \cos(\frac{n\pi x}{\ell})\}_{n \geq 1}$ in $L^2(0, \ell)$.

Finally, we remark that in mathematics one normalizes these modes to obtain an orthonormal system, while in engineering it is common to omit these constants and simply consider the complete system $\{1, \cos x, \cos 2x, \dots, \sin x, \sin 2x, \dots\}$. The basic orthogonality relations are:

$$\int_0^\ell \sin^2\left(\frac{n\pi x}{\ell}\right) dx = \frac{\ell}{2}, \quad \int_0^\ell \cos^2\left(\frac{n\pi x}{\ell}\right) dx = \frac{\ell}{2}, \quad \text{and} \quad \int_{-\ell}^\ell \exp\left(\frac{n\pi i x}{\ell}\right) dx = \begin{cases} 2\ell & n = 0, \\ 0 & n \neq 0. \end{cases}$$

Sine Fourier Series on $(0, \ell)$:

$$\varphi(x) = \sum_{n=1}^{+\infty} B_n \sin \frac{n\pi x}{\ell}, \quad B_n = \frac{2}{\ell} \int_0^\ell \varphi(x) \sin \frac{n\pi x}{\ell} dx;$$

Cosine Fourier Series on $(0, \ell)$:

$$\varphi(x) = \frac{A_0}{2} + \sum_{n=1}^{+\infty} A_n \cos \frac{n\pi x}{\ell}, \quad A_n = \frac{2}{\ell} \int_0^\ell \varphi(x) \cos \frac{n\pi x}{\ell} dx;$$

Fourier Series on $(-\ell, +\ell)$:

$$\varphi(x) = \frac{A_0}{2} + \sum_{n=1}^{+\infty} \left(A_n \cos \frac{n\pi x}{\ell} + B_n \sin \frac{n\pi x}{\ell} \right), \quad A_n = \frac{1}{\ell} \int_{-\ell}^{+\ell} \varphi(x) \cos \frac{n\pi x}{\ell} dx, \quad B_n = \frac{1}{\ell} \int_{-\ell}^{+\ell} \varphi(x) \sin \frac{n\pi x}{\ell} dx,$$

and $A_0/2$ represents the average value of φ on the interval $(-\ell, +\ell)$.

Complex Fourier Series on $(-\ell, +\ell)$:

$$\varphi(x) = \sum_{n=-\infty}^{+\infty} c_n e^{\frac{n\pi i x}{\ell}}, \quad c_n = \frac{1}{2\ell} \int_{-\ell}^{+\ell} \varphi(x) e^{-\frac{n\pi i x}{\ell}} dx,$$

$$A_n = c_n + c_{-n}, \quad B_n = i(c_n - c_{-n}), \quad A_0 = 2c_0.$$

³A basic link between a function and its frequencies is Parseval's identity (2.6), which states that the energy $\|f\|_{L^2}^2$ equals the sum of the energies of its Fourier modes.

Remark 2.23. The trigonometric family is only one possible orthonormal decomposition for square-integrable functions on a compact domain. Other examples, used in various applications, include the *Haar system*, the *wavelets* and *Legendre polynomials*.

Remark 2.24 (Convergence of the Fourier series). In addition to convergence in L^2 norm, the following results hold:

- If $f \in C^1$, then its Fourier series converges uniformly almost everywhere (see [8] for this and further results);
- If f and f' are piecewise continuous, then its Fourier series converges pointwise, except at points of discontinuity. If x_0 is a point of discontinuity, then

$$\sum_{n=-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt \right) e^{inx_0} = \frac{1}{2} \left[\lim_{x \rightarrow x_0^+} f(x) + \lim_{x \rightarrow x_0^-} f(x) \right].$$

Thus, at a jump discontinuity, the series converges to the average of the limits from the right and from the left. Moreover, the partial sums do not converge uniformly, but oscillate in an interval that contains the point of discontinuity. The width of the interval where the oscillations occur shrinks to zero, but the size of the oscillations does not. In fact, at the large n limit, the magnitude of the oscillations is approximately 9% of the jump in f at the jump discontinuity. This behaviour is called the *Gibbs phenomenon*, and it is analyzed in [10].

- The Fourier series of L^2 functions converges pointwise almost everywhere (a.e.).

Example 2.25 (Fourier series of x^2 and the Basel problem). Consider the function $f(x) = x^2$ defined on the interval $[-\pi, \pi]$ and extended periodically to $\mathbb{R}$. Since f is an even function, its Fourier sine coefficients vanish, i.e., $b_n = 0$ for all $n \geq 1$. We compute the remaining coefficients as follows:

- For $n = 0$:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{2\pi^2}{3}.$$

- For $n \geq 1$, using integration by parts twice:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos(nx) dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx = \frac{4(-1)^n}{n^2}.$$

Thus, the Fourier series of $f(x)$ is given by:

$$x^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos(nx). \quad (2.8)$$

Since f is continuous and piecewise C^1 on $[-\pi, \pi]$, the series converges pointwise everywhere. Plugging $x = \pi$ into (2.8),

$$\pi^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(n\pi) \implies \frac{2\pi^2}{3} = 4 \sum_{n=1}^{\infty} \frac{1}{n^2}$$

Dividing both sides by 4, we deduce the well-known result for the Basel problem:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Remark 2.26 (Fourier series in $\mathbb{R}^n$). If $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is 2π periodic in each variable, the multi-variable Fourier series is given by

$$f(x) = \sum_{k \in \mathbb{Z}^n} \hat{f}(k) e^{ik \cdot x}, \quad x = (x_1, \dots, x_n) \in \mathbb{R}^n.$$

The Fourier coefficients are given by the relation

$$\hat{f}(k) = \frac{1}{(2\pi)^n} \int_{[-\pi, \pi]^n} f(x) e^{-ik \cdot x} dx.$$

They satisfy *Parseval identity*:

$$\|f\|_{L^2}^2 = (2\pi)^n \sum_{k \in \mathbb{Z}^n} |\hat{f}(k)|^2.$$

These conclusions follow from the following:

Theorem 2.27. *The set $\mathcal{F} = \left\{ \frac{1}{(2\pi)^{n/2}} e^{ik \cdot x} : k \in \mathbb{Z}^n \right\}$ is a Hilbert basis of $L^2([-\pi, \pi]^n, \lambda^n, \mathbb{C})$.*

2.3 Fourier Coefficients

In some contexts, the interest in Fourier series shifts from the study of the partial sums to the study of the *coefficients*. For this reason, we denote the coefficients in [Complex Fourier Series](#) by $\hat{f}(n)$, where $n \in \mathbb{Z}$. In this way, for every $f \in L^1([-\pi, \pi])$ we obtain

$$\hat{f}(n) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

Note that L^2 is not needed to *define* the coefficients, but it is needed to obtain the L^2 -convergence

$$f = \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{inx} \quad \text{in } L^2([-\pi, \pi]),$$

and Parseval's identity (i.e. 2.6)

$$\|f\|_{L^2}^2 = 2\pi \sum_{k \in \mathbb{Z}} |\hat{f}(k)|^2 \quad \text{and} \quad \|f\|_{L^1} \geq 2\pi |\hat{f}(k)|, \quad \forall k \in \mathbb{Z}.$$

Following [8], let us formally compute the derivative of a Fourier series:

$$f'(x) = \frac{d}{dx} \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{ikx} = \sum_{k \in \mathbb{Z}} ik \hat{f}(k) e^{ikx}.$$

Hence, a reasonable claim is that, under suitable regularity assumptions on f , one has $\widehat{f'}(k) = ik \hat{f}(k)$. This is indeed the case:

Proposition 2.28. *Let $f \in C^1(\mathbb{R}, \mathbb{C})$ be a 2π -periodic function. Then*

$$\widehat{f'}(k) = ik \hat{f}(k), \quad k \in \mathbb{Z}.$$

Proof.

The case $k = 0$ is immediate, since

$$\widehat{f'}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) dx = \frac{f(\pi) - f(-\pi)}{2\pi} = 0 = i \cdot 0 \cdot \hat{f}(0).$$

Fix $k \neq 0$. Integrating by parts, we have

$$\begin{aligned} \hat{f}(k) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \frac{d}{dx} \left(-\frac{1}{ik} e^{-ikx} \right) dx \\ &= -\frac{1}{2\pi} \frac{f(x)}{ik} e^{-ikx} \Big|_{x=-\pi}^{\pi} + \frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) \frac{1}{ik} e^{-ikx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) \frac{1}{ik} e^{-ikx} dx. \end{aligned}$$

The boundary term vanishes since $f(\pi) = f(-\pi)$ and $e^{-ik\pi} = e^{ik\pi}$ for $k \in \mathbb{Z}$. Therefore

$$\hat{f}(k) = \frac{1}{ik} \widehat{f'}(k),$$

which is equivalent to $\widehat{f'}(k) = ik \hat{f}(k)$.

More generally, by iterating the same argument one obtains

$$\widehat{f^{(m)}}(k) = (ik)^m \hat{f}(k), \quad m = 0, \dots, M,$$

whenever f is a 2π -periodic function of class C^M . □

2.4 Convolutions and Kernels

In this section we study the theory of convolutions. This material will be useful later when discussing Fourier transforms and when proving the completeness of the trigonometric system $\left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\}$. Convolution is a natural operation on *locally compact abelian* (LCA) groups. We will not develop the general theory here. We only record four important examples:

- $(\mathbb{R}^n, +)$, the Euclidean space endowed with vector addition and Lebesgue measure λ^n ;
- $(\mathbb{T}^n, +) = ((\mathbb{R}/2\pi\mathbb{Z})^n, +)$, the n -torus endowed with addition modulo $2\pi\mathbb{Z}$ and Lebesgue measure (so that $\lambda^n(\mathbb{T}^n) = (2\pi)^n < \infty$). A function $f : \mathbb{T}^n \rightarrow \mathbb{C}$ can be identified with a function $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{C}$ which is 2π -periodic in each coordinate;
- $(\mathbb{Z}^n, +)$, the lattice endowed with vector addition, the discrete topology, and counting measure;
- $((\mathbb{Z}/N\mathbb{Z})^n, +)$, which can be identified with $\{0, \dots, N-1\}^n$ endowed with addition modulo N , the discrete topology, and counting measure.

As we will see later, each of these groups gives rise to a corresponding Fourier transform. In this section we prove results that hold for all the examples above (with the obvious modifications in the discrete cases). Nevertheless, we will work in detail on a space X which is either $\mathbb{R}^n$ or $\mathbb{T}^n$, since in these cases the statements are less trivial and the proofs are more geometrically meaningful thanks to the Euclidean metric structure. Henceforth, X will always denote either $\mathbb{R}^n$ or $\mathbb{T}^n$.

In practice, you may think of X as $\mathbb{R}^n$, with the caveat that in the torus case everything is understood modulo 2π (equivalently, functions are 2π -periodic in each coordinate) and the reference measure satisfies $\lambda^n(X) < \infty$ when $X = \mathbb{T}^n$, whereas $\lambda^n(X) = \infty$ when $X = \mathbb{R}^n$.

We follow [11], [12], [9] and [8].

2.4.1 Convolutions

Definition 2.29 (Convolution on a locally compact abelian group). Let $(X, +)$ be a locally compact abelian group, where $X = \mathbb{R}^n$ or $X = \mathbb{T}^n$. For $f, g \in L^1(X, \mathbb{C})$, define their *convolution* as

$$(f * g)(x) = \int_X f(x - y)g(y)dy.$$

Remark 2.30. When working on $\mathbb{Z}^n$ or $(\mathbb{Z}/N\mathbb{Z})^n$, convolution becomes a sum:

$$(f * g)(x) = \sum_{y \in G} f(y)g(x - y).$$

Loosely speaking, the convolution of two functions is a weighted average of one, with the weights provided by the other. The following lemma justifies why convolution is well-defined for integrable functions:

Lemma 2.31. *If $f, g \in L^1(X)$, where $X = \mathbb{R}^n$ or $X = \mathbb{T}^n$, then $f * g$ is well-defined and satisfies the estimate*

$$\|f * g\|_{L^1} \leq \|f\|_{L^1} \|g\|_{L^1}.$$

Proof.

Since the function $|f(x - y)| |g(y)|$ is measurable and nonnegative for every $y \in X$, then, by Fubini-Tonelli,

$$\begin{aligned} \|f * g\|_{L^1} &= \int_X |(f * g)(x)| dx = \int_X \left| \int_X f(x - y)g(y)dy \right| dx \\ &\leq \int_X \int_X |f(x - y)| |g(y)| dy dx = \int_X |g(y)| \left(\int_X |f(x - y)| dx \right) dy \\ &= \int_X |g(y)| \|f\|_{L^1} dy = \|f\|_{L^1} \|g\|_{L^1}. \end{aligned}$$

□

Remark 2.32. Since $\lambda(\mathbb{T}^n) < \infty$, by Theorem 1.101 it holds $L^2(\mathbb{T}^n) \subset L^1(\mathbb{T}^n)$. Therefore, Lemma 2.31 applies in particular to $f, g \in L^2(\mathbb{T}^n)$. Conversely, since $\lambda(\mathbb{R}^n) = \infty$, we no longer have $L^2(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$, hence Lemma 2.31 does not apply directly to $L^2(\mathbb{R}^n)$. Nevertheless, if $f, g \in L^2(\mathbb{R}^n)$, then the integral defining $(f * g)(x)$ is finite for every $x \in \mathbb{R}^n$ by Cauchy-Schwarz, and one has $\|f * g\|_\infty \leq \|f\|_2 \|g\|_2$.

We observe that $(X, +)$ forms a group, and the convolution $*$: $L^1(X) \times L^1(X) \rightarrow L^1(X)$ is a new operator on L^1 functions. The next proposition records the algebraic properties of convolution:

Proposition 2.33 (Algebraic properties of convolution). *Let $f, g, h \in L^1(X, \mathbb{C})$, where $X = \mathbb{R}^n$ or $X = \mathbb{T}^n$. Then:*

- i) *Distributive*: $f * (g + h) = f * g + f * h$;
 ii) *Commutative*: $f * g = g * f$;
 iii) *Associative*: $f * (g * h) = (f * g) * h$;

Proof.

(i). For a.e. $x \in X$,

$$\begin{aligned} (f * (g + h))(x) &= \int_X f(x - y) (g + h)(y) dy = \int_X f(x - y) g(y) dy + \int_X f(x - y) h(y) dy \\ &= (f * g)(x) + (f * h)(x). \end{aligned}$$

(ii). For a.e. $x \in X$, perform the change of variables $y \mapsto x - y$ and use translation invariance of λ :

$$(f * g)(x) = \int_X f(x - y) g(y) dy = \int_X f(y) g(x - y) dy = (g * f)(x).$$

(iii). Fix $x \in X$. By Fubini-Tonelli,

$$\begin{aligned} (f * (g * h))(x) &= \int_X f(x - y) (g * h)(y) dy = \int_X f(x - y) \left(\int_X g(y - z) h(z) dz \right) dy \\ &= \int_X \left(\int_X f(x - y) g(y - z) dy \right) h(z) dz = \int_X \left(\int_X f([x - z] - [y - z]) g(y - z) \right) h(z) dz \\ &= \int_X (f * g)(x - z) h(z) dz = ((f * g) * h)(x). \end{aligned}$$

The application of Fubini-Tonelli is justified as in Lemma 2.31. $\square$

Remark 2.34. Together with Lemma 2.31 and the completeness of $L^1(X)$, Proposition 2.33 shows that $(L^1(X), *, \|\cdot\|_{L^1})$ is a commutative Banach algebra.

Another lemma that will be used later is the following:

Lemma 2.35 (Young). *If $f \in L^2(X)$, $g \in L^1(X)$, where $X = \mathbb{R}^n$ or $X = \mathbb{T}^n$, then $f * g \in L^2(X)$ and*

$$\|f * g\|_{L^2} \leq \|f\|_{L^2} \|g\|_{L^1}. \quad (2.9)$$

Proof.

For every $x \in X$, by Cauchy-Schwarz,

$$|(f * g)(x)|^2 = \left| \int_X f(y) g(x - y) dy \right|^2 \leq \left(\int_X |f(y)|^2 |g(x - y)| dy \right) \left(\int_X |g(x - y)| dy \right).$$

Since $\int_X |g(x - y)| dy = \|g\|_{L^1}$, integrating in x and applying Fubini-Tonelli yields

$$\begin{aligned} \|f * g\|_{L^2}^2 &\leq \|g\|_{L^1} \int_X \int_X |f(y)|^2 |g(x - y)| dy dx \\ &= \|g\|_{L^1} \int_X |f(y)|^2 \left(\int_X |g(x - y)| dx \right) dy = \|g\|_{L^1}^2 \|f\|_{L^2}^2, \end{aligned}$$

$\square$

Lemma 2.36 (Regularization by Convolution). *Let $f \in L^1(\mathbb{R}^n, \mathbb{C})$ and $g \in C_c^\infty(\mathbb{R}^n, \mathbb{C})$. Then $(f * g) \in C^\infty(\mathbb{R}^n, \mathbb{C})$ and*

$$D^\alpha(f * g) = f * (D^\alpha g), \quad \forall \alpha \in \mathbb{N}^n.$$

*Moreover, if f is compactly supported, then $(f * g) \in C_c^\infty(\mathbb{R}^n, \mathbb{C})$.*

Proof.

We first show the claim for the first-order partial derivatives. Let e_i be the i -th standard basis vector.

We compute:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{(f * g)(x + he_i) - (f * g)(x)}{h} &= \lim_{h \rightarrow 0} \int_{\mathbb{R}^n} f(y) \frac{g(x + he_i - y) - g(x - y)}{h} dy \\ &= \int_{\mathbb{R}^n} f(y) \lim_{h \rightarrow 0} \frac{g(x - y + he_i) - g(x - y)}{h} dy = \int_{\mathbb{R}^n} f(y) \partial_i g(x - y) dy = (f * \partial_i g)(x), \end{aligned}$$

where the exchange of limit and integral is justified by the Dominated Convergence Theorem (1.80), using the mean value theorem to bound the difference quotient as follows:

$$\left| f(y) \frac{g(x + he_i - y) - g(x - y)}{h} \right| \leq |f(y)| \|\partial_i g\|_{L^\infty}.$$

Since $\partial_i g \in C_c^\infty$, the bounding function is in L^1 . By induction on the multi-index length $|\alpha|$, one shows the claim for every $\alpha \in \mathbb{N}^n$. Assume now that f is compactly supported. Equivalently, assume that $\text{supp}(f) \cup \text{supp}(g) \subseteq B_R(0)$ for some $R > 0$. We show that $\text{supp}(f * g) \subseteq B_{2R}(0)$. For that purpose, we show that $(f * g)(x) = 0$ for every $x \in \mathbb{R}^n \setminus B_{2R}(0)$. For every $y \in B_R(0)$ and $x \notin B_{2R}(0)$, the reverse triangle inequality yields $\|x - y\| \geq \|x\| - \|y\| > 2R - R = R$. Hence:

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y) g(x - y) dy = \int_{B_R(0)} f(y) g(x - y) dy = 0,$$

since $g(x - y) = 0$ whenever $\|x - y\| > R$. □

We conclude this section by defining convolutions of finite measures defined on Borel sets $B \in \mathcal{B}(\mathbb{R}^n)$.

Definition 2.37 (Convolution of finite measures). Let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ be a measurable space, and let $\mu, \nu : \mathcal{B}(\mathbb{R}^n) \rightarrow [0, \infty)$ be finite measures on $\mathbb{R}^n$. We define their convolution $\mu * \nu$ by:

$$(\mu * \nu)(B) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \mathbf{1}_B(x + y) d\mu(x) d\nu(y), \quad \text{for all } B \in \mathcal{B}(\mathbb{R}^n).$$

Equivalently, for every bounded Borel-measurable function φ , we have:

$$\int_{\mathbb{R}^n} \varphi(z) d(\mu * \nu)(z) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \varphi(x + y) d\mu(x) d\nu(y).$$

2.4.2 Gaussian Kernels and Approximate Identities

Definition 2.38 (Kernel). Let $(X, \mathcal{B}(X), \lambda)$ be a measure space, where $X = \mathbb{R}^n$ or $X = \mathbb{T}^n$, and λ is the corresponding Lebesgue measure. A *kernel* is an integrable function $K : X \rightarrow \mathbb{C}$ such that

$$\int_X K d\lambda = 1.$$

Definition 2.39 (Approximate identity). A family of kernels $(K_n)_{n \in \mathbb{N}}$ is an *approximate identity* if $K_n \geq 0$ and, for every $\delta > 0$,

$$\lim_{n \rightarrow \infty} \int_{\|x\| \geq \delta} |K_n(x)| d\lambda(x) = 0.$$

The elements of an approximate identity are called *good kernels*.

Remark 2.40. Since $K \geq 0$, we have $\|K\|_{L^1} = \int_X |K| d\lambda = \int_X K d\lambda = 1$.

Good kernels act as Dirac delta functions in the limit, concentrating their mass around $x = 0$. Furthermore, the conditions $K_n \geq 0$ and $\int_X K_n d\lambda = 1$ suggest an interpretation of K_n as a Radon-Nikodym density. Essentially, *good kernels* are a family of density functions that converge, as $n \rightarrow \infty$, to the Dirac measure δ_0 when acting on a continuous function via convolution. The following theorem, whose proof is largely taken from [11], clarifies this intuition:

Theorem 2.41 (Dirac Property). If $(K_n)_{n \in \mathbb{N}}$ is an approximate identity over $X = \mathbb{R}^n$ or $X = \mathbb{T}^n$ and $f \in C_c(X)$ is a continuous function with compact support, then $(K_n * f)$ converges uniformly to f as $n \rightarrow \infty$. In other words,

$$\lim_{n \rightarrow \infty} \sup_{x \in X} |(f * K_n)(x) - f(x)| = 0.$$

Proof.

Since $\int_X K_n d\lambda = 1$, then

$$f(x) = \int_X K_n(y) f(x) dy.$$

Hence,

$$(K_n * f)(x) - f(x) = \int_X K_n(y) f(x - y) dy - \int_X K_n(y) f(x) dy = \int_X K_n(y) [f(x - y) - f(x)] dy.$$

We now split the integral into two regions: $V_\delta = \{y \in X : \|y\| < \delta\}$ and its complement V_δ^c :

$$\begin{aligned} |(K_n * f)(x) - f(x)| &\leq \int_{\|y\| < \delta} K_n(y) \cdot |f(x - y) - f(x)| dy + \int_{\|y\| \geq \delta} K_n(y) \cdot |f(x - y) - f(x)| dy \\ &= I_1 + I_2. \end{aligned}$$

In order to bound I_1 and I_2 , we use two different consequences of the assumption $f \in C_c(X)$.

- On the one hand, by Heine-Cantor theorem (cf. Remark 2.47), f is uniformly continuous. Fixing $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x - y) - f(x)| < \varepsilon, \quad \forall y \in V_\delta = \{y \in X : \|y\| < \delta\}.$$

Choosing V_δ in terms of ε , we deduce

$$I_1 = \int_{\|y\| < \delta} K_n(y) \cdot |f(x - y) - f(x)| dy < \varepsilon \int_{\|y\| < \delta} K_n(y) dy = \varepsilon.$$

- On the other hand, by Heine-Borel theorem and the image of a compact set through a continuous function (cf. Remark 2.47), f is bounded. Thus, there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in X$. We deduce

$$\begin{aligned} I_2 &= \int_{\|y\| \geq \delta} K_n(y) \cdot |f(x - y) - f(x)| dy \leq \int_{\|y\| \geq \delta} K_n(y) \cdot (|f(x - y)| + |f(x)|) dy \\ &\leq 2M \int_{\|y\| \geq \delta} K_n(y) dy. \end{aligned}$$

Taking the supremum over x , the limit as $n \rightarrow \infty$ (which makes $2M \int_{\|y\| \geq \delta} K_n(y) dy \rightarrow 0$) and letting $\varepsilon \rightarrow 0$, we deduce

$$\lim_{n \rightarrow \infty} \|(K_n * f) - f\|_\infty \leq \varepsilon \implies \lim_{n \rightarrow \infty} \|(K_n * f) - f\|_\infty = 0.$$

□

Corollary 2.42 (Convergence on C_c). *If $(K_n)_{n \in \mathbb{N}}$ is an approximate identity on $X = \mathbb{R}^n$ or $X = \mathbb{T}^n$ and $f \in C_c(X)$, then $(f * K_n)$ converges uniformly to f as $n \rightarrow \infty$. In particular,*

$$\lim_{n \rightarrow \infty} (f * K_n)(x) = f(x), \quad \forall x \in X,$$

and, for every $1 \leq p < \infty$,

$$\lim_{n \rightarrow \infty} \|f * K_n - f\|_{L^p(X)} = 0.$$

Proof.

The uniform convergence follows directly from Theorem 2.41. The pointwise convergence for every $x \in X$ is an immediate consequence. It remains to show that $\|f * K_n - f\|_{L^p(X)} \rightarrow 0$ for every $1 \leq p < \infty$.

If $X = \mathbb{T}^n$, then $\lambda^n(\mathbb{T}^n) < \infty$ and, by Theorem 1.101,

$$\lim_{n \rightarrow \infty} \|f * K_n - f\|_{L^p(\mathbb{T}^n)} \leq \lim_{n \rightarrow \infty} \lambda^n(\mathbb{T}^n)^{1/p} \|f * K_n - f\|_\infty = 0.$$

If $X = \mathbb{R}^n$, let $\text{supp}(f) \subseteq B(0, R_0)$. Choose $R > R_0$ and split

$$\|f * K_n - f\|_{L^p(\mathbb{R}^n)}^p \leq \int_{B(0, R)} |(f * K_n)(x) - f(x)|^p dx + \int_{\mathbb{R}^n \setminus B(0, R)} |(f * K_n)(x)|^p dx. \quad (2.10)$$

The first term is bounded by

$$\int_{B(0, R)} |(f * K_n)(x) - f(x)|^p dx \leq \lambda^n(B(0, R)) \|f * K_n - f\|_\infty^p,$$

which tends to 0 as $n \rightarrow \infty$ by Theorem 2.41.

The second term is bounded by Hölder's inequality: for every $x \in \mathbb{R}^n$,

$$|(f * K_n)(x)| = \left| \int_{\mathbb{R}^n} f(x-y) K_n(y) dy \right| \leq \left(\int_{\mathbb{R}^n} |f(x-y)|^p K_n(y) dy \right)^{1/p} \underbrace{\left(\int_{\mathbb{R}^n} K_n(y) dy \right)^{1-1/p}}_{=1}$$

Since $\int_{\mathbb{R}^n} K_n dy = 1$, we obtain

$$|(f * K_n)(x)|^p \leq \int_{\mathbb{R}^n} |f(x-y)|^p K_n(y) dy.$$

Integrating over $\mathbb{R}^n \setminus B(0, R)$ and applying Tonelli yields

$$\begin{aligned} \int_{\mathbb{R}^n \setminus B(0, R)} |(f * K_n)(x)|^p dx &\leq \int_{\mathbb{R}^n \setminus B(0, R)} \int_{\mathbb{R}^n} |f(x-y)|^p K_n(y) dy dx \\ &= \int_{\mathbb{R}^n} K_n(y) \left(\int_{\mathbb{R}^n \setminus B(0, R)} |f(x-y)|^p dx \right) dy \end{aligned}$$

Recall that $\text{supp}(f) \subseteq B(0, R_0)$. The inner integral integrates x such that $\|x\| \geq R$. On the other hand, the integral is nonzero only where $\|x-y\| \leq R_0$. By the triangle inequality,

$$\|y\| \geq \|x\| - \|x-y\| \geq R - R_0.$$

Therefore,

$$\begin{aligned} \int_{\mathbb{R}^n \setminus B(0, R)} |(f * K_n)(x)|^p dx &\leq \int_{\mathbb{R}^n} K_n(y) \left(\int_{\mathbb{R}^n \setminus B(0, R)} |f(x-y)|^p dx \right) dy \\ &\leq \int_{\|y\| \geq R-R_0} K_n(y) \|f\|_{L^p(\mathbb{R}^n)}^p dy \\ &= \|f\|_{L^p(\mathbb{R}^n)}^p \int_{\|y\| > R-R_0} K_n(y) dy. \end{aligned}$$

Since $R > R_0$, the last integral converges to 0 as $n \rightarrow \infty$ by the defining property of an approximate identity.

Combining these two steps in (2.10) gives $\|f * K_n - f\|_{L^p(\mathbb{R}^n)} \rightarrow 0$ as $n \rightarrow \infty$. $\square$

As a byproduct, this corollary shows that good kernels act as Dirac delta functions in the limit: for every $f \in C_c(X)$, the convolution $f * K_n$ converges to f uniformly on X (hence pointwise everywhere), and also in $L^1(X)$ as $n \rightarrow \infty$.

Lemma 2.43 (Gaussian integral). *It holds*

$$\int_{-\infty}^{+\infty} e^{-x^2/2} dx = \sqrt{2\pi}.$$

Proof.

Let $I := \int_{-\infty}^{+\infty} e^{-x^2/2} dx$. Then

$$I^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dx dy.$$

Perform a change of variables to polar coordinates,

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}.$$

Then

$$I^2 = \int_0^{2\pi} \int_0^\infty e^{-r^2/2} r dr d\theta = 2\pi \int_0^\infty e^{-r^2/2} r dr = -2\pi [e^{-r^2/2}]_0^{+\infty} = 2\pi.$$

Hence $I = \sqrt{2\pi}$. □

Example 2.44 (Heat Kernel). The *Heat kernel*, also known as the *Gaussian kernel*, is a fundamental tool in analysis due to its remarkable properties; it is widely used in PDEs, probability theory, and signal processing. It is defined on $\mathbb{R}^d$ as:

$$K_n(x) = \frac{1}{(4\pi t)^{d/2}} e^{-\frac{\|x\|^2}{4t}} = \frac{1}{(4\pi t)^{d/2}} e^{-\frac{\sum_{i=1}^d x_i^2}{4t}}, \quad t = 1/n > 0.$$

Let us consider, for simplicity, the case $d = 1$:

$$K_n(x) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}}.$$

Clearly $K_n \geq 0$. Moreover, by Lemma 2.43, the total mass is one. Indeed, by performing the substitution $y = x/\sqrt{4t}$:

$$\int_{\mathbb{R}} K_n d\lambda = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}} dx \stackrel{y=x/\sqrt{4t}}{=} \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{+\infty} e^{-y^2/2} \cdot \sqrt{2t} dy = \frac{1}{\sqrt{4\pi t}} \sqrt{2\pi} \cdot \sqrt{2t} = 1.$$

Finally, we verify the concentration property at the origin. For any $\delta > 0$:

$$\lim_{n \rightarrow \infty} \int_{|x| \geq \delta} K_n dx = \lim_{t \rightarrow 0^+} \int_{|x| \geq \delta} \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}} dx.$$

By using again the substitution $y = x/\sqrt{4t}$, the limits of integration become $|y| \geq \delta/\sqrt{4t}$. Since $\delta/\sqrt{4t} \rightarrow \infty$ as $t \rightarrow 0^+$, we have:

$$\lim_{t \rightarrow 0^+} \frac{1}{\sqrt{\pi}} \int_{|y| \geq \frac{\delta}{2\sqrt{t}}} e^{-y^2} dy = 0,$$

confirming that $\{K_n\}_{n \in \mathbb{N}}$ is a good kernel (approximate identity). As a consequence, by Theorem 2.41,

$$\lim_{n \rightarrow \infty} \|K_n * f - f\|_\infty = 0, \quad \forall f \in C_c(\mathbb{R}^n).$$

Corollary 2.45 (Sub-convergence on L^p). *If $(K_n)_{n \in \mathbb{N}}$ is an approximate identity on $X = \mathbb{R}^n$ or $X = \mathbb{T}^n$ and $f \in L^p(X)$, then there exists a subsequence $(K_{n_k})_{k \in \mathbb{N}}$ such that $(f * K_{n_k})_{k \in \mathbb{N}}$ converges pointwise λ^n -a.e. to f as $n \rightarrow \infty$.*

Proof.

Fix $\varepsilon > 0$. By Theorem 1.98, there exists $g \in C_c(X, \mathbb{C})$ such that $\|f - g\|_{L^1} < \varepsilon$. Using Lemma 2.31 and the triangle inequality,

$$\begin{aligned} \|f * K_n - f\|_{L^p} &\leq \|f * K_n - g * K_n\|_{L^p} + \|g * K_n - g\|_{L^p} + \|g - f\|_{L^p} \\ &\leq \|(f - g) * K_n\|_{L^p} + \|g * K_n - g\|_{L^p} + \|g - f\|_{L^p} \\ &\leq \|f - g\|_{L^p} \|K_n\|_{L^p} + \|g * K_n - g\|_{L^p} + \|g - f\|_{L^p} \\ &\leq \varepsilon \cdot 1 + \|g * K_n - g\|_{L^p} + \varepsilon = 2\varepsilon + \|g * K_n - g\|_{L^p}. \end{aligned}$$

Since $g \in C_c(X, \mathbb{C})$, Corollary 2.42 yields $\lim_{n \rightarrow \infty} \|g * K_n - g\|_{L^p} = 0$. Taking the limit superior as

$n \rightarrow 0^+$ gives

$$\limsup_{n \rightarrow \infty} \|f * K_n - f\|_{L^p} \leq 2\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, this proves that $\lim_{n \rightarrow 0^+} \|f * K_n - f\|_{L^p} = 0$, meaning $f * K_n \rightarrow f$ in $L^p(X)$. By Lemma 1.96, for any sequence $\delta_k \rightarrow 0^+$, there exists a subsequence $(\delta_{k_j})_{j \in \mathbb{N}}$ such that, λ^n -a.e.,

$$\lim_{j \rightarrow \infty} (f * K_{\delta_{k_j}})(x) = f(x).$$

□

2.5 Completeness of the Trigonometric System

In this section, we discuss several approaches to show that the set

$$\left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\} = \left\{ \frac{1}{\sqrt{2\pi}}, \frac{e^{iz}}{\sqrt{2\pi}}, \frac{e^{-iz}}{\sqrt{2\pi}}, \frac{e^{2iz}}{\sqrt{2\pi}}, \frac{e^{-2iz}}{\sqrt{2\pi}}, \dots \right\}$$

is a complete orthonormal system for the space $L^2([-\pi, \pi], \lambda, \mathbb{C})$, i.e., its span is dense in L^2 . Following [8], we will prove that for every $f \in L^2([-\pi, \pi], \mathbb{C})$ and $\varepsilon > 0$, there exist complex polynomials $p(z)$ and $q(z)$ such that

$$\|f(t) - p(e^{it}) - q(e^{-it})\|_{L^2} \leq \varepsilon.$$

The approaches presented to establish this result are particularly instructive and can be found in [8], [9], [12], and [11]. These methods generally fall into two categories: those based on the *Stone-Weierstrass theorem* ([8], [12]) and those employing *kernel-based methods* ([12], [11], [9]).

Reduction to continuous functions with compact support. We recall the following important theorem, which is a corollary of Theorem 1.100.

Theorem 2.46 (Density of C_c in L^p). *Let X be a locally compact Hausdorff space, and let μ be a regular Borel measure on X such that $\mu(K) < \infty$ for every compact set K . For every $f \in L^p(X, \mathbb{C}, \mu)$, there exists a sequence $(f_n)_{n \in \mathbb{N}} \subseteq C_c(X, \mathbb{C})$ such that $f_n \rightarrow f$ in $L^p(X, \mathbb{C}, \mu)$, namely*

$$\|f_n - f\|_{L^p} \xrightarrow{n \rightarrow \infty} 0,$$

which is equivalent to

$$\lim_{n \rightarrow \infty} \int_X |f_n - f|^p d\mu = 0.$$

Remark 2.47. If X is a topological space, the *support* of a function $f : X \rightarrow \mathbb{C}$ is defined as $\text{supp } f := \{x \in X : f(x) \neq 0\}$, that is, the set of points $x \in X$ that have a neighbourhood where f is *not* identically zero. We say that f has *compact support* if $\text{supp } f$ is compact. If $X = \mathbb{R}^n$, the *Heine-Borel theorem* states that $\text{supp } f$ is compact if and only if it is closed and bounded. Intuitively, this means that a function with compact support on $\mathbb{R}^n$ "lives" in a finite region of space, vanishing identically outside of a sufficiently large ball. Functions in $C_c(\mathbb{R}^n)$ inherit two fundamental topological properties. The first one is the Heine-Cantor theorem: they are uniformly continuous. The second is boundedness: since the image of the compact support $f(\text{supp } f)$ is a compact subset of $\mathbb{C}$, then it is bounded by the Heine-Borel theorem.

In other words, Theorem 2.46 lets us approximate (in L^p sense) any L^p -integrable function on an open subset $\Omega \subseteq \mathbb{R}^n$ by a sequence of continuous functions with compact support. In [12] it is observed that one can strengthen this result by requiring the approximating functions to be also C^∞ . This is achieved by a sequence of L^p approximations on the interval $[-\pi, \pi] \subseteq \mathbb{R}$ (and the proof extends verbatim to open sets $\Omega \subseteq \mathbb{R}^n$):

- Using Lemma 1.52, we can approximate measurable functions by simple functions;
- We approximate simple functions by step functions.⁴

⁴Observe that simple functions have the form $\sum_{i=1}^n \alpha_i \mathbf{1}_{A_i}$ for measurable sets $(A_i)_{i=1}^n$, while step functions have the form $\sum_{j=1}^m \beta_j \mathbf{1}_{(x_{j-1}, x_j]}$, where $-\pi = x_0 < x_1 < \dots < x_m = \pi$ is a partition of $[-\pi, \pi]$. In other words, simple functions are constant on measurable sets (possibly including isolated points), while step functions are more regular because they are constant on open intervals.

- We approximate step functions with C^∞ functions by smoothing their discontinuities. Specifically, it suffices to modify the functions within a δ -neighborhood of each jump (where δ is arbitrarily small) using a bump function or a mollifying convolution with approximate identity kernel and Corollary 2.42 (see [12] or Proposition 1.4 of [8]). For instance, the function $f(x) = e^{-\frac{1}{x(\delta-x)}}$ is an ideal candidate for this purpose because it is infinitely flat at the boundary points $x = 0$ and $x = \delta$, yet C^∞ . As $\delta \rightarrow 0$, the L^p error converges to zero.
- Finally, we use the same smoothing idea to assume, without loss of generality, that the C^∞ approximating functions have compact support, e.g. by enforcing $f(-\pi) = f(\pi) = 0$.

It is noteworthy that iterative approximations within a $\varepsilon > 0$ error can be easily combined by bounding the total error via the triangle inequality. Hence, an iterative approximation scheme produces the desired density result.

The Stone-Weierstrass Approach. In this approach, we do not need to require the density of C^∞ functions with compact support, but only the density of C^0 functions with compact support (Theorem 2.46). It is easy to see that this implies the density of C_{per} functions, i.e., continuous functions such that $f(-\pi) = f(\pi)$ (equivalently, continuous functions over $\mathbb{T}$). The density of the trigonometric polynomials is then deduced thanks to the Stone-Weierstrass theorem:

Theorem 2.48 (Stone-Weierstrass). *Let K be a compact metric space and $\mathcal{F} \subseteq C(K, \mathbb{C})$ a family of continuous functions from K to $\mathbb{C}$. Suppose $\mathcal{F}$ satisfies the following properties:*

- i) *closed under addition: if $\varphi, \psi \in \mathcal{F}$, then $\varphi + \psi \in \mathcal{F}$;*
- ii) *closed under scalar multiplication: if $\lambda \in \mathbb{C}$ and $\varphi \in \mathcal{F}$, then $\lambda\varphi \in \mathcal{F}$;*
- iii) *closed under product: if $\varphi, \psi \in \mathcal{F}$, then $\varphi \cdot \psi \in \mathcal{F}$;*
- iv) *closed under conjugation: if $\varphi \in \mathcal{F}$, then $\overline{\varphi} \in \mathcal{F}$;*
- v) *contains the constants: $\varphi := 1 \in \mathcal{F}$;*
- vi) *separates points: if $x \neq y \in K$, there exists $\varphi \in \mathcal{F}$ such that $\varphi(x) \neq \varphi(y)$.*

Then $\mathcal{F}$ is dense in $C(K, \mathbb{C})$ with respect to the supremum norm, i.e., for every continuous function $f : K \rightarrow \mathbb{C}$ and every $\varepsilon > 0$, there exists $\varphi \in \mathcal{F}$ such that

$$\|f - \varphi\|_\infty < \varepsilon.$$

By applying Stone-Weierstrass to the compact set $K := [-\pi, \pi]$ and the family $\mathcal{F} = \left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\}$, we easily verify that all the assumptions are satisfied, except the separability of the points $-\pi$ and π : indeed, for every $f \in \mathcal{F}$, $f(-\pi) = f(\pi)$. Thus we deduce that the trigonometric polynomials are dense in $C_{\text{per}}([-\pi, \pi], \mathbb{C})$ endowed with the supremum norm. This convergence is much stronger than needed: indeed, it holds⁵

$$\text{sup norm convergence} \implies (\text{almost everywhere}) \text{ pointwise convergence} \implies L^2 \text{ convergence}.$$

Finally, since trigonometric polynomials approximate (in L^2 sense) C_{per} functions and C_{per} functions approximate any L^2 -integrable function, the completeness of the trigonometric family is proven (by triangle inequality).

Kernel Approaches. In this setting, we consider functions $f, g : \mathbb{T} \rightarrow \mathbb{C}$, that is, 2π -periodic functions on $\mathbb{R}$ or, even better, functions defined on $\mathbb{R}/2\pi\mathbb{Z}$. In this approach, we rely on the density of the C^∞ periodic functions (such that $f(-\pi) = f(\pi)$) in $L^2([-\pi, \pi], \mathbb{C})$, and we show that the trigonometric polynomials approximate these functions arbitrarily well in L^2 sense. We rely on convolutions and kernels, which are widely used on PDEs.

Thanks to Corollary 2.42, we can show the completeness of the orthonormal family $\mathcal{F} = \left\{ \frac{1}{\sqrt{2\pi}} e^{ikt} : k \in \mathbb{Z} \right\}$ by employing an approximate identity $(K_n)_{n \in \mathbb{N}}$ and proving that $K_n * f$ is a trigonometric polynomial for any $f \in C(\mathbb{T})$. Since $\mathbb{T}$ is a compact space, every continuous function $f \in C(\mathbb{T})$ is bounded and uniformly continuous, ensuring that $K_n * f \rightarrow f$ uniformly (and in L^2 norm!) as $n \rightarrow \infty$. There are different strategies:

⁵If $|f_n(x) - f(x)| < \varepsilon$ for every $x \in X \setminus \mathcal{N}$, where $\mathcal{N}$ is a null set, then

$$\left(\int_X |f_n(x) - f(x)|^p dx \right)^{1/p} \leq (\varepsilon^p \mu(X))^{1/p} = \varepsilon \sqrt[p]{\mu(X)} \xrightarrow{\varepsilon \rightarrow 0} 0.$$

- Following [11] and [9], we define the function $K_n \geq 0$ by

$$K_n(x) = c_n(1 + \cos x)^n, \quad \forall x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z},$$

where c_n is the normalization constant ensuring $\int_{\mathbb{T}} K_n d\lambda = 1$. It can be shown that $(K_n)_{n \in \mathbb{N}}$ is an approximate identity and that $K_n * f$ is a trigonometric polynomial for every $n \in \mathbb{N}$ and every continuous function $f : \mathbb{T} \rightarrow \mathbb{C}$ (see [9] for a complete proof).

- **Fejér's Kernel.** Following [12], we define the sequence of kernels $(D_n)_{n \in \mathbb{N}}$ by

$$D_n(x) := \frac{1}{2\pi} \sum_{k=-n}^n e^{ikx}, \quad x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}. \quad (2.11)$$

Moreover, define the sequence of kernels $(K_n)_{n \in \mathbb{N}}$ by

$$K_n := \frac{1}{n} \sum_{k=0}^{n-1} D_k = \frac{1}{2\pi} \sum_{k=-(n-1)}^{n-1} \left(1 - \frac{|k|}{n}\right) e^{ikx}, \quad x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}. \quad (2.12)$$

Then $(D_n)_{n \in \mathbb{N}}$ is a family known as *Dirichlet kernels*, while $(K_n)_{n \in \mathbb{N}}$ is a family known as *Fejér's kernels*. It is clear that $(D_n)_{n \in \mathbb{N}}$ is a family of trigonometric polynomials. Moreover, one can show that $D_n * f$ is also trigonometric polynomial, for every $n \in \mathbb{N}$ and every continuous function $f : \mathbb{T} \rightarrow \mathbb{C}$. However, Dirichlet kernels might be negative, hence they are not approximate identities. Conversely, Fejér's kernels are approximate identities. Being Cesaro sums of Dirichlet kernels, we deduce that they are trigonometric polynomials, as well as $f * K_n$ for every $n \in \mathbb{N}$ (see [12]). This concludes the proof of completeness.

- **Dirichlet Kernel.** The use of Dirichlet kernels, defined in (2.11), is the most intuitive approach once we realize that the density of $\mathcal{F} = \left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\}$ in $C_{\text{per}}([-\pi, \pi])$ is equivalent to

$$\lim_{n \rightarrow \infty} \left\| f - \sum_{k=-n}^n \left\langle f, \frac{1}{\sqrt{2\pi}} e^{ik\cdot} \right\rangle \frac{1}{\sqrt{2\pi}} e^{ik\cdot} \right\|_{L^2} = 0, \quad \forall f \in C_{\text{per}}([-\pi, \pi]).$$

Now observe that, for every $n \in \mathbb{N}$,

$$\begin{aligned} \sum_{k=-n}^n \left\langle f, \frac{1}{\sqrt{2\pi}} e^{ik\cdot} \right\rangle \frac{1}{\sqrt{2\pi}} e^{ikx} &= \sum_{k=-n}^n \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ikx} f(y) e^{-iky} dy \\ &= \int_{-\pi}^{\pi} \sum_{k=-n}^n \frac{1}{2\pi} e^{ik(x-y)} f(y) dy = D_n * f(x). \end{aligned}$$

Therefore, we only need to prove

$$\lim_{n \rightarrow \infty} \left\| f - D_n * f \right\|_{L^2} = 0, \quad \forall f \in C_{\text{per}}([-\pi, \pi]).$$

Unfortunately, as we already observed, Dirichlet kernels are not *good* kernels, since they oscillate wildly in a δ -neighbourhood of 0: it can be proved that, for every $\delta > 0$,

$$\lim_{n \rightarrow \infty} \int_{-\delta}^{+\delta} |D_n| d\lambda = \infty.$$

Therefore, any proof in the spirit of Theorem 2.41 will be inconclusive. Thankfully, one can extract directly L^2 convergence of $D_n * f$ to f if we assume a smoother regularity for f , such as C^1 or C^∞ . More details in [12].

- **Bernstein polynomials.** The last method we mention for proving completeness uses the *Bernstein Polynomials* defined by

$$B_n(f)(x) = \sum_{i=1}^n f\left(\frac{i}{n}\right) \binom{n}{i} x^i (1-x)^{n-i}, \quad \forall f \in C^0([0, 1], \mathbb{C}).$$

These polynomials give a constructive approach for proving the *Weierstrass theorem*, which states that polynomials are dense (with respect to the supremum norm) in the $C^0([-\pi, \pi])$ space. Even if Bernstein polynomials are not strictly speaking convolutions of f , they act in a similar way, and it is possible to draw a probabilistic connection between kernels, densities and Bernstein polynomials.

2.6 Fourier Transform

The *Fourier transform* is the most widely used Fourier-analytic tool in probability. It can be seen as a continuous analogue of the Fourier series. Indeed, if the Fourier series has the form $f(x) = \sum_{n=-\infty}^{\infty} \hat{f}(n)e^{inx}$, the Fourier transform replaces the discrete frequency variable $n \in \mathbb{Z}$ by a continuous one $t \in \mathbb{R}$, and the sum becomes an integral: $f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(t)e^{itx} dt$. Alternatively, one can see the shift from a discrete setting to a continuous setting in the coefficient formula $\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)e^{-inx} dx$, which extends the domain from $[-\pi, \pi]$ to $\mathbb{R}$, and the frequency index extends from $\mathbb{Z}$ to $\mathbb{R}$: $\hat{f}(t) = \int_{\mathbb{R}} f(x)e^{-itx} dx$.

Following [8], one can make the discrete-to-continuous passage more precise. Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a smooth function with compact support. Fix $\delta > 0$ small enough so that f is supported in $[-\pi/\delta, \pi/\delta]$. Set $z := \delta x$. For $z \in [-\pi, \pi]$, define $f_{\delta}(z) := f(\frac{z}{\delta})$. Then f_{δ} is supported in $[-\pi, \pi]$ (indeed it vanishes near $\pm\pi$), hence it admits a Fourier series on $\mathbb{T}$:

$$\begin{aligned} f(x) &= f_{\delta}(\delta x) = \sum_{k \in \mathbb{Z}} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f_{\delta}(y)e^{-iky} dy \right) e^{ik\delta x} \stackrel{y=\delta u}{=} \sum_{k \in \mathbb{Z}} \frac{\delta}{2\pi} \left(\int_{-\pi/\delta}^{\pi/\delta} f(u)e^{-i\delta k u} du \right) e^{i\delta k x} \\ &= \sum_{k \in \mathbb{Z}} \frac{\delta}{2\pi} \hat{f}(\delta k) e^{i\delta k x} = \frac{1}{2\pi} \sum_{k \in \mathbb{Z}} \delta \hat{f}(\delta k) e^{i\delta k x}, \end{aligned}$$

where $\hat{f}(t) := \int_{\mathbb{R}} f(u)e^{-itu} du$ and we used $(f) \subseteq [-\pi/\delta, \pi/\delta]$ to replace the truncated integral with the integral over $\mathbb{R}$. By letting $\delta \rightarrow 0$, this expression writes as

$$f(x) = \lim_{\delta \rightarrow 0} \frac{1}{2\pi} \sum_{k \in \mathbb{Z}} \delta \hat{f}(\delta k) e^{i\delta k x} = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(t)e^{itx} dt,$$

where the last passage uses the definition of the Riemann-sum over $\mathbb{R}$. The justification of this last step is one of the most delicate in analysis, because the interval grows simultaneously as the terms change. For that, we need a different and more abstract viewpoint. Let us start from an *informal* definition of the Fourier transform.

Definition 2.49 (Fourier transform). Let $f : \mathbb{R}^n \rightarrow \mathbb{C}$. We define the *Fourier transform* by

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-i\xi \cdot x} dx, \quad \xi \in \mathbb{R}^n. \quad (2.13)$$

Similarly, we define the *inverse Fourier transform* by

$$f(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(\xi)e^{i\xi \cdot x} d\xi, \quad x \in \mathbb{R}^n. \quad (2.14)$$

Remark 2.50. It is clear that Definition 2.49 has two issues. 1) It is not well-defined if $f(\cdot)e^{-i\xi \cdot (\cdot)}$ is not integrable. 2) Equation (2.14) is an identity (an inversion formula) that is proved starting from (2.13), rather than a definition valid in full generality.

In order to fully capture these issues, we shall define the Fourier transform in a more abstract way. In general, the Fourier transform can be defined as an operator on locally compact abelian groups. We shall not pursue this approach (see, e.g., the last pages of [13]), but it is important to note that the notion of Fourier transform takes different forms depending on the specific setting. In these notes we focus on the following ones:

- i) **Periodic setting.** Here the Fourier transform is a map $\mathcal{F} : L^2(\mathbb{T}^n) \rightarrow \ell^2(\mathbb{Z}^n)$: it maps every square-integrable function $f : \mathbb{T}^n \rightarrow \mathbb{C}$ (i.e., 2π -periodic in each coordinate) to a square-summable sequence (Theorem 2.14) of coefficients $\hat{f} : \mathbb{Z}^n \rightarrow \mathbb{C}$. This setting is the one described in 2.26. Essentially, the Fourier transform takes a function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ that is 2π -periodic in each component and, assuming $f \in L^2$, associates the function $\hat{f}$ on $\mathbb{Z}^n$ given by

$$\hat{f}(k) = \frac{1}{(2\pi)^n} \int_{[-\pi, \pi]^n} f(x)e^{-ik \cdot x} dx.$$

Therefore, in this case the Fourier transform coincides with Fourier series, and the previous discussions apply. We remark that one can also define the Fourier transform for L^1 , C^0 , or C^∞ functions (with suitable assumptions).

- ii) **Continuous setting.** Here the Fourier transform is an operator from functions on $\mathbb{R}^n$ to functions on $\mathbb{R}^n$. Typical examples include:
- $\mathcal{F} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$;
 - $\mathcal{F} : L^1(\mathbb{R}^n) \rightarrow C_0(\mathbb{R}^n)$ (continuous and vanishing at infinity);
 - $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$ (Schwartz $\rightarrow$ Schwartz).
- iii) **Finite discrete (DFT).** Here the Fourier transform is an operator from functions on $\{0, \dots, N-1\}^n$ to functions on $\{0, \dots, N-1\}^n$. We will study this transform in Section 5.
- iv) **Measures.** Here the Fourier transform is an operator from finite measures on $\mathbb{R}^n$ to bounded and uniformly continuous functions on $\mathbb{R}^n$, via

$$\hat{\mu}(t) := \int_{\mathbb{R}^n} e^{-it \cdot x} \mu(dx), \quad t \in \mathbb{R}^n.$$

If μ is a probability measure, then $\hat{\mu} = \varphi_X$ is the *characteristic function*.

2.6.1 Fourier transform on L^1

We start our discussion from the continuous setting, by first treating $\mathcal{F}$ as an operator on $L^1(\mathbb{R}^n)$ functions. We denote by $C_0(\mathbb{R}^n, \mathbb{C})$ the space of continuous functions $f : \mathbb{R}^n \rightarrow \mathbb{C}$ such that $\lim_{\|x\| \rightarrow \infty} |f(x)| = 0$. We follow [8].

Theorem 2.51 (Riemann-Lebesgue Lemma). *For every $f \in L^1(\mathbb{R}^n, \mathbb{C})$, we have $\hat{f} \in C_0(\mathbb{R}^n, \mathbb{C})$. Moreover,*

$$\|\hat{f}\|_{L^\infty} \leq \|f\|_{L^1}.$$

Proof.

It holds, for every $\xi \in \mathbb{R}^n$,

$$|\hat{f}(\xi)| = \left| \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx \right| \leq \int_{\mathbb{R}^n} |f(x) e^{-ix \cdot \xi}| dx = \int_{\mathbb{R}^n} |f(x)| dx = \|f\|_{L^1},$$

since $|f(x)| = |f(x) e^{-ix \cdot \xi}|$. We conclude that $\|\hat{f}\|_{L^\infty} \leq \|f\|_{L^1}$ and $\hat{f}$ is well-defined and bounded.

Next, we show that $\hat{f}$ is continuous. Take a sequence $(\xi_n)_{n \in \mathbb{N}} \subseteq \mathbb{R}^n$ converging to $\xi \in \mathbb{R}^n$. We show that the sequence $(\hat{f}(\xi_n))_{n \in \mathbb{N}} \subseteq \mathbb{C}$ converges to $\hat{f}(\xi) \in \mathbb{C}$. This holds by the Dominated Convergence Theorem (1.80):

$$\lim_{n \rightarrow \infty} \hat{f}(\xi_n) = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi_n} dx = \int_{\mathbb{R}^n} \lim_{n \rightarrow \infty} f(x) e^{-ix \cdot \xi_n} dx = \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx = \hat{f}(\xi).$$

Indeed, $f(x) e^{-ix \cdot \xi_n} \rightarrow f(x) e^{-ix \cdot \xi}$ by continuity, and $|f(x) e^{-ix \cdot \xi_n}| \leq |f(x)|$ from previous discussion, so Theorem 1.80 applies.

Finally, we prove that $\lim_{\|\xi\| \rightarrow \infty} |\hat{f}(\xi)| = 0$. Observe that this holds if $f \in C_c^\infty(\mathbb{R}^n, \mathbb{C})$: take $R > 0$ such that $\text{supp}(f) \subseteq B(0, R)$. Integrating by parts on $B(0, R)$ yields

$$\begin{aligned} \hat{f}(\xi) &= \int_{\mathbb{R}^n} f(x) e^{-i\xi \cdot x} dx = \int_{B(0, R)} f(x) e^{-i\xi \cdot x} dx \\ &= \frac{i}{\|\xi\|^2} \int_{\partial B(0, R)} f(x) e^{-i\xi \cdot x} \left(\frac{\xi \cdot x}{R} \right) dS(x) - \frac{i}{\|\xi\|^2} \int_{B(0, R)} (\xi \cdot \nabla f)(x) e^{-i\xi \cdot x} dx \\ &= 0 - i \int_{B(0, R)} \frac{(\xi \cdot \nabla f)(x)}{\|\xi\|^2} e^{-i\xi \cdot x} dx. \end{aligned}$$

where the boundary term vanishes because $f \equiv 0$ on $\partial B(0, R)$ by our choice of R . Letting $\|\xi\| \rightarrow \infty$, we get

$$\begin{aligned} |\hat{f}(\xi)| &= \left| -i \int_{B(0, R)} \frac{(\xi \cdot \nabla f)(x)}{\|\xi\|^2} e^{-i\xi \cdot x} dx \right| \leq \left| \int_{B(0, R)} \frac{(\xi \cdot \nabla f)(x)}{\|\xi\|^2} dx \right| \\ &\leq \left| \int_{B(0, R)} \frac{\|\xi\| |\nabla f(x)|}{\|\xi\|^2} dx \right| = \left| \frac{1}{\|\xi\|} \int_{B(0, R)} |\nabla f(x)| dx \right| = \frac{C}{\|\xi\|} \xrightarrow{\|\xi\| \rightarrow \infty} 0. \end{aligned}$$

We conclude $\lim_{\|\xi\| \rightarrow \infty} |\hat{f}(\xi)| = 0$.

Fix now $f \in L^1(\mathbb{R}^n, \mathbb{C})$ and $\varepsilon > 0$. By Theorem 2.46 and the subsequent discussion on how to strengthen its result by requiring the approximating functions to be also C^∞ , we know that $C_c^\infty(\mathbb{R}^n, \mathbb{C})$ is dense in $L^1(\mathbb{R}^n, \mathbb{C})$. Therefore, there exists $g \in C_c^\infty(\mathbb{R}^n, \mathbb{C})$ such that $\|f - g\|_{L^1} \leq \varepsilon$. Then,

$$\begin{aligned} \limsup_{\|\xi\| \rightarrow \infty} |\hat{f}(\xi)| &\leq \limsup_{\|\xi\| \rightarrow \infty} |\hat{g}(\xi)| + \sup_{\xi \in \mathbb{R}^n} |\hat{f}(\xi) - \hat{g}(\xi)| = \limsup_{\|\xi\| \rightarrow \infty} |\hat{g}(\xi)| + \sup_{\xi \in \mathbb{R}^n} |\widehat{(g - f)}(\xi)| \\ &= \limsup_{\|\xi\| \rightarrow \infty} |\hat{g}(\xi)| + \|g - f\|_{L^1} = \|g - f\|_{L^1} \leq \varepsilon. \end{aligned}$$

This concludes the proof. $\square$

Remark 2.52. With the lens of functional analysis, the previous theorem tells us that the Fourier transform $\mathcal{F}$ can be defined as an operator $\mathcal{F} : L^1(\mathbb{R}^n, \mathbb{C}) \rightarrow C_0(\mathbb{R}^n, \mathbb{C})$. Moreover, its *operator norm* satisfies

$$\|\mathcal{F}\|_{L^1 \rightarrow C_0} \leq 1.$$

Therefore, $\mathcal{F}$ is a non-expansive operator, bounded (hence continuous).

Theorem 2.53 (Fourier transform and convolution). Let $f, g \in L^2(\mathbb{T}^n)$ and $\varphi, \psi \in L^1(\mathbb{R}^n)$. Let $\xi \in \mathbb{R}^n$ and $k \in \mathbb{Z}$. Then:

$$\mathcal{F}(f * g)(k) = (2\pi)^n \mathcal{F}(f)(k) \cdot \mathcal{F}(g)(k) \quad \text{and} \quad \mathcal{F}(\varphi * \psi)(\xi) = (2\pi)^n \mathcal{F}(\varphi)(\xi) \cdot \mathcal{F}(\psi)(\xi). \quad (2.15)$$

Proof.

Since $\lambda(\mathbb{T}^n) < \infty$, we have $L^2(\mathbb{T}^n) \subset L^1(\mathbb{T}^n)$, hence $f, g \in L^1(\mathbb{T}^n)$ and $f(x - y)g(y)$ is integrable on $\mathbb{T}^n \times \mathbb{T}^n$. By Fubini-Tonelli,

$$\begin{aligned} \mathcal{F}(f * g)(k) &= \widehat{f * g}(k) = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} (f * g)(x) e^{-ik \cdot x} dx = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} \int_{\mathbb{T}^n} f(x - y) g(y) e^{-ik \cdot x} dy dx \\ &= \int_{\mathbb{T}^n} \left(\frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} f(x - y) e^{-ik \cdot x} dx \right) g(y) dy \\ &= \int_{\mathbb{T}^n} e^{-ik \cdot y} \left(\frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} f(x - y) e^{-ik \cdot (x - y)} dx \right) g(y) dy \\ &= \hat{f}(k) \int_{\mathbb{T}^n} g(y) e^{-ik \cdot y} dy = \hat{f}(k) (2\pi)^n \hat{g}(k) = (2\pi)^n \mathcal{F}(f)(k) \cdot \mathcal{F}(g)(k). \end{aligned}$$

The same computation applies on $\mathbb{R}^n$ for $\varphi, \psi \in L^1(\mathbb{R}^n)$, with k replaced by ξ and integrals over $\mathbb{T}^n$ replaced by integrals over $\mathbb{R}^n$, yielding $\mathcal{F}(\varphi * \psi)(\xi) = (2\pi)^n \mathcal{F}(\varphi)(\xi) \mathcal{F}(\psi)(\xi)$. $\square$

Remark 2.54 (Gelfand transform). The fact that the computations leading to (2.15) look the same on $\mathbb{T}^n$ and on $\mathbb{R}^n$ is not a coincidence. Indeed, Proposition 2.33 and Lemma 2.31 show that $(L^1, *, +)$ is a commutative Banach algebra. Equation (2.15) states that the Fourier transform $\mathcal{F}$ is an algebra homomorphism: it sends convolution into pointwise multiplication. Hence, similar results will hold for any commutative Banach algebra. This perspective is closely related to Gelfand theory: in general, commutative Banach algebras admit representations in which products become pointwise products of complex-valued functions. The Fourier transform provides exactly such a representation. Finally, it is important to remark that $\mathcal{F}$ is not, in general, a homeomorphism (or an isomorphism) on L^1 . Stronger results hold in L^2 (Plancherel, where the Fourier transform is an isometry) or C^* -algebras (Gelfand-Naimark theorem).

We conclude this section with some algebraic rules for the Fourier transform.

Lemma 2.55 (Translation and modulation). Let $f \in L^1(\mathbb{R}^n, \mathbb{C})$ and let $a \in \mathbb{R}^n$. Define

$$(\tau_a f)(x) := f(x - a), \quad (M_a f)(x) := e^{ia \cdot x} f(x).$$

Then, for every $\xi \in \mathbb{R}^n$,

$$\widehat{\tau_a f}(\xi) = e^{-i\xi \cdot a} \hat{f}(\xi), \quad \text{and} \quad \widehat{M_a f}(\xi) = \hat{f}(\xi - a). \quad (2.16)$$

Proof.

The result follows from a change of variables $u = x - a$:

$$\begin{aligned}\widehat{\tau_a f}(\xi) &= \int_{\mathbb{R}^n} f(x-a) e^{-i\xi \cdot x} dx = \int_{\mathbb{R}^n} f(u) e^{-i\xi \cdot (u+a)} du = e^{-i\xi \cdot a} \int_{\mathbb{R}^n} f(u) e^{-i\xi \cdot u} du = e^{-i\xi \cdot a} \hat{f}(\xi). \\ \widehat{M_a f}(\xi) &= \int_{\mathbb{R}^n} e^{ia \cdot x} f(x) e^{-i\xi \cdot x} dx = \int_{\mathbb{R}^n} f(x) e^{-i(\xi-a) \cdot x} dx = \hat{f}(\xi-a).\end{aligned}$$

□

Lemma 2.56 (Scaling). *Let $f \in L^1(\mathbb{R}^n, \mathbb{C})$ and let $c \in \mathbb{R} \setminus \{0\}$. Define $f_c(x) := f(cx)$. Then, for every $\xi \in \mathbb{R}^n$,*

$$\widehat{f_c}(\xi) = \frac{1}{|c|^n} \hat{f}\left(\frac{\xi}{c}\right).$$

Proof.

The result follows from a change of variables $u = cx$:

$$\widehat{f_c}(\xi) = \int_{\mathbb{R}^n} f(cx) e^{-i\xi \cdot x} dx = \frac{1}{|c|^n} \int_{\mathbb{R}^n} f(u) e^{-i\xi \cdot (u/c)} du = \frac{1}{|c|^n} \int_{\mathbb{R}^n} f(u) e^{-i(\xi/c) \cdot u} du = \frac{1}{|c|^n} \hat{f}\left(\frac{\xi}{c}\right).$$

□

Lemma 2.57 (Differentiation). *Let $f \in C_c^1(\mathbb{R}^n, \mathbb{C})$ and assume that $f \in L^1(\mathbb{R}^n, \mathbb{C})$ and $\partial_j f \in L^1(\mathbb{R}^n, \mathbb{C})$ for some $j \in \{1, \dots, n\}$. Then, for every $\xi \in \mathbb{R}^n$,*

$$\widehat{\partial_j f}(\xi) = i\xi_j \hat{f}(\xi).$$

Proof.

Fix $\xi \in \mathbb{R}^n$. Since $\partial_j f \in L^1$, then $\widehat{\partial_j f}(\xi)$ is well-defined. Let us integrate with Fubini-Tonelli only the j -th coordinate by integration by parts (observe that the boundary term vanishes because f has compact support). Writing $x = (x_j, x')$ and $\xi = (\xi_j, \xi')$,

$$\begin{aligned}\widehat{\partial_j f}(\xi) &= \int_{\mathbb{R}^{n-1}} \left(\int_{\mathbb{R}} \partial_j f(x_j, x') e^{-i\xi_j x_j} dx_j \right) e^{-i\xi' \cdot x'} dx' \\ &= \int_{\mathbb{R}^{n-1}} \left(i\xi_j \int_{\mathbb{R}} f(x_j, x') e^{-i\xi_j x_j} dx_j \right) e^{-i\xi' \cdot x'} dx' \\ &= i\xi_j \int_{\mathbb{R}^n} f(x) e^{-i\xi \cdot x} dx = i\xi_j \hat{f}(\xi).\end{aligned}$$

□

Lemma 2.58 (Fourier transform of a gaussian). *Let $\mu \in \mathbb{R}^n$ and $\sigma > 0$. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined as*

$$f(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{\|x-\mu\|^2}{2\sigma^2}}.$$

Then

$$\widehat{f}(t) = e^{-i\mu \cdot t} e^{-\frac{\sigma^2 \|t\|^2}{2}}.$$

Moreover, if $n = 1$, $\mu = 0$ and $\sigma^2 = 1$, the function is an eigenvector of the Fourier transform.

Proof.

We start from the last result. Let us consider the function

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

We compute

$$\widehat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2/2} e^{-ix\xi} dx = e^{-\xi^2/2} \int_{\mathbb{R}} \frac{e^{-(x+i\xi)^2/2}}{\sqrt{2\pi}} dx.$$

Consider a contour integration of the complex function $\varphi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}$ through the rectangle with vertices $A = -R$, $B = R$, $C = R + i\xi$, $D = -R + i\xi$.

- The side $\overline{AB}$ yields

$$I_1 := \int_{-R}^R \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx$$

- The side $\overline{BC}$ yields

$$I_2 := \int_0^\xi \frac{e^{-(R+ix)^2/2}}{\sqrt{2\pi}} i dx;$$

- The side $\overline{CD}$ yields

$$I_3 := \int_R^{-R} \frac{e^{-(x+i\xi)^2/2}}{\sqrt{2\pi}} dx;$$

- The side $\overline{DA}$ yields

$$I_4 := \int_\xi^0 \frac{e^{-(-R+ix)^2/2}}{\sqrt{2\pi}} i dx.$$

Since $\varphi : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic, by Theorem A.7,

$$I_1 + I_2 + I_3 + I_4 = 0.$$

A trivial change of variable shows that $I_2 = -I_4$, hence $I_1 = -I_3$. Letting $R \rightarrow \infty$, we have that $I_1 = 1$ (by theorem 2.43) and

$$\int_{\mathbb{R}} \frac{e^{-(x+i\xi)^2/2}}{\sqrt{2\pi}} dx = - \lim_{R \rightarrow \infty} \int_R^{-R} \frac{e^{-(x+i\xi)^2/2}}{\sqrt{2\pi}} dx = - \lim_{R \rightarrow \infty} I_3 = \lim_{R \rightarrow \infty} I_1 + I_2 + I_4 = \lim_{R \rightarrow \infty} I_1 = 1.$$

We conclude that $\hat{f}(\xi) = e^{-\xi^2/2}$. Since

$$\mathcal{F}f(\xi) = \sqrt{2\pi}f(\xi),$$

then $f(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ is an eigenvector of the Fourier transform.

If $\mu \neq 0$ or $\sigma^2 \neq 1$, let us define $h(x) := f(x/\sigma)$. By Lemma 2.56, $\hat{h}(\xi) = \sigma \hat{f}(\sigma\xi) = \sigma e^{-\sigma^2\xi^2/2}$. The general 1D Gaussian is given by $g(x) = \frac{1}{\sigma}h(x - \mu) = \frac{1}{\sigma}(\tau_\mu h)(x)$. Using Lemma 2.55, we deduce:

$$\hat{g}(\xi) = \frac{1}{\sigma} \widehat{\tau_\mu h}(\xi) = \frac{1}{\sigma} e^{-i\mu\xi} \hat{h}(\xi) = \frac{1}{\sigma} e^{-i\mu\xi} \sigma e^{-\sigma^2\xi^2/2} = e^{-i\mu\xi} e^{-\sigma^2\xi^2/2}.$$

For the n -dimensional case (isotropic gaussian), we can factor the gaussian into a product of 1D standard Gaussians as follows (by Fubini):

$$\begin{aligned} \hat{f}(t) &= \int_{\mathbb{R}^n} \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{\|x-\mu\|^2}{2\sigma^2}} e^{-it \cdot x} dx \stackrel{y=x-\mu}{=} \int_{\mathbb{R}^n} \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{\|y\|^2}{2\sigma^2}} e^{-it \cdot (y+\mu)} dy \\ &\stackrel{z=y/\sigma}{=} e^{-it \cdot \mu} \int_{\mathbb{R}^n} \frac{1}{(2\pi)^{n/2}} e^{-\frac{\|z\|^2}{2}} e^{-i(\sigma t) \cdot z} dz = e^{-it \cdot \mu} \int_{\mathbb{R}^n} \frac{1}{(2\pi)^{n/2}} e^{-\frac{\sum_{i=1}^n |z_i|^2}{2}} e^{-i\sigma \sum_{i=1}^n t_i z_i} dz \\ &= e^{-it \cdot \mu} \prod_{i=1}^n \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-\frac{|z_i|^2}{2}} e^{-i\sigma t_i z_i} dz = e^{-it \cdot \mu} \prod_{i=1}^n e^{-\frac{\sigma^2 |t_i|^2}{2}} = e^{-it \cdot \mu} e^{-\frac{\sigma^2 \|t\|^2}{2}}. \end{aligned}$$

□

2.6.2 Fourier Transform on L^2

The Fourier transform is initially defined on L^1 , where it produces a continuous function vanishing at infinity in the frequency domain. However, in applications it is typically more useful to extend it to L^2 (since $\lambda^n(\mathbb{R}^n) = \infty$, one does not have $L^2(\mathbb{R}^n) \subseteq L^1(\mathbb{R}^n)$, as shown in Example 1.102). The main reason is that, in quantum physics and signal processing, we often want to study the energy of a signal (a function, or a

density), which is measured by the L^2 norm. In the L^2 theory, the Fourier transform admits a canonical extension to a unitary operator: it preserves the L^2 norm and maps L^2 onto itself (Plancherel's theorem). In this sense, the time domain and the frequency domain become geometrically equivalent.

Definition 2.59 (Fourier anti-transform). Let $f \in L^1(\mathbb{R}^n, \mathbb{C})$. We define its Fourier anti-transform as

$$\mathcal{F}^*(f)(x) = \int_{\mathbb{R}^n} f(\xi) e^{i\xi \cdot x} d\xi.$$

Remark 2.60. If $f \in L^1(\mathbb{R}^n, \mathbb{C})$, then $\mathcal{F}^*(f)(x) = \mathcal{F}(f)(-x)$. Therefore, $\mathcal{F}^* : L^1(\mathbb{R}^n, \mathbb{C}) \rightarrow C_0(\mathbb{R}^n, \mathbb{C})$ satisfies $\|\mathcal{F}^* f\|_{L^\infty} \leq \|f\|_{L^1}$. Moreover, the inverse Fourier transform can be written as $(2\pi)^{-n} \mathcal{F}^*$.

Theorem 2.61 (Fourier inversion theorem). Suppose that $f, \mathcal{F}f \in L^1(\mathbb{R}^n, \mathbb{C})$. Then

$$\mathcal{F}\mathcal{F}^* f = \mathcal{F}^* \mathcal{F} f = (2\pi)^n f \quad (2.17)$$

λ^n -a.e. on $\mathbb{R}^n$.

Proof.

We prove that $\mathcal{F}^* \mathcal{F} f = (2\pi)^n f$ λ^n -a.e.; the identity $\mathcal{F}\mathcal{F}^* f = (2\pi)^n f$ follows from an analogous argument.

Define the Gaussian kernel

$$K_\delta(z) := \frac{1}{(2\pi\delta^2)^{n/2}} e^{-\frac{\|z\|^2}{2\delta^2}}, \quad z \in \mathbb{R}^n.$$

Observe that, for every $\xi \in \mathbb{R}^n$,

$$\lim_{\delta \rightarrow 0^+} e^{-\delta^2 \|\xi\|^2 / 2} = 1.$$

Therefore, we compute:

$$\begin{aligned} (\mathcal{F}^* \mathcal{F} f)(x) &= \int_{\mathbb{R}^n} \hat{f}(\xi) e^{i\xi \cdot x} d\xi \\ &= \int_{\mathbb{R}^n} \hat{f}(\xi) e^{i\xi \cdot x} \left(\lim_{\delta \rightarrow 0^+} e^{-\delta^2 \|\xi\|^2 / 2} \right) d\xi \\ &\stackrel{1}{=} \lim_{\delta \rightarrow 0^+} \int_{\mathbb{R}^n} e^{-\delta^2 \|\xi\|^2 / 2} \hat{f}(\xi) e^{i\xi \cdot x} d\xi \\ &\stackrel{2}{=} \lim_{\delta \rightarrow 0^+} \int_{\mathbb{R}^n} e^{-\delta^2 \|\xi\|^2 / 2} \left(\int_{\mathbb{R}^n} f(y) e^{-i\xi \cdot y} dy \right) e^{i\xi \cdot x} d\xi \\ &\stackrel{3}{=} \lim_{\delta \rightarrow 0^+} \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} e^{-\delta^2 \|\xi\|^2 / 2} e^{i\xi \cdot (x-y)} d\xi \right) dy \\ &\stackrel{4}{=} \lim_{\delta \rightarrow 0^+} \int_{\mathbb{R}^n} f(y) \left(\left(\frac{2\pi}{\delta^2} \right)^{n/2} e^{-\frac{\|x-y\|^2}{2\delta^2}} \right) dy \\ &\stackrel{5}{=} \lim_{\delta \rightarrow 0^+} \int_{\mathbb{R}^n} f(y) \cdot (2\pi)^n \left[\frac{1}{(2\pi\delta^2)^{n/2}} e^{-\frac{\|x-y\|^2}{2\delta^2}} \right] dy \\ &\stackrel{6}{=} (2\pi)^n \lim_{\delta \rightarrow 0^+} \int_{\mathbb{R}^n} f(y) \cdot K_\delta(x-y) dy \\ &\stackrel{7}{=} (2\pi)^n \lim_{\delta \rightarrow 0^+} (f * K_\delta)(x). \end{aligned}$$

Step (1) follows from the Dominated Convergence Theorem (1.80). The hypotheses hold since $\hat{f} \in L^1(\mathbb{R}^n)$ and, for every $\xi \in \mathbb{R}^n$, $x \in \mathbb{R}^n$, and $\delta > 0$,

$$\left| e^{-\delta^2 \|\xi\|^2 / 2} \hat{f}(\xi) e^{i\xi \cdot x} \right| \leq |\hat{f}(\xi)| \in L^1(\mathbb{R}^n).$$

Step (2) is the definition of $\hat{f}(\xi)$. Step (3) relies on the Fubini-Tonelli theorem: the integrand is absolutely integrable since

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(y)| e^{-\delta^2 \|\xi\|^2 / 2} dy d\xi = \|f\|_{L^1} \int_{\mathbb{R}^n} e^{-\delta^2 \|\xi\|^2 / 2} d\xi < \infty$$

Step (4) applies Lemma 2.58 to K_δ , Step (5) is an algebraic rearrangement, Step (6) uses the definition of K_δ , and Step (7) follows from the definition of convolution (see 2.29). Finally, by Corollary 2.45, for any sequence $\delta_k \rightarrow 0^+$, there exists a subsequence $(\delta_{k_j})_{j \in \mathbb{N}}$ such that, λ^n -a.e.,

$$\lim_{j \rightarrow \infty} (f * K_{\delta_{k_j}})(x) = f(x).$$

On the other hand, from the previous steps we already know that the full limit $\lim_{\delta \rightarrow 0^+} (f * K_\delta)(x)$ exists everywhere and equals $(2\pi)^{-n}(\mathcal{F}^* \mathcal{F}f)(x)$. Since the limit of the subsequence must coincide with the limit of the full sequence λ^n -a.e., we conclude that

$$(\mathcal{F}^* \mathcal{F}f)(x) = (2\pi)^n f(x) \quad \lambda^n\text{-a.e.},$$

concluding the proof. $\square$

Theorem 2.62 (Plancherel). *If $f \in L^1 \cap L^2(\mathbb{R}^n, \mathbb{C})$, then $\mathcal{F}f \in L^2(\mathbb{R}^n, \mathbb{C})$ and the following identity holds:*

$$\|\mathcal{F}f\|_{L^2}^2 = (2\pi)^n \|f\|_{L^2}^2. \quad (2.18)$$

Proof.

Assume first that $\mathcal{F}f \in L^1 \cap L^2(\mathbb{R}^n, \mathbb{C})$. Then, by the Fubini-Tonelli theorem and Equation (2.17),

$$\begin{aligned} \|\mathcal{F}f\|_{L^2}^2 &= \langle \mathcal{F}f, \mathcal{F}f \rangle = \int_{\mathbb{R}^n} \mathcal{F}f(\xi) \overline{\mathcal{F}f(\xi)} d\xi = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(x) e^{-i\xi \cdot x} dx \right) \overline{\mathcal{F}f(\xi)} d\xi \\ &= \int_{\mathbb{R}^n} f(x) \left(\int_{\mathbb{R}^n} e^{-i\xi \cdot x} \overline{\mathcal{F}f(\xi)} d\xi \right) dx = \int_{\mathbb{R}^n} f(x) \overline{\int_{\mathbb{R}^n} \mathcal{F}f(\xi) e^{i\xi \cdot x} d\xi} dx \\ &= \langle f, \mathcal{F}^* \mathcal{F}f \rangle = \langle f, (2\pi)^n f \rangle = (2\pi)^n \|f\|_{L^2}^2. \end{aligned}$$

Hence

$$\|\mathcal{F}f\|_{L^2}^2 = (2\pi)^n \|f\|_{L^2}^2. \quad (2.19)$$

Unfortunately, we haven't proved yet that $\mathcal{F}f \in L^2(\mathbb{R}^n, \mathbb{C})$. Therefore, we work on a mollified version of f and then argue with a limiting procedure. As in Theorem 2.61, for $\delta > 0$ define the Gaussian kernel

$$K_\delta(z) := \frac{1}{(2\pi\delta^2)^{n/2}} e^{-\frac{\|z\|^2}{2\delta^2}}, \quad z \in \mathbb{R}^n,$$

and the mollification $f_\delta := f * K_\delta$. The regularized version f_δ satisfies the assumptions for (2.19):

- $f_\delta \in L^1(\mathbb{R}^n, \mathbb{C})$. Since K_δ is an approximate identity as in Definition 2.39, we have $\|K_\delta\|_{L^1} = 1$. Therefore, by Lemma 2.31,

$$\|f_\delta\|_{L^1} = \|f * K_\delta\|_{L^1} \leq \|f\|_{L^1} \|K_\delta\|_{L^1} = \|f\|_{L^1} < \infty.$$

- $f_\delta \in L^2(\mathbb{R}^n, \mathbb{C})$. Applying (2.9) with $f \in L^2(\mathbb{R}^n, \mathbb{C})$ and $K_\delta \in L^1(\mathbb{R}^n, \mathbb{C})$ gives

$$\|f_\delta\|_{L^2} \leq \|f\|_{L^2} \|K_\delta\|_{L^1} = \|f\|_{L^2} < \infty. \quad (2.20)$$

- $\hat{f}_\delta = \mathcal{F}f_\delta \in L^1 \cap L^2(\mathbb{R}^n, \mathbb{C})$. By Equation (2.15) and Lemma 2.58, it holds that

$$\mathcal{F}f_\delta(\xi) = \mathcal{F}(f * K_\delta)(\xi) = \mathcal{F}f(\xi) \cdot \mathcal{F}K_\delta(\xi) = e^{-\frac{\delta^2 \|\xi\|^2}{2}} \mathcal{F}f(\xi).$$

Since $f \in L^1(\mathbb{R}^n, \mathbb{C})$, then by Theorem 2.51 $\|\hat{f}\|_{L^\infty} \leq \|f\|_{L^1} < \infty$. It follows that

$$\int_{\mathbb{R}^n} \left| e^{-\frac{\delta^2 \|\xi\|^2}{2}} \mathcal{F}f(\xi) \right| d\xi \leq \|\mathcal{F}f\|_\infty \int_{\mathbb{R}^n} e^{-\frac{\delta^2 \|\xi\|^2}{2}} d\xi < \infty \implies \hat{f}_\delta \in L^1(\mathbb{R}^n, \mathbb{C}),$$

and

$$\int_{\mathbb{R}^n} \left| e^{-\frac{\delta^2 \|\xi\|^2}{2}} \mathcal{F}f(\xi) \right|^2 d\xi \leq \|\mathcal{F}f\|_\infty^2 \int_{\mathbb{R}^n} e^{-\delta^2 \|\xi\|^2} d\xi < \infty \implies \hat{f}_\delta \in L^2(\mathbb{R}^n, \mathbb{C}).$$

Hence, by (2.19),

$$\|\hat{f}_\delta\|_{L^2}^2 = (2\pi)^n \|f_\delta\|_{L^2}^2 \quad (2.21)$$

or

$$\|\hat{f}(\xi)e^{-\frac{\|\xi\|^2\delta^2}{2}}\|_{L^2}^2 = (2\pi)^n \|f * K_\delta\|_{L^2}^2.$$

By Corollary 2.45, for any sequence $\delta_k \rightarrow 0^+$, there exists a subsequence $(\delta_{k_j})_{j \in \mathbb{N}}$ such that, λ^n -a.e.,

$$\lim_{j \rightarrow \infty} (f * K_{\delta_{k_j}})(x) = f(x).$$

Up to extracting a further subsequence, we may assume $\delta_{k_j} \downarrow 0$.

- By Fatou's lemma 1.78,

$$\begin{aligned} \liminf_{j \rightarrow \infty} \|f_{\delta_{k_j}}\|_{L^2}^2 &= \liminf_{j \rightarrow \infty} \int_{\mathbb{R}^n} |(f * K_{\delta_{k_j}})(x)|^2 dx \\ &\geq \int_{\mathbb{R}^n} \liminf_{j \rightarrow \infty} |(f * K_{\delta_{k_j}})(x)|^2 dx = \int_{\mathbb{R}^n} |f(x)|^2 dx = \|f\|_{L^2}^2. \end{aligned}$$

- By Lemma 2.35,

$$\|f_{\delta_{k_j}}\|_{L^2}^2 = \|f * K_{\delta_{k_j}}\|_{L^2}^2 \leq \|f\|_{L^2}^2 \|K_{\delta_{k_j}}\|_{L^1}^2 = \|f\|_{L^2}^2.$$

Hence

$$\limsup_{j \rightarrow \infty} \|f_{\delta_{k_j}}\|_{L^2}^2 \leq \|f\|_{L^2}^2.$$

- Combining these inequalities, we deduce

$$\lim_{j \rightarrow \infty} \|f_{\delta_{k_j}}\|_{L^2}^2 = \|f\|_{L^2}^2. \quad (2.22)$$

- By the monotone convergence theorem (Theorem 1.50), since the sequence of functions $\xi \mapsto |\hat{f}(\xi)|^2 e^{-\delta_{k_j}^2 \|\xi\|^2}$ increases pointwise to $|\hat{f}(\xi)|^2$ as $j \rightarrow \infty$, we can compute:

$$\lim_{j \rightarrow \infty} \|\hat{f}_{\delta_{k_j}}\|_{L^2}^2 = \lim_{j \rightarrow \infty} \int_{\mathbb{R}^n} |\hat{f}(\xi)|^2 e^{-\delta_{k_j}^2 \|\xi\|^2} d\xi = \int_{\mathbb{R}^n} |\hat{f}(\xi)|^2 d\xi = \|\hat{f}\|_{L^2}^2. \quad (2.23)$$

Hence

$$\|\hat{f}\|_{L^2}^2 \stackrel{2.23}{=} \lim_{j \rightarrow \infty} \|\hat{f}_{\delta_{k_j}}\|_{L^2}^2 \stackrel{2.21}{=} (2\pi)^n \|f_{\delta_{k_j}}\|_{L^2}^2 \stackrel{2.22}{=} (2\pi)^n \|f\|_{L^2}^2,$$

concluding the proof. $\square$

Plancherel theorem is important because it allows us to meaningfully extend $\mathcal{F}$ and $\mathcal{F}^*$ on $L^2(\mathbb{R}^n, \mathbb{C})$, where they become automorphisms⁶. Moreover, one operator is the adjoint of the other, that is, for every $f, g \in L^2(\mathbb{R}^n, \mathbb{C})$,

$$\begin{aligned} \langle \mathcal{F}f, g \rangle &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(x) e^{-i\xi \cdot x} dx \right) \overline{g(\xi)} d\xi \\ &= \int_{\mathbb{R}^n} f(x) \left(\int_{\mathbb{R}^n} e^{-i\xi \cdot x} \overline{g(\xi)} d\xi \right) dx = \int_{\mathbb{R}^n} f(x) \overline{\left(\int_{\mathbb{R}^n} g(\xi) e^{i\xi \cdot x} d\xi \right)} dx \\ &= \langle f, \mathcal{F}^*g \rangle. \end{aligned}$$

Moreover, Theorem 2.61 can be extended to $L^2(\mathbb{R}^n, \mathbb{C})$:

$$\mathcal{F}^* \mathcal{F} f = \mathcal{F} \mathcal{F}^* f = (2\pi)^n f, \quad \forall f \in L^2(\mathbb{R}^n, \mathbb{C}).$$

Remark 2.63. Beyond L^1 and L^2 , it is often convenient to define the Fourier transform on smoother function spaces. A typical limitation of L^1 and L^2 is that they contain functions that may be too irregular for differential calculus. However, the Fourier transform has a fundamental compatibility with differentiation: it

⁶To be formal, for every $f \in L^2(\mathbb{R}^n, \mathbb{C})$, one defines $f_r(x) := f(x) \mathbf{1}_{\|x\| \leq r} \in L^1 \cap L^2(\mathbb{R}^n, \mathbb{C})$, and then sets $\mathcal{F}f(\xi) := \lim_{r \rightarrow \infty} \hat{f}_r(\xi)$. To show that $\mathcal{F}$ is well defined, it suffices to show that, for every $f \in L^1 \cap L^2(\mathbb{R}^n, \mathbb{C})$, the sequence $(\hat{f}_r)_{r \in \mathbb{N}}$ is Cauchy, and then use the completeness of $L^2(\mathbb{R}^n, \mathbb{C})$.

turns derivatives into multiplication by $i\xi$. For this reason, analysts often work on the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ (smooth functions with rapid decay). On $\mathcal{S}$, the Fourier transform is an automorphism: it maps $\mathcal{S}$ bijectively onto itself, continuously. This is the natural setting for defining tempered distributions, including the Dirac delta.

2.7 Discrete Fourier Transform

In this section we study Fourier transforms for functions with *discrete* domain. We distinguish:

- the *discrete-time Fourier transform* (DTFT), acting on functions on $\mathbb{Z}^n$;
- the *discrete Fourier transform* (DFT), acting on functions on $(\mathbb{Z}/N\mathbb{Z})^n$, which coincides with $\{0, \dots, N-1\}^n$, with operations intended modulo N in each component.

2.7.1 Discrete Time Fourier Transform

Definition 2.64 (p -summability). Let $\varphi : \mathbb{Z}^n \rightarrow \mathbb{C}$ be a sequence, commonly denoted by $\varphi = (\varphi_k)_{k \in \mathbb{Z}^n}$. For any $p \in [1, \infty)$, define

$$\|\varphi\|_{\ell^p} := \left(\sum_{k \in \mathbb{Z}^n} |\varphi_k|^p \right)^{1/p}.$$

We say that φ is *p -summable* if $\|\varphi\|_{\ell^p} < \infty$, and we denote by $\ell^p(\mathbb{Z}^n)$ the space of p -summable sequences.

Remark 2.65. The notion of ℓ^p -summability is a special case of L^p -integrability (Definition 1.86) when the measure space is $(\mathbb{Z}^n, 2^{\mathbb{Z}^n}, \#)$, i.e. $\mathbb{Z}^n$ with the counting measure. Unlike general measure spaces, the counting measure admits no non-empty null sets. Hence, two sequences that agree $\#$ -a.e. are in fact identical everywhere, and there is no need to quotient out by null sets.

When endowed with the pointwise operations

$$(\varphi + \psi)_k := \varphi_k + \psi_k \quad \text{and} \quad (\lambda\varphi)_k := \lambda\varphi_k, \quad \forall \varphi, \psi \in \ell^p(\mathbb{Z}^n), \lambda \in \mathbb{R},$$

the space $\ell^p(\mathbb{Z}^n)$ is a vector space, by Proposition 1.95.

We recall that the Fourier transform of a function on the n -torus $\mathbb{T}^n$ can be written as

$$\mathcal{F}f(\xi) = \hat{f}(\xi) := \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} f(x) e^{-i\xi \cdot x} dx, \quad \xi \in \mathbb{Z}^n.$$

By Parseval's identity (2.6),

$$(2\pi)^n \sum_{\xi \in \mathbb{Z}^n} |\hat{f}(\xi)|^2 = \|f\|_{L^2(\mathbb{T}^n)}^2,$$

so that, for $f \in L^2(\mathbb{T}^n)$,

$$\|\mathcal{F}f\|_{\ell^2}^2 = \frac{1}{(2\pi)^n} \|f\|_{L^2}^2, \quad \forall f \in L^1 \cap L^2(\mathbb{T}^n, \mathbb{C}). \quad (2.24)$$

This is precisely the Plancherel Theorem (2.18) over the compact space $\mathbb{T}^n$. Since $\lambda^n(\mathbb{T}^n) < \infty$, we have $L^2(\mathbb{T}^n) \subseteq L^1(\mathbb{T}^n)$ by Theorem 1.101, and therefore $\mathcal{F} : L^2(\mathbb{T}^n) \rightarrow \ell^2(\mathbb{Z}^n)$ is a well-defined bounded linear operator

$$\mathcal{F} : L^2(\mathbb{T}^n) \rightarrow \ell^2(\mathbb{Z}^n).$$

We can also naturally define the inverse operation, often called the Fourier anti-transform or inverse Fourier transform.

Definition 2.66 (Fourier anti-transform). Let $c \in \ell^2(\mathbb{Z}^n, \mathbb{C})$ be a sequence of complex coefficients. We define its inverse Fourier transform as:

$$\mathcal{F}^*(c)(x) = \sum_{\xi \in \mathbb{Z}^n} c(\xi) e^{i\xi \cdot x}.$$

This function is well-defined as an element of $L^2(\mathbb{T}^n, \mathbb{C})$ by Proposition 2.15.

Proposition 2.67 (Fourier inversion theorem). If $f \in L^2(\mathbb{T}^n, \mathbb{C})$ and $c \in \ell^2(\mathbb{Z}^n, \mathbb{C})$, then

$$\mathcal{F}^* \mathcal{F} f = f \quad \lambda^n\text{-a.e. on } \mathbb{T}^n, \quad \mathcal{F} \mathcal{F}^* c = c \quad \text{in } \ell^2(\mathbb{Z}^n).$$

Proof.

The identity $\mathcal{F}^* \mathcal{F} f = f$ is precisely the L^2 -Fourier series expansion associated with the completeness of the trigonometric system proved in Section 2.5. The identity $\mathcal{F} \mathcal{F}^* c = c$ follows by uniqueness of Fourier coefficients in ℓ^2 (2.18). $\square$

Thanks to this inversion property, one may regard $\mathcal{F}^*$ as the Fourier transform on $\ell^2(\mathbb{Z}^n)$, namely the map

$$\mathcal{F}_{\mathbb{Z}^n} : \ell^2(\mathbb{Z}^n) \rightarrow L^2(\mathbb{T}^n), \quad (\mathcal{F}_{\mathbb{Z}^n} \varphi)(x) := \sum_{k \in \mathbb{Z}^n} \varphi(k) e^{ik \cdot x}, \quad (2.25)$$

which is the discrete-time Fourier transform (DTFT) in the ℓ^2 setting. Dual properties hold by exchanging the roles of $\mathcal{F}$ and $\mathcal{F}^*$.

2.7.2 Discrete Fourier Transform

The discrete Fourier transform plays a central role in applied sciences due to its computational efficiency (via the well-known *fast Fourier transform* algorithm) and its ubiquitous applications in studying the spectrum of a signal.

The DFT acts on functions on the finite group $(\mathbb{Z}/N\mathbb{Z})^n$. From a numerical perspective, it can be seen as a finite approximation of the DTFT (over $\mathbb{Z}^n$) or, with an even higher approximation, of the Fourier transform over $L^2(\mathbb{R}^n)$. From an abstract point of view, however, it is a different operator with unique properties.

The norm ℓ^p is still defined by

$$\|\varphi\|_{\ell^p} := \left(\sum_{k \in \{0, \dots, N-1\}^n} |\varphi(k)|^p \right)^{1/p},$$

and the sum is finite for every $p \in [1, \infty)$ (because it is a sum of $N^n < \infty$ complex numbers). Hence, $\ell^p((\mathbb{Z}/N\mathbb{Z})^n) \cong \mathbb{C}^{N^n}$ as a vector space, because any vector in $\mathbb{C}^{N^n}$ has finite ℓ^p norm. Moreover, $\mathbb{C}^{N^n}$ is naturally isomorphic to (the tensor product of) n copies of $\mathbb{C}^N$, and the n -dimensional Fourier basis is obtained by taking the tensor product of the 1D basis functions. Hence, all of its properties (Plancherel, orthogonality...) can be trivially extended on $\ell^2((\mathbb{Z}/N\mathbb{Z})^n)$ from $\ell^2(\mathbb{Z}/N\mathbb{Z})$. To simplify the notation, we focus from now on on the case $n = 1$ and write $\mathbb{Z}_N := \mathbb{Z}/N\mathbb{Z}$. Functions in $\ell^2(\mathbb{Z}_N)$ are represented as complex vectors $x = (x_0, \dots, x_{N-1}) \in \mathbb{C}^N$. Finally, we remark that $\mathbb{C}^N$ is a finite dimensional Hilbert space, with inner product given by

$$\langle x, y \rangle := \sum_{i=0}^{N-1} x_i \overline{y_i}, \quad \forall x, y \in \mathbb{C}^N.$$

Unfortunately, the trigonometric system $\mathcal{F} = \left\{ \frac{1}{\sqrt{2\pi}} e^{ikz} : k \in \mathbb{Z} \right\}$ is no longer a basis for this space. Therefore, we start by defining a suitable Hilbert basis. One option is to use the standard basis $\{e_k : (e_k)_n = \delta_{kn}, k, n \in \{0, 1, \dots, N-1\}\}$. With this option, the Fourier transform would trivially be the identity operator on $\mathbb{C}^N$. Let us introduce the *harmonic basis*:

Lemma 2.68. *If $\omega := e^{-2\pi i/N}$ is a primitive N -th root of unity, then*

$$\sum_{k=0}^{N-1} \omega^{kj} = \begin{cases} N & \text{if } N \mid j, \\ 0 & \text{otherwise.} \end{cases}$$

Proof.

If $N \mid j$, then $j = N\ell$ for some $\ell \in \mathbb{Z}$. Hence

$$\omega^{kj} = (\omega^j)^k = (e^{-2\pi i(N\ell)/N})^k = (e^{-2\pi i\ell})^k = 1.$$

so the sum equals N .

Otherwise, $\omega^j \neq 1$ (since ω is primitive, that would imply $N \mid j$). Thus the sum is a geometric series:

$$\sum_{k=0}^{N-1} \omega^{kj} = \frac{1 - (\omega^j)^N}{1 - \omega^j} = \frac{1 - \omega^{Nj}}{1 - \omega^j} = 0,$$

because $\omega^{Nj} = (\omega^N)^j = 1^j = 1$ and the denominator is nonzero. $\square$

Theorem 2.69 (Discrete Fourier basis). *The system*

$$\mathcal{F} := \left\{ \frac{1}{\sqrt{N}}, \frac{\omega^1}{\sqrt{N}}, \frac{\omega^2}{\sqrt{N}}, \dots, \frac{\omega^{(N-1)}}{\sqrt{N}} \mid \omega = e^{-2\pi i/N} \right\}, \quad \text{where} \quad \omega^{k\cdot} = (\omega^{k\cdot 0} \quad \omega^{k\cdot 1} \quad \dots \quad \omega^{k\cdot (N-1)})^T$$

is a Hilbert basis for $\mathbb{C}^N$.

Proof.

Let $v_k \in \mathbb{C}^N$ be the vector with components $(v_k)_j := \frac{1}{\sqrt{N}} \omega^{kj}$. Then, for $k, \ell \in \{0, \dots, N-1\}$,

$$\langle v_k, v_\ell \rangle = \frac{1}{N} \sum_{j=0}^{N-1} \omega^{(k-\ell)j}.$$

By Lemma 2.68, this equals 1 if $k = \ell$ and 0 otherwise. Hence the family is orthonormal. Since it contains N nonzero vectors in an N -dimensional vector space $\mathbb{C}^N$, it follows that $\mathcal{F}$ is a basis for $\mathbb{C}^N$. Since $\mathbb{C}^N$ is finite dimensional, every basis is complete. $\square$

Definition 2.70 (Discrete Fourier transform (DFT)). Let $x = (x_0, \dots, x_{N-1}) \in \mathbb{C}^N$ and let $\omega := e^{-2\pi i/N}$. The *discrete Fourier transform* of x is the vector $\hat{x} = (\hat{x}_0, \dots, \hat{x}_{N-1}) \in \mathbb{C}^N$ defined by

$$(\mathcal{F}x)(k) := \hat{x}_k := \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \omega^{kj}, \quad k = 0, \dots, N-1.$$

As we can see, since there are no issues of convergence, the discrete Fourier transform is an automorphism of $\mathbb{C}^N$. Therefore, one can define its inverse transform $\mathcal{F}^* : \ell^2(\mathbb{Z}_N) \rightarrow \ell^2(\mathbb{Z}_N)$ as

$$\mathcal{F}^*y(j) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} y_k \omega^{-kj},$$

Proposition 2.71. *Let $x, y \in \mathbb{C}^N$. Then $\mathcal{F}^*$ is the adjoint operator of $\mathcal{F}$, and the following identities hold:*

$$\mathcal{F}\mathcal{F}^*x = \mathcal{F}^*\mathcal{F}x = x \quad \text{and} \quad \|\hat{x}\|_{\ell^2}^2 = \|x\|_{\ell^2}^2. \quad (2.26)$$

Moreover, the transform of their convolution satisfies:

$$(\widehat{x * y})_k = \sqrt{N} \hat{x}_k \hat{y}_k, \quad \text{for all } k \in \{0, \dots, N-1\}.$$

Proof.

We first show that $\mathcal{F}^*$ is the adjoint operator of $\mathcal{F}$:

$$\begin{aligned} \langle \mathcal{F}x, y \rangle &= \sum_{k=0}^{N-1} \left[\frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \omega^{kj} \right] \bar{y}_k = \sum_{j=0}^{N-1} x_j \left[\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \bar{y}_k \omega^{-kj} \right] \\ &= \sum_{j=0}^{N-1} x_j \left[\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} y_k \omega^{-kj} \right] = \langle x, \mathcal{F}^*y \rangle. \end{aligned}$$

Next, we show that $\mathcal{F}\mathcal{F}^*x = \mathcal{F}^*\mathcal{F}x = x$. For every $k \in \{0, \dots, N-1\}$, we compute:

$$(\mathcal{F}\mathcal{F}^*x)_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} (\mathcal{F}^*x)_j \omega^{kj} = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \left[\frac{1}{\sqrt{N}} \sum_{\ell=0}^{N-1} x_\ell \omega^{-\ell j} \right] \omega^{kj} = \frac{1}{N} \sum_{\ell=0}^{N-1} x_\ell \sum_{j=0}^{N-1} \omega^{j(k-\ell)}.$$

To evaluate the inner sum, we recall the orthogonality property of the roots of unity (Lemma 2.68):

$$\sum_{j=0}^{N-1} \omega^{j(k-\ell)} = \begin{cases} N & \text{if } k \equiv \ell \pmod{N}, \\ 0 & \text{otherwise.} \end{cases}$$

Since both k and ℓ range from 0 to $N-1$, the condition $k \equiv \ell \pmod{N}$ is satisfied if and only if $k = \ell$,

meaning only the term $\ell = k$ remains in the outer sum:

$$(\mathcal{F}\mathcal{F}^*x)_k = \frac{1}{N}x_k \cdot N = x_k.$$

A perfectly symmetric argument proves that $\mathcal{F}^*\mathcal{F}x = x$, confirming that $\mathcal{F}^* = \mathcal{F}^{-1}$ and thus the DFT is a unitary operator on $\ell^2(\mathbb{Z}_N) \cong \mathbb{C}^N$.

As a direct consequence, the Discrete Fourier Transform is an isometry. For every $x \in \mathbb{C}^N$, we have:

$$\|\mathcal{F}x\|_{\ell^2}^2 = \langle \mathcal{F}x, \mathcal{F}x \rangle = \langle x, \mathcal{F}^*\mathcal{F}x \rangle = \langle x, x \rangle = \|x\|_{\ell^2}^2.$$

Finally, for every $x, y \in \mathbb{C}^N$ and every $k \in \{0, \dots, N-1\}$, we obtain:

$$\begin{aligned} \widehat{(x * y)}_k &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} (x * y)_j \omega^{kj} = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \left[\sum_{\ell=0}^{N-1} x_\ell y_{j-\ell} \right] \omega^{kj} = \sum_{\ell=0}^{N-1} x_\ell \left[\frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} y_{j-\ell} \omega^{kj} \right] \\ &= \sum_{\ell=0}^{N-1} x_\ell \left[\frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} y_{j-\ell} \omega^{k(j-\ell)} \right] \omega^{k\ell} = \hat{y}_k \sum_{\ell=0}^{N-1} x_\ell \omega^{k\ell} = \sqrt{N} \hat{x}_k \hat{y}_k. \end{aligned}$$

The emergence of the $\sqrt{N}$ factor is a direct and necessary consequence of choosing the unitary normalization $1/\sqrt{N}$ for the transform operator. $\square$

Remark 2.72 (The Fast Fourier Transform). The Discrete Fourier Transform (DFT) is a fundamental tool in numerical analysis and signal processing, as it allows one to decompose a finite sequence into its frequency components, revolutionizing scientific computing. The true power of the DFT lies in its computational efficiency. While a naive matrix-vector multiplication to evaluate the DFT requires $\mathcal{O}(N^2)$ operations, the *Fast Fourier Transform* (FFT) algorithm reduces this complexity to $\mathcal{O}(N \log N)$.

Pontryagin Duality and Symmetry: The Big Picture. To conclude our analysis of the Fourier transform, we observe the profound topological symmetries between the spaces on which it acts. From an abstract algebraic perspective, the Fourier transform is a unitary representation operator between locally compact abelian groups and their duals. Pontryagin duality establishes a perfect inverse relationship between the topology of the temporal/spatial domain and the topology of the frequency spectrum, as summarized in the following table:

Group G (time/space)	Dual group $\widehat{G}$ (frequencies)	Transform
$\mathbb{T}^n$ (compact)	$\mathbb{Z}^n$ (discrete)	Fourier series: $L^2(\mathbb{T}^n) \leftrightarrow \ell^2(\mathbb{Z}^n)$
$\mathbb{Z}^n$ (discrete)	$\mathbb{T}^n$ (compact)	DTFT: $\ell^2(\mathbb{Z}^n) \leftrightarrow L^2(\mathbb{T}^n)$
$\mathbb{R}^n$ (non-compact)	$\mathbb{R}^n$ (non-compact)	Fourier integral: $L^2(\mathbb{R}^n) \leftrightarrow L^2(\mathbb{R}^n)$
$\mathbb{Z}_N^n$ (finite)	$\mathbb{Z}_N^n$ (finite)	DFT: $\mathbb{C}^{N^n} \leftrightarrow \mathbb{C}^{N^n}$

3 Probability Foundations

3.1 A Toy Example

Probability theory historically derives from gambling. It developed to quantify how likely it was for a certain event to happen. We often like to mentally formulate the concept of probability as "what portion of a rectangle is colored". This intuition dates back to middle school. And it seems to work extraordinarily well, as the following examples clarify:

Example 3.1. Let A be the set $\{3, 6, 9, 12\}$ and B be the set $\{2, 4, 6, 8, 10, 12\}$. Furthermore, let $\Omega = \{i \in \mathbb{N}_+ : i \leq 12\} = \{1, \dots, 12\}$. Given a random draw from Ω ,

- The probability of drawing an element of A is $|A|/|\Omega| = 4/12 = 1/3$;
- The probability of drawing an element of B is $|B|/|\Omega| = 6/12 = 1/2$;
- The probability of drawing an element of $A \cap B$ is $|A \cap B|/|\Omega| = 2/12 = 1/6$;
- The probability of drawing an element of $A \cup B$ is $(|A| + |B| - |A \cap B|)/|\Omega| = (4 + 6 - 2)/12 = 2/3$.

Example 3.2. Let C be the square of side 2 centered at the origin of the Cartesian axes, D the circle of radius 1 centered at $(0, 0)$, and Ω the square of side 4 centered at the origin of the Cartesian axes. Given a random point p drawn inside Ω ,

- The probability that $p \in C$ is $\mathcal{A}_C/\mathcal{A}_\Omega = 4/16 = 1/4$;
- The probability that $p \in D$ is $\mathcal{A}_D/\mathcal{A}_\Omega = \pi/16$;
- The probability that $p \in C \setminus D$ is $(\mathcal{A}_C - \mathcal{A}_D)/\mathcal{A}_\Omega = (4 - \pi)/16$.

Here, $\mathcal{A}_X$ denotes the area of the figure X .

I hope no one doubts the validity of the two previous examples, nor that the probability of drawing an element of $A \cup B$ or that $p \in C \setminus D$ is calculated precisely through simple addition and subtraction operations. Yet, the validity of such operations remains unjustified as long as we do not define a rigorous framework that allows us to understand what a probability actually is. It relies on Measure Theory: fortunately, Chapter 1 contains everything we need.

The explanation is more or less summarized in the phrase "Probability is a measure of the possibility that an event will occur", or in the phrase "Probability is a quantification, a measure, of the uncertainty". Pay attention to the two key words: measure and event. In Example 3.1, the family of events considered is $(E_i)_{i \leq 12} = \{\text{element } i \text{ is drawn}\}$, while in Example 3.2, the family of events considered is $(F_\alpha)_{\alpha \subseteq \mathbb{R}^2} = \{\text{point } p \text{ is drawn in region } \alpha \subseteq \Omega\}$. The measure, instead, is our choice! Indeed, we have not specified whether certain elements of Ω are more likely to be drawn than others, or if some regions of Ω are more likely than others. In the absence of information, a mathematician assigns to every event the *uniform* measure most appropriate to the context.⁷

- In Example 3.1, the chosen uniform measure is $\Pr(E_i) = 1/|\Omega| = 1/12$, for every $i \in \{1, \dots, 12\}$.
- In Example 3.2, the chosen uniform measure is $\Pr(F_\alpha) = \mathcal{A}_\alpha/\mathcal{A}_\Omega$, where $\mathcal{A}_\alpha$ denotes the area of the figure $\alpha \subseteq \mathbb{R}^2$.

By treating an event like a set and a probability measure like the measure of a set, a fundamental property is *finite additivity*: for two events that cannot occur simultaneously, the probability of their union is the sum of their individual probability measures. In the case of an intersection, the measure of their intersection is subtracted. Observe that in Example 3.1, the event "drawing an element of A " is equivalent to the union $E_3 \cup E_6 \cup E_9 \cup E_{12}$, thus

$$\Pr(\{\text{drawing an element of } A\}) = \Pr(E_3) + \Pr(E_6) + \Pr(E_9) + \Pr(E_{12}) = \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = \frac{1}{3}.$$

Similarly, it is possible to formally justify all other steps. As for Example 3.2, the chosen uniform measure is such that all results are valid by definition.

⁷Remarkably, there is a beautiful justification *even* for this choice: it is the so-called *Maximum Entropy Principle*, a standard result of *information theory* [14].

3.2 Probability Spaces, Events, and Probability Measures

Building on the previous example, we are ready for the formal construction of probability.

Definition 3.3 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, \Pr)$ where

- i) Ω is a nonempty set (the *sample space*);
- ii) $\mathcal{F} \subseteq 2^\Omega$ is a σ -algebra on Ω (the *event σ -algebra*);
- iii) $\Pr : \mathcal{F} \rightarrow [0, 1]$ is a measure with total mass 1, i.e., $\Pr(\Omega) = 1$ (the *probability measure*).

We call *event* a measurable set, i.e., a set $A \in \mathcal{F}$. The *probability* of A is the number $\Pr(A) \in [0, 1]$.

Remark 3.4. It is typical for mathematicians to start a formal framework by defining a reference space. A space is often, though not always, formed by 1) a universe set Ω , 2) a family of subsets, and 3) one or more operations or objects on the elements of Ω or 2^Ω . Classic examples include a probability space $(\Omega, \mathcal{F}, \Pr)$, a measure space $(X, \mathcal{A}, \mu)$, an inner product space $(V, \langle \cdot, \cdot \rangle)$, or a topological space (X, τ) .

Example 3.5. A coin is tossed until Heads comes up. If H denotes an occurrence of Heads and T an occurrence of Tails, then Ω can be represented as follows:

$$\Omega = \{H, TH, TTH, TTTH, \dots, TTT \dots\}.$$

To better fix ideas, we repeat here the properties that define a probability measure (cf. 1.10):

Definition 3.6 (Probability measure). A *probability measure* $\Pr : \mathcal{F} \rightarrow [0, 1]$ on $(\Omega, \mathcal{F})$ is a function satisfying the following properties:

- i) $\Pr(A) \geq 0$ for all $A \in \mathcal{F}$;
- ii) $\Pr(\Omega) = 1$;
- iii) if $(A_n)_{n \in \mathbb{N}}$ is a sequence of pairwise disjoint sets in $\mathcal{F}$, then

$$\Pr\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \Pr(A_n).$$

Remark 3.7. Property iii) is known as *countable additivity* or σ -additivity. In some contexts, σ -additivity is replaced with a finitely additive version ($\Pr(A \cup B) = \Pr(A) + \Pr(B)$ for every disjoint $A, B \in \mathcal{F}$). Some authors call these functions *finitely additive* probability measures, *normalized positive charges* or, sometimes, simply *probabilities* (e.g., in the subjective approach). These are typically used to extend the notion of probability measures on algebras (which are not necessarily σ -algebras). A setting where this notion is useful is that of the *Anscombe-Aumann* theorem in decision theory.

Proposition 3.8 (Basic properties). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space. Then:

- i) $\Pr(\emptyset) = 0$.
- ii) If $A \in \mathcal{F}$, then $\Pr(A^c) = 1 - \Pr(A)$.
- iii) (Monotonicity) If $A, B \in \mathcal{F}$ and $A \subseteq B$, then $\Pr(A) \leq \Pr(B)$ and $\Pr(B \setminus A) = \Pr(B) - \Pr(A)$.
- iv) (Principle of Inclusion-Exclusion:) If $A, B \in \mathcal{F}$ then $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$.
- v) (Monotone Continuity:)
 - i) (Continuity from below) For every increasing sequence of events $A_1 \subseteq A_2 \subseteq \dots$ (i.e., $A_n \uparrow A$),

$$\lim_{n \rightarrow \infty} \Pr(A_n) = \Pr\left(\bigcup_{n=1}^{\infty} A_n\right) = \Pr(A).$$

- ii) (Continuity from above) For every decreasing sequence of events $A_1 \supseteq A_2 \supseteq \dots$ (i.e., $A_n \downarrow A$),

$$\lim_{n \rightarrow \infty} \Pr(A_n) = \Pr\left(\bigcap_{n=1}^{\infty} A_n\right) = \Pr(A).$$

- vi) Subadditivity: for every sequence $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$, $\Pr\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} \Pr(A_n)$.

Proof.

(i).

$$\Pr(\Omega) + \Pr(\emptyset) = \Pr(\Omega \cup \emptyset) = \Pr(\Omega) \implies \Pr(\emptyset) = 0.$$

(ii). Since $A \cap A^c = \emptyset$ and $A \cup A^c = \Omega$, by additivity,

$$\Pr(A) + \Pr(A^c) = \Pr(A \cup A^c) = \Pr(\Omega) = 1 \implies \Pr(A^c) = 1 - \Pr(A).$$

(iii). If $A \subseteq B$, write $B = A \cup (B \setminus A)$ with disjoint union. Then

$$\Pr(B) = \Pr(A) + \Pr(B \setminus A) \geq \Pr(A),$$

since $\Pr(B \setminus A) \geq 0$.

(iv), (v) and (vi) follow from Proposition 1.12 and countable additivity. $\square$

Definition 3.9 (Limits of a sequence of events). For a sequence of events $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$, we define

$$\begin{aligned} \liminf_{n \rightarrow \infty} A_n &:= \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k = \lim_{n \rightarrow \infty} \bigcap_{k=n}^{\infty} A_k = \{\omega \in \Omega \mid \exists n \geq 1 \forall k \geq n : \omega \in A_k\} = \{A_n \text{ eventually}\} \\ \limsup_{n \rightarrow \infty} A_n &:= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \lim_{n \rightarrow \infty} \bigcup_{k=n}^{\infty} A_k = \{\omega \in \Omega \mid \forall n \geq 1 \exists k \geq n : \omega \in A_k\} = \{A_n \text{ infinitely often}\}. \end{aligned}$$

We say that $(A_n)_{n \in \mathbb{N}}$ admits a limit if

$$\liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n,$$

in which case the limit is

$$\lim_{n \rightarrow \infty} A_n = \liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n.$$

Example 3.10. Tossing a fair coin infinitely many times can be represented with the sample space $\Omega = \{0, 1\}^\infty$, where 0 represents tails and 1 represents heads. Let $A_n = \{\text{heads at toss } n\}$. Then, the event "finitely many heads" corresponds to the event "tails eventually" (i.e., $\liminf A_n^c$). We can compute its probability as follows:

$$\begin{aligned} \Pr(\text{"Finitely many heads"}) &= \Pr(\liminf_{n \rightarrow \infty} A_n^c) = \Pr\left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c\right) \\ &= \Pr\left(\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right)^c\right) = \Pr\left((\limsup_{n \rightarrow \infty} A_n)^c\right) \\ &= 1 - \Pr(\text{"Heads infinitely often"}). \end{aligned}$$

Alternatively, using the monotone continuity of probability from below on the nested intersections, we can write:

$$\Pr(\liminf_{n \rightarrow \infty} A_n^c) = \Pr\left(\lim_{n \rightarrow \infty} \bigcap_{k=n}^{\infty} A_k^c\right) = \lim_{n \rightarrow \infty} \Pr\left(\bigcap_{k=n}^{\infty} A_k^c\right) = 0.$$

In general, De Morgan's laws naturally extend to limits of sets, yielding the duality:

$$\left(\liminf_{n \rightarrow \infty} A_n\right)^c = \limsup_{n \rightarrow \infty} A_n^c \quad \text{and} \quad \left(\limsup_{n \rightarrow \infty} A_n\right)^c = \liminf_{n \rightarrow \infty} A_n^c.$$

Proposition 3.11. For every sequence of events $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$, it holds

$$\Pr(\liminf_{n \rightarrow \infty} A_n) \leq \liminf_{n \rightarrow \infty} \Pr(A_n) \leq \limsup_{n \rightarrow \infty} \Pr(A_n) \leq \Pr(\limsup_{n \rightarrow \infty} A_n)$$

and, if A_n admits a limit, then

$$\Pr(\lim_{n \rightarrow \infty} A_n) = \lim_{n \rightarrow \infty} \Pr(A_n).$$

Proof.

It holds,

$$\Pr(\limsup_{n \rightarrow \infty} A_n) = \Pr\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = \lim_{n \rightarrow \infty} \Pr\left(\bigcup_{k=n}^{\infty} A_k\right) = \limsup_{n \rightarrow \infty} \Pr\left(\bigcup_{k=n}^{\infty} A_k\right) \geq \limsup_{n \rightarrow \infty} \Pr(A_n).$$

By complementing,

$$\begin{aligned} \Pr(\liminf_{n \rightarrow \infty} A_n) &= \Pr\left(\left(\limsup_{n \rightarrow \infty} A_n^c\right)^c\right) = 1 - \Pr(\limsup_{n \rightarrow \infty} A_n^c) \\ &\leq 1 - \limsup_{n \rightarrow \infty} \Pr(A_n^c) = \liminf_{n \rightarrow \infty} (1 - \Pr(A_n^c)) = \liminf_{n \rightarrow \infty} \Pr(A_n). \end{aligned}$$

Finally,

$$\liminf_{n \rightarrow \infty} \Pr(A_n) \leq \limsup_{n \rightarrow \infty} \Pr(A_n)$$

holds because of the properties of sequences of numbers. Setting $\Pr(\lim_{n \rightarrow \infty} A_n) = L$, then

$$L = \Pr(\liminf_{n \rightarrow \infty} A_n) \leq \liminf_{n \rightarrow \infty} \Pr(A_n) \leq \limsup_{n \rightarrow \infty} \Pr(A_n) \leq \Pr(\limsup_{n \rightarrow \infty} A_n) = L.$$

□

3.3 Examples of Probability Measures

In this section we revisit Section 3.1 by formalizing that intuitive framework with our new tools from measure theory. The moral is the following: a probability measure is "context-dependent". The choice of $(\Omega, \mathcal{F}, \Pr)$ depends on what outcomes are considered possible (Ω), which questions are observable ($\mathcal{F}$), and how uncertainty is quantified ($\Pr$).

- i) Example 3.1 is an instance of a *uniform model on a finite set*: $\Omega = \{1, \dots, n\}$, $\mathcal{F} = 2^\Omega$ and $\Pr(A) = |A|/n$, for every $A \subseteq \Omega$;
- ii) A natural generalization of Example 3.1 is a (*purely*) *atomic probability model*: a probability measure supported on countably many *atoms* (here, singletons). Let Ω be finite or countable, $\mathcal{F} = 2^\Omega$, and choose weights $(p_\omega)_{\omega \in \Omega}$ with $p_\omega \geq 0$ and $\sum_{\omega \in \Omega} p_\omega = 1$. Define

$$\Pr(A) = \sum_{\omega \in A} p_\omega, \quad A \subseteq \Omega.$$

This gives a probability space that models a countably (or finitely) supported random outcome.

- iii) Example 3.2 is more similar in spirit to a continuous model, where $\Omega = \mathbb{R}^d$, $\mathcal{F} = \mathcal{B}(\mathbb{R}^d)$, $f : \mathbb{R}^d \rightarrow [0, \infty)$ is a measurable function (called *density*) such that $\int_{\mathbb{R}^d} f d\lambda^d = 1$, and

$$\Pr(A) = \int_A f(x) d\lambda^d(x), \quad A \in \mathcal{B}(\mathbb{R}^d).$$

For instance, if $d = 1$, the standard normal density is

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad x \in \mathbb{R},$$

and it induces a probability measure via $\Pr(A) = \int_A \varphi d\lambda$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. In Example 3.2, the density is uniform on $[-2, 2]^2$:

$$\Pr(A) = \frac{1}{16} \int_A d\lambda^2$$

- iv) Independent experiments, such as tossing a die multiple times, can be modeled via product measures. Given probability spaces $(\Omega_1, \mathcal{F}_1, \Pr_1)$ and $(\Omega_2, \mathcal{F}_2, \Pr_2)$, On $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$ we define the product measure $\Pr_1 \otimes \Pr_2$ as

$$(\Pr_1 \otimes \Pr_2)(A \times B) = \Pr_1(A) \Pr_2(B), \quad A \in \mathcal{F}_1, B \in \mathcal{F}_2.$$

Product measures reappear as the natural setting for sequences of independent trials and, later, for path spaces in stochastic processes.

We observe that, in all these models, events with probability 0 are common. These play an important role in probability theory. We say that A is a *null event* if $\Pr(A) = 0$, and we say that A is an *almost sure event* if $\Pr(A) = 1$.

3.4 Independence and Conditional Probability

In this section we define a foundational concept in probability theory, that of independence. While it is formally defined using the tools of measure theory, independence is specific to probability theory: it is a modeling assumption, and it is one of the key features that distinguishes probability from general analysis. In practice, independence is an assumption one makes to compute probabilities effectively. Its role in probability is similar to that of continuity for real analysis: the first question one should ask when hearing a theorem in the field of probability or statistics is "are we assuming independence?".

Definition 3.12 (Independence of events). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $\{A_i\}_{i \in I} \subseteq \mathcal{F}$ be a family of events. We say that such a family is a family of *independent* events if, for every $n \in \mathbb{N}$ and every choice of *distinct* indices $i_1, \dots, i_n \in I$,

$$\Pr(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_n}) = \Pr(A_{i_1}) \Pr(A_{i_2}) \cdot \dots \cdot \Pr(A_{i_n}). \quad (3.1)$$

In particular, if $A, B \in \mathcal{F}$, we say that A and B are *independent* if

$$\Pr(A \cap B) = \Pr(A) \Pr(B).$$

Remark 3.13. I have always found (3.1) slightly annoying to remember. At first sight, it may seem more natural to require a single identity such as

$$\Pr\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \Pr(A_i).$$

However, this formulation is only meaningful when I is finite or countable. If I is infinite, both expressions become problematic: the intersection $\bigcap_{i \in I} A_i$ need not be measurable when I is uncountable, and an infinite product $\prod_{i \in I} \Pr(A_i)$ is not canonically defined in general. For this reason, the standard definition tests independence only on *finite* subfamilies with *distinct* indices.

When I is finite, this condition is equivalent to the identity involving $\bigcap_{i \in J} A_i$, with $J \subseteq I$, since one can always pad the intersection by adding trivial events equal to Ω . The finite-subfamily formulation is therefore the correct extension to arbitrary index sets.

Definition 3.14 (Pairwise independence). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $\{A_i\}_{i \in I} \subseteq \mathcal{F}$ be a family of events. We say that such a family is a family of *pairwise independent* events if, for every choice of indices $i, j \in I$, $i \neq j$,

$$\Pr(A_i \cap A_j) = \Pr(A_i) \Pr(A_j).$$

It is clear that, if $|I| = 2$, then independence and pairwise independence are equivalent.

Example 3.15. A fair coin is tossed twice. Let A = "Heads at first toss", B = "Heads at second toss", C = "equal outcomes". Then $\{A, B, C\}$ are pairwise independent but not independent events. Indeed,

$$\begin{aligned} \Pr(A) &= \Pr(B) = \Pr(C) = \frac{1}{2} \\ \Pr(A \cap B) &= \Pr(\{HH\}) = \frac{1}{4} = \Pr(A) \Pr(B) \\ \Pr(A \cap C) &= \Pr(\{HH\}) = \frac{1}{4} = \Pr(A) \Pr(C) \\ \Pr(B \cap C) &= \Pr(\{HH\}) = \frac{1}{4} = \Pr(B) \Pr(C). \\ \Pr(A \cap B \cap C) &= \Pr(\{HH\}) = \frac{1}{4} \neq \Pr(A) \Pr(B) \Pr(C) = \frac{1}{8} \end{aligned}$$

Intuitively, knowing A and B completely determines C (if A and B happen, it's HH , so C must happen), destroying independence.

Proposition 3.16 (Stability under complements). *If A and B are independent, then the pairs*

$$(A^c, B), (A, B^c), (A^c, B^c)$$

are also independent.

Proof.

Assume $\Pr(A \cap B) = \Pr(A) \Pr(B)$. Then

$$\Pr(A^c \cap B) = \Pr(B) - \Pr(A \cap B) = \Pr(B) - \Pr(A) \Pr(B) = (1 - \Pr(A)) \Pr(B) = \Pr(A^c) \Pr(B).$$

The other cases follow analogously, using $\Pr(B^c) = 1 - \Pr(B)$ and the inclusion-exclusion identity. $\square$

Remark 3.17. As one can show, null events and almost sure events are always independent of any other event.

We next define the concept of conditional probability. In itself, this concept can be viewed both as a useful notation and as a convenient quantity for inference and practical applications.

Definition 3.18 (Conditional Probability). Let A and B be events such that $\Pr(B) > 0$. Then the *conditional probability of A given B* is defined as

$$\Pr(A \mid B) := \frac{\Pr(A \cap B)}{\Pr(B)}$$

Lemma 3.19 (Law of total probability). Let $\{B_i\}_{i \in I} \subseteq \mathcal{F}$ be a countable or finite family of pairwise disjoint events such that $\bigcup_{i \in I} B_i = \Omega$ and $\Pr(B_i) > 0$ for every $i \in I$. Then, for every event $A \in \mathcal{F}$,

$$\Pr(A) = \sum_{i \in I} \Pr(A \mid B_i) \Pr(B_i).$$

Proof. Since $A = \bigcup_{i \in I} (A \cap B_i)$, then $\Pr(A) = \sum_{i \in I} \Pr(A \cap B_i) = \sum_{i \in I} \Pr(A \mid B_i) \Pr(B_i)$. $\square$

Proposition 3.20 (Chain rule for probabilities). Let $A_1, \dots, A_n$ be events such that $\Pr(\bigcap_{k < j} A_k) > 0$ for all j . Then

$$\Pr(A_1 \cap \dots \cap A_n) = \prod_{j=1}^n \Pr(A_j \mid A_1 \cap \dots \cap A_{j-1}).$$

Proof.

The proof follows from the definition of conditional probability and an iterated expansion:

$$\begin{aligned} \Pr\left(\bigcap_{j=1}^n A_j\right) &= \Pr\left(A_n \mid \bigcap_{k=1}^{n-1} A_k\right) \Pr\left(\bigcap_{k=1}^{n-1} A_k\right) \\ &= \Pr\left(A_n \mid \bigcap_{k=1}^{n-1} A_k\right) \Pr\left(A_{n-1} \mid \bigcap_{k=1}^{n-2} A_k\right) \dots \Pr(A_2 \mid A_1) \Pr(A_1) \\ &= \prod_{j=1}^n \Pr\left(A_j \mid \bigcap_{k=1}^{j-1} A_k\right), \end{aligned}$$

where for $j = 1$ we adopt the convention $\bigcap_{k=1}^0 A_k = \Omega$, so that $\Pr(A_1 \mid \Omega) = \Pr(A_1)$. $\square$

Lemma 3.21 (Independence of two events). If A and B are events such that $\Pr(A) > 0$ and $\Pr(B) > 0$, then A and B are independent if and only if

$$\Pr(A \mid B) = \Pr(A) \quad \text{or, equivalently,} \quad \Pr(B \mid A) = \Pr(B).$$

Proof. Just combine Definitions 3.12 and 3.18. $\square$

In other words, A is independent of B if knowing whether or not B occurred does not change our assessment of the probability of A . Moreover, this property is symmetric: knowing that A occurred does not affect the probability of B . We can view this in an alternative way using Lemma 3.19: observe that, for every event B such that $0 < \Pr(B) < 1$:

$$\Pr(A) = \Pr(A \mid B) \Pr(B) + \Pr(A \mid B^c) \Pr(B^c). \quad (3.2)$$

If $\Pr(A \mid B) = \Pr(A)$, then the equation implies $\Pr(A \mid B^c) = \Pr(A)$ as well. This means that neither the occurrence nor the non-occurrence of B influences A .

Theorem 3.22 (Bayes). Let $\{B_i\}_{i \in I} \subseteq \mathcal{F}$ be a countable or finite family of pairwise disjoint events such that $\bigcup_{i \in I} B_i = \Omega$ and $\Pr(B_i) > 0$ for every $i \in I$. Then, for every event $A \in \mathcal{F}$ such that $\Pr(A) > 0$ and every $i \in I$,

$$\Pr(B_i | A) = \frac{\Pr(A | B_i) \Pr(B_i)}{\sum_{j \in I} \Pr(A | B_j) \Pr(B_j)}, \quad \forall i \in I.$$

Proof. It holds

$$\Pr(B_i | A) = \frac{\Pr(A \cap B_i)}{\Pr(A)} = \frac{\Pr(A | B_i) \Pr(B_i)}{\sum_{j \in I} \Pr(A | B_j) \Pr(B_j)}.$$

□

Remark 3.23 (Bayes' rule: priors and posteriors). A convenient interpretation of Bayes' formula is in terms of *prior* and *posterior* probabilities. The numbers $\Pr(B_i)$ represent the prior belief that hypothesis B_i holds before observing the evidence A . The conditional probabilities $\Pr(A | B_i)$ are the *likelihoods* of observing A under each hypothesis. Bayes' rule then updates priors into posteriors:

$$\Pr(B_i | A) = \frac{\Pr(A | B_i) \Pr(B_i)}{\sum_{j \in I} \Pr(A | B_j) \Pr(B_j)}.$$

The denominator $\Pr(A) = \sum_{j \in I} \Pr(A | B_j) \Pr(B_j)$ is the *evidence* (or *marginal likelihood*), i.e. the total probability of observing A under the model. This update mechanism is the basic ingredient of Bayesian statistics.

Remark 3.24 (Conditional independence). Independence is so important in applied mathematical modeling, especially in statistics, that when it does not hold one often tries to suitably modify the setting so that it does. A standard solution is *conditional independence*. Sometimes, a family of events becomes independent "once conditioned on the right thing".

To make this idea concrete, think of two thermometers A and B placed in the same room. If we look only at their readings, they typically move together: when A reports a high temperature, B is also more likely to report a high temperature. So the two observations are not independent.

However, this dependence is not because the devices "communicate", but because they share a common driving factor: the true (unobserved) room temperature T . Each instrument outputs the same underlying signal T plus its own measurement noise. Once we condition on T - that is, once we fix the actual temperature of the room - the temperature observed on the two thermometers will be independent, depending only on the sensor errors. Equivalently, after we know T , learning the value of A provides no additional information about B beyond what T already tells us.

Mathematically, conditional independence of events A and B given an event C (with $\Pr(C) > 0$) occurs whenever $\Pr(A \cap B | C) = \Pr(A | C) \Pr(B | C)$. Of course, this notion can be suitably generalized to more events. Conditional independence does not imply independence and, vice versa, independence does not imply conditional independence.

3.5 Random Variables

Random variables are the central objects in probability and statistics. Formally, a random variable is a measurable function from a probability space into a measurable space (typically $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$). With some exceptions (Example 3.51), random variables are typically discrete or continuous, and it is often convenient to treat these cases separately. We will nevertheless aim for a unified picture whenever possible. Indeed, discrete random variables are often easier to visualize, but they lack the smooth structure of the continuous case, so arguments can become cumbersome. Meanwhile, continuous models can be less intuitive at first, yet they come with a rich analytic structure and often lead to more elegant and conceptual results.

Definition 3.25 (Random variable). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space. A *random variable* X is a measurable map $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e., $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$.

Example 3.26 (Roll of a die). Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ be the space of outcomes of the roll of a fair six-sided die. The corresponding model (see Section 3.3) dictates $\mathcal{F} = 2^\Omega$ and $\Pr(\{\omega\}) = 1/6$ for every $\omega \in \Omega$. We define the function $X : \Omega \rightarrow \mathbb{R}$ such that

$$X(\omega) = \begin{cases} 1 & \text{if } \omega \text{ is even} \\ 0 & \text{if } \omega \text{ is odd} \end{cases}.$$

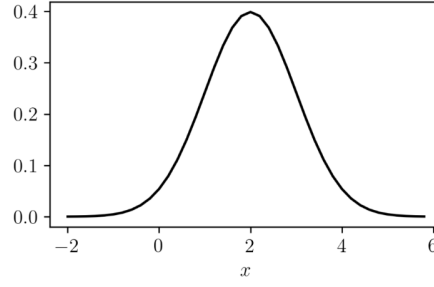


Figure 1: Gaussian Distribution

Here, $X(\omega)$ denotes the value taken by X if the outcome of the roll is ω . Then, for every $S \in \mathcal{B}(\mathbb{R})$,

$$X^{-1}(S) = \begin{cases} \emptyset & \text{if } 0 \notin S, 1 \notin S \\ \{2, 4, 6\} & \text{if } 0 \notin S, 1 \in S \\ \{1, 3, 5\} & \text{if } 0 \in S, 1 \notin S \\ \Omega & \text{if } 0 \in S, 1 \in S \end{cases}.$$

Therefore, X is a random variable.

Example 3.27 (Gaussian random variable). One can think of random variables as numerical quantities that depend on a specific outcome, and whose values can be more or less plausible. For example, if X denotes the height of an Italian male of 25 years old, then the value of X is not determined a priori, and will depend on the specific person selected. However, some values are far more frequent than others. The following graph 1 gives a visual idea of how certain values of X are unlikely (the tails), while others are very common (the peak). Observe how the notion of random variable is getting *dangerously* close to the realm of statistics.

Remark 3.28 (Notation for events). In probability theory, we frequently deal with events defined by the values of random variables. To simplify notation, we adopt the following conventions for subsets of Ω :

- $(X \in A) := \{\omega \in \Omega : X(\omega) \in A\} = X^{-1}(A)$;
- $(X = x, b \leq Y \leq c) := \{\omega \in \Omega : X(\omega) = x \text{ and } Y(\omega) \in [b, c]\} = X^{-1}(\{x\}) \cap Y^{-1}([b, c])$;
- $(X \in A, Y \in B) := (X \in A) \cap (Y \in B) = ((X, Y) \in A \times B)$.

Remark 3.29. Thanks to Proposition 1.28 and Example 1.29, one can equivalently say that $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a random variable if, for every $a \in \mathbb{R}$, $(X \leq a) \in \mathcal{F}$.

To better understand the likelihood of a certain value of a random variable, it is necessary to relate the random variable with the probability measure of the space. Before doing that, let us generalize temporarily the notion of random variable.

Definition 3.30 (Random element). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and $(E, \mathcal{E})$ a measurable space. A (general) *random variable* is a $\mathcal{E}$ -measurable map

$$X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$$

i.e., $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{E}$. If $E = \mathbb{R}^d$ and $\mathcal{E} = \mathcal{B}(\mathbb{R}^d)$, we speak of a (real-valued) *random vector*.

Definition 3.31 (Law of a random element). Let $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (E, \mathcal{E})$ be a random element. The *law* (or *distribution*) of X is the probability measure $\Pr_X$ on $(E, \mathcal{E})$, defined by the pushforward

$$\Pr_X(B) := X_{\#} \Pr(B) = \Pr(\{X \in B\}) = \Pr(X^{-1}(B)) = \Pr(\{\omega \in \Omega : X(\omega) \in B\}), \quad B \in \mathcal{E}.$$

In other words, we call *distribution* of X the probability measure $\mu_X = \Pr \circ X^{-1}$ on $(E, \mathcal{E})$.

After choosing a map X and a probability measure $\Pr$ on $(\Omega, \mathcal{F})$, it is possible to measure sets in $\mathcal{E}$ through $\Pr_X = \Pr \circ X^{-1}$. The way it is done depends on whether the random element is discrete, singular continuous or absolutely continuous (see Section 1.8) with respect to a canonical measure on E (typically λ if $E \cong \mathbb{R}^k$).

Definition 3.32 (Support for a random element). The support of a random element $X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$, where (E, τ_E) is a topological space, is the support of its distribution $\Pr_X$, given by

$$x \in \text{supp}(\Pr_X) \iff \forall U \ni x \text{ open, } \Pr_X(U) > 0.$$

Remark 3.33 (Terminology). Formally, a *random element* is a measurable map $X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$. When $(E, \mathcal{E}) = (\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$, one speaks of a *real-valued random vector*, where $X(\omega) = (X_1(\omega), \dots, X_k(\omega))$. In practice, we usually omit "real-valued" and simply say *random vector*, even though the term could in principle refer to $(\mathbb{C}^k, \mathcal{B}(\mathbb{C}^k))$. When $k = 1$, we speak of a *random variable*, implicitly meaning *real-valued*. Occasionally, the literature uses the phrase *random variable* even when $k > 1$, whenever the dimension plays no role and the arguments are identical to the one-dimensional case. In these notes, I will try to be consistent, yet some ambiguities may arise; the intended meaning should be clear from the context.

Proposition 3.34 (Stability of random vectors). *Let $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ be random vectors, $(X_n)_{n \in \mathbb{N}}$ be a sequence of random vectors, and let $\phi : \mathbb{R}^k \rightarrow \mathbb{R}^j$ be a measurable function. Then:*

- i) $Z = \phi(X)$ is a random vector (in $\mathbb{R}^j$);
- ii) $X + Y$ is a random vector;
- iii) The scalar product $X \cdot Y$ is a random variable;
- iv) If the sequence $(X_n)_{n \in \mathbb{N}}$ converges pointwise to X , then X is a random vector.

Moreover, if $k = 1$ (i.e., X, Y and X_n are random variables), then:

- v) XY , $\max(X, Y)$ and $\min(X, Y)$ are random variables;
- vi) $X_+ = \max(X, 0)$, $X_- = \max(-X, 0)$ and $|X|$ are random variables;
- vii) $\sup_{n \in \mathbb{N}} X_n$, $\inf_{n \in \mathbb{N}} X_n$, $\limsup_{n \rightarrow \infty} X_n$ and $\liminf_{n \rightarrow \infty} X_n$ are random variables (valued in $\overline{\mathbb{R}}$).

Proof.

(i) holds by Lemma 1.27; (ii), (iii), (v) and (vi) hold by Proposition 1.31; (iv) and (vii) hold by Lemma 1.34. $\square$

3.6 Law of a Random Variable

In this section, we expand the notion of law of a random variable by restricting on real-valued random variables. From now on, $(\Omega, \mathcal{F}, \Pr)$ denotes a probability space, λ^k denotes the k -th dimensional Lebesgue measure on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$, and random variables are $\mathcal{B}(\mathbb{R}^k)$ -measurable functions $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$.

Definition 3.35 (Discrete Probability Measure). A probability measure $\Pr$ on $(\Omega, \mathcal{F})$ is discrete if there exists a countable set of measurable points (called *atoms*) $S = \{x_n\}_{n \in \mathbb{N}} \subseteq \Omega$ such that $\Pr(S) = 1$ and $\Pr(\{x_n\}) > 0$ for every $n \in \mathbb{N}$. In this case, $\Pr$ can be written as a convex combination of Dirac measures (see Example 1.14):

$$\Pr = \sum_{n=1}^{\infty} p_n \delta_{x_n},$$

where $p_n = \Pr(\{x_n\}) > 0$ and $\sum_{n=1}^{\infty} p_n = 1$.

Definition 3.36 (Support). The *support* of a probability measure $\Pr$, denoted by $\text{supp}(\Pr)$, is intuitively the smallest closed set containing all the "probability mass".

- If $\Pr$ is a discrete measure, the support is the set of atoms: $\text{supp}(\Pr) = \{\omega \in \Omega : \Pr(\{\omega\}) > 0\}$.
- If Ω is a topological space (like $\mathbb{R}^d$), $\text{supp}(\Pr)$ is defined as the set of all points x for which every open neighborhood has positive probability. Equivalently, it is the intersection of all closed sets C such that $\Pr(C) = 1$.

Definition 3.37 (Discrete random variable). A real-valued random variable X is called *discrete* if its law is supported on a countable set, i.e., there exists a countable set $S = \{x_i\}_{i \in I} \subseteq \mathbb{R}$ such that $\Pr_X(S) = \Pr(X \in S) = 1$.⁸

Example 3.38 (Law of a discrete random variable). Combining Definitions 3.31 and 3.37, we deduce that the law $\Pr_X$ of a discrete random variable X is given by

$$\Pr_X(B) = \Pr(\{X \in B\}) = \sum_{i \in I} \Pr(X = x_i) \mathbf{1}_B(x_i), \quad B \in \mathcal{B}(\mathbb{R}).$$

⁸Equivalently, $\Pr_X$ is an *atomic* probability measure with at most countably many atoms.

As we see, the set S together with the probabilities $\Pr(X = x_i)$ describe the probability distribution of X . Therefore, we can encompass this information in a function as follows:

Definition 3.39 (Probability Mass function). The *probability mass function* (PMF) of a discrete random variable $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a function $p_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$p_X(x) = \Pr(X = x).$$

We often write $p(x)$ in place of p_X , when the context is clear.

From the above definition, the following properties are trivial:

- i) $p(x) = \sum_{i \in I} \Pr(X = x_i) \mathbf{1}_{\{x_i\}}(x)$;
- ii) $0 \leq p(x) \leq 1$ for every $x \in \mathbb{R}$;
- iii) $p(x) = 0$ if $x \notin S$, where S is the set of atoms (support) of X .
- iv) $\sum_{i \in I} p(x_i) = 1$.
- v) For every $B \in \mathcal{B}(\mathbb{R})$, $\Pr(X \in B) = \sum_{x \in B \cap S} p(x)$.

Remark 3.40. Note that, to simplify notation, we typically write $\Pr(X \in B)$ instead of the more formal set-theoretic notation $\Pr(\{\omega \in \Omega : X(\omega) \in B\})$.

Definition 3.41 (Continuous random variable). A real-valued random variable X is called (*absolutely continuous*) if its law satisfies, for every $B \in \mathcal{B}(\mathbb{R})$,

$$\lambda(B) = 0 \implies \Pr(X \in B) = 0.$$

In other words, its law $\Pr_X$ is absolutely continuous with respect to Lebesgue measure λ on $\mathbb{R}$: $\Pr_X \ll \lambda$.

Remark 3.42. From now on, we will simply say that X is a *continuous random variable*. This shorthand should be understood as: " X is a real-valued random variable whose law $\Pr_X$ is absolutely continuous with respect to the Lebesgue measure λ ."

Example 3.43 (Law of a continuous random variable). Combining Definition 3.31, Definition 3.41 and Theorem 1.109, we deduce that the Law can be described through a λ -a.e. unique measurable function $f : \mathbb{R} \rightarrow [0, \infty)$ such that

$$\Pr_X(B) = \Pr(X \in B) = \int_B f(x) d\lambda(x) = \int_B f(x) dx, \quad B \in \mathcal{B}(\mathbb{R}).$$

In particular, for an interval (a, b) ,

$$\Pr(a < X < b) = \int_{(a,b)} f(x) d\lambda = \int_a^b f(x) dx.$$

In the context of measure theory, f is known as *Radon-Nikodym derivative* or *density*, denoted $f = \frac{d\Pr_X}{d\lambda}$. In the context of probability theory, it is known as *probability density function*:

Definition 3.44 (Probability Density Function). Let X be a continuous random variable. A *probability density function* of X is *any* measurable function $f_X : \mathbb{R} \rightarrow [0, \infty)$ such that

$$\Pr(X \in B) = \int_B f_X(x) d\lambda(x), \quad B \in \mathcal{B}(\mathbb{R}). \quad (3.3)$$

We often write $f(x)$ in place of f_X when the context is clear.

Remark 3.45. The existence of f_X for every continuous random variable is guaranteed by the *Radon-Nikodym Theorem* (1.109), and it is unique λ -a.e. Furthermore, there always exists a non-negative version of f_X . Consequently, we conventionally treat non-negative functions as PDFs and, whenever possible, we select the "simplest" representative, typically one that is continuous or has the fewest possible discontinuities. In general, any function f such that $f(x) \geq 0$ for all $x \in \mathbb{R}$ and $\int_{-\infty}^{+\infty} f(x) dx = 1$ serves as a probability density function for some random variable X .

Example 3.46 (PDFs are only unique λ -a.e.). Let X be uniformly distributed on $[0, 1]$. Define

$$f(x) := \mathbf{1}_{[0,1]}(x), \quad g(x) := \mathbf{1}_{[0,1]}(x) + \mathbf{1}_{\{0\}}(x).$$

Then f and g are both nonnegative and satisfy

$$\int_{\mathbb{R}} f(x) dx = \int_{\mathbb{R}} g(x) dx = 1.$$

Moreover, for every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$\int_B g(x) dx = \int_B f(x) dx + \int_B \mathbf{1}_{\{0\}}(x) dx = \int_B f(x) dx,$$

since $\lambda(\{0\}) = 0$. Hence f and g define the same law via (3.3), although $1 = f(0) \neq g(0) = 2$. In particular, $f = g$ λ -a.e.

Example 3.47 (Continuous Random Vectors). Let $(X, Y) : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$ be a random vector whose law $\Pr_{(X,Y)}$ is absolutely continuous with respect to the two-dimensional Lebesgue measure λ^2 . Then, by Theorem 1.109, (X, Y) admits a *joint probability density function* $f_{X,Y} : \mathbb{R}^2 \rightarrow [0, \infty)$ such that

$$\Pr((X, Y) \in B) = \iint_B f_{X,Y}(x, y) d\lambda^2(x, y), \quad B \in \mathcal{B}(\mathbb{R}^2).$$

Example 3.48 (A mixed random variable). The distinction between discrete and absolutely continuous random variables does not properly capture every possible behaviour for the law of a random variable. A common instance is that of a mixed law. Consider the following random variable:

$$X = \begin{cases} 0 & \text{with probability } 1/3 \\ Y & \text{with probability } 2/3 \end{cases},$$

where Y is a continuous random variable with probability density function $f(x) = e^{-x} \mathbf{1}_{(0, \infty)}(x)$, and Y is independent of the first choice determining X . Then the law of X is the *mixture*

$$\Pr_X(B) = \frac{1}{3} \mathbf{1}_B(0) + \frac{2}{3} \int_{B \cap (0, \infty)} e^{-x} dx,$$

which captures an atomic spike at zero, and a distributed distribution of $(0, \infty)$. There are, however, even wilder distributions, which are called *singular probability measures*.

Singular probability measures. As Theorem 1.110 clarifies, every probability measure on $\mathbb{R}^k$ can be decomposed into an absolutely continuous component and a singular component:

$$\Pr_X(A) = \Pr_{ac}(A) + \Pr_s(A) = \int_A f d\lambda^k + \Pr_s(A).$$

Let $D = \{x \in \mathbb{R}^k : \Pr_X(\{x\}) > 0\}$. Then D is at most countable, as the next lemma shows:

Lemma 3.49. *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space, and let $\{B_t\}_{t \in T} \subseteq \mathcal{F}$ be a family of pairwise disjoint events such that $\Pr(B_t) > 0$ for every $t \in T$. Then T is at most countable.*

Remark 3.50. The same argument applies, with minor modifications, to any σ -finite measure space.

Proof.

Let $|\cdot|$ denote the cardinality of a set. For every $n \in \mathbb{N}$, let

$$T_n := \{t \in T : \Pr(B_t) > 1/n\}.$$

Then $T = \bigcup_{n=1}^{\infty} T_n$. Hence

$$|T| = \left| \bigcup_{n=1}^{\infty} T_n \right| \leq \sum_{n=1}^{\infty} |T_n|.$$

If each T_n is finite, it follows that T is at most countable. In fact, we can prove that $|T_n| < n$. By utilizing the disjointness of the sets and the finite additivity of probability:

$$1 = \Pr(\Omega) \geq \Pr\left(\bigcup_{t \in T_n} B_t\right) = \sum_{t \in T_n} \Pr(B_t) > \sum_{t \in T_n} \frac{1}{n} = \frac{|T_n|}{n},$$

concluding the proof. □

Since D is at most countable, let us denote it as $D = \{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}^k$. The restriction of our random variable's law to D defines a *discrete* distribution with probability mass function $p_X(x) = \Pr(X = x)$. For every Borel set $A \in \mathcal{B}(\mathbb{R}^k)$, we can rigorously define a *discrete component* of $\Pr_X$ as:

$$\Pr_d(A) = \sum_{x \in A \cap D} p_X(x) = \Pr_s(A \cap D) \leq \Pr_s(A) \leq \Pr_X(A) \leq 1.$$

Consequently, every probability measure $\Pr_X$ on $\mathbb{R}^k$ can be uniquely decomposed as

$$\Pr_X = \Pr_{ac} + \Pr_d + \Pr_{sc}, \quad \Pr_X(A) = \int_A f d\lambda^k + \sum_{x \in A \cap D} p_X(x) + \Pr_{sc}(A),$$

where $\Pr_{sc}$ is the *singular continuous* part (i.e., singular with respect to λ^k and atom-free).

Example 3.51 (Nontrivial singular continuous distribution). Let $(X, Y) : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2), \lambda^2)$ be a random vector such that $X + Y = 1$ almost surely and

$$\Pr(X \in B) = \int_B \mathbf{1}_{[0,1]}(x) d\lambda(x), \quad B \in \mathcal{B}(\mathbb{R}),$$

i.e. X is uniformly distributed on $[0, 1]$ (and $Y = 1 - X$). The law $\Pr_{(X,Y)}$ is not discrete, since

$$\Pr_{(X,Y)}(\{(x, y)\}) = 0 \quad \text{for every } (x, y) \in \mathbb{R}^2.$$

On the other hand, $\Pr_{(X,Y)} \perp \lambda^2$, because $\Pr_{(X,Y)}$ is supported on the line $L := \{(x, y) \in \mathbb{R}^2 : x + y = 1\}$, and $\lambda^2(L) = 0$. More generally, the same phenomenon occurs for random vectors in $\mathbb{R}^k$ whose law is supported on a smooth manifold of dimension $< k$. For further examples, see [5].

3.7 Cumulative Distribution Function

The summary of Section 3.6 is that, given a probability measure $\Pr : \mathcal{F} \rightarrow [0, 1]$ on a measurable space $(\Omega, \mathcal{F})$ and a random element $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (E, \mathcal{E})$, we can define a measure on $(E, \mathcal{E})$ called the *law* of X , denoted by $\Pr_X$ and defined via the pushforward $\Pr_X = X_{\#} \Pr$.

Focusing on random vectors $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$, it is possible to uniquely decompose $\Pr_X$ into a discrete part $\Pr_d$, an absolutely continuous part $\Pr_{ac}$, and a singular continuous part $\Pr_{sc}$. From now on, we will restrict our analysis to random vectors without a singular component, as they lack a tractable analytic structure.

Furthermore, to characterize $\Pr_d$ and $\Pr_{ac}$, we can employ two specifically defined functions. The first is the *probability mass function* p_X , defined in Definition 3.39, such that $\Pr_d(A) = \sum_{x \in A \cap D} p_X(x)$ for every $A \in \mathcal{B}(\mathbb{R}^k)$, where D is the support of the discrete distribution. The second is the *probability density function* f_X , whose existence is guaranteed by the Radon-Nikodym Theorem, such that $\Pr_{ac}(A) = \int_A f_X d\lambda^k$ for every $A \in \mathcal{B}(\mathbb{R}^k)$.

There is, however, another function widely used in probability and statistics to describe the law of a random vector taking values in $\mathbb{R}^k$, called the *cumulative distribution function* (CDF).

Definition 3.52 (Cumulative distribution function of a random vector). Let $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ be a random vector. The *cumulative distribution function* (CDF) of X is the function $F : \mathbb{R}^k \rightarrow [0, 1]$ defined by

$$F(x) = \Pr(X_1 \leq x_1, \dots, X_k \leq x_k), \quad x = (x_1, \dots, x_k) \in \mathbb{R}^k.$$

Example 3.53. To compute the probability of a multi-dimensional rectangle $R = \prod_{i=1}^k (a_i, b_i]$, one uses the principle of inclusion-exclusion on the CDF. For instance, if $k = 2$ and $R = (a_1, b_1] \times (a_2, b_2]$, then

$$\Pr(a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2) = F(b_1, b_2) - F(a_1, b_2) - F(b_1, a_2) + F(a_1, a_2).$$

If $k = 3$ and $R = (a_1, b_1] \times (a_2, b_2] \times (a_3, b_3]$, the formula involves $2^3 = 8$ terms:

$$\begin{aligned} \Pr(X \in R) &= F(b_1, b_2, b_3) - F(a_1, b_2, b_3) - F(b_1, a_2, b_3) - F(b_1, b_2, a_3) \\ &\quad + F(a_1, a_2, b_3) + F(a_1, b_2, a_3) + F(b_1, a_2, a_3) - F(a_1, a_2, a_3). \end{aligned}$$

Let us now restrict our focus to real-valued random variables $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Definition 3.54 (Cumulative Distribution Function). Let X be a random variable. The *cumulative distribution function* (CDF) of X is the function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(x) = \Pr(X \leq x), \quad x \in \mathbb{R}.$$

Proposition 3.55 (Basic properties of the CDF). For any real-valued random variable X , the CDF F_X satisfies:

- i) $\Pr(X > x) = 1 - F_X(x)$ and $\Pr(x < X \leq y) = F_X(y) - F_X(x)$, for every $x, y \in \mathbb{R}$;
- ii) F_X is non-decreasing;
- iii) F_X is right-continuous;
- iv) $\lim_{t \rightarrow -\infty} F_X(t) = 0$ and $\lim_{t \rightarrow +\infty} F_X(t) = 1$;
- v) $\Pr(X = x) = F_X(x) - F_X(x^-)$, for every $x \in \mathbb{R}$.

Proof.

(i). For every $x, y \in \mathbb{R}$, $\Pr(X > x) = \Pr((X \leq x)^c) = 1 - \Pr(X \leq x) = 1 - F_X(x)$ and $\Pr(x < X \leq y) = \Pr((X \leq y) \setminus (X \leq x)) = \Pr(X \leq y) - \Pr(X \leq x) = F_X(y) - F_X(x)$.

(ii). To show that F_X is non-decreasing, pick $x < y$. Then $(X \leq x) \subseteq (X \leq y)$, thus

$$F_X(x) = \Pr(X \leq x) \leq \Pr(X \leq y) = F_X(y).$$

(iii), (iv) and (v). Since F_X is monotonic, we can compute the following four limits:

$$\begin{aligned} \lim_{h \rightarrow 0^+} F_X(x+h) &= \lim_{n \rightarrow \infty} F_X\left(x + \frac{1}{n}\right) = \lim_{n \rightarrow \infty} \Pr\left(X \leq x + \frac{1}{n}\right) \\ &= \Pr\left(\bigcap_{n \in \mathbb{N}} \left(X \leq x + \frac{1}{n}\right)\right) = \Pr(X \leq x) = F_X(x). \end{aligned}$$

$$\lim_{t \rightarrow -\infty} F_X(t) = \lim_{n \rightarrow +\infty} \Pr(X \leq -n) = \Pr\left(\bigcap_{n \in \mathbb{N}} (X \leq -n)\right) = \Pr(\emptyset) = 0$$

$$\lim_{t \rightarrow +\infty} F_X(t) = \lim_{n \rightarrow +\infty} \Pr(X \leq n) = \Pr\left(\bigcup_{n \in \mathbb{N}} (X \leq n)\right) = \Pr(\Omega) = 1.$$

$$\begin{aligned} F_X(x^-) &= \lim_{h \rightarrow 0^+} F_X(x-h) = \lim_{n \rightarrow \infty} \Pr\left(X \leq x - \frac{1}{n}\right) = \Pr\left(\bigcup_{n \in \mathbb{N}} \left(X \leq x - \frac{1}{n}\right)\right) \\ &= \Pr(X < x) = F_X(x) - \Pr(X = x). \end{aligned}$$

□

Remark 3.56. By definition,

$$F_X(x) = (X_{\#} \Pr)((-\infty, x]).$$

Hence, the law $\Pr_X$ uniquely determines the CDF F_X . The converse, that the CDF uniquely determines the law, is a fundamental result in measure theory, following from the *Carathéodory extension theorem* (see Appendix A.1 of [15]). Intuitively, since

$$\Pr(X \in (a, b]) = F_X(b) - F_X(a), \quad \forall a < b,$$

the probabilities are specified on sufficiently many Borel sets, guaranteeing the uniqueness of the measure. Consequently, we say that X and Y have the same distribution (or are equal in law), and write $X \stackrel{d}{=} Y$, if and only if $F_X(x) = F_Y(x)$ for all $x \in \mathbb{R}$.

Remark 3.57. The CDF F_X is always increasing and right-continuous. If X is discrete, then F_X is a step function, with jumps of size $\Pr(X = x)$ exactly at the atoms x . If X admits a continuous density, then F_X is continuous, and it is differentiable at every point where the PDF is continuous.

Proposition 3.58 (Relationship between PMF, PDF and CDF). Let p_X and F_X denote, respectively, the PMF and the CDF of a discrete random variable X with support $S = \{x_i\}_{i \in I}$, and let f_Y and F_Y denote, respectively, the PDF and the CDF of a continuous random variable Y . Then, the following relations hold:

- i) For every $i \in I$, $p_X(x_i) = F_X(x_i) - F_X(x_i^-)$;
- ii) For every $x \in \mathbb{R}$, $F_X(x) = \sum_{x_i \leq x} p_X(x_i)$;
- iii) For every $y \in \mathbb{R}$ where f_Y is continuous, we have $f_Y(y) = F_Y'(y)$;
- iv) For every $y \in \mathbb{R}$, $F_Y(y) = \Pr(Y \leq y) = \Pr(Y < y) = \int_{-\infty}^y f_Y(t) dt$.

Proof.

(i). By Proposition 3.55, we deduce $p_X(x_i) = \Pr(X = x_i) = F_X(x_i) - F_X(x_i^-)$.

(ii). By definition of the CDF,

$$F_X(x) = \Pr(X \leq x) = \sum_{x_i \leq x} \Pr(X = x_i) = \sum_{x_i \leq x} p_X(x_i).$$

(iii). Fix $y \in \mathbb{R}$ such that f_Y is continuous at y . For $h \neq 0$,

$$\frac{F_Y(y+h) - F_Y(y)}{h} = \frac{1}{h} \int_y^{y+h} f_Y(t) dt.$$

By the Fundamental Theorem of Calculus (and the continuity of f_Y at y),

$$F_Y'(y) = \lim_{h \rightarrow 0} \frac{F_Y(y+h) - F_Y(y)}{h} = f_Y(y).$$

(iv). Since Y is a continuous random variable, its law satisfies $\Pr_Y \ll \lambda$. Therefore $\Pr(Y = y) = \Pr_Y(\{y\}) = 0$ for every $y \in \mathbb{R}$, because $\lambda(\{y\}) = 0$. This implies $\Pr(Y \leq y) = \Pr(Y < y) + \Pr(Y = y) = \Pr(Y < y)$. Finally, by (3.3) applied to $B = (-\infty, y]$,

$$F_Y(y) = \Pr(Y \leq y) = \int_{-\infty}^y f_Y(t) dt.$$

□

Remark 3.59 (Why the term "density"). If X has PDF f_X and f_X is continuous at x , then for a small radius $r > 0$ one has

$$\Pr(X \in B_r(x)) = \int_{B_r(x)} f_X(t) dt \approx \text{Vol}(B_r(x)) \cdot f_X(x).$$

Hence $f_X(x)$ describes how probability mass concentrates locally around x , directly analogous to a mass density in physics.

Example 3.60. Let X be a continuous random variable with PDF given by $f_X(x) = -\log(\theta)\theta^x \mathbf{1}_{(0,\infty)}(x)$ for $\theta \in (0, 1)$. Define

$$N := \lfloor X \rfloor.$$

We show that N is a discrete random variable with PMF given by

$$p_N(n) = (1 - \theta)\theta^n \quad \text{for } n = 0, 1, \dots$$

and CDF given by

$$F_N(x) = 1 - \theta^{\lfloor x \rfloor + 1} \mathbf{1}_{[0,\infty)}(x).$$

Direct computation shows that, for every $n = 0, 1, \dots$,

$$\Pr(N = n) = \Pr(\lfloor X \rfloor = n) = \int_n^{n+1} -\log(\theta)\theta^x dx = -\left(e^{\log(\theta)(n+1)} - e^{\log(\theta)n}\right) = \theta^n - \theta^{n+1} = (1 - \theta)\theta^n$$

and

$$\Pr(N \leq n) = \sum_{x=0}^n (1 - \theta)\theta^x = (1 - \theta) \frac{1 - \theta^{n+1}}{1 - \theta} = 1 - \theta^{n+1}.$$

Moreover, it is clear that if $n \leq x < n + 1$, then

$$\Pr(N \leq x) = \Pr(\lfloor X \rfloor \leq x) = \Pr(\lfloor X \rfloor \leq n) = \Pr(N \leq n).$$

Finally, if $x < 0$, then $\Pr(N \leq x) = 0$, as expected.

We conclude this section by defining a legitimate inverse function of the CDF:

Definition 3.61 (Quantile function). Let F be a distribution function on $\mathbb{R}$. Define its *quantile function*

$$F^{-1}(u) := \inf\{x \in \mathbb{R} : F(x) \geq u\}, \quad u \in (0, 1].$$

Example 3.62. If $F(x) = (1 - e^{-\lambda x})\mathbf{1}_{(0, \infty)}(x)$, then the corresponding quantile function is given by $F^{-1}(t) = -\log(1 - t)/\lambda$.

Lemma 3.63. *The quantile function is increasing and left-continuous. If F is strictly increasing, then $F^{-1}(F(x)) = x$.*

Proof.

Take $0 < u_1 \leq u_2 \leq 1$. Since $\{x \in \mathbb{R} : F(x) \geq u_2\} \subseteq \{x \in \mathbb{R} : F(x) \geq u_1\}$, then

$$F^{-1}(u_1) = \inf\{x \in \mathbb{R} : F(x) \geq u_1\} \leq \inf\{x \in \mathbb{R} : F(x) \geq u_2\} = F^{-1}(u_2).$$

Let $u \in (0, 1]$ and take a sequence $u_1 \leq u_2 \leq \dots \leq u$ converging to u as $n \rightarrow \infty$. Let $L := \lim_{n \rightarrow \infty} F^{-1}(u_n)$, which exists since F^{-1} is increasing, and satisfies $L \leq F^{-1}(u)$. Suppose by contradiction that $L < F^{-1}(u)$. Then, for every $n \in \mathbb{N}$,

$$F^{-1}(u_n) \leq L \iff u_n \leq F(L) \xrightarrow{n \rightarrow \infty} u \leq F(L).$$

Hence $L \in \{y \in \mathbb{R} : F(y) \geq u\}$, and thus $F^{-1}(u) \leq L$. This contradicts our previous assumption that $L < F^{-1}(u)$. Therefore, we must have $L = F^{-1}(u)$, which proves left-continuity.

Fix $x \in \mathbb{R}$. Clearly, $x \in \{y \in \mathbb{R} : F(y) \geq F(x)\}$, which implies $F^{-1}(F(x)) \leq x$. If F is strictly increasing, for any $y < x$ we have $F(y) < F(x)$. Hence no $y < x$ can belong to the set $\{y \in \mathbb{R} : F(y) \geq F(x)\}$. In particular, x is the infimum of $\{y \in \mathbb{R} : F(y) \geq F(x)\}$, or $F^{-1}(F(x)) = x$. $\square$

Lemma 3.64 (Quantile transform). *Let U be a continuous random variable whose PDF is given by $f_U(x) = \mathbf{1}_{(0, 1]}(x)$. Let F be a CDF function on $\mathbb{R}$. Then $X := F^{-1}(U)$ has cumulative distribution function F .*

Proof.

Fix $x \in \mathbb{R}$. We need to show that

$$\Pr(F^{-1}(U) \leq x) = F(x) \iff \{F^{-1}(U) \leq x\} = \{U \leq F(x)\}.$$

If $U(\omega) \leq F(x)$, then

$$F^{-1}(U(\omega)) = \inf\{y : F(y) \geq U(\omega)\} \leq x.$$

Hence, the second event implies the first.

Conversely, if $F^{-1}(U(\omega)) \leq x$, then $F(x + \varepsilon) \geq U(\omega)$ for every $\varepsilon > 0$. Passing to the limit as $\varepsilon \rightarrow 0^+$, we conclude $F(x) \geq U(\omega)$ by right continuity. Hence, the first event implies the second. $\square$

Lemma 3.64 provides a straightforward method for generating a random variable with distribution F whenever the inverse distribution function F^{-1} can be computed explicitly. Moreover, the same construction will be used in the proof of Lemma 6.19.

3.8 Independent Random Variables

In section 3.4 we defined independence in the context of probability measures. In this section we extend this notion to random variables, which are the central objects of probability and statistics.

Definition 3.65 (Marginal law). Let $(X, Y) : (\Omega, \mathcal{F}, \Pr) \rightarrow (E_X \times E_Y, \mathcal{E}_X \otimes \mathcal{E}_Y)$ be a random vector with law $\Pr_{X, Y}$. We call it the *joint law* of X and Y . The *marginal laws* of X and Y are the probability measures on $(E_X, \mathcal{E}_X)$ and $(E_Y, \mathcal{E}_Y)$ defined by

$$\Pr_X = \Pr_{X, Y} \circ \pi_{E_X}^{-1} \quad \Pr_Y = \Pr_{X, Y} \circ \pi_{E_Y}^{-1},$$

where $\pi_{E_X} : E_X \times E_Y \rightarrow E_X$ and $\pi_{E_Y} : E_X \times E_Y \rightarrow E_Y$ are the coordinate projections. In other words, $\Pr_X$ and $\Pr_Y$ satisfy, for every $A \in \mathcal{E}_X$ and $B \in \mathcal{E}_Y$,

$$\Pr_X(A) = \Pr_{X, Y}(A \times E_Y), \quad \Pr_Y(B) = \Pr_{X, Y}(E_X \times B). \quad (3.4)$$

Remark 3.66. It follows from the definition that if (X, Y) is a discrete random vector, then X and Y are discrete random variables. Likewise, if (X, Y) is an absolutely continuous random vector with respect to the Lebesgue measure λ^2 , then X and Y are absolutely continuous. The converse does not hold. Example 3.51 defines a random vector whose marginal laws are absolutely continuous, but the joint is singular.

Definition 3.67 (Joint PMF and joint PDF). Let $(X, Y) : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R} \times \mathbb{R}, \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}))$ be a random vector with law $\Pr_{X,Y}$.

- i) If (X, Y) is discrete (i.e., it is supported on a countable set of points), we define the *joint probability mass function* (PMF) of X and Y by

$$p_{X,Y}(x, y) = \Pr_{X,Y}(\{(x, y)\}) = \Pr(X = x, Y = y), \quad \forall (x, y) \in \mathbb{R}^2.$$

- ii) If (X, Y) is absolutely continuous, we define the *joint probability density function* (PDF) of X and Y by

$$f_{X,Y} := \frac{d\Pr_{X,Y}}{d\lambda^2},$$

where $\frac{d\Pr_{X,Y}}{d\lambda^2}$ is the Radon-Nikodym derivative. In other words, for every $B \in \mathcal{B}(\mathbb{R}^2)$,

$$\Pr_{X,Y}(B) = \Pr((X, Y) \in B) = \iint_B f_{X,Y} dx dy.$$

Proposition 3.68 (Marginal Distributions).

- i) Let $p_{X,Y}(x, y)$, $p_X(x)$ and $p_Y(y)$ be probability mass functions of (X, Y) , X and Y , respectively. Then

$$p_X(x) = \sum_y p_{X,Y}(x, y) \quad \text{and} \quad p_Y(y) = \sum_x p_{X,Y}(x, y),$$

where the sums are taken over the possible values of Y and of X , respectively.

- ii) Let $f_{X,Y}(x, y)$, $f_X(x)$ and $f_Y(y)$ be probability density functions of (X, Y) , X and Y , respectively. Then

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dx$$

Proof.

- (i). Since (X, Y) is discrete, its support $S \subseteq \mathbb{R}^2$ is countable and

$$\Pr((X, Y) \in B) = \sum_{(x,y) \in S \cap B} \Pr(X = x, Y = y) = \sum_{(x,y) \in S \cap B} p_{X,Y}(x, y), \quad B \in \mathcal{B}(\mathbb{R}^2).$$

In particular, taking $B_X = \{x\} \times \mathbb{R}$ or $B_Y = \mathbb{R} \times \{y\}$, we deduce

$$p_X(x) = \Pr(X = x) = \Pr((X, Y) \in B_X) = \sum_y p_{X,Y}(x, y),$$

and

$$p_Y(y) = \Pr(Y = y) = \Pr((X, Y) \in B_Y) = \sum_x p_{X,Y}(x, y).$$

- (ii). Similarly, for every Borel set $B \in \mathcal{B}(\mathbb{R}^2)$, we have

$$\Pr((X, Y) \in B) = \iint_B f_{X,Y}(x, y) dx dy.$$

In particular, taking $B_X = (-\infty, x] \times \mathbb{R}$, we obtain

$$F_X(x) = \Pr(X \leq x) = \Pr((X, Y) \in B_X) = \int_{-\infty}^x \int_{-\infty}^{+\infty} f_{X,Y}(u, v) dv du,$$

and differentiating with respect to x (at points where F_X is differentiable) yields

$$f_X(x) = F'_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, v) dv.$$

The identity for f_Y is analogous. □

Remark 3.69 (Marginals do not determine the joint distribution). Proposition 3.68 implies that the marginal distributions are uniquely determined by the joint distribution. However, the converse is generally false: the marginal distributions do not uniquely determine the joint distribution unless additional structural assumptions, such as independence, are imposed.

Definition 3.70 (Independence of random elements). Let $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (E_X, \mathcal{E}_X)$ and $Y : (\Omega, \mathcal{F}, \Pr) \rightarrow (E_Y, \mathcal{E}_Y)$ be random elements. Let $\Pr_X$, $\Pr_Y$ and $\Pr_{X,Y}$ be the laws of X , Y and (X, Y) , respectively. We say that X and Y are *independent* if

$$\Pr_{X,Y} = \Pr_X \otimes \Pr_Y \quad \text{on } (E_X \times E_Y, \mathcal{E}_X \otimes \mathcal{E}_Y).$$

We state the following characterization without proof. It says that independence of X and Y is already determined by checking the factorization on rectangles $A \times B$ with $A \in \mathcal{E}_X$ and $B \in \mathcal{E}_Y$. Its proof mimics the uniqueness principle underlying the construction of the product measure (1.6).

Lemma 3.71 (Characterization of independence on rectangles). *Random elements X and Y are independent if and only if, for every $A \in \mathcal{E}_X$ and every $B \in \mathcal{E}_Y$,*

$$\Pr(X \in A, Y \in B) = \Pr(X \in A) \Pr(Y \in B). \quad (3.5)$$

More generally, a family $\{X_i\}_{i \in I}$ of random elements is independent if and only if, for every finite subset $J \subseteq I$ (cf. Remark 3.13) and every choice of measurable sets $\{A_j\}_{j \in J}$ (where $A_j \in \mathcal{E}_j$),

$$\Pr\left(\bigcap_{j \in J} (X_j \in A_j)\right) = \prod_{j \in J} \Pr(X_j \in A_j).$$

Proposition 3.72 (Independence of Discrete Random Variables). *Let X and Y be discrete random variables with supports $S_X = \{x_i\}_{i \in I}$ and $S_Y = \{y_j\}_{j \in J}$. The following are equivalent:*

i) X and Y are independent;

ii) for every $i \in I$ and every $j \in J$,

$$\Pr(X = x_i, Y = y_j) = \Pr(X = x_i) \Pr(Y = y_j); \quad (3.6)$$

iii) for every $i \in I$ and every $j \in J$,

$$p_{X,Y}(x_i, y_j) = p_X(x_i) p_Y(y_j).$$

Proof.

($i \Rightarrow ii$). If X and Y are independent, setting $A = \{x_i\}$ and $B = \{y_j\}$ yields

$$\Pr(X = x_i, Y = y_j) = \Pr(X \in A, Y \in B) = \Pr(X \in A) \Pr(Y \in B) = \Pr(X = x_i) \Pr(Y = y_j).$$

($ii \Rightarrow i$). Conversely, suppose that (3.6) holds. Then for every $A \subseteq \mathbb{R}$ and $B \subseteq \mathbb{R}$,

$$\begin{aligned} \Pr((X \in A) \cap (Y \in B)) &= \sum_{\substack{x_i \in A \\ y_j \in B}} \Pr(X = x_i, Y = y_j) = \sum_{\substack{x_i \in A \\ y_j \in B}} \Pr(X = x_i) \Pr(Y = y_j) \\ &= \sum_{x_i \in A} \Pr(X = x_i) \sum_{y_j \in B} \Pr(Y = y_j) = \Pr(X \in A) \Pr(Y \in B). \end{aligned}$$

so X and Y are independent.

($ii \Leftrightarrow iii$). This equivalence is immediate from the definitions of the joint and marginal PMFs. □

Proposition 3.73 (Independence of Continuous Random Variables). *Let X and Y be continuous random variables. The following are equivalent:*

i) X and Y are independent;

ii) For every $x, y \in \mathbb{R}$,

$$\Pr(X \leq x, Y \leq y) = \Pr(X \leq x) \Pr(Y \leq y);$$

iii) If $f_{X,Y} : \mathbb{R}^2 \rightarrow [0, \infty)$ is the joint PDF of (X, Y) , there exist measurable functions $g, h : \mathbb{R} \rightarrow [0, \infty)$ such that, for every $x, y \in \mathbb{R}$,

$$f_{X,Y}(x, y) = g(x)h(y) \quad \text{for } \lambda^2\text{-a.e. } (x, y).$$

In particular, if (iii) holds, then $g \propto f_X$ λ -a.e. and $h \propto f_Y$ λ -a.e.

Proof.

(i $\Rightarrow$ ii). If X and Y are independent, then (3.5) holds for every $A, B \in \mathcal{B}(\mathbb{R})$. Applying this with $A = (-\infty, x]$ and $B = (-\infty, y]$ yields (ii).

(ii $\Rightarrow$ i). Thanks to Lemma 3.71, it is enough to show that the factorization

$$\Pr(X \in A, Y \in B) = \Pr(X \in A)\Pr(Y \in B), \quad \forall A, B \in \mathcal{B}(\mathbb{R}),$$

holds. The assumption in (ii) verifies this identity only for sets of the form $(-\infty, x]$ and $(-\infty, y]$. The extension from these generating classes to the full Borel σ -algebra is carried out by the same σ -algebra closure argument used in Proposition 1.28 (and Example 1.29): one introduces the class of sets for which the identity holds, proves that it is a σ -algebra, and concludes because it contains a family of sets that generates $\mathcal{B}(\mathbb{R})$.

(i $\Rightarrow$ iii). Assume that X and Y are independent, and let $f_{X,Y}$ be a joint PDF, i.e.

$$\Pr((X, Y) \in C) = \iint_C f_{X,Y}(x, y) dx dy, \quad C \in \mathcal{B}(\mathbb{R}^2).$$

By Remark 3.66, both X and Y are absolutely continuous. By Theorem 1.109, there exist measurable functions $f_X, f_Y : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \rightarrow \mathbb{R}$, unique λ -a.e., such that

$$\Pr_X(A) = \int_A f_X d\lambda, \quad \Pr_Y(A) = \int_A f_Y d\lambda, \quad \forall A \in \mathcal{B}(\mathbb{R}).$$

Then, by independence and Tonelli's theorem, for every $C \in \mathcal{B}(\mathbb{R}^2)$,

$$\begin{aligned} \iint_C f_{X,Y}(x, y) dx dy &= \Pr_{X,Y}(C) = \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_C(x, y) \Pr_X(dx) \Pr_Y(dy) \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_C(x, y) f_X(x) dx f_Y(y) dy = \iint_C f_X(x) f_Y(y) dx dy. \end{aligned}$$

Subtracting the two integrals and applying linearity we deduce that, for every $C \in \mathcal{B}(\mathbb{R}^2)$,

$$\iint_C (f_{X,Y}(x, y) - f_X(x) f_Y(y)) dx dy = 0.$$

By Corollary 1.60, this implies

$$[f_{X,Y} - f_X f_Y = 0 \text{ } \lambda^2\text{-a.e.}] \iff [f_{X,Y} = f_X f_Y \text{ } \lambda^2\text{-a.e.}].$$

Setting $g = f_X$ and $h = f_Y$ we deduce (iii).

(iii $\Rightarrow$ ii). Assume that $f_{X,Y}(x, y) = g(x)h(y)$ λ^2 -a.e., with $g, h \geq 0$. Set

$$G := \int_{\mathbb{R}} g(x) dx \in [0, \infty], \quad H := \int_{\mathbb{R}} h(y) dy \in [0, \infty].$$

By Tonelli's theorem,

$$1 = \iint_{\mathbb{R}^2} g(x)h(y) dx dy = \left(\int_{\mathbb{R}} g(x) dx \right) \left(\int_{\mathbb{R}} h(y) dy \right) = GH.$$

Hence $G, H \in (0, \infty)$. Now, by Proposition 3.68 and Tonelli,

$$\begin{aligned} f_X(x) &= \int_{\mathbb{R}} f_{X,Y}(x, y) dy = g(x) \int_{\mathbb{R}} h(y) dy = Hg(x) \quad \lambda\text{-a.e.}, \\ f_Y(y) &= \int_{\mathbb{R}} f_{X,Y}(x, y) dx = h(y) \int_{\mathbb{R}} g(x) dx = Gh(y) \quad \lambda\text{-a.e.} \end{aligned}$$

Using $GH = 1$ we conclude

$$f_{X,Y}(x, y) = g(x)h(y) = \frac{f_X(x)}{H} \frac{f_Y(y)}{G} = f_X(x)f_Y(y) \quad \lambda^2\text{-a.e.}$$

Since the joint density is the product of the marginals, then, for every $(x^*, y^*) \in \mathbb{R}^2$,

$$\begin{aligned} \Pr(X \leq x^*, Y \leq y^*) &= \int_{-\infty}^{x^*} \int_{-\infty}^{y^*} f_{X,Y}(x, y) dx dy = \int_{-\infty}^{x^*} \int_{-\infty}^{y^*} f_X(x) f_Y(y) dx dy \\ &= \left(\int_{-\infty}^{x^*} f_X(x) dx \right) \left(\int_{-\infty}^{y^*} f_Y(y) dy \right) = \Pr(X \leq x^*) \Pr(Y \leq y^*). \end{aligned}$$

□

Proposition 3.74 (Stability of Independence). *If X and Y are real-valued independent random variables and $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ are measurable functions, then $g(X)$ and $h(Y)$ are independent.*

Proof.

Direct computation shows that, for every pair of events $A, B \in \mathcal{B}(\mathbb{R})$,

$$\begin{aligned} \Pr(g(X) \in A, h(Y) \in B) &\stackrel{\dagger}{=} \Pr(X \in g^{-1}(A), Y \in h^{-1}(B)) = \Pr(X \in g^{-1}(A)) \Pr(Y \in h^{-1}(B)) \\ &= \Pr(g(X) \in A) \Pr(h(Y) \in B), \end{aligned}$$

where $\dagger$ uses the measurability assumption to guarantee that $g^{-1}(A), h^{-1}(B) \in \mathcal{B}(\mathbb{R})$ (by Proposition 3.34). □

Example 3.75 (Transformations preserve independence). Proposition 3.74 becomes immediately intuitive when considering a real-world scenario. Let Y be a random variable representing the "average height of the Italian population", and let X be the "temperature in degrees Celsius tomorrow in Milan". If we apply a deterministic transformation $g(x)$ so that $g(X)$ represents the same temperature measured in degrees Fahrenheit, it is evident that the independence of Y from X is perfectly equivalent to the independence of Y from $g(X)$. A deterministic transformation merely rescales or maps the numerical values without altering the underlying source of randomness; therefore, it cannot artificially induce or destroy statistical dependence.

3.9 Transformations of Random Variables

In this section, we study how to compute the laws of transformations of random variables.

Theorem 3.76 (Change of Variables Formula). *Let $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ be a continuous random vector with probability density function f_X . Let $U \subseteq \mathbb{R}^k$ be an open set such that $\Pr(X \in U) = 1$. Let $V \subseteq \mathbb{R}^k$ be an open set and let $T : U \rightarrow V$ be a C^1 -diffeomorphism.⁹*

Then, the random vector $Y = T(X)$ is continuous and its probability density function f_Y is given by:

$$f_Y(y) = f_X(T^{-1}(y)) |\det J_{T^{-1}}(y)| \cdot \mathbf{1}_V(y).$$

In particular, if $T(x) = Ax + b$ where $A \in \mathbb{R}^{k \times k}$ is invertible and $b \in \mathbb{R}^k$, then $V = \mathbb{R}^k$ and:

$$f_Y(y) = f_X(A^{-1}(y - b)) \frac{1}{|\det A|}.$$

Proof.

Let $B \subseteq V$ be an arbitrary Borel set. We compute the probability that $Y = T(X) \in B$ by expressing it in terms of X :

$$\Pr(Y \in B) = \Pr(T(X) \in B) = \Pr(X \in T^{-1}(B)).$$

Since X admits a density f_X and $\Pr(X \in U) = 1$, we can write this probability as an integral over U :

$$\Pr(X \in T^{-1}(B)) = \int_{T^{-1}(B)} f_X(x) dx.$$

⁹That is, T is a bijection, T is continuously differentiable, and its inverse $T^{-1} : V \rightarrow U$ is continuously differentiable.

We now apply the change of variables formula 1.114 and Proposition 1.117 using the substitution $x = T^{-1}(y)$. The Jacobian matrix of this transformation is $J_{T^{-1}}(y)$. The volume element transforms as $dx = |\det J_{T^{-1}}(y)| dy$. As x ranges over the set $T^{-1}(B)$, the variable y ranges over B . Thus, the integral becomes:

$$\int_{T^{-1}(B)} f_X(x) dx = \int_B f_X(T^{-1}(y)) |\det J_{T^{-1}}(y)| dy.$$

Combining these equalities, we obtain:

$$\Pr(Y \in B) = \int_B (f_X(T^{-1}(y)) |\det J_{T^{-1}}(y)|) dy.$$

Since this equality holds for every Borel set $B \subseteq V$ (and the density is zero outside V), the integrand is by definition the probability density function of Y .

The affine case follows trivially from the general result. $\square$

Next, we study the sum of random variables. Before proceeding, let us recall Definition 2.37:

Definition 3.77 (Convolution of finite measures). Let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ be a measurable space, and let $\mu, \nu : \mathcal{B}(\mathbb{R}^n) \rightarrow [0, \infty)$ be finite measures on $\mathbb{R}^n$. We define their convolution $\mu * \nu$ by:

$$(\mu * \nu)(B) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \mathbf{1}_B(x+y) d\mu(x) d\nu(y), \quad \text{for all } B \in \mathcal{B}(\mathbb{R}^n).$$

Equivalently, for every bounded Borel-measurable function φ , we have:

$$\int_{\mathbb{R}^n} \varphi(z) d(\mu * \nu)(z) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \varphi(x+y) d\mu(x) d\nu(y).$$

Theorem 3.78 (Law of the Sum of Independent Random Variables). Let X and Y be independent random variables, and let $Z = X + Y$. Then the law of Z , denoted by $\Pr_Z$, is given by the convolution of the laws of X and Y :

$$\Pr_Z = \Pr_X * \Pr_Y.$$

Proof.

Let $s : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the summation map $(x, y) \mapsto x + y$. By Proposition 3.34, $X + Y$ is a random variable, and its law is given by:

$$\Pr_Z(B) = \Pr(X + Y \in B) = \Pr((X, Y) \in s^{-1}(B)) = \Pr_{X,Y}(s^{-1}(B)), \quad \forall B \in \mathcal{B}(\mathbb{R}).$$

By independence, the joint law $\Pr_{X,Y}$ factorizes. Thus, for every $B \in \mathcal{B}(\mathbb{R})$:

$$\Pr_{X,Y}(s^{-1}(B)) = (\Pr_X \otimes \Pr_Y)(\{(x, y) \in \mathbb{R}^2 : x + y \in B\}) = \int_{\mathbb{R}^2} \mathbf{1}_B(x+y) d(\Pr_X \otimes \Pr_Y)(x, y).$$

By comparing this with Definition 2.37, we recognize that this integral matches exactly the definition of the convolution of $\Pr_X$ and $\Pr_Y$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Since this equality holds for every $B \in \mathcal{B}(\mathbb{R})$, we conclude that:

$$\Pr_Z = \Pr_X * \Pr_Y,$$

which completes the proof. $\square$

In practice, it is often sufficient to consider the sum of two discrete or two continuous random variables. The following results generalize this concept to variables that are not necessarily independent. The theorems and proofs are largely adapted from [16].

Theorem 3.79 (Sum of discrete random variables). Let X and Y be discrete random variables with joint PMF $p_{X,Y}(x, y)$. Then the PMF of $Z = X + Y$ is:

$$p_Z(z) = \sum_x p_{X,Y}(x, z-x),$$

where the sum is taken over the set of possible values of X . In particular, if X and Y are independent, then:

$$p_Z(z) = \sum_x p_X(x) p_Y(z-x) = (p_X * p_Y)(z).$$

Proof.

For every z , we can write:

$$p_Z(z) = \Pr(X + Y = z) = \sum_{x+y=z} \Pr(X = x, Y = y) = \sum_x \Pr(X = x, Y = z - x) = \sum_x p_{X,Y}(x, z - x).$$

If X and Y are independent, this becomes:

$$p_Z(z) = \sum_x p_{X,Y}(x, z - x) = \sum_x p_X(x) p_Y(z - x).$$

□

Theorem 3.80 (Sum of continuous random variables). *Let X and Y be continuous random variables with joint PDF $f_{X,Y}(x, y)$. Then the random variable $Z = X + Y$ is continuous and its probability density function is:*

$$f_Z(z) = \int_{\mathbb{R}} f_{X,Y}(x, z - x) dx = \int_{\mathbb{R}} f_{X,Y}(z - y, y) dy.$$

In particular, if X and Y are independent with PDFs f_X and f_Y , then Z has probability density function:

$$f_Z(z) = \int_{\mathbb{R}} f_X(x) f_Y(z - x) dx = (f_X * f_Y)(z).$$

Proof.

Fix $z \in \mathbb{R}$ and consider the C^1 -diffeomorphism $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by:

$$T(x, y) = (x + y, y), \quad \text{with inverse} \quad T^{-1}(z, y) = (z - y, y).$$

Its Jacobian determinant satisfies $|\det J_{T^{-1}}(z, y)| = 1$. By the change of variables formula:

$$\Pr(Z \in B) = \iint_{\mathbb{R}^2} \mathbf{1}_B(z) f_{X,Y}(z - y, y) dz dy = \int_B \left(\int_{\mathbb{R}} f_{X,Y}(z - y, y) dy \right) dz.$$

Hence Z is absolutely continuous and its PDF is:

$$f_Z(z) = \int_{\mathbb{R}} f_{X,Y}(z - y, y) dy = \int_{\mathbb{R}} f_{X,Y}(x, z - x) dx,$$

where the second identity follows by the substitution $x = z - y$.

If X and Y are independent, then $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ for λ^2 -a.e. (x, y) , so:

$$f_Z(z) = \int_{\mathbb{R}} f_X(z - y) f_Y(y) dy = (f_X * f_Y)(z),$$

i.e., the density of $X + Y$ is the convolution of the marginal densities.

□

Corollary 3.81. *Let $(X_i)_{i \leq n}$ be a sequence of independent and identically distributed discrete random variables with PMF p_X , and let $(Y_j)_{j \leq n}$ be a sequence of independent and identically distributed continuous random variables with PDF f_Y . Let $S_X := X_1 + \dots + X_n$ and $S_Y := Y_1 + \dots + Y_n$. Then*

$$p_{S_X} = (p_X * p_X * \dots * p_X) = (p_X^{*n}) \quad \text{and} \quad f_{S_Y} = (f_Y * f_Y * \dots * f_Y) = (f_Y^{*n}),$$

where $*n$ denotes the n -fold convolution.

Proof. Just iteratively apply Theorems 3.79 and 3.80.

□

4 Expectation and Moments

In this section, we develop the framework for integrating random variables. Most results presented here are not new; they are straightforward probabilistic reformulations of the theorems already established in Sections 1.4, 1.5, 1.6, and 1.7.

4.1 Expectation

We begin with a summary table 1 outlining the probabilistic analogs of the measure-theoretic concepts we have previously encountered and, in particular, the analogy $\int_X f d\mu \longleftrightarrow \mathbb{E}[X]$. Just as the Lebesgue integral is invariant modulo null sets, the expectation is invariant modulo null events. Let us rigorously recall their definitions.

Measure Theory	Probability Theory
Measure Space $(X, \mathcal{F}, \mu)$	Probability Space $(\Omega, \mathcal{F}, \Pr)$
Measure $\mu(A) \in [0, \infty]$	Probability measure $\Pr(A) \in [0, 1]$
Measurable function $f : X \rightarrow \overline{\mathbb{R}}$	Random variable $X : \Omega \rightarrow \mathbb{R}$
Integral $\int_X f d\mu$	Expectation $\mathbb{E}[X]$
Almost everywhere (a.e.)	Almost surely (a.s.)
L^p space	L^p space
Norm $\ f\ _p = (\int f ^p)^{1/p}$	Moment $(\mathbb{E}[X ^p])^{1/p}$

Table 1: Analogies between Lebesgue integration and probability expectation

Definition 4.1 (Null event). Given a probability space $(\Omega, \mathcal{F}, \Pr)$, an event $\mathcal{N} \in \mathcal{F}$ is a *null event* if $\Pr(\mathcal{N}) = 0$.

Definition 4.2 (Almost sure). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $P(\omega)$ be a property depending on $\omega \in \Omega$. We say that P holds *almost surely* (abbreviated a.s.) if there exists a null event $\mathcal{N} \in \mathcal{F}$ such that $P(\omega)$ holds for each $\omega \in \mathcal{N}^c$.

Definition 4.3 (Expectation). Let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a random variable.

- If $X \geq 0$ almost surely, its *expectation* (or *mean*) is defined as

$$\mathbb{E}[X] := \int_{\Omega} X(\omega) d\Pr(\omega) \in [0, +\infty]. \quad (4.1)$$

- If X satisfies $\mathbb{E}[|X|] < \infty$, writing $X = X^+ - X^- = \max(X, 0) - \max(-X, 0)$, we define

$$\mathbb{E}[X] := \mathbb{E}[X^+] - \mathbb{E}[X^-].$$

Random variables satisfying $\mathbb{E}[|X|] < \infty$ are called *integrable*. The class of integrable random variables is denoted as $\mathcal{L}^1(\Omega, \Pr)$, $\mathcal{L}^1(\Omega)$, $\mathcal{L}^1(\Pr)$ or simply $\mathcal{L}^1$.

Theorem 4.4 (Expectation of a transformation). Let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a random variable and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function. Suppose that $g(X) \in \mathcal{L}^1$, and let $\Pr_X$ be the law of X . Then

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) d\Pr_X(x). \quad (4.2)$$

Proof.

By the Pushforward Theorem (Theorem 1.114),

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega)) d\Pr(\omega) = \int_{\mathbb{R}} g(x) d(X_{\#} \Pr)(x) = \int_{\mathbb{R}} g(x) d\Pr_X(x).$$

□

Example 4.5 (Expectation of a discrete random variable). Let us specialize Equation 4.2 to the case of a discrete random variable with support $S = \{x_i\}_{i \in I}$, PMF p_X , and $g \equiv \text{id}_{\mathbb{R}}$. In this case, the probability measure is purely atomic and the integral reduces to a sum over the elements of the support S . By definition, $\Pr(X(\omega) = x_i) = p_X(x_i)$, so the expectation attains the familiar weighted average form:

$$\mathbb{E}[X] = \sum_{i \in I} x_i p_X(x_i). \quad (4.3)$$

Example 4.6 (Expectation of a continuous random variable). Let us specialize Equation 4.2 to the case of a continuous random variable with PDF f_X and $g \equiv \text{id}_{\mathbb{R}}$. In this case, the law satisfies $\Pr_X \ll \lambda$ and, by the Radon-Nikodym Theorem 1.109 and Equation 3.3, we deduce $d\Pr_X = f_X d\lambda$. Therefore,

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) dx. \quad (4.4)$$

It represents the average of the possible values of X , weighted by the probability density function.

Remark 4.7. As a consequence, integrability for a continuous random variable is equivalent to the condition

$$\int_{\mathbb{R}} |x| f_X(x) dx < \infty.$$

Example 4.8 (Undefined Expectation). Consider a random variable X following the Cauchy distribution, whose PDF is $f(x) = \frac{1}{\pi(1+x^2)}$. Then:

$$\mathbb{E}[X^+] = \int_0^{\infty} \frac{x}{\pi(1+x^2)} dx = \infty \quad \text{and} \quad \mathbb{E}[X^-] = \int_{-\infty}^0 \frac{|x|}{\pi(1+x^2)} dx = \infty.$$

Since both components evaluate to infinity, the difference $\infty - \infty$ is mathematically undefined, and we conclude that the expectation of X does not exist.

Example 4.9 (Almost Sure Equality). Let $X \sim U([0, 1])$ be a uniform random variable on $[0, 1]$. Define a new random variable Y such that $Y(\omega) = X(\omega)$ for all ω where $X(\omega) \neq 0.5$, and $Y(\omega) = 100$ if $X(\omega) = 0.5$. Since $\Pr(X = 0.5) = 0$ (because X is continuous), we have $X = Y$ almost surely. Consequently:

$$\mathbb{E}[X] = \int_0^1 x dx = \frac{1}{2} = \mathbb{E}[Y],$$

demonstrating that modifying a random variable on a set of probability zero does not alter its expectation.

Remark 4.10. The mathematical expectation rigorously formalizes the intuitive notion of an average. For a discrete random variable, the expectation is precisely the weighted average of its possible values, weighted by their probabilities of occurring. For a continuous random variable, the mean value of a function over a domain is typically defined as its integral divided by the total volume of the domain. In probability, since the total probability mass (the "volume") is exactly 1, the expectation naturally simplifies to just the integral, reinforcing its physical interpretation as the center of mass of the probability distribution.

Following Section 1.4, we collect a few foundational properties of expectation. Their proofs are immediate once the corresponding results are established for the Lebesgue integral.

Theorem 4.11 (Linearity of the Expectation). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let X, Y be real-valued integrable random variables. If α, β are real-valued constants, then $Z = \alpha X + \beta Y$ is integrable, and*

$$\mathbb{E}[\alpha X + \beta Y] = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y].$$

Proof.

By (4.1) and Theorem 1.46,

$$\begin{aligned} \mathbb{E}[\alpha X + \beta Y] &= \int_{\Omega} (\alpha X(\omega) + \beta Y(\omega)) d\Pr(\omega) \\ &= \alpha \int_{\Omega} X(\omega) d\Pr(\omega) + \beta \int_{\Omega} Y(\omega) d\Pr(\omega) = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y]. \end{aligned}$$

□

Corollary 4.12. *Let X be a real-valued random variable and let $\alpha \in \mathbb{R} \setminus \{0\}$, $\beta \in \mathbb{R}$. Then $Z = \alpha X + \beta$ is integrable if and only if X is, and*

$$\mathbb{E}[Z] = \mathbb{E}[\alpha X + \beta] = \alpha \mathbb{E}[X] + \beta.$$

Proof.

Take $Y(\omega) := 1$ for every $\omega \in \Omega$. Then, observe that $\mathbb{E}[X] = (\mathbb{E}[Z] - \beta)/\alpha$, hence $X \in \mathcal{L}^1$ if and only if $Z \in \mathcal{L}^1$. □

Theorem 4.13 (Monotonicity of the expectation). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let X, Y be real-valued integrable random variables. If $X \leq Y$ a.s., then*

$$\mathbb{E}[X] \leq \mathbb{E}[Y].$$

Proof.

Let $Z := Y - X$. Then $Z \geq 0$ almost surely. Since $d\Pr(\omega) \geq 0$ for every $\omega \in \Omega$ and $Z(\omega) \geq 0$ almost surely, by (4.1) and Theorem 1.45:

$$0 \leq \int_{\Omega} Z(\omega) d\Pr(\omega) = \int_{\Omega} (Y - X)(\omega) d\Pr(\omega) = \int_{\Omega} Y(\omega) d\Pr(\omega) - \int_{\Omega} X(\omega) d\Pr(\omega).$$

Hence $\mathbb{E}[X] \leq \mathbb{E}[Y]$. □

Proposition 4.14. *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let X, Y be random variables. If $X \leq Y$ a.s. and $\mathbb{E}[Y] < \infty$, then*

$$\mathbb{E}[X] = \mathbb{E}[Y] \iff [X = Y \text{ almost surely}].$$

Proof.

This is a probabilistic reformulation of Corollary 1.62. □

Theorem 4.15 (Factorization of expectation under independence). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let X, Y be integrable random variables. If they are independent, then XY is integrable and*

$$\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y].$$

Proof.

Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $g(x, y) = xy$. Let $\Pr_X$, $\Pr_Y$ and $\Pr_{X,Y}$ be the marginals and joint distributions of X and Y . By independence, $\Pr_{(X,Y)} = \Pr_X \otimes \Pr_Y$. Hence, by Tonelli's theorem,

$$\mathbb{E}[|XY|] = \mathbb{E}[|g(X, Y)|] = \int_{\mathbb{R}^2} |g(x, y)| d\Pr_{X,Y}(x, y) = \left(\int_{\mathbb{R}} |x| d\Pr_X(x) \right) \left(\int_{\mathbb{R}} |y| d\Pr_Y(y) \right) < \infty.$$

Therefore, XY is integrable. By Fubini's theorem,

$$\mathbb{E}[XY] = \int_{\mathbb{R}^2} xy d\Pr_{X,Y}(x, y) = \left(\int_{\mathbb{R}} x d\Pr_X(x) \right) \left(\int_{\mathbb{R}} y d\Pr_Y(y) \right) = \mathbb{E}[X] \mathbb{E}[Y].$$

□

The following example is taken from [16]:

Example 4.16 (Counterexample to the converse of Theorem 4.15). Let $Z : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$ be a discrete random vector with PMF given by

$$p_Z = \frac{1}{4}(\delta_{(-1,0)} + \delta_{(0,-1)} + \delta_{(0,1)} + \delta_{(1,0)}),$$

as shown in Table 2. Let $X = \pi_1(Z)$ and $Y = \pi_2(Z)$ be the coordinate random variables of Z , so that $Z = (X, Y)$. Then X and Y are not independent. For instance,

$$\Pr(Z = (-1, -1)) = p_Z(-1, -1) = 0 \neq \frac{1}{16} = \frac{1}{4} \cdot \frac{1}{4} = \Pr(X = -1) \Pr(Y = -1).$$

However,

$$\begin{aligned}\mathbb{E}[X] \mathbb{E}[Y] &= \left[-1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \right] \cdot \left[-1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \right] = 0 \cdot 0 \\ &= 0 = \left[\sum_{i \in \{-1, 0, 1\}} \sum_{j \in \{-1, 0, 1\}} ij \cdot p_Z(i, j) \right] = \mathbb{E}[XY].\end{aligned}$$

Therefore, the factorization of the expectation is valid even if X and Y are not independent.

$X \setminus Y$	-1	0	1	$p_X(x)$
-1	0	1/4	0	1/4
0	1/4	0	1/4	1/2
1	0	1/4	0	1/4
$p_Y(y)$	1/4	1/2	1/4	1

Table 2: Joint and marginal PMFs for the vector $Z = (X, Y)$.

Expectation inherits its main convergence properties from Lebesgue integration. We recall the standard results here in probabilistic language. We do not need to prove them because they are standard measure-theoretic results rephrased in a probabilistic setting.

Theorem 4.17 (Monotone Convergence (MCT)). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space, $(X_n)_{n \in \mathbb{N}}$ be an increasing sequence of nonnegative random variables that converge to $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ for almost every ω , i.e.,*

$$\Pr(\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1.$$

Then X is a random variable and

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[\lim_{n \rightarrow \infty} X_n] = \mathbb{E}[X].$$

Lemma 4.18 (Fatou's lemma). *Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of nonnegative random variables. Then*

$$\mathbb{E} \left[\liminf_{n \rightarrow \infty} X_n \right] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n].$$

Theorem 4.19 (Dominated Convergence (DCT)). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $(X_n)_{n \in \mathbb{N}}$ be a sequence of integrable random variables. Assume that $X_n \rightarrow X$ almost surely for some random variable $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e.,*

$$\Pr(\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1.$$

Assume that there exists an integrable random variable $Y \in L^1(\Omega)$ that, for every $n \in \mathbb{N}$,

$$|X_n(\omega)| \leq Y(\omega) \quad \text{almost surely for all } n \in \mathbb{N}.$$

Then X is integrable and

$$\mathbb{E}[X_n] \rightarrow \mathbb{E}[X] \quad \text{as } n \rightarrow \infty.$$

The next result is a useful identity for computing the expectation of a random variable. Remarkably, it will be useful in the proof of Proposition 6.39.

Proposition 4.20 (Layer cake representation). *Let X be a real-valued continuous random variable, let $f(x)$ be its PDF and $F(x)$ be its CDF. Assuming integrability,*

$$\int_0^{+\infty} x f(x) dx = \int_0^{+\infty} (1 - F(x)) dx \quad \text{and} \quad \int_{-\infty}^0 x f(x) dx = - \int_{-\infty}^0 F(x) dx.$$

More generally, if X is an integrable random variable, then

$$\mathbb{E}[X^+] = \int_0^{+\infty} (1 - F(t)) dt \quad \mathbb{E}[X^-] = \int_{-\infty}^0 F(t) dt$$

Proof.

We prove the first result in an elementary way, and then prove the more general result with a measure-theoretic framework. Rewrite x and $-x$ using indicator integrals:

$$\int_{-\infty}^0 xf(x)dx = \int_{-\infty}^0 \left[\int_x^0 (-1)dt \right] f(x)dx \quad \text{and} \quad \int_0^{+\infty} xf(x)dx = \int_0^{\infty} \left[\int_0^x 1dt \right] f(x)dx.$$

Let us define $\mathcal{A}$ and $\mathcal{B}$ as the regions of integration of the double integral in the $x-t$ plane. We observe that

$$\mathcal{A} = \{(x, t) \in \mathbb{R}^2 : x \leq 0, x \leq t \leq 0\} = \{(x, t) \in \mathbb{R}^2 : t \leq 0, x \leq t\}$$

and

$$\mathcal{B} = \{(x, t) \in \mathbb{R}^2 : x \geq 0, 0 \leq t \leq x\} = \{(x, t) \in \mathbb{R}^2 : t \geq 0, x \geq t\}.$$

Therefore, by interchanging the order of integration via Tonelli's Theorem,

$$-\int_{-\infty}^0 \left[\int_x^0 1dt \right] f(x)dx = -\int_{-\infty}^0 \left[\int_{-\infty}^t f(x)dx \right] dt = \int_{-\infty}^0 -F(t)dt,$$

and

$$\int_0^{\infty} \left[\int_0^x 1dt \right] f(x)dx = \int_0^{\infty} \left[\int_t^{\infty} f(x)dx \right] dt = \int_0^{\infty} (1 - F(t))dt.$$

In general, writing $X^+ = \int_0^{+\infty} \mathbf{1}_{\{X^+ > t\}} dt$, taking the expectation on both sides, and applying Tonelli's theorem yields

$$\mathbb{E}[X^+] = \mathbb{E} \left[\int_0^{+\infty} \mathbf{1}_{\{X^+ > t\}} dt \right] = \int_0^{+\infty} \mathbb{E}[\mathbf{1}_{\{X^+ > t\}}] dt = \int_0^{+\infty} \Pr(X > t) dt = \int_0^{+\infty} (1 - F(t)) dt.$$

Similarly, writing

$$X^- = \int_0^{+\infty} \mathbf{1}_{\{X^- > t\}} dt = \int_0^{+\infty} \mathbf{1}_{\{X < -t\}} dt = \int_{-\infty}^0 \mathbf{1}_{\{X < t\}} dt,$$

taking the expectation on both sides, and applying Tonelli's theorem, we obtain

$$\mathbb{E}[X^-] = \mathbb{E} \left[\int_{-\infty}^0 \mathbf{1}_{\{X < t\}} dt \right] = \int_{-\infty}^0 \mathbb{E}[\mathbf{1}_{\{X < t\}}] dt = \int_{-\infty}^0 \Pr(X < t) dt = \int_{-\infty}^0 F(t) dt.$$

□

4.2 Moments and L^p -integrability

Thanks to the notion of *expectation*, it is possible to define a *norm* for random variables, as much as we did in section 1.7:

Definition 4.21 (p -Integrability). Let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a random variable. For any $p \in [1, \infty)$, let

$$\|X\|_p := (\mathbb{E}[|X|^p])^{1/p}.$$

Moreover, for $p = \infty$ let

$$\|X\|_{\infty} := \operatorname{ess\,sup}_{\omega \in \Omega} |X(\omega)|.$$

For any $p \in [1, \infty]$, we say that X is p -integrable if $\|X\|_p < \infty$, and we denote by $\mathcal{L}^p(\Omega)$ the space of p -integrable random variables.

Remark 4.22. For historical reasons, we call *integrable* a 1-integrable random variable, and *square integrable* a 2-integrable random variable.

In order to construct a vector space, let us consider the following equivalence relation on $\mathcal{L}^p(\Omega)$:

$$X \sim Y \iff X \stackrel{\text{a.s.}}{=} Y.$$

We denote by $L^p(\Omega)$ the quotient space $\mathcal{L}^p(\Omega)/\sim$. It is easy to see that $\sim$ is an equivalence relation and $L^p(\Omega)$, endowed with the operations

$$[X] + [Y] := [X + Y] \quad \text{and} \quad \lambda[X] := [\lambda X], \quad \forall X, Y \in \mathcal{L}^p(\Omega), \lambda \in \mathbb{R},$$

is a vector space. To simplify notation, we write $X \in L^p(\Omega)$ in place of $[X] \in L^p(\Omega)$. For every $p \in [1, \infty]$, we refer to $L^p(\Omega)$ as the L^p -space on Ω .

In other words, $L^p(\Omega)$ is the space of equivalence classes of p -integrable random variables modulo almost sure equivalence.

Definition 4.23 (Moment). Let X be a random variable and suppose $X \in L^p$ for some $p \in [1, \infty]$. Then $\mathbb{E}[|X|^p]$ is finite and, in particular, $\mathbb{E}[X^p]$ is finite. In this case,

- $\mathbb{E}[X^p]$ is the p -th *moment* of the random variable X .
- $\mathbb{E}[|X|^p]$ is the p -th *absolute moment* of the random variable X .
- $\mathbb{E}[(X - \mathbb{E}[X])^p]$ is the p -th *central moment* of the random variable X .

Proposition 4.24 (Moment Comparison). If X is a random variable and $X \in L^p$ for some $p \in [1, \infty]$, then $X \in L^q$ for every $q \leq p$. Moreover,

$$\|X\|_q \leq \|X\|_p.$$

Equivalently, for $1 \leq q < p < \infty$,

$$\mathbb{E}[|X|^q] \leq (\mathbb{E}[|X|^p])^{q/p}.$$

Proof. Apply Theorem 1.101 to the finite measure space $(\Omega, \mathcal{F}, \text{Pr})$, where $\text{Pr}(\Omega) = 1$. Then $\|X\|_q \leq \text{Pr}(\Omega)^{1/q-1/p} \|X\|_p = \|X\|_p$. $\square$

Proposition 4.25 (Existence of central moments). Fix $p \in \mathbb{Z}_+$ and assume $\mu := \mathbb{E}[X]$ is finite. Then

$$\mathbb{E}[|X|^p] < \infty \iff \mathbb{E}[|X - \mu|^p] < \infty.$$

More precisely, the bounds

$$\mathbb{E}[|X - \mu|^p] \leq 2^{p-1}(\mathbb{E}[|X|^p] + |\mu|^p), \quad \mathbb{E}[|X|^p] \leq 2^{p-1}(\mathbb{E}[|X - \mu|^p] + |\mu|^p)$$

hold.

Proof.

Throughout, note that $|X|^p$ and $|X - \mu|^p$ are nonnegative random variables, so their expectations are well-defined (possibly infinity). Applying (1.9) with $a = X - \mathbb{E}[X]$ and $b = \mathbb{E}[X] = \mu$:

$$|(X - \mu) + \mu|^p \leq 2^{p-1}(|X - \mu|^p + |\mu|^p) \implies \mathbb{E}[|X|^p] \leq 2^{p-1}(\mathbb{E}[|X - \mu|^p] + |\mu|^p).$$

Similarly, applying (1.9) with $a = X$ and $b = -\mu$ yields

$$|X - \mu|^p = |X + (-\mu)|^p \leq 2^{p-1}(|X|^p + |\mu|^p),$$

and therefore

$$\mathbb{E}[|X - \mu|^p] \leq 2^{p-1}(\mathbb{E}[|X|^p] + |\mu|^p).$$

In particular, if $\mathbb{E}[|X|^p] < \infty$ then $\mathbb{E}[|X|] < \infty$ by Proposition 4.24, so μ is finite and the last inequality gives $\mathbb{E}[|X - \mu|^p] < \infty$. Conversely, if μ is finite and $\mathbb{E}[|X - \mu|^p] < \infty$, then the first inequality gives $\mathbb{E}[|X|^p] < \infty$. $\square$

4.3 Classical Inequalities

Theorem 4.26 (Markov inequality). Let $(\Omega, \mathcal{F}, \text{Pr})$ be a probability space and let X be a real-valued random variable and let $r > 0$ be such that $\mathbb{E}[|X|^r] < \infty$. Then, for every $k > 0$,

$$\text{Pr}(|X| \geq k) \leq \frac{\mathbb{E}[|X|^r]}{k^r}.$$

Proof.

$A := \{|X| \geq k\} \in \mathcal{F}$. On A we have $|X|^r \geq k^r$, hence

$$\mathbb{E}[|X|^r] \geq k^r \mathbb{E}[\mathbf{1}_A] = k^r \Pr(|X| \geq k).$$

To understand the first inequality, let $Y := X \cdot \mathbf{1}_A$. On A , we have $|Y|^r \geq k^r$. Then $\mathbb{E}[|X|^r] \geq \mathbb{E}[|Y|^r] \geq k^r \mathbb{E}[\mathbf{1}_A]$. The final equality follows from the fact that $\mathbb{E}[\mathbf{1}_A] = \Pr(A) = \Pr(|X| \geq k)$. Dividing by k^r gives the claim. $\square$

Theorem 4.27 (Chebyshev inequality). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $X \in L^2$ and set $\mu := \mathbb{E}[X]$ and $\sigma^2 := \text{Var}(X) = \mathbb{E}[(X - \mu)^2]$. Then, for every $k > 0$,*

$$\Pr(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Proof.

Apply Markov's inequality (Theorem 4.26) with $r = 2$ to the random variable $Y := X - \mu$:

$$\Pr(|X - \mu| \geq k) = \Pr(|Y| \geq k) \leq \frac{\mathbb{E}[|Y|^2]}{k^2} = \frac{\mathbb{E}[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2}.$$

$\square$

The following theorem is the probabilistic version of Theorem 1.91:

Theorem 4.28 (Jensen's Inequality). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})) \in L^1(\Omega)$ be an integrable random variable. If $g : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function and $g(X) \in L^1(\Omega)$, then*

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)].$$

The following theorem is the probabilistic version of Theorem 1.93:

Theorem 4.29 (Hölder inequality). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $p, q \in (1, \infty)$ satisfy*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then, for any $X \in L^p(\Omega)$ and $Y \in L^q(\Omega)$, we have that $XY \in L^1(\Omega)$ and

$$\mathbb{E}[|XY|] \leq \|X\|_p \|Y\|_q.$$

Theorem 4.30 (Cauchy-Schwarz Inequality). *If X and Y are real-valued square-integrable random variables, then*

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2], \tag{4.5}$$

with equality holding if and only if $Y = tX$ almost surely for some $t \in \mathbb{R}$ (or vice versa). In particular, if $X, Y \in L^2$, then $XY \in L^1$.

Proof.

The inequality directly follows from the well-known Cauchy-Schwarz inequality for generic inner product spaces, $\langle X, Y \rangle^2 \leq \|X\|^2 \|Y\|^2$, or, alternatively, it arises as an immediate corollary of Hölder's inequality (Theorem 4.29) by setting $p = q = 2$.

An elementary algebraic proof proceeds as follows: if $Y = 0$ almost surely, then $\mathbb{E}[XY] = 0$ and the conclusion holds trivially. Assume henceforth that $\mathbb{E}[Y^2] > 0$.

For every $t \in \mathbb{R}$, the random variable $(X - tY)^2$ is inherently non-negative and integrable, so

$$0 \leq \mathbb{E}[(X - tY)^2] = t^2 \mathbb{E}[Y^2] - 2t \mathbb{E}[XY] + \mathbb{E}[X^2].$$

Because this quadratic polynomial in t is universally non-negative, its discriminant must be non-positive:

$$(-2 \mathbb{E}[XY])^2 - 4 \mathbb{E}[Y^2] \mathbb{E}[X^2] \leq 0,$$

which immediately simplifies to the desired Cauchy-Schwarz inequality.

As for the equality case, it is attained if and only if there exists a scalar t such that the expected value is exactly zero, which demands $(X - tY)^2 = 0$, meaning $tX - Y = 0$ almost surely. $\square$

Remark 4.31. Observe that if $Y = tX$ almost surely for some $t \in \mathbb{R}$ and X, Y are absolutely continuous with respect to the Lebesgue measure λ , then the vector (X, Y) cannot admit a joint density with respect to λ^2 , as explained in Example 3.51. Hence, in settings where (X, Y) is jointly continuous, the Cauchy-Schwarz inequality is strict.

4.4 Variance and Covariance

We have already observed that the L^p space of random variables, after a suitable identification up to almost sure equivalence, is a vector space endowed with a norm. Moreover, this vector space is complete (Banach) (cf. 1.97). Among Banach spaces, a class of particular interest is known as *Hilbert spaces*. These are spaces where the norm is induced by an inner product. An inner product structure is particularly useful because it allows us to define the notion of orthogonality and, in general, provides a richer structure, enabling tools such as Fourier decomposition. Among the L^p Banach spaces of random variables, the only Hilbert space is the L^2 space of square-integrable random variables. In this section, we study the space L^2 .

Definition 4.32 (Hilbert space). A *Hilbert space* is a Banach space $(H, \|\cdot\|)$ such that there exists an inner product $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{F}$ (where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) that induces the norm, i.e.,

$$\|x\| = \sqrt{\langle x, x \rangle} \quad \forall x \in H.$$

We state the following theorem without proof (see [17]).

Theorem 4.33 (Jordan-von Neumann). Let $(H, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$. Then the norm $\|\cdot\|$ is induced by an inner product if and only if it satisfies the parallelogram law

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \forall x, y \in H. \quad (4.6)$$

Moreover, when this holds, the inner product is uniquely determined by the norm and is given by

$$\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2). \quad (4.7)$$

Theorem 4.34 (Space of square-integrable random variables). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $1 \leq p \leq \infty$. Then the normed space $(L^p(\Omega), \|\cdot\|_p)$ is a Hilbert space if and only if $p = 2$. In the case $p = 2$, the unique inner product inducing $\|\cdot\|_2$ is

$$\langle X, Y \rangle := \mathbb{E}[XY]. \quad (4.8)$$

Proof.

The first part of the proof is a consequence of Theorem 4.33: we only need to show that the parallelogram law (4.6) holds if and only if $p = 2$.

Let $\Omega = \{1, 2\}$, with $\Pr(\{1\}) = \Pr(\{2\}) = \frac{1}{2}$ and define the random variables

$$X := \mathbf{1}_{\{1\}} \quad Y := \mathbf{1}_{\{2\}}.$$

Then $X, Y \in L^p$ for every $p \in [1, \infty]$ and

$$\|Y\|_p = \|X\|_p = (\mathbb{E}[|X|^p])^{1/p} = \left(\int_{\Omega} X(\omega) d\Pr(\omega) \right)^{1/p} = \left(\sum_{i \in I} |x_i|^p p_X(x_i) \right)^{1/p} = \left(1^p \cdot \frac{1}{2} \right)^{1/p} = 2^{-1/p}$$

if $p < \infty$, while $\|Y\|_{\infty} = \|X\|_{\infty} = 1$.

Let us compute $\|X + Y\|_p$ and $\|X - Y\|_p$ in order to verify (4.6). We observe that

$$X + Y \equiv 1, \quad X - Y = \begin{cases} 1 & \text{if } \omega = 1; \\ -1 & \text{if } \omega = 2. \end{cases}$$

Hence, for every $p \in [1, \infty]$,

$$\|X + Y\|_p = \|1\|_p = (\mathbb{E}[1^p])^{1/p} = 1, \quad \|X - Y\|_p = \|1\|_p = (\mathbb{E}[1^p])^{1/p} = 1.$$

- If $p \in [1, \infty)$, then (4.6) applied to X and Y yields

$$\begin{aligned}\|X + Y\|_p^2 + \|X - Y\|_p^2 &= 2\|X\|_p^2 + 2\|Y\|_p^2 \\ 1^2 + 1^2 &= 2 \cdot 2^{-2/p} + 2 \cdot 2^{-2/p} \\ 2 &= 4 \cdot 2^{-2/p} \\ 1 - 2/p &= 0 \\ p &= 2\end{aligned}$$

- If $p = \infty$, then (4.6) applied to X and Y yields

$$\begin{aligned}\|X + Y\|_\infty^2 + \|X - Y\|_\infty^2 &\stackrel{?}{=} 2\|X\|_\infty^2 + 2\|Y\|_\infty^2 \\ 1^2 + 1^2 &\stackrel{?}{=} 2 \cdot 1^2 + 2 \cdot 1^2 \\ 2 &\neq 4\end{aligned}$$

We conclude that L^p is Hilbert only if $p = 2$. To see that L^2 is a Hilbert space we can exploit linearity of the expectation:

$$\|X + Y\|_2^2 + \|X - Y\|_2^2 = \mathbb{E}[(X + Y)^2] + \mathbb{E}[(X - Y)^2] = 2\mathbb{E}[X^2] + 2\mathbb{E}[Y^2] = 2\|X\|_2^2 + 2\|Y\|_2^2.$$

Here, we are tacitly assuming that if $X, Y \in L^2$, then $XY \in L^1$. This is indeed true by the Cauchy-Schwarz inequality 4.30.

As for the second part of the proof, it suffices to exploit Equation (4.7):

$$\langle X, Y \rangle := \frac{1}{4}(\|X + Y\|_2^2 - \|X - Y\|_2^2) = \frac{1}{4}(\mathbb{E}[(X + Y)^2] - \mathbb{E}[(X - Y)^2]) = \frac{1}{4}(4\mathbb{E}[XY]) = \mathbb{E}[XY].$$

This is a well-defined inner product by the Cauchy-Schwarz inequality 4.30. $\square$

The remarkable consequence of the previous results is that we can now treat square-integrable random variables effectively as vectors in a Hilbert space L^2 . In particular, we can define the notion of an "angle" between random variables. Two random variables will be (perfectly) correlated if one is directly proportional to the other, meaning they are linearly dependent. This allows us to bridge two concepts coming from different fields into a unified framework: linear dependence from linear algebra, which states that two vectors are "similar" if they are linearly dependent, and correlation from statistics, which states that two random variables are "similar" if their distributions are highly correlated. We are therefore able to provide a measure of the correlation between random variables simply by computing the angle between them.

What we seek is a formula analogous to the one in the Euclidean plane, namely

$$\cos \theta = \frac{\langle X, Y \rangle}{\|X\| \|Y\|} = \frac{x_1 y_1 + x_2 y_2}{\sqrt{x_1^2 + x_2^2} \sqrt{y_1^2 + y_2^2}}.$$

Observe that, to capture the true relationship between variables independently of constant shifts, our measure must be invariant under translation. Geometrically, this implies that to obtain a robust notion of angle, we must first center the vectors at the origin. The proper way to achieve this in the L^2 space of random variables is to subtract its expectation from each random variable, so that the translated variables all have zero expectation.

Definition 4.35 (Mean-zero random variable). Let X be a real-valued square-integrable random variable. We denote by $\tilde{X}$ its *centered* (*mean-zero*) version, defined by

$$\tilde{X} := X - \mathbb{E}[X].$$

Definition 4.36 (Correlation coefficient). Let X and Y be real-valued square-integrable random variables. The correlation coefficient of X and Y is defined as

$$\rho(X, Y) := \cos(\angle(\tilde{X}, \tilde{Y})) = \frac{\langle \tilde{X}, \tilde{Y} \rangle}{\|\tilde{X}\|_2 \|\tilde{Y}\|_2},$$

whenever the denominator is nonzero.

Remark 4.37. We observe that $\rho(X, Y) \in [-1, 1]$ by Theorem 4.30.

Remark 4.38 (Cosine similarity). The geometric interpretation of correlation as the cosine of an angle is fundamental in machine learning, where *cosine similarity* is a metric that discerns vector orientation independently of magnitude.

The inner product and the norm are so useful when restricted to mean-zero random variables that they are given the special names of *covariance* and *variance*.

Definition 4.39 (Covariance). Let X, Y be real-valued square-integrable random variables. The *covariance* between X and Y is

$$\text{Cov}(X, Y) := \langle \tilde{X}, \tilde{Y} \rangle = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)],$$

where $\mu_X := \mathbb{E}[X]$ and $\mu_Y := \mathbb{E}[Y]$.

Definition 4.40 (Variance). Let X be a real-valued square-integrable random variable. Its second central moment is called the *variance*, denoted by $\text{Var}(X)$ (also $V(X)$, σ_X^2 , or simply σ^2):

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

It corresponds to the square of the L^2 -norm of the mean-zero version of X :

$$\text{Var}(X) = \|\tilde{X}\|_2^2.$$

We call *standard deviation* the square root of the variance:

$$\sigma_X = \sqrt{\text{Var}(X)} = \|X - \mathbb{E}[X]\|_2 = \|\tilde{X}\|_2.$$

Example 4.41 (Variance of a discrete random variable). Let X be a discrete real-valued random variable with support $S = \{x_i\}_{i \in I}$, expected value μ_X , and PMF p_X . The variance is the weighted sum of squared deviations from the mean:

$$\text{Var}(X) = \sum_{i \in I} (x_i - \mu_X)^2 p_X(x_i) = \left(\sum_{i \in I} x_i^2 p_X(x_i) \right) - \mu_X^2.$$

This quantity measures the dispersion or "spread" of the probability mass around μ_X . Since each term $(x_i - \mu_X)^2 p_X(x_i)$ is nonnegative (and $p_X(x_i) > 0$ for $x_i \in S$), the sum is zero if and only if $x_i = \mu_X$ for every $i \in I$. Equivalently, $\text{Var}(X) = 0$ if and only if $\Pr(X = \mu_X) = 1$, meaning $X = \mu_X$ almost surely (i.e., the distribution is entirely concentrated at a single, deterministic point).

Example 4.42 (Variance of a continuous random variable). Let X be a continuous real-valued random variable with expected value μ_X and PDF f_X . By applying the Radon-Nikodym theorem as in Example 4.6, the variance becomes a Riemann integral:

$$\text{Var}(X) = \int_{-\infty}^{+\infty} (x - \mu_X)^2 f_X(x) dx = \left(\int_{-\infty}^{+\infty} x^2 f_X(x) dx \right) - \mu_X^2.$$

Geometrically, this integral computes the continuous analog of the moment of inertia from classical mechanics: it measures the spread of the probability density f_X around its center of mass μ_X .

Furthermore, if the integral were zero, since $(x - \mu_X)^2 f_X(x) \geq 0$, we would have $(x - \mu_X)^2 f_X(x) = 0$ for Lebesgue-a.e. x . Hence $f_X(x) = 0$ for Lebesgue-a.e. $x \neq \mu_X$, which is impossible for a valid probability density function (which must integrate to 1 over $\mathbb{R}$). Therefore, if X admits a PDF and is not degenerate, its variance is strictly positive: $\text{Var}(X) > 0$.

Remark 4.43 (Correlation coefficient). Operatively, it is more convenient to define the correlation coefficient in terms of variance and covariance:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

It represents the linear correlation between two square-integrable random variables.

Proposition 4.44 (Properties of variance and covariance). Let X, Y and Z be square-integrable random variables and $a, b, c, d \in \mathbb{R}$.

$$i) \text{ Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

$$ii) \operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y].$$

$$iii) \operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

$$iv) \operatorname{Cov}(aX + b, cY + d) = ac \operatorname{Cov}(X, Y).$$

$$v) \operatorname{Cov}(X, X) = \operatorname{Var}(X).$$

$$vi) \operatorname{Cov}(X, Y) = \operatorname{Cov}(Y, X).$$

$$vii) \operatorname{Cov}(aX + bY, Z) = a \operatorname{Cov}(X, Z) + b \operatorname{Cov}(Y, Z).$$

$$viii) \operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2 \operatorname{Cov}(X, Y).$$

$$ix) \operatorname{Var}(X - Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) - 2 \operatorname{Cov}(X, Y).$$

$$x) |\operatorname{Cov}(X, Y)| \leq \sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}.$$

Proof.

The first seven properties directly follow from Theorem 4.11 (linearity of the expectation):

$$\begin{aligned} \operatorname{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] + \mathbb{E}[\mu^2] - 2 \mathbb{E}[X\mu] = \mathbb{E}[X^2] + \mu^2 - 2\mu \mathbb{E}[X] = \mathbb{E}[X^2] + \mu^2 - 2\mu^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2. \end{aligned}$$

$$\begin{aligned} \operatorname{Cov}(X, Y) &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] + \mathbb{E}[\mu_X \mu_Y] - \mathbb{E}[X \mu_Y] - \mathbb{E}[Y \mu_X] \\ &= \mathbb{E}[XY] + \mu_X \mu_Y - \mu_Y \mathbb{E}[X] - \mu_X \mathbb{E}[Y] = \mathbb{E}[XY] + \mu_X \mu_Y - \mu_Y \mu_X - \mu_X \mu_Y \\ &= \mathbb{E}[XY] - \mu_X \mu_Y = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]. \end{aligned}$$

$$\begin{aligned} \operatorname{Var}(aX + b) &= \mathbb{E}[(aX + b - \mathbb{E}[aX + b])^2] = \mathbb{E}[(aX + b - a\mu_X - b)^2] = \mathbb{E}[a^2(X - \mu_X)^2] \\ &= a^2 \mathbb{E}[(X - \mu_X)^2] = a^2 \operatorname{Var}(X). \end{aligned}$$

$$\begin{aligned} \operatorname{Cov}(aX + b, cY + d) &= \mathbb{E}[(aX + b - \mathbb{E}[aX + b])(cY + d - \mathbb{E}[cY + d])] \\ &= \mathbb{E}[(aX + b - a\mu_X - b)(cY + d - c\mu_Y - d)] = \mathbb{E}[ac(X - \mu_X)(Y - \mu_Y)] \\ &= ac \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = ac \operatorname{Cov}(X, Y). \end{aligned}$$

$$\operatorname{Cov}(X, X) = \mathbb{E}[(X - \mu_X)(X - \mu_X)] = \mathbb{E}[(X - \mu_X)^2] = \operatorname{Var}(X).$$

$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[(Y - \mu_Y)(X - \mu_X)] = \operatorname{Cov}(Y, X).$$

$$\begin{aligned} \operatorname{Cov}(aX + bY, Z) &= \mathbb{E}[(aX + bY - \mathbb{E}[aX + bY])(Z - \mu_Z)] = \mathbb{E}[(aX + bY - a\mu_X - b\mu_Y)(Z - \mu_Z)] \\ &= \mathbb{E}[a(X - \mu_X)(Z - \mu_Z) + b(Y - \mu_Y)(Z - \mu_Z)] = a \operatorname{Cov}(X, Z) + b \operatorname{Cov}(Y, Z). \end{aligned}$$

Of course, combining the last two properties, we also deduce

$$\operatorname{Cov}(X, aY + bZ) = a \operatorname{Cov}(X, Y) + b \operatorname{Cov}(X, Z).$$

Observe that if $X, Y \in L^2$, then $X \pm Y \in L^2$, since L^2 is a vector space (or by Minkowski's inequality).

Thanks to the previous property,

$$\begin{aligned} \operatorname{Var}(X + Y) &= \operatorname{Cov}(X + Y, X + Y) = \operatorname{Cov}(X, X + Y) + \operatorname{Cov}(Y, X + Y) \\ &= \operatorname{Cov}(X, X) + \operatorname{Cov}(Y, Y) + \operatorname{Cov}(X, Y) + \operatorname{Cov}(Y, X) \\ &= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2 \operatorname{Cov}(X, Y), \end{aligned}$$

and

$$\begin{aligned} \operatorname{Var}(X - Y) &= \operatorname{Cov}(X - Y, X - Y) = \operatorname{Cov}(X, X - Y) - \operatorname{Cov}(Y, X - Y) \\ &= \operatorname{Cov}(X, X) + \operatorname{Cov}(Y, Y) - \operatorname{Cov}(X, Y) - \operatorname{Cov}(Y, X) \\ &= \operatorname{Var}(X) + \operatorname{Var}(Y) - 2 \operatorname{Cov}(X, Y). \end{aligned}$$

Applying Theorem 4.30 to $X' := X - \mathbb{E}[X]$ and $Y' := Y - \mathbb{E}[Y]$, we deduce

$$\mathbb{E}[X'Y']^2 \leq \mathbb{E}[X'^2] \mathbb{E}[Y'^2] \iff \operatorname{Cov}(X, Y)^2 \leq \operatorname{Var}(X) \operatorname{Var}(Y),$$

thus

$$|\operatorname{Cov}(X, Y)| \leq \sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}.$$

□

Remark 4.45. Property (x) of Proposition 4.44 implies that the correlation coefficient $\rho(X, Y)$ takes values between -1 and 1 . Therefore, it is a unit-free measure of the linear correlation. The meaning of its values is the following:

- If $|\rho(X, Y)| \approx 1$, then X and Y have a strong linear correlation, with the extreme case given by $\rho(X, Y) = \pm 1$, in which case we can write $Y = \alpha X + \beta$ for some β (which is a translational constant) and $\alpha < 0$ if $\rho(X, Y) < 0$ and $\alpha > 0$ if $\rho(X, Y) > 0$.
- If $\rho(X, Y) \approx 0$, then X and Y have low correlation.

Observe that ρ does not encode nonlinear correlations. For instance, if $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is an absolutely continuous random variable with respect to the Lebesgue measure, with PDF

$$f_X(x) = \frac{1}{2} \mathbf{1}_{[-1,1]}(x),$$

and $Y = X^2$, then $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = \mathbb{E}[X^3] = 0$, meaning $\rho = 0$, even if the two functions are quadratically correlated. The *maximal correlation* gives a measure of the correlation $\rho_{\max} \in [0, 1]$ between variables regardless of the correlation rule. It is defined as the following optimization problem:

$$\rho_{\max} = \sup_{\psi, \varphi} \mathbb{E}[\psi(X)\varphi(Y)] \quad \text{sub} \quad \mathbb{E}[\psi(X)] = \mathbb{E}[\varphi(Y)] = 0, \quad \mathbb{E}[\psi^2(X)] = \mathbb{E}[\varphi^2(Y)] = 1,$$

where the conditions on the first and second moment of $\psi(X)$ and $\varphi(Y)$ fix the scale and constants. Applying Lagrange-multipliers, we can retrieve a closed form solution in the form of a *functional nonlinear integral equation*. Even if analytically appealing, in practice this is hard to solve. Hence, there exist numerical algorithms such as *alternating conditional expectation* (ACE) that find the maximal correlation by iterative smoothing.

Corollary 4.46. *If X and Y are square-integrable and uncorrelated (meaning $\text{Cov}(X, Y) = 0$), then $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$.*

Proof. It holds $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2 \text{Cov}(X, Y) = \text{Var}(X) + \text{Var}(Y)$. $\square$

Corollary 4.47. *If X and Y are square-integrable and independent, then $\text{Cov}(X, Y) = 0$ and $\rho(X, Y) = 0$, meaning X and Y are uncorrelated. Moreover, $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$.*

Proof.

If X and Y are integrable and independent, by Theorem 4.15, then $\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = 0$. Since $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]$, we conclude that two independent random variables are uncorrelated. Since $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2 \text{Cov}(X, Y)$ and $\text{Cov}(X, Y) = 0$ for independent random variables, then $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$.

Observe that uncorrelated random variables need not be independent. For example, if X is a random variable with PDF $f_X = \mathbf{1}_{[-1,1]}/2$ and $Y = X^2$, then

$$\mathbb{E}[X] = \int_{-1}^1 x dx = 0, \quad \mathbb{E}[Y] = \int_{-1}^1 x^2 dx = \frac{2}{3}, \quad \mathbb{E}[XY] = \int_{-1}^1 x^3 dx = 0.$$

Therefore, $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = 0 - 0 \cdot \frac{2}{3} = 0$, even if X and Y are clearly quadratically correlated. $\square$

Example 4.48. It follows from Corollary 4.47 that independent random variables are uncorrelated. The converse does not hold. A counterexample is the random variables $X = \pi_1(Z)$ and $Y = \pi_2(Z)$ given in Example 4.15, which was earlier used to show that $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$ can hold for a pair of non-independent random variables. Then X and Y are uncorrelated but non-independent.

5 Generating functions

In this section, we examine the *moment generating function*, the *probability generating function*, and the *characteristic function*. These can be viewed as infinite series whose coefficients enable us to deduce important structural properties of the law of a random variable.

5.1 Moment Generating Function

Moments can be conveniently encoded into functions called *moment generating functions*. The process is similar to that of generating functions for sequences. To understand the process, we recall the concept of generating function.

First, we recall that a *formal power series* is an infinite sum of the form

$$\sum_{n=0}^{\infty} a_n x^n$$

that is considered independently from any notion of convergence. In fact, it is not regarded as a function of x , and the focus is on the coefficients $(a_n)_{n \in \mathbb{N}}$ that form a sequence. Of course, in case of convergence, the formal power series can be rewritten. For example, if $|x| < 1$, then

$$1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

Definition 5.1 (Generating function). Let $(a_n)_{n \in \mathbb{N}}$ be a real-valued sequence. The *generating function* corresponding to $(a_n)_{n \in \mathbb{N}}$ is given by

$$\sum_{n \in \mathbb{N}} a_n x^n.$$

Example 5.2 (Fibonacci sequence). The *Fibonacci sequence* is defined recursively as follows:

$$F_n := \begin{cases} 1 & \text{if } n = 1 \\ 1 & \text{if } n = 2 \\ F_{n-1} + F_{n-2} & \text{if } n \geq 3 \end{cases}.$$

Its corresponding generating function is given by

$$s(x) = \sum_{n=1}^{\infty} F_n x^n = \frac{x}{1-x-x^2}, \quad |x| < 1/\varphi = \frac{2}{1+\sqrt{5}}.$$

This can be proved by multiplying by $1 - x - x^2$ on both sides and using $F_n - F_{n-1} - F_{n-2} = 0$:

$$\begin{aligned} (1 - x - x^2)s(x) &= \sum_{n=1}^{\infty} (1 - x - x^2)F_n x^n = \sum_{n=1}^{\infty} F_n x^n - \sum_{n=1}^{\infty} F_n x^{n+1} - \sum_{n=1}^{\infty} F_n x^{n+2} \\ &= \sum_{n=1}^{\infty} F_n x^n - \sum_{n=2}^{\infty} F_{n-1} x^n - \sum_{n=3}^{\infty} F_{n-2} x^n = F_1 x + F_2 x^2 - F_1 x^2 + \sum_{n=3}^{\infty} (F_n - F_{n-1} - F_{n-2}) x^n \\ &= x + x^2 - x^2 = x. \end{aligned}$$

To recover the coefficients from the closed form $x/(1 - x - x^2)$, the common trick is to use Taylor series and common expansions such as

$$\frac{1}{1 \pm x} = 1 \mp x + x^2 \mp x^3 + \cdots$$

For example, we can easily recover the first few terms of the Fibonacci sequence:

$$\begin{aligned} \frac{x}{1 - (x + x^2)} &= x \cdot [1 + (x + x^2) + (x + x^2)^2 + (x + x^2)^3 + \cdots] \\ &= x \cdot [1 + x + x^2 + x^2 + 2x^3 + x^4 + x^3 + 3x^4 + 3x^5 + x^6 + x^4 \cdots] \\ &= x \cdot [1 + x + 2x^2 + 3x^3 + 5x^4 + \cdots] \\ &= x + x^2 + 2x^3 + 3x^4 + 5x^5 + \cdots \end{aligned}$$

Example 5.3 (Negative binomial identity). Fix $k \in \mathbb{N}$ and consider the sequence

$$a_n := \binom{n+k-1}{k-1}, \quad n \in \mathbb{N}.$$

This sequence counts the number of k -tuples $(m_1, \dots, m_k) \in \mathbb{N}^k$ such that $m_1 + \dots + m_k = n$ (a classic "stars and bars" combinatorial problem, cf. [18]). Let us compute its generating function as a *formal power series*. For a single nonnegative integer variable $m_i \in \mathbb{N}$, the generating function of the constant sequence $(1, 1, 1, \dots)$ is the standard geometric series:

$$\sum_{m_i=0}^{\infty} x^{m_i} = 1 + x + x^2 + \dots = \frac{1}{1-x}, \quad \forall x \in \mathbb{C} : |x| < 1.$$

When considering k independent choices $m_1, \dots, m_k \in \mathbb{N}$, the generating function that records their sum $m_1 + \dots + m_k$ is simply the product of their individual generating functions:

$$\left(\sum_{m_1=0}^{\infty} x^{m_1} \right) \cdots \left(\sum_{m_k=0}^{\infty} x^{m_k} \right) = \left(\frac{1}{1-x} \right)^k = \frac{1}{(1-x)^k}.$$

Expanding this product as a Cauchy product of formal power series (i.e., a discrete convolution, cf. [19] and Remark 2.30), the coefficient of x^n counts exactly the number of solutions to $m_1 + \dots + m_k = n$. Hence, we obtain the identity:

$$\frac{1}{(1-x)^k} = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^n, \quad |x| < 1. \quad (5.1)$$

Generating functions are used in probability to recover moments:

Definition 5.4 (Moment generating function). The *moment generating function* (MGF) of a real-valued random variable X is the function $M_X : I \rightarrow (0, \infty)$ defined by

$$M_X(t) := \mathbb{E}[e^{tX}], \quad t \in I, \quad (5.2)$$

where $I \subseteq \mathbb{R}$ is the interval of convergence of the expectation: $I := \{t \in \mathbb{R} : \mathbb{E}[e^{tX}] < \infty\}$. We say that the MGF *exists* if $0 \in \text{int}(I)$, that is, there exists $h > 0$ such that $(-h, h) \subseteq I$.

Example 5.5 (MGF for discrete random variables). For discrete random variables, Equation (5.2) becomes

$$M_X(t) = \sum_{x \in S} e^{tx} \Pr(X = x) = \sum_{x \in S} e^{tx} p_X(x)$$

Example 5.6 (MGF for continuous random variables). For continuous random variables, Equation (5.2) becomes

$$M_X(t) = \int_{\mathbb{R}} e^{tx} f_X(x) d\lambda(x).$$

Proposition 5.7 (Properties of the MGF). Let X be a random variable and assume its MGF M_X exists. Then

i) $M_X(t) > 0$ for every t in the domain of $M_X(t)$, and $M_X(0) = 1$;

ii) For any $a, b \in \mathbb{R}$, the MGF of $Y := aX + b$ is

$$M_Y(t) = e^{bt} M_X(at)$$

whenever both sides are finite.

iii) Its domain I is convex, and M_X is convex in I ;

Proof.

(i). By direct computation,

$$M_X(0) = \mathbb{E}[e^0] = \mathbb{E}[1] = 1.$$

By monotonicity of the expectation, $e^{tX} > 0 \implies \mathbb{E}[e^{tX}] > 0$.

(ii). Suppose $a, b \in \mathbb{R}$. Then

$$M_{aX+b}(t) = \mathbb{E}[e^{t(aX+b)}] = \mathbb{E}[e^{atX} \cdot e^{bt}] = e^{bt} \cdot \mathbb{E}[e^{atX}] = e^{bt} M_X(at).$$

(iii). Finally, in order to prove that I is convex, let us suppose that $M_X(t_1)$ and $M_X(t_2)$ exist for some $t_1 \neq t_2$. We will prove that, for every $\alpha \in [0, 1]$, also $M_X(\alpha t_1 + (1 - \alpha)t_2)$ exists. Moreover, since we should also prove that M_X is convex in I , we already know that the inequality

$$M_X(\alpha t_1 + (1 - \alpha)t_2) \leq \alpha M_X(t_1) + (1 - \alpha)M_X(t_2) \quad (5.3)$$

is expected to hold. But since $\alpha M_X(t_1) + (1 - \alpha)M_X(t_2) < \infty$, this inequality yields as a byproduct the convexity of I . As a consequence, we only need to show that (5.3) holds true. Let us rewrite the inequality by making the expectations explicit:

$$\mathbb{E}[e^{(\alpha t_1 + (1 - \alpha)t_2)X}] \leq \alpha \mathbb{E}[e^{t_1 X}] + (1 - \alpha) \mathbb{E}[e^{t_2 X}] = \mathbb{E}[\alpha e^{t_1 X} + (1 - \alpha)e^{t_2 X}].$$

It is clear that a sufficient condition is the inequality to hold pointwise, i.e., for every $\omega \in \Omega$,

$$e^{(\alpha t_1 + (1 - \alpha)t_2)X(\omega)} \leq \alpha e^{t_1 X(\omega)} + (1 - \alpha)e^{t_2 X(\omega)}.$$

Setting $x := t_1 X(\omega)$ and $y := t_2 X(\omega)$, this sufficient condition becomes

$$e^{\alpha x + (1 - \alpha)y} \leq \alpha e^x + (1 - \alpha)e^y,$$

which is trivially valid by definition of convexity of the exponential map $t \mapsto e^t$. $\square$

The following proof is inspired by [20] and [5]:

Theorem 5.8 (Moments from the MGF). *Assume that M_X exists in a neighborhood of 0, say $(-h, h)$. Then X has finite absolute moments of all orders. Moreover, $M_X(t)$ admits an absolutely convergent Taylor expansion*

$$M_X(t) = \sum_{n=0}^{\infty} \frac{\mathbb{E}[X^n]}{n!} t^n, \quad t \in (-h, h). \quad (5.4)$$

In particular, $\mathbb{E}[X^n] = M_X^{(n)}(0)$ for all $n \geq 0$.

Proof.

(5.4) is equivalent to

$$\sum_{n=0}^{\infty} \frac{\mathbb{E}[X^n]}{n!} t^n \stackrel{?}{=} M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\sum_{n=0}^{\infty} \frac{(tX)^n}{n!}\right].$$

As we see, this boils down to exchanging series and expectation. Such swap can be justified via Theorem 1.80. In what follows, we verify that the theorem can be used.

- Fix $t \in (-h, h)$ and choose t_0 such that $|t| < t_0 < h$. Since $t_0 \in (-h, h)$, we have $\mathbb{E}[e^{t_0 X}] < \infty$ and $\mathbb{E}[e^{-t_0 X}] < \infty$ by hypothesis. As a consequence, also $\mathbb{E}[e^{t_0 |X|}] < \infty$. Indeed, $e^{t_0 |X|} \leq e^{t_0 X} + e^{-t_0 X}$ holds pointwise, hence

$$\mathbb{E}[e^{t_0 |X|}] \leq \mathbb{E}[e^{t_0 X}] + \mathbb{E}[e^{-t_0 X}] < \infty.$$

Moreover, for every $k \in \mathbb{N}$ we have $e^{t_0 |X|} \geq (t_0 |X|)^k / k!$ pointwise, hence

$$\mathbb{E}[|X|^k] \leq \frac{k!}{t_0^k} \mathbb{E}[e^{t_0 |X|}] < \infty.$$

In particular, X has finite absolute moments of all orders.

- Next, we find a dominating integrable random variable for the partial sums. For every $n \in \mathbb{N}$,

$$\left| \sum_{k=0}^n \frac{X^k}{k!} t^k \right| \leq \sum_{k=0}^n \frac{|X|^k}{k!} |t|^k \leq \sum_{k=0}^{\infty} \frac{|X|^k}{k!} |t|^k = e^{|t||X|} \leq e^{t_0 |X|}.$$

Since $\mathbb{E}[e^{t_0 |X|}] < \infty$, Theorem 1.80 applies.

Therefore,

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{(tX)^k}{k!}\right] = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!} t^k, \quad t \in (-h, h).$$

Finally, since M_X agrees on $(-h, h)$ with the power series $\sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!} t^k$ (which has positive radius of convergence), we may differentiate term-by-term and obtain

$$M_X^{(n)}(0) = \mathbb{E}[X^n], \quad n \geq 0.$$

□

Proposition 5.9 (Convolution property). *Let $X_1, \dots, X_n$ be independent random variables whose MGF's all exist on some neighbourhood I of 0. Then the MGF of $S_n := X_1 + \dots + X_n$ is well-defined on I and it is given by*

$$M_{S_n}(t) = M_{X_1}(t) \cdots M_{X_n}(t), \quad \forall t \in I.$$

Proof.

By definition and independence (4.15),

$$M_{S_n}(t) = \mathbb{E}[e^{t(X_1 + \dots + X_n)}] = \mathbb{E}[e^{tX_1} \cdots e^{tX_n}] = \mathbb{E}[e^{tX_1}] \cdots \mathbb{E}[e^{tX_n}] = M_{X_1}(t) \cdots M_{X_n}(t).$$

□

5.2 Probability Generating Function

Definition 5.10 (Probability generating function). Let X be a random variable taking values in $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, and let $\mathbb{D} := \{s \in \mathbb{C} : |s| \leq 1\}$. The *probability generating function* (pgf) of X is the function $G_X : \mathbb{D} \rightarrow \mathbb{C}$ defined by

$$G_X(s) := \mathbb{E}[s^X] = \sum_{x=0}^{\infty} p_X(x) s^x,$$

where $p_X(x) = \Pr(X = x)$ is the probability mass function (PMF) of X .

Since $|s^X| \leq 1$ for all $|s| \leq 1$, we have $\mathbb{E}[|s^X|] \leq \mathbb{E}[|1|] = 1$. Hence, the pgf is always well-defined on $\mathbb{D}$, and the series is absolutely convergent there.

Proposition 5.11 (Properties of the probability generating function). *Let X be a random variable taking values in $\mathbb{N}_0$, and let G_X be its probability generating function. Then:*

i) $G_X(1) = 1$;

ii) *The PGF uniquely determines the distribution of X . The PMF can be recovered via the Taylor coefficients of $G_X(s)$ at the origin:*

$$\Pr(X = k) = \frac{G_X^{(k)}(0)}{k!}, \quad k \in \mathbb{N}_0. \quad (5.5)$$

iii) *If $X \in L^n(\Omega)$, then the n -th left derivative at 1 exists and*

$$G_X^{(n)}(1^-) := \lim_{s \rightarrow 1^-} G_X^{(n)}(s) = \mathbb{E}[X(X-1) \cdots (X-n+1)].$$

iv) *Whenever both sides are finite, it holds that*

$$M_X(t) = G_X(e^t).$$

Proof.

(i). By direct computation, $G_X(1) = \mathbb{E}[1^X] = \mathbb{E}[1] = 1$.

(ii). Since $G_X(s) = \sum_{k=0}^{\infty} p_X(k) s^k$ is a power series with radius of convergence $R \geq 1$, it is analytic inside the open unit disk $\mathbb{D}$. Hence, it coincides with its Taylor expansion on $\mathbb{D}$, and we may differentiate

term-by-term n times:

$$\begin{aligned} G_X^{(n)}(s) &= \frac{d^n}{ds^n} \sum_{k=0}^{\infty} p_X(k) s^k = \sum_{k=0}^{\infty} p_X(k) \frac{d^n s^k}{ds^n} = \sum_{k=n}^{\infty} k(k-1)\dots(k-n+1) p_X(k) s^{k-n} \\ &= \sum_{k=n}^{\infty} \frac{k!}{(k-n)!} p_X(k) s^{k-n}. \end{aligned}$$

Evaluating at $s = 0$, all terms in the sum vanish except for the one where $k = n$, which yields

$$G_X^{(n)}(0) = n! p_X(n) \implies \Pr(X = n) = \frac{G_X^{(n)}(0)}{n!}.$$

Since all probabilities are uniquely determined by the derivatives at zero, the PGF uniquely identifies the distribution of X .

(iii). Consider the series

$$G_X^{(n)}(s) = \sum_{k=n}^{\infty} \frac{k!}{(k-n)!} p_X(k) s^{k-n}, \quad s \in [0, 1).$$

Since for $s \in [0, 1)$ all terms in the series are non-negative, we can apply the Monotone Convergence Theorem (1.50) to deduce

$$\begin{aligned} G_X^{(n)}(1^-) &:= \lim_{s \rightarrow 1^-} G_X^{(n)}(s) = \lim_{s \rightarrow 1^-} \sum_{k=n}^{\infty} \frac{k!}{(k-n)!} p_X(k) s^{k-n} \\ &= \sum_{k=n}^{\infty} \frac{k!}{(k-n)!} p_X(k) \lim_{s \rightarrow 1^-} s^{k-n} = \sum_{k=n}^{\infty} \frac{k!}{(k-n)!} p_X(k) \\ &= \mathbb{E}[X(X-1)\dots(X-n+1)]. \end{aligned}$$

(iv). By direct substitution, whenever both sides are finite, we compute:

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}[(e^t)^X] = G_X(e^t).$$

□

Proposition 5.12 (Convolution property). Let $X_1, \dots, X_n$ be independent discrete $\mathbb{N}_0$ -valued random variables. Then, for every $|s| \leq 1$,

$$G_{X_1+\dots+X_n}(s) = G_{X_1}(s) \cdot \dots \cdot G_{X_n}(s).$$

Proof.

By independence and iterating Theorem 4.15,

$$G_{X_1+\dots+X_n}(s) = \mathbb{E}[s^{X_1} \cdot \dots \cdot s^{X_n}] = \mathbb{E}[s^{X_1}] \cdot \dots \cdot \mathbb{E}[s^{X_n}] = G_{X_1}(s) \cdot \dots \cdot G_{X_n}(s).$$

□

Example 5.13 (Sum of i.i.d. discrete random variables). Let $X_1, \dots, X_n$ be independent and identically distributed random variables taking values in $\{0, 1, \dots, \ell\}$ with PMF p_X . Let $S := X_1 + \dots + X_n$. By Corollary 3.81, the PMF p_S of S is the n -fold convolution p_X^{*n} . The direct computation of p_S through successive convolutions is however computationally expensive. Indeed, for each $k = 1, \dots, n-1$ we need to compute $p_X^{*(k+1)}$ from p_X^{*k} , and the total number of multiplications has order $\mathcal{O}(n^2 \ell^2)$. We can improve this by leveraging the PGF and its connection to the Discrete Fourier Transform (Definition 2.70).

By definition, S takes values in $\{0, 1, \dots, n\ell\}$. Let $N := n\ell + 1$. By Proposition 5.12, $G_S(s) = [G_X(s)]^n$, which implies that

$$[G_X(s)]^n = \sum_{k=0}^{N-1} p_S(k) s^k.$$

To efficiently find the coefficients $(p_S(k))_{k=0}^{N-1}$, we evaluate $G_S(s)$ at the N -th roots of unity. Let $\omega = e^{-2\pi i/N}$. For every $m = 0, \dots, N-1$, evaluating both expressions yields:

$$G_S(\omega^m) = \sum_{k=0}^{N-1} p_S(k) \omega^{mk} \quad \text{and} \quad G_S(\omega^m) = [G_X(\omega^m)]^n.$$

Comparing this with Definition 2.70, we immediately recognize that $G_S(\omega^m)$ and $G_X(\omega^m)$ are the Fourier transforms of p_S and p_X , respectively:

$$G_S(\omega^m) = \sqrt{N}(\mathcal{F}p_S)(m) \quad \text{and} \quad G_X(\omega^m) = \sqrt{N}(\mathcal{F}p_X)(m).$$

Equating the two expressions for $G_S(\omega^m)$, we obtain:

$$\sqrt{N}(\mathcal{F}p_S)(m) = \left[\sqrt{N}(\mathcal{F}p_X)(m) \right]^n = N^{n/2}[(\mathcal{F}p_X)(m)]^n.$$

Finally, to recover the PMF of S , we divide by $\sqrt{N}$ and apply the inverse DFT ($\mathcal{F}^*$) to this sequence. Denoting by $(\mathcal{F}p_X)^n$ the element-wise n -th power of the vector $\mathcal{F}p_X$, we get:

$$p_S = \mathcal{F}^* \left(N^{\frac{n-1}{2}} (\mathcal{F}p_X)^n \right).$$

This approach, when implemented via the Fast Fourier Transform (FFT), reduces the computational complexity to $\mathcal{O}(N \log N) = \mathcal{O}(n\ell \log(n\ell))$ (see Remark 2.72).¹⁰ By using the DTFT (2.25) rather than the DFT (Definition 2.70) one can generalize this procedure for i.i.d. random variables taking values in $\mathbb{Z}^n$.

5.3 Characteristic function

One of the most important results about the MGF is that it uniquely determines the distribution. This uniqueness property is closely related to the *rigidity of holomorphic functions* (Theorem A.4). This is a property that holds in $\mathbb{C}$ but does not generally hold in $\mathbb{R}$. Therefore, to analyze the uniqueness properties of generating functions, we extend their domain to the complex plane: depending on the context, this analytic extension is known as the *Laplace transform* or, for discrete variables, the *Z-transform*. Alternatively, the theoretical analysis becomes even cleaner when introducing a new function, which is the Fourier transform of the distribution of a random variable: it is the *characteristic function*.

Throughout, we work on the additive group $(\mathbb{R}, +)$ and we exploit the Fourier-analytic framework developed in Section 2. The multidimensional case $\mathbb{R}^n$ is completely analogous.

Definition 5.14 (Convolution of finite measures). Let μ, ν be finite Borel measures on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Their *convolution* $\mu * \nu$ is the finite Borel measure defined by

$$(\mu * \nu)(B) := \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_B(x+y) \mu(dx) \nu(dy), \quad B \in \mathcal{B}(\mathbb{R}).$$

Equivalently, for every bounded Borel-measurable function $\psi : \mathbb{R} \rightarrow \mathbb{C}$,

$$\int_{\mathbb{R}} \psi(z) (\mu * \nu)(dz) = \int_{\mathbb{R}} \int_{\mathbb{R}} \psi(x+y) \mu(dx) \nu(dy). \quad (5.6)$$

Definition 5.15 (Fourier transform of a finite measure). Let μ be a finite Borel measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Its *Fourier transform* is the bounded function $\hat{\mu} : \mathbb{R} \rightarrow \mathbb{C}$ defined by

$$\hat{\mu}(t) := \int_{\mathbb{R}} e^{-itx} \mu(dx), \quad t \in \mathbb{R}.$$

Proposition 5.16. Let μ be a finite Borel measure on $\mathbb{R}$. Then:

- i) *Boundedness.* $|\hat{\mu}(t)| \leq \mu(\mathbb{R})$ for all $t \in \mathbb{R}$;
- ii) *Uniform continuity.* $\hat{\mu}$ is uniformly continuous on $\mathbb{R}$.
- iii) *Convolution property.* $\widehat{\mu * \nu}(t) = \hat{\mu}(t) \hat{\nu}(t)$, $t \in \mathbb{R}$.

¹⁰More generally, if $X_1, \dots, X_n$ are independent with supports in $\{0, \dots, \ell_1\}, \dots, \{0, \dots, \ell_n\}$, letting $L := \ell_1 + \dots + \ell_n$ and $m := L + 1$, we have

$$\Pr(X_1 + \dots + X_n = z) = \frac{1}{m} \sum_{k=0}^{m-1} \left(\prod_{j=1}^n G_{X_j}(\omega^k) \right) \omega^{-kz}, \quad z = 0, 1, \dots, L.$$

Proof.

(i). By the triangle inequality,

$$|\hat{\mu}(t)| = \left| \int_{\mathbb{R}} e^{-itx} \mu(dx) \right| \leq \int_{\mathbb{R}} |e^{-itx}| \mu(dx) = \int_{\mathbb{R}} 1 \mu(dx) = \mu(\mathbb{R}).$$

(ii). For $h \in \mathbb{R}$,

$$\begin{aligned} \sup_{t \in \mathbb{R}} |\hat{\mu}(t+h) - \hat{\mu}(t)| &= \sup_{t \in \mathbb{R}} \left| \int_{\mathbb{R}} (e^{-i(t+h)x} - e^{-itx}) \mu(dx) \right| \\ &= \sup_{t \in \mathbb{R}} \left| \int_{\mathbb{R}} e^{-itx} (e^{-ihx} - 1) \mu(dx) \right| \leq \int_{\mathbb{R}} |e^{-ihx} - 1| \mu(dx). \end{aligned}$$

Since $|e^{-ihx} - 1| \leq 2$ and $|e^{-ihx} - 1| \rightarrow 0$ pointwise as $h \rightarrow 0$, the Dominated Convergence Theorem (4.19) yields

$$\lim_{h \rightarrow 0} \int_{\mathbb{R}} |e^{-ihx} - 1| \mu(dx) = 0,$$

which proves uniform continuity.

(iii). By Definition 5.15 and (5.6) with $\psi(z) = e^{-itz}$,

$$\begin{aligned} \widehat{\mu * \nu}(t) &= \int_{\mathbb{R}} e^{-itz} (\mu * \nu)(dz) = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-it(x+y)} \mu(dx) \nu(dy) \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-itx} e^{-ity} \mu(dx) \nu(dy) = \left(\int_{\mathbb{R}} e^{-itx} \mu(dx) \right) \left(\int_{\mathbb{R}} e^{-ity} \nu(dy) \right) = \hat{\mu}(t) \hat{\nu}(t), \end{aligned}$$

where the factorization is justified by Fubini-Tonelli, since $|e^{-it(x+y)}| = 1$ and μ, ν are finite. $\square$

Definition 5.17 (Characteristic function). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable with law $\Pr_X$. The *characteristic function* (CF) of X is

$$\varphi_X(t) := \widehat{\Pr_X}(t) = \int_{\mathbb{R}} e^{-itx} \Pr_X(dx) = \mathbb{E}[e^{-itX}], \quad t \in \mathbb{R}.$$

Example 5.18 (CF of a discrete random variable). If X is a discrete random variable with support $S_X = \{x_i\}_{i \in I}$, where I is a countable index set, then its probability measure is given by $\Pr_X = \sum_{i \in I} p_X(x_i) \delta_{x_i}$, and

$$\widehat{\Pr_X}(t) = \int_{\mathbb{R}} e^{-itx} \Pr_X(dx) = \sum_{i \in I} p_X(x_i) e^{-itx_i}.$$

In particular, if X is integer-valued, then

$$\varphi_X(t) = \sum_{n \in \mathbb{Z}} p_X(n) e^{-itn},$$

which is 2π -periodic. We recognize this as a Fourier series expansion. Therefore, by the orthogonality of the trigonometric system, we can recover the probabilities as

$$p_X(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \varphi_X(t) e^{itk} dt.$$

Example 5.19 (CF of an absolutely continuous random variable). If X is an absolutely continuous random variable with respect to Lebesgue measure, then $\Pr_X(dx) = f_X(x)dx$, where $f_X \in L^1(\mathbb{R})$ is a Radon-Nikodym density. Then

$$\varphi_X(t) = \widehat{\Pr_X}(t) = \int_{\mathbb{R}} e^{-itx} f_X(x) dx,$$

i.e., $\widehat{\Pr_X}$ coincides with the Fourier transform of the density f_X . In particular, by the Riemann-Lebesgue Lemma 2.51, $\varphi_X \in C_0(\mathbb{R})$. If in addition $\varphi_X \in L^1(\mathbb{R})$ (equivalently, under the assumptions of Theorem 2.61 applied to f_X), we can reconstruct f_X from φ_X as

$$f_X(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi_X(t) e^{ixt} dt.$$

Example 5.20 (CF of a Gaussian distribution). We say that a random variable $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ has a *normal (Gaussian) distribution*, written $X \sim \mathcal{N}(\mu, \sigma^2)$, if it is absolutely continuous with respect to Lebesgue measure λ and its PDF is given by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

where $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ are parameters. By Example 5.19, it holds that

$$\varphi_X(t) = \widehat{f}(t).$$

By Lemma 2.58, we have

$$\varphi_X(t) = e^{-i\mu t} e^{-\frac{\sigma^2 t^2}{2}}.$$

Proposition 5.21 (Basic properties). *Let X, Y be real-valued random variables. Then:*

- i) $\varphi_X(0) = 1$ and $|\varphi_X(t)| \leq 1$ for all $t \in \mathbb{R}$;
- ii) φ_X is uniformly continuous on $\mathbb{R}$;
- iii) $\varphi_X(-t) = \overline{\varphi_X(t)}$ for all $t \in \mathbb{R}$;
- iv) If X and Y are independent, then $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$;
- v) If $a, b \in \mathbb{R}$, then $\varphi_{aX+b}(t) = e^{-itb}\varphi_X(at)$.

Proof.

(i). $\varphi_X(0) = \mathbb{E}[1] = 1$. Since $|e^{-itX}| = 1$, then

$$|\varphi_X(t)| = |\mathbb{E}[e^{-itX}]| \leq \mathbb{E}[|e^{-itX}|] = 1.$$

In particular, φ_X is well-defined for every $t \in \mathbb{R}$.

(ii). This is Proposition 5.16 applied to $\mu = \Pr_X$.

(iii). Since $\overline{e^{-itX}} = e^{itX}$,

$$\varphi_X(-t) = \mathbb{E}[e^{itX}] = \mathbb{E}[\overline{e^{-itX}}] = \overline{\mathbb{E}[e^{-itX}]} = \overline{\varphi_X(t)}.$$

(iv). By independence (cf. 4.15),

$$\varphi_{X+Y}(t) = \mathbb{E}[e^{-it(X+Y)}] = \mathbb{E}[e^{-itX}e^{-itY}] = \mathbb{E}[e^{-itX}] \mathbb{E}[e^{-itY}] = \varphi_X(t)\varphi_Y(t).$$

(v). By direct computation,

$$\varphi_{aX+b}(t) = \mathbb{E}[e^{-it(aX+b)}] = \mathbb{E}[e^{-iatX}] e^{-itb} = e^{-itb}\varphi_X(at).$$

□

Theorem 5.22 (Uniqueness of the characteristic function). *The Fourier transform is injective on Borel probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$: if μ, ν are such measures and*

$$\hat{\mu}(t) = \hat{\nu}(t), \quad \forall t \in \mathbb{R},$$

then $\mu = \nu$. Equivalently, if $\varphi_X = \varphi_Y$, then X and Y have the same distribution: $F_X = F_Y$ and we write $X \stackrel{d}{=} Y$.

Proof.

Let $f \in C_c^\infty(\mathbb{R})$. By the Fourier inversion theorem 2.61, $f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(t) e^{itx} dt$. Integrating with respect to μ and applying Fubini's theorem (which is justified since $\hat{f} \in L^1(\mathbb{R})$ and μ is a finite measure), we obtain:

$$\int_{\mathbb{R}} f(x) \mu(dx) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(t) \left(\int_{\mathbb{R}} e^{itx} \mu(dx) \right) dt = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(t) \hat{\mu}(-t) dt.$$

Since $\hat{\mu} = \hat{\nu}$ by assumption, the right-hand side is identical for ν , yielding

$$\int_{\mathbb{R}} f \, d\mu = \int_{\mathbb{R}} f \, d\nu, \quad \forall f \in C_c^\infty(\mathbb{R}).$$

Using the density of $C_c^\infty(\mathbb{R})$ in $C_c(\mathbb{R})$ with respect to $\|\cdot\|_\infty$ (see Theorem 2.46 and the subsequent discussion), the equality extends to all $f \in C_c(\mathbb{R})$:

$$\int_{\mathbb{R}} f \, d\mu = \int_{\mathbb{R}} f \, d\nu, \quad \forall f \in C_c(\mathbb{R}).$$

By the Riesz-Markov-Kakutani representation theorem (see 2.3 of [3]), finite regular Borel measures are uniquely determined by their integrals against continuous compactly supported functions, which concludes the proof that $\mu = \nu$. Finally, $\mu = \nu$ if and only if $F_X = F_Y$ by Remark 3.56. $\square$

The following Corollary and its proof are largely taken from [20]:

Corollary 5.23 (Uniqueness of the MGF). *Suppose that X and Y are two real-valued random variables whose MGFs are well-defined and equal on some neighbourhood $(-h, h)$ of $t = 0$. Then $F_X = F_Y$.*

Proof.

Define the complex-valued functions

$$\Phi_X(z) := \mathbb{E}[e^{zX}], \quad \Phi_Y(z) := \mathbb{E}[e^{zY}],$$

on the vertical strip $S := \{z \in \mathbb{C} : \Re z \in (-h, h)\}$. We first show that Φ_X is holomorphic on S . Fix $z_0 = a_0 + ib_0 \in S$ and choose $r > 0$ such that $[a_0 - r, a_0 + r] \subset (-h, h)$. To prove that Φ_X is analytic at z_0 , we follow the same power series expansion strategy used in Theorem 5.8. For any $z \in B_r(z_0)$, we can write

$$e^{zX} = e^{z_0X} e^{(z-z_0)X} = e^{z_0X} \sum_{n=0}^{\infty} \frac{(z-z_0)^n X^n}{n!}.$$

We justify exchanging the series and the expectation via the Dominated Convergence Theorem. Indeed, the series of the absolute values is uniformly bounded by the random variable

$$\sum_{n=0}^{\infty} \left| e^{z_0X} \frac{(z-z_0)^n X^n}{n!} \right| = e^{a_0X} e^{|z-z_0||X|} \leq e^{a_0X} e^{r|X|}.$$

This dominating function is integrable because it can be bounded by $e^{(a_0+r)X} + e^{(a_0-r)X}$, and both terms have finite expectations since $a_0 \pm r \in (-h, h)$. Therefore, the Dominated Convergence Theorem applies, yielding

$$\Phi_X(z) = \sum_{n=0}^{\infty} \frac{\mathbb{E}[e^{z_0X} X^n]}{n!} (z-z_0)^n, \quad \forall z \in B_r(z_0).$$

Since Φ_X admits a convergent power series expansion around every $z_0 \in S$, it is analytic, and therefore holomorphic on S (for a fully detailed proof, see [20]). The exact same argument applies to Φ_Y .

By assumption, $\Phi_X(t) = \Phi_Y(t)$ for all real $t \in (-h, h)$. Since the real interval $(-h, h)$ contains an accumulation point in S , the Identity Theorem (Theorem A.4) implies that $\Phi_X = \Phi_Y$ on the entire strip S . In particular, their restrictions to the imaginary axis $i\mathbb{R}$ (which is fully contained in S since $0 \in (-h, h)$) coincide. This gives the equality of the corresponding characteristic functions:

$$\varphi_X(t) = \Phi_X(-it) = \Phi_Y(-it) = \varphi_Y(t), \quad \forall t \in \mathbb{R}.$$

By Theorem 5.22, it follows that $F_X = F_Y$. $\square$

The following proof is largely taken from [15].

Theorem 5.24 (Moments and differentiation). *Let X be a real-valued random variable and let $k \in \mathbb{N}^+$. If $\mathbb{E}[|X|^k] < \infty$, then φ_X is k -times continuously differentiable and, for every $t \in \mathbb{R}$,*

$$\varphi_X^{(k)}(t) = \mathbb{E}[(-iX)^k e^{-itX}].$$

In particular,

$$\varphi_X^{(k)}(0) = (-i)^k \mathbb{E}[X^k].$$

Proof.

Lemma 5.25. Let $(S, \mathcal{A}, \mu)$ be a measure space. Let $\delta > 0$ and let $f : B_\delta(y) \times S \rightarrow \mathbb{C}$ be a measurable function defined on $B_\delta(y) \times S = (y - \delta, y + \delta) \times S$. If the following conditions hold:

- i) $u(x) = \int_S f(x, t) \mu(dt)$ and $\int_S |f(x, t)| \mu(dt) < \infty$ for every $x \in B_\delta(y)$;
- ii) For every $t \in S$, $\frac{\partial f}{\partial x}(x, t)$ exists and is a continuous function of x ;
- iii) $v(x) = \int_S \frac{\partial f}{\partial x}(x, t) \mu(dt)$ is continuous at $x = y$;
- iv) $\int_S \int_{-\delta}^{\delta} \left| \frac{\partial f}{\partial x}(y + \theta, t) \right| d\theta \mu(dt) < \infty$;

then $u'(y) = v(y)$.

Proof.

For h such that $0 < |h| < \delta$, it holds that

$$\begin{aligned} u'(y) &= \lim_{h \rightarrow 0} \frac{u(y+h) - u(y)}{h} \stackrel{(i)}{=} \lim_{h \rightarrow 0} \frac{1}{h} \int_S [f(y+h, t) - f(y, t)] \mu(dt) \\ &\stackrel{(ii)}{=} \lim_{h \rightarrow 0} \frac{1}{h} \int_S \left[\int_0^h \frac{\partial f}{\partial x}(y + \theta, t) d\theta \right] \mu(dt) \\ &\stackrel{+(iv)}{=} \lim_{h \rightarrow 0} \frac{1}{h} \int_0^h \left[\int_S \frac{\partial f}{\partial x}(y + \theta, t) \mu(dt) \right] d\theta \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \int_0^h v(y + \theta) d\theta \stackrel{(iii)}{=} v(y) = \int_S \frac{\partial f}{\partial x}(y, t) \mu(dt), \end{aligned}$$

where $(\dagger)$ is Fubini's theorem. □

We prove the theorem by induction on $m \in \{1, \dots, k\}$. Let $\Pr_X$ be the law of X , so that $\varphi_X(t) = \int_{\mathbb{R}} e^{-itx} \Pr_X(dx)$. For the base case, fix $y \in \mathbb{R}$ and set $f(t, x) = e^{-itx}$. We verify the hypotheses of Lemma 5.25 for an arbitrary $\delta > 0$:

- i) *Integrability:* $\int_{\mathbb{R}} |e^{-itx}| \Pr_X(dx) = \int_{\mathbb{R}} 1 \Pr_X(dx) = 1 < \infty$.
- ii) *Derivative:* For each fixed x , $\frac{\partial f}{\partial t}(t, x) = -ixe^{-itx}$, which is continuous in t .
- iii) *Continuity of v :* We define $v(t) := \int_{\mathbb{R}} (-ix)e^{-itx} \Pr_X(dx)$. Since $|-ixe^{-itx}| = |x|$, and $\mathbb{E}[|X|] \leq 1 + \mathbb{E}[|X|^k] < \infty$ by hypothesis, the integrand is bounded by the $\Pr_X$ -integrable function $|x|$. By the Dominated Convergence Theorem (1.80), $v(t)$ is continuous for all $t \in \mathbb{R}$, and in particular at $t = y$.
- iv) *Fubini condition:* We compute

$$\int_{\mathbb{R}} \int_{-\delta}^{\delta} \left| \frac{\partial f}{\partial t}(y + \theta, x) \right| d\theta \Pr_X(dx) = \int_{\mathbb{R}} \int_{-\delta}^{\delta} |x| d\theta \Pr_X(dx) = 2\delta \int_{\mathbb{R}} |x| \Pr_X(dx) < \infty.$$

Thus, Lemma 5.25 applies, yielding $\varphi'_X(y) = v(y) = \mathbb{E}[-iXe^{-iyX}]$. Since y is arbitrary, the formula holds for all $t \in \mathbb{R}$. Similarly, one can prove the inductive step by verifying the assumptions. What changes is that $|(-ix)^{m-1}e^{-itx}| = |x|^{m-1}$ is integrable since $\mathbb{E}[|X|^{m-1}] \leq 1 + \mathbb{E}[|X|^k] < \infty$. □

Corollary 5.26. If $\mathbb{E}[|X|^2] < \infty$, then

$$\varphi_X(t) = 1 - i\mathbb{E}[X] \cdot t - \frac{\mathbb{E}[X^2]}{2} \cdot t^2 + o(t^2).$$

In particular, if $\mathbb{E}[X] = 0$ and $\text{Var}(X) = \sigma^2$, then

$$\varphi_X(t) = 1 - \frac{\sigma^2}{2} t^2 + o(t^2).$$

Proof.

By Theorem 5.24 with $k = 2$, the function φ_X is $C^2(\mathbb{R})$. Hence, it admits a Taylor expansion of the form

$$\varphi_X(t) = \varphi_X(0) + \varphi'_X(0) \cdot t + \frac{\varphi''_X(0)}{2} \cdot t^2 + o(t^2).$$

Theorem 5.24 also gives

$$\varphi_X(0) = 1, \quad \varphi'_X(0) = -i \mathbb{E}[X], \quad \varphi''_X(0) = -\mathbb{E}[X^2],$$

hence

$$\varphi_X(t) = 1 - i \mathbb{E}[X] \cdot t - \frac{\mathbb{E}[X^2]}{2} \cdot t^2 + o(t^2).$$

□

6 Convergence of Random Variables and Limit Theorems

Nowadays, probability is not about computing the probability that a coin gives heads. Rather, it is about predicting the behavior of a system in the future. That is, we want to understand how a system evolves when the number of agents (or interacting components) is large, or when the time span is long. Such questions are modeled by sequences of random variables $(X_n)_{n \in \mathbb{N}}$, where each X_n represents a random feature of the system at size n or time n . In applied mathematics we either study the limiting behaviour of such sequences ($\lim_{n \rightarrow \infty} X_n$) or that of the aggregate series ($\lim_{n \rightarrow \infty} \sum_{i=1}^n X_i$). The goal of this chapter is to develop the main limit theorems that describe these asymptotic behaviours.

6.1 Almost Sure Convergence

Definition 6.1 (Almost sure convergence). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $(X_n)_{n \in \mathbb{N}}$ be a sequence of real-valued random variables. We say that X_n *converges to X almost surely* (a.s.) and we write $X_n \xrightarrow{\text{a.s.}} X$ if

$$\Pr(\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1 \quad (6.1)$$

Lemma 6.2. Condition (6.1) is equivalent to

$$\Pr\left(\limsup_{n \rightarrow \infty} (|X_n - X| > \varepsilon)\right) = 0 \quad \forall \varepsilon > 0 \quad (6.2)$$

or

$$\Pr\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} (|X_n - X| > \varepsilon)\right) = 0.$$

Proof.

The proof follows from a series of equivalent reformulations of the same quantity.

By Definition 3.9, (6.2) is equivalent to

$$\Pr\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} (|X_n - X| > \varepsilon)\right) = 0.$$

Meanwhile, the set $C \subseteq \Omega$ of convergence in (6.1) is characterized by

$$\forall k \in \mathbb{N}, \exists m \in \mathbb{N}, \forall n \geq m : |X_n(\omega) - X(\omega)| \leq \frac{1}{k}$$

Therefore, the set $C \subseteq \Omega$ of convergence can be written as:

$$C := \{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\} = \bigcap_{k=1}^{\infty} \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \left\{ \omega \in \Omega : |X_n(\omega) - X(\omega)| \leq \frac{1}{k} \right\}.$$

Its complement is

$$C^c = \{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega)\} = \bigcup_{k=1}^{\infty} \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \left\{ \omega \in \Omega : |X_n(\omega) - X(\omega)| > \frac{1}{k} \right\}.$$

Observe that the following inequalities always hold:

$$\sum_{k=1}^{\infty} \Pr\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} (|X_n - X| > \frac{1}{k})\right) \geq \Pr\left(\bigcup_{k=1}^{\infty} \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \left\{ \omega \in \Omega : |X_n(\omega) - X(\omega)| > \frac{1}{k} \right\}\right) \geq 0 \quad (1)$$

$$\Pr\left(\bigcup_{k=1}^{\infty} \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \left\{ \omega \in \Omega : |X_n(\omega) - X(\omega)| > \frac{1}{k} \right\}\right) \geq \Pr\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} (|X_n - X| > \frac{1}{k})\right) \geq 0 \quad \forall k \in \mathbb{N}. \quad (2)$$

The first inequality holds by subadditivity of the measure, while the second holds by monotonicity of the measure. If (6.1) holds, then $\Pr(C^c) = 0$: by 2, we deduce $\Pr(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} (|X_n - X| > \frac{1}{k})) = 0$, which is equivalent to (6.2); if instead (6.2), then $\Pr(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} (|X_n - X| > \frac{1}{k})) = 0$: by 1, we deduce $\Pr(C^c) = 0$, which is equivalent to (6.1). $\square$

Almost sure convergence is the strongest convergence one might hope to deduce about random variables, because it is invariant under modifications on null sets. In measure theory and functional analysis, one also encounters stronger pointwise-type notions of convergence, listed here in order of strength:

- **Uniform convergence:** we require $\sup_{\omega \in \Omega} |X_n(\omega) - X(\omega)| \xrightarrow{n \rightarrow \infty} 0$.
- **Pointwise convergence:** we require $X_n(\omega) \rightarrow X(\omega)$ for every $\omega \in \Omega$.
- **Uniform convergence almost everywhere:** there exists a measurable set $N \in \mathcal{F}$ with $\Pr(N) = 0$ such that

$$\sup_{\omega \in \Omega \setminus N} |X_n(\omega) - X(\omega)| \xrightarrow{n \rightarrow \infty} 0$$

- **Almost uniform convergence:** for every $\delta > 0$, there exists a measurable set $N_\delta \in \mathcal{F}$ with $\Pr(N_\delta) < \delta$ such that

$$\sup_{\omega \in \Omega \setminus N_\delta} |X_n(\omega) - X(\omega)| \xrightarrow{n \rightarrow \infty} 0.$$

Neither uniform nor pointwise convergence are relevant in the context of probability. The reason is that these notions are practically impossible to achieve, since random variables are identified up to a.s. equality. As a consequence, no tangible property of random variables will provide global or pointwise information. Furthermore, while uniform convergence almost everywhere is stable modulo null sets, it is practically impossible to achieve in most stochastic settings. Consequently, we accept almost sure convergence as the strongest notion that is both meaningful and robust. It is simply pointwise convergence outside a null set, and in measure theory it is known as convergence *almost everywhere*.

As for almost uniform convergence, it is actually *equivalent* to almost sure convergence (on any finite measure space) by **Egorov's theorem** (which we do not prove). To summarize, the following are equivalent:

- $X_n \xrightarrow{a.s.} X$;
- $\Pr(\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1$;
- $\Pr(\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega)\}) = 0$;
- There exists an event $\mathcal{N} \in \mathcal{F}$ with $\Pr(\mathcal{N}) = 0$ such that $X_n(\omega) \rightarrow X(\omega)$ for every $\omega \in \Omega \setminus \mathcal{N}$.
- For every $\delta > 0$ there exists $N_\delta \in \mathcal{F}$ with $\Pr(N_\delta) < \delta$ such that

$$\sup_{\omega \in \Omega \setminus N_\delta} |X_n(\omega) - X(\omega)| \xrightarrow{n \rightarrow \infty} 0.$$

6.2 Convergence in Probability

Definition 6.3 (Convergence in probability). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $(X_n)_{n \in \mathbb{N}}$ be a sequence of real-valued random variables. We say that X_n *converges to X in probability* and we write $X_n \xrightarrow{\Pr} X$ if

$$\Pr(|X_n - X| > \varepsilon) \xrightarrow{n \rightarrow \infty} 0, \quad \forall \varepsilon > 0.$$

Equivalently,

$$\Pr(|X_n - X| \leq \varepsilon) \xrightarrow{n \rightarrow \infty} 1, \quad \forall \varepsilon > 0.$$

Convergence in probability is known as *convergence in measure* in measure theory. To understand its difference from almost sure convergence, we can rewrite it as

$$\lim_{n \rightarrow \infty} \Pr(\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) = 0, \quad \forall \varepsilon > 0.$$

As we can see, the limit is taken outside: we no longer track single trajectories.

The following theorem shows that convergence in probability defines a metrizable topology (unlike almost sure convergence):

Proposition 6.4 (Metritzation of convergence in probability). Let $L^0(\Omega, \mathcal{F}, \Pr)$ be the space of random variables identified up to almost sure equality. The function

$$d(X, Y) := \mathbb{E}[\min(1, |X - Y|)]$$

defines a metric on L^0 . Moreover, $X_n \xrightarrow{\Pr} X$ if and only if $d(X_n, X) \rightarrow 0$ as $n \rightarrow \infty$.

Proof.

First, we verify that d is a valid metric on L^0 :

- i) Since $\min(1, |X - Y|) \geq 0$, we have $d(X, Y) \geq 0$ by the monotonicity of the expectation (4.13). Moreover, $d(X, Y) = 0$ if and only if $\min(1, |X - Y|) = 0$ almost surely (by Corollary 1.59), which in turn happens if and only if $|X - Y| = 0$ almost surely. Hence, $d(X, Y) = 0$ if and only if $X = Y$ in L^0 .
- ii) Symmetry follows easily by $d(X, Y) = \mathbb{E}[\min(1, |X - Y|)] = \mathbb{E}[\min(1, |Y - X|)] = d(Y, X)$.
- iii) The function $f(t) = \min(1, t)$ is subadditive and non-decreasing for $t \geq 0$. Hence, using the standard triangle inequality for absolute values, we get:

$$\min(1, |X - Z|) \leq \min(1, |X - Y| + |Y - Z|) \leq \min(1, |X - Y|) + \min(1, |Y - Z|).$$

Taking the expectation on both sides and using linearity, we deduce the triangle inequality: $d(X, Z) \leq d(X, Y) + d(Y, Z)$.

Next, we prove the equivalence $X_n \xrightarrow{\text{Pr}} X \iff d(X_n, X) \rightarrow 0$.

Assume $X_n \xrightarrow{\text{Pr}} X$, and take $\varepsilon > 0$. Then:

$$\begin{aligned} d(X_n, X) &= \mathbb{E}[\min(1, |X_n - X|)] \\ &= \mathbb{E}[\min(1, |X_n - X|) \cdot \mathbf{1}_{|X_n - X| > \varepsilon}] + \mathbb{E}[\min(1, |X_n - X|) \cdot \mathbf{1}_{|X_n - X| \leq \varepsilon}] \\ &\leq \mathbb{E}[\mathbf{1}_{|X_n - X| > \varepsilon}] + \mathbb{E}[\varepsilon \cdot \mathbf{1}_{|X_n - X| \leq \varepsilon}] \\ &\leq \Pr(|X_n - X| > \varepsilon) + \varepsilon \cdot \mathbb{E}[\mathbf{1}_{|X_n - X| \leq \varepsilon}] \\ &\leq \Pr(|X_n - X| > \varepsilon) + \varepsilon. \end{aligned}$$

Letting $n \rightarrow \infty$, the probability term vanishes, yielding $\limsup_{n \rightarrow \infty} d(X_n, X) \leq \varepsilon$. Since $\varepsilon > 0$ is arbitrary, we deduce $d(X_n, X) \rightarrow 0$.

Conversely, assume $d(X_n, X) \rightarrow 0$, and take, without loss of generality, $0 < \varepsilon < 1$. Then $\Pr(|X_n - X| > \varepsilon) = \Pr(\min(1, |X_n - X|) > \varepsilon)$. Hence, by Markov's inequality (4.26),

$$\Pr(|X_n - X| > \varepsilon) = \Pr(\min(1, |X_n - X|) > \varepsilon) \leq \frac{1}{\varepsilon} \mathbb{E}[\min(1, |X_n - X|)] = \frac{d(X_n, X)}{\varepsilon}.$$

Letting $n \rightarrow \infty$, it follows that $\Pr(|X_n - X| > \varepsilon) \rightarrow 0$ for every $\varepsilon \in (0, 1)$. In particular, it holds for every $\varepsilon > 0$. This implies $X_n \xrightarrow{\text{Pr}} X$. $\square$

6.3 Convergence in L^p

Definition 6.5 (L^r convergence). Let $(\Omega, \mathcal{F}, \text{Pr})$ be a probability space and let $(X_n)_{n \in \mathbb{N}} \subset L^r$ be a sequence of real-valued r -integrable random variables, and let $X \in L^r$, for some $r \in [1, \infty)$. We say that X_n converges to X in r -th mean and we write $X_n \xrightarrow{L^r} X$ if

$$\mathbb{E}[|X_n - X|^r] \xrightarrow{n \rightarrow \infty} 0.$$

If $r = 1$ we say that X_n converges in mean to X .

Remark 6.6. If $r \in [1, \infty)$ and $X_n, X \in L^r$, then $\mathbb{E}[|X_n - X|^r]$ is well-defined. Indeed, by (1.9),

$$|X_n - X|^r \leq (|X_n| + |X|)^r \leq 2^{r-1}(|X_n|^r + |X|^r) \implies \mathbb{E}[|X_n - X|^r] < \infty.$$

Proposition 6.7. If $X_n \xrightarrow{L^r} X$ for some $r > 1$, then $X_n \xrightarrow{L^p} X$ for every $p \in [1, r)$.

Proof.

Since $p/r < 1$, the function $x \mapsto x^{p/r}$ is concave on $\mathbb{R}_+$, hence by Jensen's inequality,

$$\mathbb{E}[|X_n - X|^p] = \mathbb{E}[(|X_n - X|^r)^{p/r}] \leq (\mathbb{E}[|X_n - X|^r])^{p/r} \xrightarrow{n \rightarrow \infty} 0.$$

$\square$

Proposition 6.8 ($L^p \Rightarrow$ convergence of moments). If $X_n \xrightarrow{L^p} X$ for some $p \in [1, \infty)$, then $\mathbb{E}[|X_n|^p] \rightarrow \mathbb{E}[|X|^p]$.

Proof.

By Minkovski's inequality 1.94,

$$\begin{cases} \mathbb{E}[|X_n|^p]^{1/p} \leq \mathbb{E}[|X|^p]^{1/p} + \mathbb{E}[|X_n - X|^p]^{1/p} \\ \mathbb{E}[|X|^p]^{1/p} \leq \mathbb{E}[|X_n|^p]^{1/p} + \mathbb{E}[|X_n - X|^p]^{1/p}. \end{cases}$$

Together, we deduce, by definition of absolute value,

$$\limsup_{n \rightarrow \infty} \left| \mathbb{E}[|X_n|^p]^{1/p} - \mathbb{E}[|X|^p]^{1/p} \right| \leq \lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|^p]^{1/p} = 0.$$

Therefore

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n|^p]^{1/p} = \mathbb{E}[|X|^p]^{1/p}.$$

By continuity of $t \mapsto t^p$ on $[0, \infty)$, we conclude

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n|^p] = \mathbb{E}[|X|^p].$$

□

Proposition 6.9 (Uniqueness of Limits). If $X_n \xrightarrow{a.s.} X$ and $X_n \xrightarrow{a.s.} Y$, then $X = Y$ almost surely. The same uniqueness property holds for convergence in probability and convergence in L^p .

Proof.

Let

$$A = \{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\} \quad \text{and} \quad B = \{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = Y(\omega)\}.$$

Since $X_n \xrightarrow{a.s.} X$ and $X_n \xrightarrow{a.s.} Y$, then $\Pr(A) = \Pr(B) = 1$, and $A \cap B \subseteq (X = Y)$. Hence

$$\Pr(X = Y) \geq \Pr(A \cap B) = 1 - \Pr((\Omega \setminus A) \cup (\Omega \setminus B)) \geq 1 - \Pr(\Omega \setminus A) - \Pr(\Omega \setminus B) = 1.$$

Next, suppose that $X_n \xrightarrow{\Pr} X$ and $X_n \xrightarrow{\Pr} Y$. For every $\varepsilon > 0$,

$$\begin{aligned} \lim_{n \rightarrow \infty} \Pr(|X - Y| > \varepsilon) &= \lim_{n \rightarrow \infty} \Pr(|[X - X_n] + [X_n - Y]| > \varepsilon) \\ &\leq \lim_{n \rightarrow \infty} \Pr\left(\left[|X_n - X| > \frac{\varepsilon}{2}\right] \cup \left[|X_n - Y| > \frac{\varepsilon}{2}\right]\right) \\ &\leq \lim_{n \rightarrow \infty} \left(\Pr\left(|X_n - X| > \frac{\varepsilon}{2}\right) + \Pr\left(|X_n - Y| > \frac{\varepsilon}{2}\right)\right) = 0. \end{aligned}$$

Letting $\varepsilon \rightarrow 0$ we deduce $X \stackrel{a.s.}{=} Y$.

Finally, suppose that $X_n \xrightarrow{L^p} X$ and $X_n \xrightarrow{L^p} Y$ for some $p \in [1, \infty)$. Then, by Minkovski's inequality (Theorem 1.94)

$$\|X - Y\|_{L^p} = \lim_{n \rightarrow \infty} \|X - Y\|_{L^p} \leq \lim_{n \rightarrow \infty} [\|X_n - X\|_{L^p} + \|X_n - Y\|_{L^p}] = 0.$$

Since $\|X - Y\|_{L^p} = \|0\|_{L^p}$, we conclude that $X \stackrel{a.s.}{=} Y$.

□

6.4 Convergence in Distribution

Definition 6.10 (Convergence in distribution). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $(X_n)_{n \in \mathbb{N}}$ be a sequence of real-valued random variables. We say that X_n converges to X in distribution and we write $X_n \xrightarrow{d} X$ if, for every $x \in \mathbb{R}$ at which F_X is continuous,

$$F_{X_n}(x) \xrightarrow{n \rightarrow \infty} F_X(x). \quad (6.3)$$

In this case, we say that F_{X_n} converges weakly to F_X , denoted $F_{X_n} \xrightarrow{w} F_X$.

The following example is due to [21]:

Example 6.11. Let $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a random variable and let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables such that, for every $n \in \mathbb{N}$ and every $\omega \in \Omega$, $X_n(\omega) = X(\omega) + 1/n$. We expect X_n to converge in distribution to X . Let us verify this claim through their CDFs. For every $t \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} F_{X_n}(t) = \lim_{n \rightarrow \infty} \Pr(X_n \leq t) = \lim_{n \rightarrow \infty} \Pr\left(X + \frac{1}{n} \leq t\right) = \lim_{n \rightarrow \infty} \Pr\left(X \leq t - \frac{1}{n}\right) = F_X(t^-).$$

If F_X is continuous at t , then $F_{X_n}(t) \rightarrow F_X(t)$. Conversely, if F_X is discontinuous at t , then $F_{X_n}(t) \rightarrow F_X(t^-) < F_X(t)$, since a CDF exhibits a jump at its discontinuity points. This explains precisely why we must restrict such convergence to the continuity points of the limiting distribution function, as stated in Definition 6.10.

Convergence in distribution is the hardest type of convergence to understand, because it requires some knowledge of functional analysis, which we briefly provide here. We start by defining the space $C_b(\mathbb{R})$ of bounded continuous functions.

Definition 6.12 (Space of Test Functions). The *space of test functions* $(C_b(\mathbb{R}), \|\cdot\|_\infty)$ is the Banach space of bounded continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$, equipped with the supremum norm

$$\|f\|_\infty := \sup_{x \in \mathbb{R}} |f(x)|.$$

Definition 6.13 (Topological Dual). The *topological dual* of $C_b(\mathbb{R})$, denoted by $C_b(\mathbb{R})^*$, is the space of all continuous linear functionals $\ell : C_b(\mathbb{R}) \rightarrow \mathbb{R}$.

Lemma 6.14 (Dual Norm). The dual space $C_b(\mathbb{R})^*$ is a Banach space when equipped with the operator norm

$$\|\ell\|_* := \sup_{f \in C_b(\mathbb{R}), f \neq 0} \frac{|\ell(f)|}{\|f\|_\infty} = \sup_{\|f\|_\infty \leq 1} |\ell(f)|.$$

Definition 6.15 (Weak* Convergence). A sequence of functionals $(\ell_n)_{n \in \mathbb{N}} \subseteq C_b(\mathbb{R})^*$ is said to converge **weakly*** to $\ell \in C_b(\mathbb{R})^*$, denoted $\ell_n \rightharpoonup^* \ell$, if for every $f \in C_b(\mathbb{R})$,

$$\ell_n(f) \rightarrow \ell(f).$$

Definition 6.16 (Tightness). A sequence of probability measures $(\mu_n)_{n \in \mathbb{N}}$ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is *tight* if for every $\varepsilon > 0$ there exists a compact set $K_\varepsilon \subseteq \mathbb{R}$ such that

$$\mu_n(K_\varepsilon) \geq 1 - \varepsilon \quad \forall n \in \mathbb{N}.$$

Essentially, tightness guarantees that the probability mass of the measures μ_n is uniformly confined within compact sets and cannot "escape to infinity".

Proposition 6.17 (Measures as linear functionals). Every probability measure μ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ defines a positive linear functional $\ell_\mu \in C_b(\mathbb{R})^*$ by

$$\ell_\mu(f) := \int_{\mathbb{R}} f(x) d\mu(x), \quad f \in C_b(\mathbb{R}),$$

and it satisfies $\|\ell_\mu\|_* = 1$.

Definition 6.18 (Weak convergence of probability measures). A sequence of probability measures $(\mu_n)_{n \in \mathbb{N}}$ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ converges *weakly* to μ , denoted $\mu_n \Rightarrow \mu$, if for every $f \in C_b(\mathbb{R})$,

$$\int_{\mathbb{R}} f d\mu_n \rightarrow \int_{\mathbb{R}} f d\mu.$$

For the next result, we need the following lemma from [21]:

Lemma 6.19 (Skorokhod representation on $\mathbb{R}$). Let $(F_n)_{n \in \mathbb{N}}$ be distribution functions that converge weakly to F . Then there exist random variables $(Y_n)_{n \in \mathbb{N}}$ and Y on the probability space $((0, 1], \mathcal{B}((0, 1]), \lambda)$ such that $F_{Y_n} = F_n$ for all $n \in \mathbb{N}$ and $Y_n \rightarrow Y$ almost surely.

Proof.

We only provide a sketch of the proof, emphasizing the quantile construction. For a complete proof, see [21].

Step 1: Quantiles are inverses of CDFs. For $\omega \in (0, 1]$ and $n \in \mathbb{N}$, define the upper and lower quantiles (cf. Definition 3.61) by

$$Y_n^+(\omega) := \sup\{y \in \mathbb{R} : F_n(y) \leq \omega\}, \quad Y_n^-(\omega) := \sup\{y \in \mathbb{R} : F_n(y) < \omega\},$$

and define similarly Y^+ and Y^- using F . It can be shown that Y_n^- is measurable and satisfies

$$\lambda(\{\omega \in (0, 1] : Y_n^-(\omega) \leq x\}) = F_n(x), \quad x \in \mathbb{R}.$$

In particular, $F_{Y_n^-} = F_n$ for every $n \in \mathbb{N}$, and similarly $F_{Y^+} = F$.

Step 2: Quantile functions have countably many discontinuities. A quantile function is monotone increasing by Lemma 3.63. The set of discontinuity points of a monotone function is at most countable. The rough idea is to define a function Φ that picks one rational number for each jump of the function using the density of $\mathbb{Q}$ in $\mathbb{R}$. By monotonicity, these jumps are disjoint, hence Φ is injective and its image is a subset of $\mathbb{Q}$. We conclude, by injectivity, that the number of jumps must have at most the cardinality of $\mathbb{Q}$, that is, countable infinity. It follows that the sets $\text{Disc}(Y_n^+)$, $\text{Disc}(Y_n^-)$, $\text{Disc}(Y^-)$ and $\text{Disc}(Y^+)$ of discontinuity points of the quantile functions are at most countable.

Step 3: Quantile functions coincide almost everywhere. The key observation (to which most of the proof in [21] is dedicated) is that Y_n^+ is nothing but the *right limit* of Y_n^- : one can show that, for every $u \in (0, 1]$,

$$Y_n^+(u) = \inf_{\xi > u} Y_n^-(\xi) = \lim_{\xi \downarrow u} Y_n^-(\xi). \quad (6.4)$$

Consequently, if u is a continuity point of Y_n^- , then

$$Y_n^+(u) = \lim_{\xi \downarrow u} Y_n^-(\xi) = Y_n^-(u).$$

Since $\text{Disc}(Y_n^-)$ is countable, $\{Y_n^-(u) \neq Y_n^+(u)\} \subseteq \text{Disc}(Y_n^-)$ is countable. Since λ has no atoms, this implies

$$\lambda(\{u : Y_n^-(u) \neq Y_n^+(u)\}) = 0.$$

The same argument is valid for Y^+ and Y^- . Therefore, up to a λ -null set we may identify $Y_n^- = Y_n^+$ and $Y^- = Y^+$. From now on, we set

$$Y_n := Y_n^-, \quad Y := Y^-,$$

and we keep in mind that $Y_n(\omega)$ is the (generalized) inverse of F_n evaluated at ω .

Step 4: Pointwise convergence of quantiles outside a null set. We only need to show that $Y_n(\omega) \rightarrow Y(\omega)$ λ -almost everywhere. Fix $\omega \in (0, 1]$ such that $Y^-(\omega) = Y^+(\omega)$ (this holds for a.e. ω by Step 3). Let $\varepsilon > 0$. By the definition of the quantile function,

$$F(Y^-(\omega) - \varepsilon) < \omega \leq F(Y^-(\omega) + \varepsilon).$$

It can be shown that one may choose ε so that $Y^-(\omega) - \varepsilon$ and $Y^-(\omega) + \varepsilon$ are continuity points of F (since continuity points are dense). Because $F_n \rightarrow F$ weakly, convergence holds at these continuity points; thus, for all n sufficiently large,

$$F_n(Y^-(\omega) - \varepsilon) < \omega \leq F_n(Y^-(\omega) + \varepsilon).$$

By the definition of $Y_n(\omega)$ as the generalized inverse, this implies

$$Y^-(\omega) - \varepsilon < Y_n(\omega) \leq Y^-(\omega) + \varepsilon,$$

which, by letting $\varepsilon \downarrow 0$, forces $Y_n(\omega) \rightarrow Y(\omega)$. □

The following proof is due to [21].

Theorem 6.20 (Convergence in distribution and weak convergence). *Let $(\Omega, \mathcal{F}, \text{Pr})$ be a probability space and let $(X_n)_{n \in \mathbb{N}}$, X be real-valued random variables. Then X_n converges to X in distribution if and*

only if the corresponding laws converge weakly. In other words,

$$F_{X_n} \xrightarrow{w} F_X \iff \Pr_{X_n} \Rightarrow \Pr_X.$$

Equivalently,

$$F_{X_n} \xrightarrow{w} F_X \iff \mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)] \quad \forall f \in C_b(\mathbb{R}).$$

Proof.

Let F_n be the distribution function of X_n , for every $n \in \mathbb{N}$, and let F be the distribution function of X . Since $F_n \xrightarrow{w} F$, by Lemma 6.19, there exist random variables $(Y_n)_{n \in \mathbb{N}}$ and Y on the probability space $((0, 1], \mathcal{B}((0, 1]), \lambda)$ such that $F_{Y_n} = F_n$ for all $n \in \mathbb{N}$, and $Y_n \xrightarrow{a.s.} Y$. Let $f \in C_b(\mathbb{R})$ be a test function. By continuity, we have $f(Y_n) \xrightarrow{a.s.} f(Y)$ and, since f is bounded, by the Dominated Convergence Theorem 4.19:

$$\lim_{n \rightarrow \infty} \mathbb{E}[f(Y_n)] = \lim_{n \rightarrow \infty} \int_{(0,1]} f(Y_n(\omega)) d\omega = \int_{(0,1]} \lim_{n \rightarrow \infty} f(Y_n(\omega)) d\omega = \int_{(0,1]} f(Y(\omega)) d\omega = \mathbb{E}[f(Y)].$$

Conversely, suppose that, for every $f \in C_b(\mathbb{R})$, it holds that

$$\lim_{n \rightarrow \infty} \mathbb{E}[f(X_n)] = \mathbb{E}[f(X)].$$

Fix $\alpha \in \mathbb{R}$. For each $k \in \mathbb{N}$, define the functions $h_k^-, h_k^+ \in C_b(\mathbb{R})$ by

$$h_k^-(x) := \begin{cases} 1, & x \leq \alpha - \frac{1}{k}, \\ k(\alpha - x), & \alpha - \frac{1}{k} < x < \alpha, \\ 0, & x \geq \alpha, \end{cases} \quad h_k^+(x) := \begin{cases} 1, & x \leq \alpha, \\ 1 - k(x - \alpha), & \alpha < x < \alpha + \frac{1}{k}, \\ 0, & x \geq \alpha + \frac{1}{k}. \end{cases}$$

Clearly, $0 \leq h_k^\pm \leq 1$, and for every $x \in \mathbb{R}$ we have

$$h_k^-(x) \leq \mathbf{1}_{(-\infty, \alpha)}(x) \leq \mathbf{1}_{(-\infty, \alpha]}(x) \leq h_k^+(x).$$

Evaluating at $x = X_n$ and taking expectations yields, for every n, k ,

$$\mathbb{E}[h_k^-(X_n)] \leq \Pr(X_n < \alpha) \leq \Pr(X_n \leq \alpha) \leq \mathbb{E}[h_k^+(X_n)]. \quad (6.5)$$

Now take $\liminf_{n \rightarrow \infty}$ and $\limsup_{n \rightarrow \infty}$ in (6.5), and use the assumption $\mathbb{E}[h(X_n)] \rightarrow \mathbb{E}[h(X)]$ for $h = h_k^\pm$:

$$\mathbb{E}[h_k^-(X)] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[h_k^-(X_n)] \leq \limsup_{n \rightarrow \infty} \mathbb{E}[h_k^-(X_n)] \leq \mathbb{E}[h_k^-(X)].$$

Finally, we let $k \rightarrow \infty$. By construction, $h_k^-(x) \uparrow \mathbf{1}_{(-\infty, \alpha)}(x)$ pointwise in x , and $h_k^+(x) \downarrow \mathbf{1}_{(-\infty, \alpha]}(x)$ pointwise in x . Since $0 \leq h_k^\pm \leq 1$, by Theorem 4.19 we obtain

$$\begin{aligned} F_X(\alpha-) &= \Pr(X < \alpha) = \mathbb{E}[\mathbf{1}_{(-\infty, \alpha)}(X)] = \lim_{k \rightarrow \infty} \mathbb{E}[h_k^-(X)] \\ &\leq \liminf_{n \rightarrow \infty} \mathbb{E}[h_k^-(X_n)] \leq \limsup_{n \rightarrow \infty} \mathbb{E}[h_k^-(X_n)] \leq \lim_{k \rightarrow \infty} \mathbb{E}[h_k^-(X)] \\ &= \mathbb{E}[\mathbf{1}_{(-\infty, \alpha]}(X)] = \Pr(X \leq \alpha) = F_X(\alpha). \end{aligned}$$

Therefore, for every $\alpha \in \mathbb{R}$ we have

$$F_X(\alpha-) \leq \liminf_{n \rightarrow \infty} F_{X_n}(\alpha) \leq \limsup_{n \rightarrow \infty} F_{X_n}(\alpha) \leq F_X(\alpha).$$

If α is a continuity point of F_X , then $F_X(\alpha-) = F_X(\alpha)$, hence the inequalities force

$$\liminf_{n \rightarrow \infty} F_{X_n}(\alpha) = \limsup_{n \rightarrow \infty} F_{X_n}(\alpha) = F_X(\alpha),$$

that is, $F_{X_n}(\alpha) \rightarrow F_X(\alpha)$. This proves $X_n \xrightarrow{d} X$. $\square$

This equivalence is powerful but often impractical, since the condition $\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)]$ is required for every test function. The following theorem provides equivalent topological and geometric characterizations. Its proof builds upon Lemma 6.19 and Theorem 6.20, and it will be provided in future editions.

Theorem 6.21 (Portmanteau theorem). Let $(X_n)_{n \in \mathbb{N}}$ and X be real-valued random variables with laws μ_{X_n} and μ_X . The following statements are equivalent:

- i) $X_n \xrightarrow{d} X$ (equivalently, $\Pr_{X_n} \Rightarrow \Pr_X$);
- ii) $\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)]$ for all bounded continuous functions $f \in C_b(\mathbb{R})$;
- iii) $\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)]$ for all bounded, globally Lipschitz continuous functions f ;
- iv) $\limsup_{n \rightarrow \infty} \Pr(X_n \in C) \leq \Pr(X \in C)$ for all closed sets $C \subseteq \mathbb{R}$;
- v) $\liminf_{n \rightarrow \infty} \Pr(X_n \in G) \geq \Pr(X \in G)$ for all open sets $G \subseteq \mathbb{R}$;
- vi) $\Pr(X_n \in A) \rightarrow \Pr(X \in A)$ for all Borel sets A such that $\Pr(X \in \partial A) = 0$;
- vii) $F_{X_n}(x) \rightarrow F_X(x)$ for every $x \in \mathbb{R}$ at which F_X is continuous;
- viii) $\Pr_{X_n}(A) \rightarrow \Pr_X(A)$ for every Borel set A such that $\mu_X(\partial A) = 0$.

Proof. See 3.2.21 of [21]. □

6.5 Hierarchy of Convergence

In this section, fix a probability space $(\Omega, \mathcal{F}, \Pr)$ and suppose $(X_n)_{n \in \mathbb{N}}$ is a sequence of real-valued random variables. Moreover, assume X is a real-valued random variable. Roughly, one should keep in mind the chain

$$X_n \xrightarrow{a.s.} X \quad \Rightarrow \quad X_n \xrightarrow{\Pr} X \quad \Rightarrow \quad X_n \xrightarrow{d} X, \quad (6.6)$$

and (for $r \geq 1$)

$$X_n \xrightarrow{L^r} X \quad \Rightarrow \quad X_n \xrightarrow{\Pr} X \quad \Rightarrow \quad X_n \xrightarrow{d} X. \quad (6.7)$$

None of the reverse implications holds in general.

Theorem 6.22 (Almost sure $\Rightarrow$ in probability). If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{\Pr} X$.

Proof.

Fix $\varepsilon > 0$ and define $A_n = \{|X_n - X| \leq \varepsilon\}$. Consider the random variable

$$\mathbf{1}_{A_n} = \begin{cases} 1 & \text{if } |X_n(\omega) - X(\omega)| \leq \varepsilon, \\ 0 & \text{if } |X_n(\omega) - X(\omega)| > \varepsilon. \end{cases}$$

Since $X_n(\omega) \rightarrow X(\omega)$ for almost every $\omega \in \Omega$, we have $\mathbf{1}_{A_n} \rightarrow 1$ for almost every $\omega \in \Omega$. Moreover, $|\mathbf{1}_{A_n}| \leq 1$ almost surely for every $n \in \mathbb{N}$, hence by Theorem 4.19,

$$\lim_{n \rightarrow \infty} \mathbb{E}[\mathbf{1}_{A_n}] = \mathbb{E}[1] = 1.$$

On the other hand,

$$\mathbb{E}[\mathbf{1}_{A_n}] = \int_{\Omega} \mathbf{1}_{A_n} d\Pr = \int_{A_n} d\Pr = \Pr(A_n) = \Pr(\{|X_n - X| \leq \varepsilon\}),$$

which is exactly $X_n \xrightarrow{\Pr} X$. □

Proposition 6.23 (In probability $\Rightarrow$ in distribution). If $X_n \xrightarrow{\Pr} X$, then $X_n \xrightarrow{d} X$.

Proof.

Let x be a continuity point of F_X . For any $\varepsilon > 0$, we have the inclusions

$$\begin{aligned} (X \leq x - \varepsilon) &\subseteq (X_n \leq x) \cup (|X_n - X| > \varepsilon), \\ (X_n \leq x) &\subseteq (X \leq x + \varepsilon) \cup (|X_n - X| > \varepsilon). \end{aligned}$$

Taking probabilities,

$$\begin{aligned} F_X(x - \varepsilon) &\leq F_{X_n}(x) + \Pr(|X_n - X| > \varepsilon), \\ F_{X_n}(x) &\leq F_X(x + \varepsilon) + \Pr(|X_n - X| > \varepsilon), \end{aligned}$$

hence

$$F_X(x - \varepsilon) - \Pr(|X_n - X| > \varepsilon) \leq F_{X_n}(x) \leq F_X(x + \varepsilon) + \Pr(|X_n - X| > \varepsilon).$$

Since $X_n \xrightarrow{\Pr} X$, we have $\Pr(|X_n - X| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$. Therefore,

$$F_X(x - \varepsilon) \leq \liminf_{n \rightarrow \infty} F_{X_n}(x) \leq \limsup_{n \rightarrow \infty} F_{X_n}(x) \leq F_X(x + \varepsilon).$$

Since $\varepsilon > 0$ was arbitrary, we can let $\varepsilon \rightarrow 0^+$ and use the continuity of F_X at x :

$$F_X(x) \leq \liminf_{n \rightarrow \infty} F_{X_n}(x) \leq \limsup_{n \rightarrow \infty} F_{X_n}(x) \leq F_X(x).$$

Hence $F_{X_n}(x) \rightarrow F_X(x)$, i.e. $X_n \xrightarrow{d} X$. □

The converse is true if X is a constant:

Proposition 6.24. *If $X_n \xrightarrow{d} c$, where c is almost surely a constant function, then $X_n \xrightarrow{\Pr} c$.*

Proof.

Fix $\varepsilon > 0$. Since $c - \varepsilon$ and $c + \varepsilon$ are continuity points of F_c , by $X_n \xrightarrow{d} c$ we get

$$\begin{aligned} \Pr(X_n \leq c - \varepsilon) &= F_{X_n}(c - \varepsilon) \rightarrow F_c(c - \varepsilon) = 0, \\ \Pr(X_n \leq c + \varepsilon) &= F_{X_n}(c + \varepsilon) \rightarrow F_c(c + \varepsilon) = 1. \end{aligned}$$

Thus $\Pr(X_n > c + \varepsilon) = 1 - \Pr(X_n \leq c + \varepsilon) \rightarrow 0$, and

$$\Pr(|X_n - c| > \varepsilon) \leq \Pr(X_n \leq c - \varepsilon) + \Pr(X_n > c + \varepsilon) \rightarrow 0.$$

□

Proposition 6.25 ($L^r \Rightarrow$ in probability). *If $X_n \xrightarrow{L^r} X$ for some $r \geq 1$, then $X_n \xrightarrow{\Pr} X$.*

Proof.

Fix $\varepsilon > 0$. By Markov's inequality applied to the nonnegative random variable $|X_n - X|^r$,

$$\Pr(|X_n - X| > \varepsilon) = \Pr(|X_n - X|^r > \varepsilon^r) \leq \frac{\mathbb{E}[|X_n - X|^r]}{\varepsilon^r} \xrightarrow{n \rightarrow \infty} 0.$$

□

If the convergence in probability is fast enough, it can yield almost sure convergence:

Lemma 6.26 (Borel–Cantelli). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and suppose $(X_n)_{n \in \mathbb{N}}$ is a sequence of real-valued random variables. Moreover, assume X is a real-valued random variable. If, for every $\varepsilon > 0$,*

$$\sum_{n=1}^{\infty} \Pr(|X_n - X| > \varepsilon) < \infty,$$

then $X_n \xrightarrow{a.s.} X$. Moreover, if the events $(\{|X_n - X| > \varepsilon\})_{n \in \mathbb{N}}$ are mutually independent for every $\varepsilon > 0$, the converse also holds.

Proof.

We recall that almost sure convergence is equivalent to (6.2):

$$\Pr\left(\limsup_{n \rightarrow \infty} \{|X_n - X| > \varepsilon\}\right) = 0.$$

In turn, this is equivalent to showing

$$\Pr \left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{|X_n - X| > \varepsilon\} \right) = 0,$$

or, by continuity from above of the measure (1.30),

$$\lim_{m \rightarrow \infty} \Pr \left(\bigcup_{n=m}^{\infty} \{|X_n - X| > \varepsilon\} \right) = 0. \quad (6.8)$$

By countable subadditivity,

$$\Pr \left(\bigcup_{n=m}^{\infty} \{|X_n - X| > \varepsilon\} \right) \leq \sum_{n=m}^{\infty} \Pr(|X_n - X| > \varepsilon),$$

and the right-hand side tends to 0 as $m \rightarrow \infty$ because the series converges. This proves the first claim.

For the converse, we argue by contrapositive. Assume $\sum_{n=1}^{\infty} \Pr(|X_n - X| > \varepsilon) = \infty$ and that the events $\{|X_n - X| > \varepsilon\}$ are independent. We show that

$$\Pr \left(\limsup_{n \rightarrow \infty} \{|X_n - X| > \varepsilon\} \right) = 1.$$

Equivalently, using (6.8), it suffices to show

$$\lim_{m \rightarrow \infty} \Pr \left(\bigcap_{n=m}^{\infty} \{|X_n - X| \leq \varepsilon\} \right) = 0.$$

If $\Pr(|X_n - X| > \varepsilon) = 1$ for some n , then the intersection has probability 0 for all $m \leq n$ and we are done. Otherwise, using independence and $\log(1 - x) \leq -x$ for $x \in (0, 1)$,

$$\begin{aligned} \Pr \left(\bigcap_{n=m}^{\infty} \{|X_n - X| \leq \varepsilon\} \right) &= \prod_{n=m}^{\infty} \Pr(|X_n - X| \leq \varepsilon) \\ &= \prod_{n=m}^{\infty} (1 - \Pr(|X_n - X| > \varepsilon)) \\ &= \exp \left(\sum_{n=m}^{\infty} \log(1 - \Pr(|X_n - X| > \varepsilon)) \right) \\ &\leq \exp \left(- \sum_{n=m}^{\infty} \Pr(|X_n - X| > \varepsilon) \right) \xrightarrow{m \rightarrow \infty} 0, \end{aligned}$$

since the tail sums diverge to $+\infty$. This concludes the proof. $\square$

Corollary 6.27 (Subsequence principle). *Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables and X be a random variable. The following hold:*

- i) *If $X_n \xrightarrow{\text{Pr}} X$, then there exists a subsequence $(X_{n_k})_{k \in \mathbb{N}}$ such that $X_{n_k} \xrightarrow{a.s.} X$ as $k \rightarrow \infty$.*
- ii) *$X_n \xrightarrow{\text{Pr}} X$ if and only if every subsequence $(X_{n_m})_{m \in \mathbb{N}}$ contains a further sub-subsequence $(X_{n_{m_k}})_{k \in \mathbb{N}}$ such that $X_{n_{m_k}} \xrightarrow{a.s.} X$ as $k \rightarrow \infty$.*

Proof.

We begin with the first statement. Since $X_n \xrightarrow{\text{Pr}} X$, we can extract a subsequence $(X_{n_k})_{k \in \mathbb{N}}$ such that $n_1 < n_2 < \dots < n_k < \dots$ and, for every $k \in \mathbb{N}$,

$$\Pr \left(|X_{n_k} - X| > \frac{1}{k} \right) < \frac{1}{2^k}.$$

For every $\varepsilon > 0$, there exists $K \in \mathbb{N}$ such that $1/K < \varepsilon$. Hence, for every $k \geq K$, we have $\frac{1}{k} \leq \frac{1}{K} < \varepsilon$

and

$$\sum_{k=1}^{\infty} \Pr(|X_{n_k} - X| > \varepsilon) \leq K + \sum_{k=K}^{\infty} \Pr\left(|X_{n_k} - X| > \frac{1}{k}\right) \leq K + \sum_{k=K}^{\infty} \frac{1}{2^k} < \infty.$$

Then by the Borel-Cantelli lemma 6.26, $X_{n_k} \xrightarrow{a.s.} X$, proving the first statement.

For the second statement, assume first that $X_n \xrightarrow{\Pr} X$, and let $(X_{n_m})_{m \in \mathbb{N}}$ be an arbitrary subsequence. Then $X_{n_m} \xrightarrow{\Pr} X$ as $m \rightarrow \infty$. Hence, there exists a further sub-subsequence $X_{n_{m_k}} \xrightarrow{a.s.} X$ as $k \rightarrow \infty$, by the previous point.

For the converse implication, suppose by contradiction that $X_n \not\xrightarrow{\Pr} X$. Then there exist $\varepsilon > 0$, $\delta > 0$ and a subsequence $(X_{n_m})_{m \in \mathbb{N}}$ such that

$$\Pr(|X_{n_m} - X| > \varepsilon) \geq \delta \quad \forall m \in \mathbb{N}.$$

By hypothesis, $(X_{n_m})_{m \in \mathbb{N}}$ admits a further sub-subsequence $(X_{n_{m_k}})_{k \in \mathbb{N}}$ converging almost surely to X .

However, since $X_{n_{m_k}} \xrightarrow{a.s.} X$, Theorem 6.22 implies that $X_{n_{m_k}} \xrightarrow{\Pr} X$, which forces $\Pr(|X_{n_{m_k}} - X| > \varepsilon) \rightarrow 0$ as $k \rightarrow \infty$. This contradicts the lower bound $\delta > 0$. $\square$

6.6 Further Convergence Results

In this section we provide several advanced convergence results. Complete proofs will be provided in future editions.

The following proof is inspired by [21] and [16].

Theorem 6.28 (Continuous mapping). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and (S, d_S) and (T, d_T) be metric spaces. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random elements $X_n : (\Omega, \mathcal{F}, \Pr) \rightarrow (S, \mathcal{B}(S))$ and $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (S, \mathcal{B}(S))$ a random element.*

Let $g : (S, \mathcal{B}(S)) \rightarrow (T, \mathcal{B}(T))$ be a Borel measurable function which is continuous outside a set $D_g \subseteq S$. Then:

- i) If $X_n \xrightarrow{a.s.} X$ and $\Pr(X \in D_g) = 0$, then $g(X_n) \xrightarrow{a.s.} g(X)$.*
- ii) If $X_n \xrightarrow{\Pr} X$, $\Pr(X \in D_g) = 0$ and $\Pr_X$ is tight on S , then $g(X_n) \xrightarrow{\Pr} g(X)$.*
- iii) If $X_n \xrightarrow{d} X$ and $\Pr(X \in D_g) = 0$, then $g(X_n) \xrightarrow{d} g(X)$.*

Proof.

(i). Let $A = (X_n \rightarrow X)$ and $B = (X \notin D_g)$. Then $\Pr(A) = \Pr(B) = 1$. Hence

$$\Pr(A \cap B) = 1 - \Pr((\Omega \setminus A) \cup (\Omega \setminus B)) \geq 1 - \Pr(\Omega \setminus A) - \Pr(\Omega \setminus B) = 1.$$

Moreover, if $\omega \in A \cap B$, then $X_n(\omega) \rightarrow X(\omega)$ and $X(\omega) \notin D_g$, so g is continuous at $X(\omega)$ and therefore $g(X_n(\omega)) \rightarrow g(X(\omega))$. Thus $A \cap B \subseteq (g(X_n) \rightarrow g(X))$, and

$$1 = \Pr(A \cap B) \leq \Pr(g(X_n) \rightarrow g(X)),$$

which implies $\Pr(g(X_n) \rightarrow g(X)) = 1$.

(ii). Fix $\varepsilon > 0$ and $\delta > 0$. By tightness of $\Pr_X$ and since $\Pr(X \in D_g) = 0$, choose a compact set $K \subseteq S \setminus D_g$ such that

$$\Pr(X \in K) \geq 1 - \delta.$$

Since g is continuous on the compact set K , it is uniformly continuous there: there exists $\eta > 0$ such that

$$x, y \in K, \quad d_S(x, y) < \eta \implies d_T(g(x), g(y)) < \varepsilon.$$

Thus, for every n ,

$$\begin{aligned} \Pr(d_T(g(X_n), g(X)) > \varepsilon) &\leq \Pr(X \notin K) + \Pr(d_T(g(X_n), g(X)) > \varepsilon, X \in K) \\ &\leq \Pr(X \notin K) + \Pr(d_S(X_n, X) \geq \eta, X \in K) \\ &\leq \Pr(X \notin K) + \Pr(d_S(X_n, X) \geq \eta) \\ &\leq \delta + \Pr(d_S(X_n, X) \geq \eta), \end{aligned}$$

where the second inequality follows from uniform continuity on K : on the event $(X \in K)$, if $d_S(X_n, X) < \eta$ then $d_T(g(X_n), g(X)) < \varepsilon$. Finally, since $X_n \xrightarrow{\text{Pr}} X$, we have $\Pr(d_S(X_n, X) \geq \eta) \rightarrow 0$ as $n \rightarrow \infty$. Hence

$$\limsup_{n \rightarrow \infty} \Pr(d_T(g(X_n), g(X)) > \varepsilon) \leq \delta.$$

Since $\delta > 0$ is arbitrary, this implies $\Pr(d_T(g(X_n), g(X)) > \varepsilon) \rightarrow 0$, i.e. $g(X_n) \xrightarrow{\text{Pr}} g(X)$.

(iii). By Lemma 6.19, there exist random variables $(Y_n)_{n \in \mathbb{N}}$ and Y on the probability space $((0, 1], \mathcal{B}((0, 1]), \lambda)$ such that $Y_n \xrightarrow{\text{a.s.}} Y$, $Y_n \stackrel{d}{=} X_n$ for every $n \in \mathbb{N}$ and $Y \stackrel{d}{=} X$. Fix $f \in C_b(\mathbb{R})$. Then clearly $D_{f \circ g} \subseteq D_g$, so

$$\Pr(Y \in D_{f \circ g}) \leq \Pr(Y \in D_g) = \Pr(X \in D_g) = 0.$$

Therefore, by point (i) applied to $(Y_n)_{n \in \mathbb{N}}$ and Y , and on the function $f \circ g$, we conclude that $(f \circ g)(Y_n) \xrightarrow{\text{a.s.}} (f \circ g)(Y)$. Moreover, $f \circ g$ is bounded and $Y_n \stackrel{d}{=} X_n$ for all $n \in \mathbb{N}$. It follows by Theorem 4.19 and continuity almost everywhere of $f \circ g$ that

$$\lim_{n \rightarrow \infty} \mathbb{E}[f(g(X_n))] = \lim_{n \rightarrow \infty} \mathbb{E}[(f \circ g)(Y_n)] = \mathbb{E}[(f \circ g)(\lim_{n \rightarrow \infty} Y_n)] = \mathbb{E}[(f \circ g)(Y)] = \mathbb{E}[f(g(X))].$$

This holds for any $f \in C_b(\mathbb{R})$, so by Theorem 6.20 we conclude that $g(X_n) \xrightarrow{d} g(X)$. $\square$

Computing the limit of a sequence of CDFs is not always an easy task. Thankfully, the MGF and the characteristic functions are effective tools for determining the convergence in distribution. We state the following *convergence theorems* without proof.

Theorem 6.29 (Lévy's continuity theorem). Let $(X_n)_{n \in \mathbb{N}}$ be real-valued random variables with laws $\Pr_n$, and let $\varphi_n := \widehat{\Pr_n}$ be their characteristic functions.

- i) If $X_n \xrightarrow{d} X$, then $\varphi_n(t) \rightarrow \varphi_X(t)$ for every $t \in \mathbb{R}$.
- ii) Conversely, assume $\varphi_n(t) \rightarrow \varphi(t)$ for every $t \in \mathbb{R}$ and that φ is continuous at $t = 0$. Then there exists a random variable X with characteristic function φ , and $X_n \xrightarrow{d} X$.

Proof. See [15], Theorem 3.3.17. $\square$

Theorem 6.30 (Curtiss convergence theorem). Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of real-valued random variables, and let M_{X_n} be their moment generating functions (MGFs).

- i) If $X_n \xrightarrow{d} X$ and the sequence $(M_{X_n}(t))_{n \in \mathbb{N}}$ is uniformly bounded in some neighborhood $(-h, h)$ of the origin, then the MGF of X , $M_X(t)$, is well-defined on $(-h, h)$, and $M_{X_n}(t) \rightarrow M_X(t)$ for every $t \in (-h, h)$.
- ii) Conversely, assume $M_{X_n}(t)$ is well-defined in some neighborhood $(-h, h)$ of the origin and $M_{X_n}(t) \rightarrow M(t)$ for every $t \in (-h, h)$. Then there exists a unique random variable X with MGF M , and $X_n \xrightarrow{d} X$.

Proof. See [20], Theorem 3.1. $\square$

Theorem 6.31 (Slutsky). Let $(X_n)_{n \in \mathbb{N}}$ and X , and $(Y_n)_{n \in \mathbb{N}}$ and Y , be real-valued random variables defined on the same probability space. Assume that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$, and that Y is a constant $c \in \mathbb{R}$. Then:

- i) $X_n + Y_n \xrightarrow{d} X + c$;
- ii) $X_n Y_n \xrightarrow{d} cX$;
- iii) If $c \neq 0$, $X_n / Y_n \xrightarrow{d} X / c$.

Proof. See [22], Lemma 8.18. $\square$

Theorem 6.32 (Discrete convergence via PMF). Let $(X_n)_{n \in \mathbb{N}}$ and X be discrete real-valued random variables with countable support $S \subseteq \mathbb{R}$, and denote by p_{X_n} and p_X the corresponding PMFs on S . If $p_{X_n}(s) \rightarrow p_X(s)$ for every $s \in S$, then $X_n \xrightarrow{d} X$. Moreover, the converse holds if S has no accumulation points. In particular, this holds if $S = \mathbb{N}$ or $S = \mathbb{Z}$.

Proof.

Suppose $p_{X_n}(s) \rightarrow p_X(s)$ for every $s \in S$. Fix $\varepsilon > 0$, $x \in \mathbb{R}$ a point of continuity of F_X and let $S_x := \{s \in S : s \leq x\}$. Since $\sum_{s \in S} p_X(s) = 1$ and S is countable, we can choose a finite set $S_\varepsilon \subseteq S$ such that

$$\sum_{s \in S \setminus S_\varepsilon} p_X(s) < \varepsilon. \quad (6.9)$$

Since $p_{X_n}(s) \rightarrow p_X(s)$ for each $s \in S_\varepsilon$, and noting that $S_\varepsilon \cap S_x$ is finite, there exists N such that for all $n \geq N$,

$$\left| \sum_{s \in S_\varepsilon \cap S_x} (p_{X_n}(s) - p_X(s)) \right| < \varepsilon \quad (6.10)$$

and

$$\left| \sum_{s \in S_\varepsilon} p_{X_n}(s) - \sum_{s \in S_\varepsilon} p_X(s) \right| < \varepsilon. \quad (6.11)$$

In particular, for $n \geq N$,

$$\sum_{s \in S \setminus S_\varepsilon} p_{X_n}(s) = 1 - \sum_{s \in S_\varepsilon} p_{X_n}(s) \stackrel{(6.11)}{\leq} 1 - \sum_{s \in S_\varepsilon} p_X(s) + \varepsilon = \sum_{s \in S \setminus S_\varepsilon} p_X(s) + \varepsilon \stackrel{(6.9)}{<} 2\varepsilon. \quad (6.12)$$

Therefore, for all $n \geq N$,

$$\begin{aligned} |F_{X_n}(x) - F_X(x)| &= \left| \sum_{s \in S_x} p_{X_n}(s) - \sum_{s \in S_x} p_X(s) \right| \\ &\leq \left| \sum_{s \in S_\varepsilon \cap S_x} (p_{X_n}(s) - p_X(s)) \right| + \sum_{s \in S_x \setminus S_\varepsilon} p_{X_n}(s) + \sum_{s \in S_x \setminus S_\varepsilon} p_X(s) \\ &\stackrel{(6.10)}{\leq} \varepsilon + \sum_{s \in S \setminus S_\varepsilon} p_{X_n}(s) + \sum_{s \in S \setminus S_\varepsilon} p_X(s) \\ &\stackrel{(6.9)+(6.12)}{<} \varepsilon + 2\varepsilon + \varepsilon = 4\varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we conclude that $F_{X_n}(x) \rightarrow F_X(x)$.

Conversely, suppose $F_{X_n}(x) \rightarrow F_X(x)$ for every point of continuity $x \in \mathbb{R}$. Fix $s \in S$. Let B_s be an open neighbourhood of s such that $\overline{B_s} \cap S = \{s\}$. Such B_s exists for every $s \in S$ because S has no accumulation points. Moreover, it is a Borel set and

$$\begin{aligned} \Pr(X \in \partial B_s) &= \Pr(X \in \overline{B_s} \setminus \text{int}(B_s)) = \Pr(X \in \overline{B_s} \setminus B_s) = \Pr(X \in (S \cap \overline{B_s}) \setminus (S \cap B_s)) \\ &= \Pr(X \in (\{s\} \setminus \{s\})) = 0. \end{aligned}$$

Hence, by Portmanteau (Theorem 6.21, item (6)),

$$p_{X_n}(s) = \Pr(X_n = s) = \Pr(X_n \in B_s) \rightarrow \Pr(X \in B_s) = \Pr(X = s) = p_X(s),$$

concluding the proof. $\square$

6.7 Uniform Integrability

In this section, we investigate the relationship between convergence in moments and other modes of convergence for random variables.

Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space, let $(X_n)_{n \in \mathbb{N}}$ be a sequence of real-valued random variables, and let X be a real-valued random variable. Recall that, by the Dominated Convergence Theorem (4.19), almost sure convergence $X_n \xrightarrow{a.s.} X$ implies the convergence of the p -th absolute moments $\mathbb{E}[|X_n|^p] \rightarrow \mathbb{E}[|X|^p]$, provided the sequence $(|X_n|^p)_{n \in \mathbb{N}}$ is suitably dominated by an integrable random variable Y . This domination condition is strictly necessary, as the following example illustrates:

Example 6.33 (Almost sure convergence without domination). Let $\Omega = (0, 1)$ equipped with the Borel σ -algebra

$\mathcal{B}((0, 1))$ and the Lebesgue measure λ as probability measure $\Pr$. For $n \geq 1$, define

$$X_n(\omega) := n \mathbf{1}_{(0, 1/n)}(\omega).$$

Then $X_n(\omega) \rightarrow 0$ for every $\omega \in (0, 1)$, hence $X_n \rightarrow 0$ almost surely. However, for every $p \geq 1$, we have

$$\mathbb{E}[|X_n|^p] = \int_0^1 n^p \mathbf{1}_{(0, 1/n)}(\omega) d\omega = n^p \cdot \frac{1}{n} = n^{p-1}.$$

In particular,

$$\mathbb{E}[|X_n|] \equiv 1 \quad \text{but} \quad \mathbb{E}[|X_n|^p] \rightarrow +\infty \quad \text{for all } p > 1,$$

while $\mathbb{E}[|0|^p] = 0$. Thus, almost sure convergence alone does not guarantee the convergence of moments if no integrable random variable dominates $(|X_n|^p)_{n \in \mathbb{N}}$.

In essence, the implication

$$\text{Almost sure convergence} + \text{Domination} \implies \text{Convergence of moments} \quad (6.13)$$

can be simultaneously strengthened in three different aspects, yielding the following:

$$\text{Convergence in probability} + \text{Uniform integrability} \implies \text{Convergence in } L^p. \quad (6.14)$$

In this section, we formalize (6.14) and clarify precisely in what sense it provides a threefold improvement over (6.13).

Definition 6.34. A sequence of random variables $(X_n)_{n \in \mathbb{N}}$ is uniformly integrable if

$$\lim_{C \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n| > C\}}] = 0.$$

The following lemma gives a necessary and a sufficient condition for uniform integrability:

Lemma 6.35. Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $(X_n)_{n \in \mathbb{N}}$ be a sequence of integrable random variables.

i) If $(X_n)_{n \in \mathbb{N}}$ is uniformly integrable, then

$$\sup_{n \in \mathbb{N}} \mathbb{E}[|X_n|] < \infty.$$

ii) If there exists $p \in (1, \infty)$ such that $\sup_{n \in \mathbb{N}} \mathbb{E}[|X_n|^p] < \infty$, then $(X_n)_{n \in \mathbb{N}}$ is uniformly integrable.

Proof.

Assume that $(X_n)_{n \in \mathbb{N}}$ is uniformly integrable, and fix $\varepsilon > 0$. There exists $C > 0$ such that

$$\sup_{n \in \mathbb{N}} \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n| > C\}}] < \varepsilon.$$

Then

$$\sup_{n \in \mathbb{N}} \mathbb{E}[|X_n|] \leq \sup_{n \in \mathbb{N}} \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n| > C\}}] + \sup_{n \in \mathbb{N}} \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n| \leq C\}}] \leq \varepsilon + C < \infty.$$

Now assume there exists $p \in (1, \infty)$ such that $\sup_{n \in \mathbb{N}} \mathbb{E}[|X_n|^p] < \infty$. Then

$$\begin{aligned} \limsup_{C \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n| > C\}}] &= \limsup_{C \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E}\left[\frac{|X_n|^p}{|X_n|^{p-1}} \mathbf{1}_{\{|X_n| > C\}}\right] \\ &\leq \limsup_{C \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E}\left[\frac{|X_n|^p}{C^{p-1}} \mathbf{1}_{\{|X_n| > C\}}\right] \\ &\leq \limsup_{C \rightarrow \infty} \frac{1}{C^{p-1}} \sup_{n \in \mathbb{N}} \mathbb{E}[|X_n|^p] = 0. \end{aligned}$$

Therefore $(X_n)_{n \in \mathbb{N}}$ is uniformly integrable □

Proposition 6.36 (Domination implies uniform integrability). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and let $(X_n)_{n \in \mathbb{N}}$ be a sequence of integrable random variables. If there exists $Y \in L^1(\Omega)$ such that $|X_n| \leq Y$ almost surely for every $n \in \mathbb{N}$, then $(X_n)_{n \in \mathbb{N}}$ is uniformly integrable.

Proof.

We compute:

$$\begin{aligned}
 \lim_{C \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E} [|X_n| \mathbf{1}_{\{|X_n| > C\}}] &\stackrel{(1)}{\leq} \lim_{C \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E} [Y \mathbf{1}_{\{|X_n| > C\}}] \\
 &\stackrel{(2)}{\leq} \lim_{C \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E} [Y \mathbf{1}_{\{Y > C\}}] = \lim_{C \rightarrow \infty} \mathbb{E} [Y \mathbf{1}_{\{Y > C\}}] \\
 &\stackrel{(3)}{=} \mathbb{E} \left[\lim_{C \rightarrow \infty} Y \mathbf{1}_{\{Y > C\}} \right] = \mathbb{E}[0] = 0,
 \end{aligned}$$

where Step (1) uses the domination Pr-a.s. by Y for every $n \in \mathbb{N}$, Step (2) uses the set inclusion $\{|X_n| > C\} \subseteq \{Y > C\}$, Step (3) uses the Dominated Convergence Theorem 4.19 applied $Y \mathbf{1}_{\{Y > C\}} \rightarrow 0$ dominated by $Y \in L^1$. $\square$

We are now ready to state the central theorem of this section:

Theorem 6.37 (Vitali's Convergence Theorem). *Let $p \in (1, \infty)$. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables and let X be a random variable. Assume that $X_n \xrightarrow{\text{Pr}} X$ [P]. Then the following statements are equivalent:*

[UI] *The sequence of p -th absolute powers $(|X_n|^p)_{n \in \mathbb{N}}$ is uniformly integrable;*

[M] *$X_n \in L^p$ for every $n \in \mathbb{N}$, $X \in L^p$, and $\mathbb{E}[|X_n|^p] \rightarrow \mathbb{E}[|X|^p]$ as $n \rightarrow \infty$;*

[Lp] *$X_n \in L^p$ for every $n \in \mathbb{N}$, $X \in L^p$, and $X_n \xrightarrow{L^p} X$ (i.e., $\mathbb{E}[|X_n - X|^p] \rightarrow 0$).*

Proof. See Theorem 1.3.49 of [21] or Theorem 4.6.3 of [15]. $\square$

Remark 6.38. Let [AS] denote almost sure convergence ($X_n \xrightarrow{a.s.} X$), and let [Dom] denote the domination condition ($|X_n|^p \leq Y$ almost surely for some $Y \in L^1(\Omega)$ and all $n \in \mathbb{N}$). By combining Theorem 6.37 with Theorem 6.22, Proposition 6.8, and Proposition 6.36, we obtain the following logical implications. These clearly illustrate how (6.14) provides a threefold strengthening of (6.13):

$$\begin{aligned}
 [\text{Lp}] &\implies [\text{P}], [\text{UI}], [\text{M}] \\
 [\text{P}] + [\text{UI}] &\implies [\text{Lp}], [\text{M}] \\
 [\text{P}] + [\text{M}] &\implies [\text{Lp}], [\text{UI}] \\
 [\text{Dom}] &\implies [\text{UI}] \\
 [\text{AS}] &\implies [\text{P}]
 \end{aligned}$$

We can further adapt the implication $[\text{P}] + [\text{UI}] \implies [\text{Lp}], [\text{M}]$ by weakening the hypothesis from convergence in probability to convergence in distribution (denoted [Dis]). In doing so, we lose L^p convergence, but we retain the convergence of moments:

$$[\text{Dis}] + [\text{UI}] \implies [\text{M}].$$

The following proof is inspired by [23]:

Theorem 6.39. *Let $p \geq 1$. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of real-valued random variables and X be a real-valued random variable. If $X_n \xrightarrow{d} X$ and $(|X_n|^p)_{n \in \mathbb{N}}$ is uniformly integrable, then $X \in L^p$ and*

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n|^p] = \mathbb{E}[|X|^p].$$

Proof.

Step 1: $X \in L^p$. Set $Z_n := |X_n|^p \geq 0$ and $Z := |X|^p \geq 0$. Since $t \mapsto |t|^p$ is continuous, by the Continuous Mapping Theorem (Theorem 6.28) we have $Z_n \xrightarrow{d} Z$. Then

$$\mathbb{E}[Z] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[Z_n] \leq \sup_{n \in \mathbb{N}} \mathbb{E}[Z_n] < \infty,$$

where $\dagger$ follows by Portmanteau 6.21 and Fatou's lemma 1.78, while $\ddagger$ follows by Lemma 6.35. Hence, $X \in L^p$ and $Z \in L^1$.

Step 2: $\Pr(Z_n > t) \rightarrow \Pr(Z > t)$ for a.e. t . We observe that $Z_n \xrightarrow{d} Z$ is equivalent to the following:

$$\begin{aligned} F_{Z_n}(t) &\rightarrow F_Z(t) && \text{for every } t \text{ where } F_Z \text{ is continuous} \\ \Pr(Z_n \leq t) &\rightarrow \Pr(Z \leq t) && \text{for every } t \geq 0 \text{ such that } \Pr(Z = t) = 0 \\ \Pr(Z_n > t) &\rightarrow \Pr(Z > t) && \text{for every } t \geq 0 \text{ such that } \Pr(Z = t) = 0 \\ \Pr(Z_n > t) &\rightarrow \Pr(Z > t) && \text{for all but at most countably many } t \in \mathbb{R}. \end{aligned} \quad (6.15)$$

Indeed, if F_Z is discontinuous at t , then, by the right-continuity and monotonicity of F , necessarily $\Pr(Z < t) = F_Z(t^-) < F_Z(t) = \Pr(Z \leq t)$, meaning $\Pr(Z = t) > 0$. If $\mathcal{I} := \{t \in \mathbb{R} : \Pr(Z = t) > 0\}$ were uncountable, then $1 = \Pr(Z \in \mathbb{R}) \geq \Pr(Z \in \mathcal{I}) = \sum_{t \in \mathcal{I}} \Pr(Z = t) = +\infty$, a clear contradiction.

Step 3: *Removing the tails.* Fix $\varepsilon > 0$. By the uniform integrability of $(Z_n)_{n \in \mathbb{N}}$, there exists $\alpha > 0$ such that

$$\sup_{n \in \mathbb{N}} \mathbb{E}[Z_n \mathbf{1}_{Z_n > \alpha}] < \varepsilon.$$

Since $Z \in L^1$ by Step 1, we can select α large enough so that

$$\mathbb{E}[Z \mathbf{1}_{Z > \alpha}] < \varepsilon.$$

Decomposing the expectations, we have

$$\mathbb{E}[Z_n] = \mathbb{E}[Z_n \mathbf{1}_{Z_n \leq \alpha}] + \mathbb{E}[Z_n \mathbf{1}_{Z_n > \alpha}], \quad \mathbb{E}[Z] = \mathbb{E}[Z \mathbf{1}_{Z \leq \alpha}] + \mathbb{E}[Z \mathbf{1}_{Z > \alpha}].$$

Hence, up to choosing α large enough, it suffices to show that

$$\mathbb{E}[Z_n \mathbf{1}_{Z_n \leq \alpha}] \rightarrow \mathbb{E}[Z \mathbf{1}_{Z \leq \alpha}],$$

which, by Proposition 4.20, is equivalent to

$$\lim_{n \rightarrow \infty} \int_0^\alpha \Pr(Z_n > t) dt - \alpha \Pr(Z_n > \alpha) = \int_0^\alpha \Pr(Z > t) dt - \alpha \Pr(Z > \alpha) \quad (6.16)$$

(up to choosing α such that $\Pr(Z = \alpha) = 0$, which is always possible since there are at most countably many such α 's by Step 1). Equation 6.16 directly follows from the dominated convergence theorem 4.19, since $\Pr(Z_n > t) \rightarrow \Pr(Z > t)$ for almost every $t \in [0, \alpha]$ by (6.15), $t \mapsto \Pr(Z_n > t)$ is bounded by 1, and the integration interval has finite measure. Letting $\varepsilon \rightarrow 0$ and α vary accordingly, we conclude that $\lim_{n \in \mathbb{N}} |\mathbb{E}[Z_n] - \mathbb{E}[Z]| = 0$. $\square$

6.8 Laws of Large Numbers

In this section, we introduce the Law of Large Numbers (LLN) in both its weak and strong formulations. These theorems, together with the Central Limit Theorem that we will prove later, are the foundational results that justify the very existence of statistics. Almost every statistical result, no matter how simple, relies to some extent on one of these theorems. Therefore, it is crucial to introduce them rigorously.

While previously we analyzed the limit of a sequence of random variables, we now investigate the limit of a sum of random variables. In particular, we focus on arithmetic means (with weights $1/n$) and compute their limits. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of integrable random variables with *the same* expectation μ . We define

$$S_n = \sum_{i=1}^n X_i \quad \text{and} \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

We call $\bar{X}_n$ the *sample mean*. Moreover, we say that $(X_n)_{n \in \mathbb{N}}$ is a sequence of i.i.d. random variables if they are mutually independent and identically distributed.

We recall that $X \in L^1$ means it admits a finite expectation, denoted by μ . If $X \in L^2$, it admits a finite second moment and thus a finite variance (its second central moment), denoted by $0 \leq \sigma^2 < \infty$. In general, $X \in L^p$ for $p \in [1, \infty)$ means it admits a finite absolute p -th moment.

We say that $(X_n)_{n \in \mathbb{N}}$ obeys a Weak Law of Large Numbers (WLLN) if

$$\bar{X}_n \xrightarrow{\text{Pr}} \mu.$$

We say that $(X_n)_{n \in \mathbb{N}}$ obeys a Strong Law of Large Numbers (SLLN) if

$$\overline{X}_n \xrightarrow{a.s.} \mu.$$

The main challenge is to remember the assumptions, which vary for each version. The following is a list of typical assumptions:

• **WLLN:**

- (*Chebyshev*): we assume $(X_n)_{n \in \mathbb{N}} \subseteq L^2$, with the same expectation and variance for all n , and that they are uncorrelated ($\text{Cov}(X_i, X_j) = 0$ for every $i \neq j$).
- (*Khintchine*): we assume $(X_n)_{n \in \mathbb{N}} \subseteq L^1$ and that they are i.i.d.

• **SLLN:**

- (*Kolmogorov*): we assume $(X_n)_{n \in \mathbb{N}} \subseteq L^1$ and that they are i.i.d.
- (*Borel-Cantelli*): we assume $(X_n)_{n \in \mathbb{N}} \subseteq L^4$ (that is, finite fourth moment) and that they are i.i.d.

In general, the Khintchine and Kolmogorov versions are the strongest versions available. This is because the laws of large numbers, from an analytical perspective, are first-order asymptotic approximations. Therefore, it is reasonable to claim that the random variables should belong to L^1 , without the need for additional finite moments. However, there is a small exception concerning the *Chebyshev* WLLN: here, we do not assume independence or identical distributions, but only uncorrelation and identical expectation and variance.

In what follows, we prove the *Chebyshev* and *Borel-Cantelli* versions, leaving *Khintchine* and *Kolmogorov* for future work (see [15]). Interestingly, each of these four theorems employs a completely different proof technique.

Theorem 6.40 (Chebyshev WLLN). *Let $(X_n)_{n \in \mathbb{N}} \subseteq L^2$ be a sequence of uncorrelated real-valued random variables with $\text{Var}(X_n) = \sigma^2 < \infty$ and $\mathbb{E}[X_n] = \mu$ for every $n \in \mathbb{N}$. Then $\overline{X}_n \xrightarrow{L^2} \mu$ as $n \rightarrow \infty$. In particular, $\overline{X}_n \xrightarrow{\text{Pr}} \mu$ as $n \rightarrow \infty$.*

Proof.

Observe that

$$\mathbb{E}[\overline{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mu$$

and, since $X_1, \dots, X_n$ are uncorrelated, by Corollary 4.46,

$$\text{Var}[\overline{X}_n] = \text{Var}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n^2} \sum_{i=1}^n \text{Var}[X_i] = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Hence

$$\lim_{n \rightarrow \infty} \|\overline{X}_n - \mu\|_{L^2}^2 = \lim_{n \rightarrow \infty} \mathbb{E}[\|\overline{X}_n - \mathbb{E}[\overline{X}_n]\|^2] = \lim_{n \rightarrow \infty} \text{Var}[\overline{X}_n] = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0.$$

By Proposition 6.25, we conclude that $\overline{X}_n \xrightarrow{\text{Pr}} \mu$. □

The following proof is inspired by [16].

Theorem 6.41 (Borel-Cantelli SLLN). *Let $(X_n)_{n \in \mathbb{N}} \subseteq L^4$ be a sequence of i.i.d. real-valued random variables with finite fourth moment. Then $\overline{X}_n \xrightarrow{a.s.} \mu$ as $n \rightarrow \infty$.*

Proof.

If $\sigma^2 := \text{Var}(X_1) = 0$, then $X_n = \mu$ a.s. for every n , hence $\overline{X}_n = \mu$ a.s. and there is nothing to prove. Assume $\sigma > 0$.

We start from a reshuffling of (6.1):

$$\begin{aligned} \{\omega \in \Omega : \lim_{n \rightarrow \infty} \bar{X}_n(\omega) = \mu\} &= \left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (X_i(\omega) - \mu) = 0 \right\} \\ &= \left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \frac{X_i(\omega) - \mu}{\sigma} = 0 \right\} \\ &= \left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n Y_i(\omega) \right)^4 = 0 \right\} = \{\omega \in \Omega : \lim_{n \rightarrow \infty} \bar{Y}_n^4 = 0\}, \end{aligned}$$

where $Y_i := \frac{X_i - \mu}{\sigma}$ and $\bar{Y}_n := \frac{1}{n} \sum_{i=1}^n Y_i$. Thus it is enough to prove $\bar{Y}_n \xrightarrow{a.s.} 0$. By Lemma 6.26, it suffices to show that, for every $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \Pr(\bar{Y}_n^4 > \varepsilon) < \infty. \quad (6.17)$$

For every $i \in \mathbb{N}$,

$$\mathbb{E}[Y_i] = \frac{\mathbb{E}[X_i] - \mu}{\sigma} = 0, \quad \mathbb{E}[Y_i^2] = \frac{\mathbb{E}[(X_i - \mu)^2]}{\sigma^2} = \frac{\text{Var}(X_i)}{\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1.$$

Moreover, the fourth moment is finite: by Proposition 4.25 with $p = 4$, we immediately get

$$\mathbb{E}[Y_i^4] = \frac{1}{\sigma^4} \mathbb{E}[(X_i - \mu)^4] \leq \frac{2^{4-1}}{\sigma^4} (\mathbb{E}[X_i^4] + \mu^4) < \infty.$$

Let us compute the fourth moment of $\bar{Y}_n$. We denote $m_2 := \mathbb{E}[Y_i^2]$ and $m_4 := \mathbb{E}[Y_i^4]$, for every $i \in \mathbb{N}$. Now

$$\left(\sum_{i=1}^n Y_i \right)^4 = \sum_{i=1}^n Y_i^4 + 4 \sum_{i \neq j} Y_i^3 Y_j + \frac{4!}{2!2!} \sum_{i < j} Y_i^2 Y_j^2 + \frac{4!}{2!} \sum_{i \neq j \neq k} Y_i^2 Y_j Y_k + 4! \sum_{i \neq j \neq k \neq \ell} Y_i Y_j Y_k Y_\ell.$$

By linearity of the expectation and the independence of Y_i and Y_j whenever $i \neq j$, we deduce that

$$\mathbb{E} \left[\sum_{i \neq j} Y_i^3 Y_j \right] = \sum_{i \neq j} \mathbb{E}[Y_i^3] \underbrace{\mathbb{E}[Y_j]}_{=0} = 0 \quad \mathbb{E} \left[\sum_{i \neq j \neq k} Y_i^2 Y_j Y_k \right] = \sum_{i \neq j \neq k} \mathbb{E}[Y_i^2] \underbrace{\mathbb{E}[Y_j]}_{=0} \underbrace{\mathbb{E}[Y_k]}_{=0} = 0$$

and

$$\mathbb{E} \left[\sum_{i \neq j \neq k \neq \ell} Y_i Y_j Y_k Y_\ell \right] = \sum_{i \neq j \neq k \neq \ell} \mathbb{E}[Y_i] \mathbb{E}[Y_j] \mathbb{E}[Y_k] \mathbb{E}[Y_\ell] = 0.$$

Hence,

$$\mathbb{E}[\bar{Y}_n^4] = \frac{1}{n^4} \left[\sum_{i=1}^n \mathbb{E}[Y_i^4] + 6 \sum_{i < j} \mathbb{E}[Y_i^2] \mathbb{E}[Y_j^2] \right] = \frac{nm_4 + 6 \frac{n(n-1)}{2} m_2^2}{n^4} = \frac{nm_4 + 3n(n-1)m_2^2}{n^4}.$$

Let us go back to (6.17). By Markov's inequality

$$\Pr(\bar{Y}_n^4 > \varepsilon) \leq \frac{\mathbb{E}[\bar{Y}_n^4]}{\varepsilon} = \frac{nm_4 + 3n(n-1)m_2^2}{\varepsilon \cdot n^4} = \frac{m_4 + 3(n-1)m_2^2}{\varepsilon \cdot n^3}.$$

Therefore, for every $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \Pr(\bar{Y}_n^4 > \varepsilon) \leq \sum_{n=1}^{\infty} \frac{m_4 + 3(n-1)m_2^2}{\varepsilon \cdot n^3} < \infty,$$

since $\sum_{n \in \mathbb{N}} \frac{1}{n^3} < \sum_{n \in \mathbb{N}} \frac{1}{n^2} < \infty$. Hence $\bar{Y}_n \rightarrow 0$ a.s., and thus $\bar{Y}_n \rightarrow 0$ a.s. Finally, $\bar{X}_n = \mu + \sigma \bar{Y}_n \rightarrow \mu$ a.s. $\square$

6.9 Central Limit Theorem

In this section, we introduce the Central Limit Theorem (CLT). If the Law of Large Numbers is the foundational result for all of statistics, the Central Limit Theorem is quite possibly the most useful. Throughout applied mathematics, a standard strategy to study complex stochastic systems is to reduce them to the CLT, which reveals a surprising regularity hidden within chaos.

It is an astonishing theorem: by summing independent and identically distributed square-integrable random variables, no matter their distribution, the resulting distribution, appropriately normalized, always converges to

$$F(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-\frac{x^2}{2}} dx.$$

We call this distribution *standard Gaussian*. This result provides the formal mathematical justification for the privileged status of the Gaussian distribution in nature and science. As we will see, the emergence of the Gaussian relies on Taylor series and the characteristic function, which is fundamentally a Fourier transform (5.17). This is not a coincidence: in Harmonic Analysis we already showed that the standard Gaussian is an eigenfunction of the Fourier transform (2.58). In this sense, it is a universal attractor, or a fixed point, for the convolution of distributions (which, under the Fourier transform, become products via 2.53).

Lemma 6.42. *Let $(c_n)_{n \in \mathbb{N}} \subseteq \mathbb{C}$ be a sequence of complex numbers. If $\lim_{n \rightarrow \infty} c_n = c$, then*

$$\lim_{n \rightarrow \infty} \left(1 + \frac{c_n}{n}\right)^n = e^c.$$

Proof.

Use the well-known factorization

$$a^n - b^n = (a - b) \sum_{k=0}^{n-1} a^{n-1-k} b^k,$$

so

$$|a^n - b^n| \leq n \max(|a|, |b|)^{n-1} |a - b|.$$

We apply this with $a = 1 + \frac{c_n}{n}$ and $b = e^{c_n/n}$. Since $|1 + z| \leq e^{|z|}$ and $|e^z| = e^{\Re z} \leq e^{|z|}$ for every $z \in \mathbb{C}$, we get

$$\begin{aligned} \left| \left(1 + \frac{c_n}{n}\right)^n - e^{c_n} \right| &\leq n \max \left(\left|1 + \frac{c_n}{n}\right|, |e^{c_n/n}| \right)^{n-1} \left|1 + \frac{c_n}{n} - e^{c_n/n}\right| \\ &\leq n (e^{|c_n|/n})^{n-1} \left|1 + \frac{c_n}{n} - e^{c_n/n}\right| \\ &= n (e^{|c_n|/n})^{n-1} \left|1 + \frac{c_n}{n} - \sum_{i=0}^{\infty} \frac{1}{i!} \left(\frac{c_n}{n}\right)^i \right|. \end{aligned}$$

Let us study the large n limit. Take $\gamma > |c|$. Eventually, $\gamma > |c_n|$ and $|c_n/n| \leq 1$. Notice that the exponent is easily bounded: $(e^{|c_n|/n})^{n-1} = e^{|c_n| \frac{n-1}{n}} \leq e^{|c_n|} < e^\gamma$. Moreover, the $i = 0$ and $i = 1$ terms in the series exactly cancel $1 + \frac{c_n}{n}$. Hence:

$$\begin{aligned} \left| \left(1 + \frac{c_n}{n}\right)^n - e^{c_n} \right| &\leq n e^\gamma \left| \sum_{i=2}^{\infty} \frac{1}{i!} \left(\frac{c_n}{n}\right)^i \right| \leq n e^\gamma \sum_{i=2}^{\infty} \frac{1}{i!} \left|\frac{c_n}{n}\right|^i \leq n e^\gamma \left|\frac{c_n}{n}\right|^2 \sum_{i=2}^{\infty} \frac{1}{i!} \\ &\leq n e^\gamma \left|\frac{c_n}{n}\right|^2 = \frac{e^\gamma |c_n|^2}{n} \leq \frac{e^\gamma \gamma^2}{n} \xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

□

Theorem 6.43 (Central Limit Theorem). *Let $(X_n)_{n \in \mathbb{N}} \subseteq L^2$ be a sequence of i.i.d. square-integrable random variables, with expectation μ and variance $\sigma^2 > 0$ (strictly positive). Then*

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} Z \sim \mathcal{N}(0, 1), \quad n \rightarrow \infty,$$

where $\mathcal{N}(0, 1)$ is the absolutely continuous distribution with respect to Lebesgue measure λ associated with the PDF

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}.$$

Proof.

Without loss of generality, we can assume $\mu = 0$. Indeed, if $\mu \neq 0$, we can define the centered random variables $Y_i := X_i - \mu$. The sequence $(Y_n)_{n \in \mathbb{N}}$ is still i.i.d., with $\mathbb{E}[Y_1] = 0$ and $\text{Var}(Y_1) = \sigma^2$. Since our target random variable can be rewritten as

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} = \sqrt{n} \frac{\bar{Y}_n}{\sigma},$$

proving the theorem for the centered sequence $(Y_n)_{n \in \mathbb{N}}$ immediately yields the general result. Therefore, we proceed assuming $\mu = 0$.

Let $\varphi(t)$ be the characteristic function of X_1 . By Proposition 5.21,

$$\varphi_{\sqrt{n} \frac{\bar{X}_n}{\sigma}}(t) = \varphi_{\frac{X_1 + \dots + X_n}{\sigma \sqrt{n}}}(t) = \varphi\left(\frac{t}{\sigma \sqrt{n}}\right)^n.$$

By Corollary 5.26,

$$\varphi(u) = 1 - \frac{\sigma^2}{2} u^2 + o(u^2).$$

Setting $u = \frac{t}{\sigma \sqrt{n}}$,

$$\varphi\left(\frac{t}{\sigma \sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right).$$

Hence

$$\varphi_{\sqrt{n} \frac{\bar{X}_n}{\sigma}}(t) = \varphi\left(\frac{t}{\sigma \sqrt{n}}\right)^n = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \xrightarrow{n \rightarrow \infty} e^{-t^2/2},$$

where the convergence follows from Lemma 6.42. The limit $e^{-t^2/2}$ is continuous at 0, and it is the characteristic function of a Gaussian distribution $Z \sim \mathcal{N}(0, 1)$, by Example 5.20. By Lévy's continuity theorem (Theorem 6.29), we conclude $\sqrt{n} \frac{\bar{X}_n}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1)$. $\square$

Remark 6.44 (First and second-order asymptotics). Together, the Weak Law of Large Numbers and the Central Limit Theorem provide a beautiful asymptotic picture for the sample mean of i.i.d. square-integrable random variables. The WLLN guarantees that the centered sample mean vanishes:

$$\bar{X}_n - \mu \xrightarrow{\text{Pr}} 0.$$

The CLT refines this by describing the exact rate and shape of this convergence:

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} Z \sim \mathcal{N}(0, 1).$$

This implies that the fluctuations $\bar{X}_n - \mu$ decay to zero at a rate proportional to $1/\sqrt{n}$. Therefore, if we interpret the WLLN as a *first-order* approximation of the sample mean, the CLT acts as a *second-order* approximation.

7 Conditioning

7.1 Conditional expectation

In this section, we formalize some ideas introduced in Section 3.4 regarding the possession of some further information for an agent that can predict some events. We will hence formalize the concepts of conditional expectation $\mathbb{E}[X | Y]$ and conditional distribution $\Pr(X \in \cdot | Y = y)$. We recall that in probability theory, a σ -algebra is the collection of events that we can compute the probabilities of. That is, it is the collection of events of which we have information. A sub σ -algebra is used when the information we have is coarser than $\mathcal{F}$, i.e., only some sets are detectable. We are essentially evaluating the behavior of a random variable based on the restricted information provided by $\mathcal{G}$.

Example 7.1 (The σ -algebra generated by a partition). Consider a partition of Ω into a countable collection of disjoint sets $\{E_i\}_{i \in I}$ such that $\bigcup_i E_i = \Omega$. The collection $\mathcal{G}$ of all possible unions of these sets E_i forms a sub σ -algebra of the power set $\mathcal{P}(\Omega)$. This represents a state of knowledge where we know which "cell" E_i the outcome ω belongs to, but we cannot distinguish between individual points within the same cell.

Definition 7.2 (Conditional expectation). Let $X \in L^1(\mathcal{F})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub σ -algebra of $\mathcal{F}$. An integrable random variable Y is called a *conditional expectation* of X given $\mathcal{G}$ if it is $\mathcal{G}$ -measurable and

$$\int_G Y d\Pr = \int_G X d\Pr, \quad \text{for all } G \in \mathcal{G}.$$

In this case, Y is a.s. unique and is denoted by $\mathbb{E}[X | \mathcal{G}]$.

Example 7.3. Let us model the roll of a fair six-sided die. The sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$, endowed with the uniform probability measure $\Pr(\{\omega\}) = 1/6$ for all $\omega \in \Omega$. Let X be the random variable representing the value of the roll, so $X(\omega) = \omega$. Now, consider the partition of Ω into odd and even numbers: $E_1 = \{1, 3, 5\}$ and $E_2 = \{2, 4, 6\}$. As seen in Example 7.1, this partition generates a sub σ -algebra $\mathcal{G} = \{\emptyset, E_1, E_2, \Omega\}$. This $\mathcal{G}$ represents the information "we only know if the roll is odd or even". We want to find the conditional expectation $Y = \mathbb{E}[X | \mathcal{G}]$. By definition, Y must satisfy two properties:

- i) *$\mathcal{G}$ -measurability:* Since $\mathcal{G}$ is generated by $\{E_1, E_2\}$, any $\mathcal{G}$ -measurable function must be constant on the atoms of the partition. Therefore, Y must be of the form:

$$Y(\omega) = \begin{cases} c_1 & \text{if } \omega \in E_1 \\ c_2 & \text{if } \omega \in E_2 \end{cases}$$

for some constants $c_1, c_2 \in \mathbb{R}$.

- ii) *Integral equivalence:* Y should satisfy the following conditions:

$$\begin{aligned} \int_{E_1} Y d\Pr &= c_1 \Pr(E_1) = \frac{c_1}{2} = \int_{E_1} X d\Pr = \sum_{\omega \in E_1} X(\omega) \Pr(\{\omega\}) = \frac{1+3+5}{6} = \frac{3}{2} \\ \int_{E_2} Y d\Pr &= c_2 \Pr(E_2) = \frac{c_2}{2} = \int_{E_2} X d\Pr = \sum_{\omega \in E_2} X(\omega) \Pr(\{\omega\}) = \frac{2+4+6}{6} = 2. \end{aligned}$$

Thus, the conditional expectation is the random variable

$$\mathbb{E}[X | \mathcal{G}](\omega) = \begin{cases} 3 & \text{if } \omega \in \{1, 3, 5\} \\ 4 & \text{if } \omega \in \{2, 4, 6\} \end{cases}.$$

Notice how Y "averages out" the values of X over each cell of the partition.

Example 7.4. Let $(\Omega, \mathcal{F}, \Pr) = ([0, 1], \mathcal{B}([0, 1]), \lambda)$. Let $X : \Omega \rightarrow \mathbb{R}$ be the continuous random variable defined by $X(\omega) = \omega^2$. Let $E_1 = [0, 1/3)$, $E_2 = [1/3, 2/3)$, $E_3 = [2/3, 1)$, and define $\mathcal{G} = \mathcal{F}(\{E_1, E_2, E_3\}) = \{\emptyset, [0, 1/3), [1/3, 2/3), [2/3, 1), [0, 2/3), [1/3, 1), [0, 1/3) \cup [2/3, 1), [0, 1)\}$. be the σ -algebra generated by this partition. As in Example 7.3, Y must be of the form:

$$Y(\omega) = \begin{cases} c_1 & \omega \in E_1 \\ c_2 & \omega \in E_2 \\ c_3 & \omega \in E_3 \end{cases}$$

for some constants $c_1, c_2, c_3 \in \mathbb{R}$. Moreover, Y should satisfy the following conditions:

$$\begin{aligned}\int_{E_1} Y d\Pr &= c_1 \Pr(E_1) = \frac{c_1}{3} = \int_{E_1} X d\Pr = \int_0^{1/3} \omega^2 d\omega = \left[\frac{\omega^3}{3} \right]_0^{1/3} = \frac{1}{81} - 0 = \frac{1}{81} \implies c_1 = \frac{1}{27}, \\ \int_{E_2} Y d\Pr &= c_2 \Pr(E_2) = \frac{c_2}{3} = \int_{E_2} X d\Pr = \int_{1/3}^{2/3} \omega^2 d\omega = \left[\frac{\omega^3}{3} \right]_{1/3}^{2/3} = \frac{8}{81} - \frac{1}{81} = \frac{7}{81} \implies c_2 = \frac{7}{27}, \\ \int_{E_3} Y d\Pr &= c_3 \Pr(E_3) = \frac{c_3}{3} = \int_{E_3} X d\Pr = \int_{2/3}^1 \omega^2 d\omega = \left[\frac{\omega^3}{3} \right]_{2/3}^1 = \frac{27}{81} - \frac{8}{81} = \frac{19}{81} \implies c_3 = \frac{19}{27}.\end{aligned}$$

These values uniquely determine Y Pr-a.s. and satisfy $\int_G Y d\Pr = \int_G X d\Pr$ also on $[0, 2/3)$, $[1/3, 1)$ and $[0, 1)$ by additivity of the Lebesgue measure. More generally, the conditional expectation is, by definition, the Radon-Nikodym derivative of $\nu(G) := \int_G X d\Pr$ with respect to $\Pr$:

$$\mathbb{E}[X | \mathcal{G}] = \frac{d\nu}{d(\Pr|_{\mathcal{G}})}.$$

By uniqueness of the Radon-Nikodym derivative (Theorem 1.109), the conditional expectation is Pr-a.s. unique.

Proposition 7.5 (Basic properties of conditional expectation). *Let $X, Y \in L^1(\mathcal{F})$, let $\mathcal{G} \subseteq \mathcal{F}$ and $a, b, c \in \mathbb{R}$.*

- i) *Linearity:* $\mathbb{E}[aX + bY | \mathcal{G}] = a \mathbb{E}[X | \mathcal{G}] + b \mathbb{E}[Y | \mathcal{G}]$.
- ii) *For every $c \in \mathbb{R}$, $\mathbb{E}[c | \mathcal{G}] = c$.*
- iii) *Monotonicity:* if $X \geq Y$ a.s., then $\mathbb{E}[X | \mathcal{G}] \geq \mathbb{E}[Y | \mathcal{G}]$ a.s.
- iv) *Jensen:* if $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex, then

$$\varphi(\mathbb{E}[X | \mathcal{G}]) \leq \mathbb{E}[\varphi(X) | \mathcal{G}], \quad \text{Pr-a.s.}$$

- v) *Independence:* if X is independent of $\mathcal{G}$ (i.e., $\Pr(\{X \in B\} \cap G) = \Pr(X \in B) \cdot \Pr(G)$ for every $G \in \mathcal{G}$), then

$$\mathbb{E}[X | \mathcal{G}] = \mathbb{E}[X] \quad \text{Pr-a.s.}$$

Proof.

(i), (ii), (iii) and (iv) follow from similar arguments to standard expectation properties (cf. 4.11, 4.12, 4.13 and 4.28), exploiting the linearity and monotonicity of the integral over arbitrary sets $G \in \mathcal{G}$.

(v). We need to verify that the constant random variable $Y = \mathbb{E}[X]$ satisfies the definition of conditional expectation. First, being a constant, Y is trivially $\mathcal{G}$ -measurable. Second, let $G \in \mathcal{G}$. We compute the integral of Y over G :

$$\int_G \mathbb{E}[X] d\Pr = \mathbb{E}[X] \int_G 1 d\Pr = \mathbb{E}[X] \Pr(G).$$

On the other hand, since X is independent of $\mathcal{G}$, X is independent of the indicator function $\mathbf{1}_G$. Thus,

$$\int_G X d\Pr = \mathbb{E}[X \mathbf{1}_G] = \mathbb{E}[X] \mathbb{E}[\mathbf{1}_G] = \mathbb{E}[X] \Pr(G).$$

Since $\int_G \mathbb{E}[X] d\Pr = \int_G X d\Pr$ for all $G \in \mathcal{G}$, uniqueness yields $\mathbb{E}[X | \mathcal{G}] = \mathbb{E}[X]$ Pr-a.s. $\square$

Theorem 7.6 (Law of Total Expectation). *Let $\mathcal{G} \subseteq \mathcal{F}$. Let Z be $\mathcal{G}$ -measurable, and assume $X, ZX \in L^1(\mathcal{F})$. Then*

$$\mathbb{E}[ZX | \mathcal{G}] = Z \mathbb{E}[X | \mathcal{G}] \quad \text{Pr-a.s.}$$

Moreover,

$$\mathbb{E}[ZX] = \mathbb{E}[Z \mathbb{E}[X | \mathcal{G}]], \quad \mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X].$$

Proof.

Assume first that $Z = \mathbf{1}_G$ for some $G \in \mathcal{G}$. Define $Y := \mathbb{E}[X | \mathcal{G}] \mathbf{1}_G$, which is $\mathcal{G}$ -measurable. For every $H \in \mathcal{G}$, we have

$$\int_H Y d\Pr = \int_H \mathbb{E}[X | \mathcal{G}] \mathbf{1}_G d\Pr = \int_{H \cap G} \mathbb{E}[X | \mathcal{G}] d\Pr = \int_{H \cap G} X d\Pr = \int_H \mathbf{1}_G X d\Pr.$$

Hence Y satisfies the defining property of $\mathbb{E}[\mathbf{1}_G X | \mathcal{G}]$, so by uniqueness a.s.

$$\mathbb{E}[\mathbf{1}_G X | \mathcal{G}] = \mathbf{1}_G \mathbb{E}[X | \mathcal{G}] \quad \text{Pr-a.s.}$$

Using linearity, it is easy to extend the previous result to measurable simple functions of the form $Z = \sum_{j=1}^m a_j \mathbf{1}_{G_j}$ with $a_j \geq 0$ and $G_j \in \mathcal{G}$:

$$\mathbb{E}[ZX | \mathcal{G}] = \sum_{j=1}^m a_j \mathbb{E}[\mathbf{1}_{G_j} X | \mathcal{G}] = \sum_{j=1}^m a_j \mathbf{1}_{G_j} \mathbb{E}[X | \mathcal{G}] = Z \mathbb{E}[X | \mathcal{G}] \quad \text{Pr-a.s.}$$

Next, we assume $Z, X \geq 0$ and measurable. Choose $\mathcal{G}$ -measurable simple functions $(Z_n)_{n \in \mathbb{N}}$ such that $0 \leq Z_n \uparrow Z$ pointwise, as in Lemma 1.52. By the Monotone Convergence Theorem (1.50), for every $H \in \mathcal{G}$,

$$\begin{aligned} \int_H ZX d\Pr &= \int_H \lim_{n \rightarrow \infty} Z_n X d\Pr \stackrel{\text{MCT}}{=} \lim_{n \rightarrow \infty} \int_H Z_n X d\Pr = \lim_{n \rightarrow \infty} \int_H \mathbb{E}[Z_n X | \mathcal{G}] d\Pr \\ &= \lim_{n \rightarrow \infty} \int_H Z_n \mathbb{E}[X | \mathcal{G}] d\Pr \stackrel{\text{MCT}}{=} \int_H \lim_{n \rightarrow \infty} Z_n \mathbb{E}[X | \mathcal{G}] d\Pr = \int_H Z \mathbb{E}[X | \mathcal{G}] d\Pr. \end{aligned}$$

Hence $Z \mathbb{E}[X | \mathcal{G}]$ satisfies the defining property of $\mathbb{E}[ZX | \mathcal{G}]$, and by uniqueness,

$$\mathbb{E}[ZX | \mathcal{G}] = Z \mathbb{E}[X | \mathcal{G}] \quad \text{Pr-a.s.}$$

If $X = X^+ - X^-$ and $Z = Z^+ - Z^-$, it suffices to apply the previous argument to each nonnegative pair and use the linearity of conditional expectation:

$$\mathbb{E}[ZX | \mathcal{G}] = \mathbb{E}[(Z^+ - Z^-)(X^+ - X^-) | \mathcal{G}] = Z \mathbb{E}[X | \mathcal{G}] \quad \text{Pr-a.s.}$$

In the particular case where $H = \Omega \in \mathcal{G}$, we have

$$\mathbb{E}[\mathbb{E}[U | \mathcal{G}]] = \mathbb{E}[U] \quad \text{for every } U \in L^1(\mathcal{F}).$$

Setting $U = ZX$ and $U = X$, respectively, yields

$$\mathbb{E}[ZX] = \mathbb{E}[\mathbb{E}[ZX | \mathcal{G}]] = \mathbb{E}[Z \mathbb{E}[X | \mathcal{G}]], \quad \text{and} \quad \mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X].$$

□

7.2 Conditioning with respect to a random variable

Rather than conditioning with respect to a generic sub- σ -algebra, we typically work with the special case of conditioning with respect to a σ -algebra generated by a random variable.

Lemma 7.7 (σ -algebra generated by a random variable). *Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space, and $X : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ a random vector. Then*

$$\mathcal{F}(X) = \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R}^n)\}.$$

Proof.

Let $\mathcal{G} := \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R}^n)\}$. It is easy to see that it is a σ -algebra. Since X is $\mathcal{F}(X)$ -measurable, then $X^{-1}(B) \in \mathcal{F}(X)$ for any $B \in \mathcal{B}(\mathbb{R}^n)$. Conversely, we observe that X is $\mathcal{G}$ -measurable by construction. Since $\mathcal{G}$ is a σ -algebra, then $\mathcal{F}(X) \subseteq \mathcal{G}$, by definition of $\mathcal{F}(X)$. □

Example 7.8. Let $(\Omega, \mathcal{F}, \Pr) = ([0, 1], \mathcal{B}([0, 1]), \lambda)$ be a measure space, and define $X, Y : \Omega \rightarrow \mathbb{R}$ as

$$X(\omega) := \mathbf{1}_{[0, 1/2]}(\omega), \quad Y(\omega) := \omega.$$

Then X only takes the values 0 and 1, and

$$\{X = 1\} = [0, 1/2], \quad \{X = 0\} = (1/2, 1].$$

Hence the σ -algebra generated by X is

$$\mathcal{F}(X) = \{\emptyset, \Omega, [0, 1/2], (1/2, 1]\}.$$

Meanwhile, for any Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$Y^{-1}(B) = \{\omega \in [0, 1] : \omega \in B\} = B \cap [0, 1].$$

Therefore the σ -algebra generated by Y is

$$\mathcal{F}(Y) = \{B \cap [0, 1] : B \in \mathcal{B}(\mathbb{R})\} = \mathcal{B}([0, 1]).$$

In other words, observing Y reveals the full Borel information on $[0, 1]$.

Definition 7.9 (Conditional expectation given a random variable). Let $X \in L^1$ and let Y be any random variable. Let $\mathcal{F}(Y)$ be the σ -algebra generated by Y . Then the conditional expectation of X given Y is

$$\mathbb{E}[X | Y] := \mathbb{E}[X | \mathcal{F}(Y)].$$

The following is a special case of a more general theorem of *Doob-Dynkin* (See [5], [15]):

Lemma 7.10 (Regression function). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space, let $X \in L^1$ be real-valued, and let $Y : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ be a random vector. Then there exists a Borel measurable function $m : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\mathbb{E}[X | Y] = m(Y), \quad \Pr\text{-a.s.}$$

Any Borel function such that $m(Y) = \mathbb{E}[X | Y]$ $\Pr$ -a.s. is called a *regression function* of X on Y .

Remark 7.11 (Operative definition of conditional expectation). By Lemma 7.10, we can equivalently define the conditional expectation of X given Y as any Borel measurable function $m_X : (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that, for every $B \in \mathcal{B}(\mathbb{R}^n)$,

$$\int_{\{Y \in B\}} X d\Pr = \int_B m_X(y) d\Pr_Y(y). \quad (7.1)$$

7.3 Conditional probability

Having defined the conditional expectation, we would like to define the *conditional distribution* $\Pr(X \in B | Y = y)$, that is, a pointwise conditional law of X given the information about Y . By the Radon-Nikodym theorem (Theorem 1.109), for any fixed Borel set B , we can define a measurable function $y \mapsto \Pr(X \in B | Y = y)$ which is valid $\Pr_Y$ -almost everywhere. The theoretical subtlety here is that this construction is performed *one set B at a time*. In general, one cannot always choose the versions $m_B(y) := \Pr(X \in B | Y = y)$ simultaneously for all B such that, for $\Pr_Y$ -almost every y , the map $B \mapsto m_B(y)$ is a valid probability measure (i.e., countably additive) on the Borel σ -algebra.

On standard Borel spaces (such as $\mathbb{R}^n$), this simultaneous choice is possible because their Borel σ -algebras are countably generated. However, in these notes we skip this formal construction and safely treat conditional probabilities as proper probability distributions.

Definition 7.12 (Conditional probability given a σ -algebra). For any $A \in \mathcal{F}$, the *conditional probability* of A given $\mathcal{G}$ is

$$\Pr(A | \mathcal{G}) := \mathbb{E}[\mathbf{1}_A | \mathcal{G}].$$

It is $\mathcal{G}$ -measurable, takes values in $[0, 1]$ $\Pr$ -a.s., and satisfies

$$\int_G \Pr(A | \mathcal{G}) d\Pr = \Pr(A \cap G) \quad \text{for all } G \in \mathcal{G}.$$

Definition 7.13 (Conditional distribution). Let X and Y be random variables. The *conditional distribution* of X given Y is any function $y \mapsto \Pr(X \in \cdot | Y = y)$.

Remark 7.14 (Operative definition of conditional distribution). The conditional distribution $\Pr(X \in C | Y = y)$ is a function (often called a Markov kernel) such that, for every $B \in \mathcal{B}(\mathbb{R}^n)$,

$$\Pr(X \in C, Y \in B) = \int_B \Pr(X \in C | Y = y) d\Pr_Y(y). \quad (7.2)$$

Definition 7.15 (Conditional cumulative distribution function). Let X and Y be random variables. The conditional cumulative distribution function (conditional CDF) of X given $Y = y$ is defined as

$$F_{X|Y}(x | y) := \Pr(X \leq x | Y = y).$$

For almost every y , the mapping $x \mapsto F_{X|Y}(x | y)$ satisfies all the standard properties of a CDF.

Definition 7.16 (Conditional probability mass function). If X and Y are discrete random variables, the conditional probability mass function (conditional PMF) of X given $Y = y$ is defined as

$$p_{X|Y}(x | y) := \Pr(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)},$$

for any y such that the marginal PMF satisfies $p_Y(y) > 0$. Here, $p_{X,Y}$ denotes the joint PMF of (X, Y) .

Definition 7.17 (Conditional density function). If X and Y are random variables such that $\Pr_{X,Y}$ is absolutely continuous with respect to Lebesgue measure, a *conditional density* of X given Y is any version

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \quad \forall y : f_Y(y) > 0.$$

Remark 7.18. Since $f_{X|Y}(\cdot | y)$ is a true probability density for almost every y , it satisfies the standard properties:

- i) *Non-negativity:* $f_{X|Y}(x | y) \geq 0$ for almost every x .
- ii) *Normalization:* $\int_{\mathbb{R}^n} f_{X|Y}(x | y) dx = \int_{\mathbb{R}^n} \frac{f_{X,Y}(x, y)}{f_Y(y)} dx = \frac{f_Y(y)}{f_Y(y)} = 1$.

Consequently, we can compute the conditional probability of X falling in any Borel set A simply by integrating its conditional density:

$$\Pr(X \in A | Y = y) = \int_A f_{X|Y}(x | y) dx.$$

Moreover, for every bounded measurable g ,

$$\mathbb{E}[g(X) | Y = y] = \int_{\mathbb{R}^n} g(x) f_{X|Y}(x | y) dx.$$

Proposition 7.19 (Bayes formulas). Let $(\Omega, \mathcal{F}, \Pr)$ be a probability space and $X, Y : (\Omega, \mathcal{F}, \Pr) \rightarrow (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ be random variables. Assume that (X, Y) is either a jointly continuous random vector (with joint PDF $f_{X,Y}$) or a jointly discrete random vector (with joint PMF $f_{X,Y}$). Then for almost every y such that $f_Y(y) > 0$, the conditional density (or mass) function $f_{X|Y}$ exists and satisfies:

$$f_{X,Y}(x, y) = f_{X|Y}(x | y) f_Y(y), \tag{7.3}$$

$$f_{X|Y}(x | y) \propto f_{Y|X}(y | x) f_X(x), \quad \text{for } x, y \text{ such that } f_X(x) > 0, \quad f_Y(y) > 0. \tag{7.4}$$

Proof.

The proof is identical whether $f_{X,Y}$ denotes a joint PDF or a joint PMF. By the definition of conditional distribution,

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \implies f_{X,Y}(x, y) = f_{X|Y}(x | y) f_Y(y)$$

for almost every (x, y) . Moreover, whenever the conditional distribution $f_{Y|X}$ is also defined (i.e., whenever $f_X(x) > 0$), we similarly have

$$f_{X,Y}(x, y) = f_{Y|X}(y | x) f_X(x).$$

Combining the two factorizations yields, for those y with $f_Y(y) > 0$,

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{f_{Y|X}(y | x) f_X(x)}{f_Y(y)} \propto f_{Y|X}(y | x) f_X(x).$$

□

Corollary 7.20 (Independence via conditional distributions). Let X and Y be random variables as in Proposition 7.19. Then the following conditions are equivalent:

- i) X and Y are independent;
- ii) For almost every y such that $f_Y(y) > 0$, $f_{X|Y}(x | y) = f_X(x)$.
- iii) For almost every x such that $f_X(x) > 0$, $f_{Y|X}(y | x) = f_Y(y)$;

Proof.

We recall the characterization of independence via joint distributions: X and Y are independent if and only if $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ for almost all (x, y) (Proposition 3.72 or 3.73 depending on whether $f_{X,Y}$ denotes a joint PMF or a joint PDF).

($i \Rightarrow ii$). If X and Y are independent, combining Proposition 3.72 or 3.73 with Definition 7.16 or 7.17:

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x).$$

Similarly, one proves ($i \Rightarrow iii$).

($ii \Rightarrow i$). Conversely, assume that $f_{X|Y}(x | y) = f_X(x)$ for almost all x and all y with $f_Y(y) > 0$. Combining Proposition 7.19 with this assumption, we can factorize the joint distribution as:

$$f_{X,Y}(x, y) = f_{X|Y}(x | y)f_Y(y) = f_X(x)f_Y(y).$$

Notice that if $f_Y(y) = 0$, then the joint distribution $f_{X,Y}(x, y)$ must also be 0 almost everywhere, meaning the equality $0 = 0$ trivially holds. Therefore, the joint distribution factorizes globally. By Proposition 3.72 or 3.73, this implies that X and Y are independent. One proves symmetrically ($i \Rightarrow iii$) and ($iii \Rightarrow i$). $\square$

Disintegration. In applied probability, we frequently construct the joint distribution $\Pr_{(X,Y)}$ by defining the marginal law of X and the conditional law of Y given X . These operations require an underlying theory of *disintegration of measures* which we do not cover in these notes. However, the practical result allows us to seamlessly mix continuous and discrete variables, and compute any joint probability using the rule:

$$\Pr(X \in A, Y \in B) = \int_A \Pr(Y \in B | X = x) \Pr_X(dx).$$

Indeed, for every Borel sets A and B ,

$$\Pr(X \in A, Y \in B) = \iint_{A \times B} f_{X,Y}(x, y) dx dy = \int_A \left(\int_B f_{Y|X}(y | x) dy \right) f_X(x) dx.$$

In particular, if X is continuous and $Y | X = x$ is discrete, then

$$\Pr(X \in A, Y \in B) = \int_A \left(\sum_{y \in B} p_{Y|X}(y | x) \right) f_X(x) dx.$$

Analogously, if X is discrete and $Y | X = x$ is continuous, then

$$\Pr(X \in A, Y \in B) = \sum_{x \in A} \left(\int_B f_{Y|X}(y | x) dy \right) p_X(x).$$

Consider the following example from [16]:

Example 7.21. Let X be a continuous random variable with PDF $f_X(x) = e^{-x} \mathbf{1}_{(0,\infty)}(x)$, and let $Y | X = x$ be a discrete random variable with PMF $p_{Y|X}(y | x) = \frac{x^y}{y!} e^{-x} \mathbf{1}_{\mathbb{N}}(y)$. We compute $\Pr(X < 1, Y = 1)$ and $\text{Cov}(X, Y)$:

$$\begin{aligned} \Pr(X < 1, Y = 1) &= \int_0^1 \Pr(Y = 1 | X = x) f_X(x) dx = \int_0^1 \left(\frac{x^1}{1!} e^{-x} \right) e^{-x} dx \\ &= \int_0^1 x e^{-2x} dx = [-x e^{-2x} / 2]_0^1 + \int_0^1 \frac{e^{-2x}}{2} dx = \frac{1}{4} - \frac{3}{4} e^{-2}. \end{aligned}$$

As for the covariance, we compute:

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[\mathbb{E}[Y | X]] = \mathbb{E}[X] = \int_0^\infty x e^{-x} dx = [-x e^{-x}]_0^\infty + \int_0^\infty e^{-x} dx = [-e^{-x}]_0^\infty = 1 \\ \mathbb{E}[XY] &= \mathbb{E}[\mathbb{E}[XY | X]] = \mathbb{E}[X \mathbb{E}[Y | X]] = \mathbb{E}[X^2] = \int_0^\infty x^2 e^{-x} dx = [-x^2 e^{-x}]_0^\infty + 2 \int_0^\infty x e^{-x} dx = 2 \\ \text{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = 2 - 1 \cdot 1 = 1.\end{aligned}$$

7.4 Conditional variance

Rather than conditioning with respect to a generic sub- σ -algebra, in statistical learning we almost exclusively condition a square-integrable random variable $Y \in L^2$ on the realization of a random vector $X \in \mathbb{R}^n$. In this section, we focus on this specific practical case, leaving a fully abstract measure-theoretic discussion of conditional variance for future work.

Remark 7.22 (Conditioning on random variables). In measure theory, the conditional variance is formally defined with respect to a generic σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. However, in statistical learning and engineering applications, we almost exclusively condition a random variable $Y \in L^2$ on the realization of another random vector $X \in \mathbb{R}^n$. In this section, we will explicitly adopt this perspective to provide more intuitive and actionable formulas.

The conditional variance of Y given X represents the variance of the random variable Y , given the information about the value of X .

Definition 7.23 (Conditional variance). Let $(\Omega, \mathcal{F}, \text{Pr})$ be a probability space, let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ be a random vector and let $Y \in L^2(\Omega)$ be a random variable. The *conditional variance* of Y given X is defined as the random variable

$$\text{Var}(Y | X) := \mathbb{E}\left[(Y - \mathbb{E}[Y | X])^2 \mid X\right].$$

Equivalently, thanks to Lemma 7.10, we can write

$$\text{Var}(Y | X) = \mathbb{E}\left[(Y - m_Y(X))^2 \mid X\right],$$

where $m_Y(x)$ is a regression function of Y on X .

Example 7.24 (Conditional variance of a discrete random variable). Let X and Y be discrete random variables with joint probability mass function $p_{X,Y}$, and let $p_X(x) > 0$. By specializing the abstract definition to a purely atomic measure, the conditional expectation $m_Y(x) = \mathbb{E}[Y | X = x]$ reduces to a sum over the support of Y . Consequently, the conditional variance takes the form of a weighted sum of squared deviations from this conditional mean:

$$\text{Var}(Y | X = x) = \sum_y (y - m_Y(x))^2 p_{Y|X}(y | x).$$

Example 7.25 (Conditional variance of a continuous random variable). Let X and Y be continuous random variables with joint probability density function $f_{X,Y}$, and let $f_X(x) > 0$. As in Definition 7.17, they admit a conditional density function $f_{Y|X}(\cdot | x)$. Thus, the conditional expectation is $m_Y(x) = \mathbb{E}[Y | X = x] = \int_{\mathbb{R}} y f_{Y|X}(y | x) dy$, and the conditional variance becomes a standard Riemann integral:

$$\text{Var}(Y | X = x) = \int_{-\infty}^{+\infty} (y - m_Y(x))^2 f_{Y|X}(y | x) dy.$$

Just like the unconditioned variance measures the spread of the marginal density f_Y , the conditional variance measures the spread of the conditional density $f_{Y|X}(\cdot | x)$ around its mean $m_Y(x)$ (cf. Example 4.42).

Proposition 7.26 (Properties of conditional variance). Let $(\Omega, \mathcal{F}, \text{Pr})$ be a probability space, let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ be a random vector and let $Y \in L^2(\Omega)$ be a random variable. The following properties hold Pr -a.s.:

- i) $\text{Var}(Y | X) \geq 0$, and $\text{Var}(Y | X) = 0$ if and only if $Y = \mathbb{E}[Y | X]$ almost surely.
- ii) $\text{Var}(Y | X) = \mathbb{E}[Y^2 | X] - (\mathbb{E}[Y | X])^2$.

iii) For any measurable function g such that $g(X)Y \in L^2$,

$$\text{Var}(g(X)Y \mid X) = g(X)^2 \text{Var}(Y \mid X).$$

iv) For any constants $a, b \in \mathbb{R}$,

$$\text{Var}(aY + b \mid X) = a^2 \text{Var}(Y \mid X).$$

Proof.

(i). Since $(Y - \mathbb{E}[Y \mid X])^2 \geq 0$, its conditional expectation is non-negative, by the monotonicity property in Proposition 7.5. If $\text{Var}(Y \mid X) = 0$ Pr-a.s., then

$$\mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2] = \mathbb{E}[\mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 \mid X]] = \mathbb{E}[\text{Var}(Y \mid X)] = 0.$$

By Corollary 1.59, this implies $Y = \mathbb{E}[Y \mid X]$ Pr-a.s.

(ii). By the linearity property in Proposition 7.5,

$$\begin{aligned} \text{Var}(Y \mid X) &= \mathbb{E}[Y^2 - 2Y \mathbb{E}[Y \mid X] + (\mathbb{E}[Y \mid X])^2 \mid X] \\ &= \mathbb{E}[Y^2 \mid X] - 2\mathbb{E}[Y \mid X] \mathbb{E}[Y \mid X] + (\mathbb{E}[Y \mid X])^2 \\ &= \mathbb{E}[Y^2 \mid X] - (\mathbb{E}[Y \mid X])^2. \end{aligned}$$

(iii). By Theorem 7.6, setting $Z := g(X)$, then

$$\begin{aligned} \text{Var}(g(X)Y \mid X) &= \mathbb{E}[(g(X)Y - \mathbb{E}[g(X)Y \mid X])^2 \mid X] \\ &= \mathbb{E}[(g(X)Y - g(X)\mathbb{E}[Y \mid X])^2 \mid X] \\ &= \mathbb{E}[g(X)^2(Y - \mathbb{E}[Y \mid X])^2 \mid X] \\ &= g(X)^2 \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 \mid X] = g(X)^2 \text{Var}(Y \mid X). \end{aligned}$$

(iv). This follows directly from property (iii) setting $g(X) = a$ and noting that the additive constant b cancels out inside the difference $aY + b - \mathbb{E}[aY + b \mid X] = a(Y - \mathbb{E}[Y \mid X])$. $\square$

Theorem 7.27 (Variance decomposition). Let $(\Omega, \mathcal{F}, \text{Pr})$ be a probability space, let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ be a random vector and let $Y \in L^2(\Omega)$ be a random variable. If $\text{Var}(Y \mid X)$ is integrable and $\mathbb{E}[Y \mid X]$ is square-integrable, then the total variance of Y can be decomposed as:

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y \mid X)] + \text{Var}(\mathbb{E}[Y \mid X]).$$

Proof.

We compute:

$$\begin{aligned} \text{Var}(Y) &\stackrel{(1)}{=} \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \\ &\stackrel{(2)}{=} \mathbb{E}[\mathbb{E}[Y^2 \mid X]] - \left(\mathbb{E}[\mathbb{E}[Y \mid X]]\right)^2 \\ &\stackrel{(3)}{=} \mathbb{E}[\mathbb{E}[Y^2 \mid X] - (\mathbb{E}[Y \mid X])^2] + \mathbb{E}[(\mathbb{E}[Y \mid X])^2] - \left(\mathbb{E}[\mathbb{E}[Y \mid X]]\right)^2 \\ &\stackrel{(4)}{=} \mathbb{E}[\text{Var}(Y \mid X)] + \text{Var}(\mathbb{E}[Y \mid X]). \end{aligned}$$

Specifically, step (1) uses the algebraic definition of variance (property (i) of Proposition 4.44); step (2) uses the Tower Property (Theorem 7.6) twice; step (3) adds and subtracts $\mathbb{E}[(\mathbb{E}[Y \mid X])^2]$ using linearity (Theorem 4.11); and finally, step (4) uses the algebraic definition of conditional variance (property (ii) of Proposition 7.26) for the first term, and again the algebraic definition of variance (property (i) of Proposition 4.44) for the second term, applied to the random variable $Z := \mathbb{E}[Y \mid X]$. $\square$

8 Classical Distributions

In this section, we analyze several fundamental families of probability distributions.

8.1 Discrete Distributions

8.1.1 Discrete Uniform Distribution

The discrete uniform distribution is the uniform law on the integers $\{1, 2, \dots, n\}$. We write $X \sim \text{Unif}(n)$ if $X \in \{1, \dots, n\}$ and

$$p_X(x) = \frac{1}{n} \quad \forall x \in \{1, \dots, n\}.$$

The discrete uniform distribution plays a pivotal role in applied statistics because it is essentially the *only* distribution we can directly generate using a pseudo-random number generator. Since computers are entirely deterministic, true randomness cannot be algorithmically produced. However, generating random variables is a fundamental necessity in statistics. To overcome this limitation, statisticians use deterministic algorithms designed to produce sequences with robust statistical properties. We sketch here the most basic method. To generate *pseudo-random* variables with a discrete uniform distribution, we produce a sequence $\{V_n\}_{n \in \mathbb{N}}$ taking values in $\{0, \dots, N-1\}$ via a *linear congruential generator* (LCG). By fixing integer parameters (a, b, N) and an initial *seed* V_0 , we define the recurrence relation

$$V_{i+1} \equiv (aV_i + b) \pmod{N}.$$

With suitably chosen parameters, the sequence behaves as if it were uniformly distributed on $\{0, \dots, N-1\}$. Next, we rescale the sequence by defining $U_n := V_n/N$, so that $U_n \in [0, 1)$. It can be shown that the sequence $\{U_n\}_{n \in \mathbb{N}}$ approximates a sequence of i.i.d. continuous uniform random variables on $[0, 1]$ (see Lemma 8.16). From the sequences $(V_n)_{n \in \mathbb{N}}$ and $(U_n)_{n \in \mathbb{N}}$, any other distribution can be obtained via suitable transformations or procedures. For instance, if computable, one can apply the Quantile transform $F^{-1}(U)$ to the uniform as in Lemma 3.64 to generate any other distribution. This is the simplest method to generate gaussian noise.

Let us compute the relevant quantities for this distribution.

Theorem 8.1 (Properties of the discrete uniform distribution). *Let $X \sim \text{Unif}(n)$. Then X has the following properties:*

$$\begin{aligned} p_X(x) &= \frac{1}{n} \mathbf{1}_{\{1, \dots, n\}}(x), & F_X(x) &= \frac{\lfloor x \rfloor}{n} \mathbf{1}_{[0, n]}(x) + \mathbf{1}_{(n, \infty)}(x), \\ \mathbb{E}[X] &= \frac{n+1}{2}, & \text{Var}(X) &= \frac{n^2-1}{12}, & M_X(t) &= \frac{e^t(e^{tn}-1)}{n(e^t-1)} \quad (t \neq 0). \end{aligned}$$

Proof.

The PMF p_X is given by definition. It is straightforward to verify that the given formula for F_X holds. From here, we compute the moments:

$$\begin{aligned} \mathbb{E}[X] &= \sum_{x=1}^n x \frac{1}{n} = \frac{1}{n} \frac{n(n+1)}{2} = \frac{n+1}{2}, \\ \mathbb{E}[X^2] &= \sum_{x=1}^n x^2 \frac{1}{n} = \frac{1}{n} \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}, \\ \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{(n+1)[4n+2-3n-3]}{12} = \frac{n^2-1}{12}. \end{aligned}$$

Finally, the MGF is obtained via the sum of a finite geometric series:

$$M_X(t) = \sum_{x=1}^n e^{tx} \frac{1}{n} = \frac{1}{n} \sum_{x=1}^n (e^t)^x = \frac{e^t}{n} \frac{1 - (e^t)^n}{1 - e^t} = \frac{e^t(e^{tn}-1)}{n(e^t-1)}.$$

□

8.1.2 Bernoulli distribution

A Bernoulli random variable models a single yes/no outcome, such as success/failure in *one* trial. We set $X(\omega) = 1$ if the trial ω results in a success, and $X(\omega) = 0$ if the trial ω results in a failure. If $p \in (0, 1)$ is the probability of success, then the probability of failure is $1 - p$. Therefore, we write $X \sim \text{Bernoulli}(p)$ if $X \in \{0, 1\}$ and

$$\Pr(X = 1) = p, \quad \Pr(X = 0) = 1 - p.$$

Let us compute the relevant quantities for this distribution.

Theorem 8.2 (Properties of the Bernoulli distribution). *Let $X \sim \text{Bernoulli}(p)$. Then X has the following properties:*

$$\begin{aligned} p_X(x) &= p^x(1-p)^{1-x} \mathbf{1}_{\{0,1\}}(x), & F_X(x) &= (1-p)\mathbf{1}_{[0,1)}(x) + \mathbf{1}_{[1,\infty)}(x), \\ \mathbb{E}[X] &= p, & \text{Var}(X) &= p(1-p), & M_X(t) &= 1-p+pe^t \quad (t \in \mathbb{R}). \end{aligned}$$

Proof.

Since the support of $\Pr_X$ is $\{0, 1\}$, we only need to verify that $p_X(1) = p$ and $p_X(0) = 1 - p$, and this is clearly the case:

$$p_X(1) = p^1(1-p)^0 = p, \quad p_X(0) = p^0(1-p)^1 = 1-p.$$

We then compute the moments and the MGF:

$$\begin{aligned} \mathbb{E}[X] &= 1 \cdot p_X(1) + 0 \cdot p_X(0) = p_X(1) = p, \\ \mathbb{E}[X^2] &= 1^2 \cdot p_X(1) + 0^2 \cdot p_X(0) = p_X(1) = p, \\ \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = p - p^2 = p(1-p), \\ M_X(t) &= \mathbb{E}[e^{tX}] = e^{t \cdot 1} p_X(1) + e^{t \cdot 0} p_X(0) = pe^t + (1-p). \end{aligned}$$

Alternatively, one can compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ by differentiating $M_X(t)$ at 0:

$$\mathbb{E}[X] = M'_X(0) = pe^0 = p \quad \text{and} \quad \mathbb{E}[X^2] = M''_X(0) = pe^0 = p.$$

To compute $F_X(x)$, we observe that the probability mass is concentrated at the points $x = 0$ and $x = 1$. These correspond to the jump discontinuities of F_X . Since $p_X(0) = 1 - p$, we have $F_X(x) = 0$ for $x < 0$, $F_X(x) = 1 - p$ for $0 \leq x < 1$, and $F_X(x) = (1 - p) + p = 1$ for $x \geq 1$. $\square$

8.1.3 Binomial distribution

A binomial random variable counts the total number of successes in n independent Bernoulli trials. If $p \in (0, 1)$ is the probability of success for a single Bernoulli trial, the probability that the total number of successes in n independent trials is exactly x is given by

$$\binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, \dots, n,$$

where $p^x(1-p)^{n-x}$ is the probability of observing x successes and $n-x$ failures in a fixed order (by independence), and $\binom{n}{x}$ accounts for the number of possible orderings of successes and failures.

Therefore, we write $X \sim \text{Binom}(n, p)$ if $X \in \{0, 1, \dots, n\}$ and

$$p_X(x) = \Pr(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n.$$

We expect by construction that $\Pr(X \in \{0, 1, \dots, n\}) = 1$. This is easily proven analytically via the binomial theorem:

$$\sum_{x=0}^n p_X(x) = \sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x} = (p + (1-p))^n = 1^n = 1.$$

This construction immediately suggests the following representation, which we record for later use.

Lemma 8.3 (Binomial distribution as a sum of i.i.d. Bernoulli). *Let $X_1, \dots, X_n$ be i.i.d. random variables with $X_i \sim \text{Bernoulli}(p)$ and set $S_n = \sum_{i=1}^n X_i$. Then $S_n \sim \text{Binom}(n, p)$.*

Proof.

First approach (probabilistic). By definition, a $\text{Binom}(n, p)$ random variable models the total number of successes in n independent trials, where each trial has a success probability p . If we define $Y_i = \mathbf{1}_{\{\text{success in the } i\text{-th trial}\}}$, then $Y_1, \dots, Y_n$ are inherently i.i.d. Bernoulli(p) random variables. Their sum $Y = \sum_{i=1}^n Y_i$ counts the total number of successes, meaning $Y \sim \text{Binom}(n, p)$ by definition. Since the sequence $(X_1, \dots, X_n)$ is identical in law to the sequence of indicators $(Y_1, \dots, Y_n)$, their respective sums must also share the same probability distribution. Thus, $S_n \sim \text{Binom}(n, p)$.

Second approach (analytical). We proceed via moment generating functions. By Proposition 5.9, the MGF of the sum of independent random variables is the product of their individual MGFs:

$$M_{S_n}(t) = \prod_{i=1}^n M_{X_i}(t) = \prod_{i=1}^n (pe^t + 1 - p) = (pe^t + 1 - p)^n.$$

On the other hand, the MGF of a random variable $X \sim \text{Binom}(n, p)$ is computed directly via the binomial theorem:

$$M_X(t) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=0}^n \binom{n}{x} (e^t p)^x (1-p)^{n-x} = (e^t p + 1 - p)^n.$$

Since $M_{S_n}(t) = M_X(t)$ for all $t \in \mathbb{R}$, Corollary 5.23 guarantees that S_n and X share the exact same distribution, confirming that $S_n \sim \text{Binom}(n, p)$. $\square$

Theorem 8.4 (Properties of the binomial distribution). Let $X \sim \text{Binom}(n, p)$. Then X has the following properties:

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x} \mathbf{1}_{\{0, \dots, n\}}(x),$$

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1-p), \quad M_X(t) = (1-p + pe^t)^n \quad (t \in \mathbb{R}).$$

The CDF F_X does not admit a simple closed form.

Proof.

The PMF p_X is given by definition. The jumps of the CDF correspond to the points in the support of p_X . To compute the expectation and the variance, there are at least four instructive approaches.

First approach (algebraic). Let us compute the moments directly:

$$\begin{aligned} \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} &= \sum_{x=1}^n \frac{n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x} = n \sum_{x=1}^n \frac{(n-1)!}{(x-1)!(n-x)!} p^x (1-p)^{n-x} \\ &= n \sum_{x=0}^{n-1} \binom{n-1}{x} p^{x+1} (1-p)^{n-1-x} = np \sum_{x=0}^{n-1} \binom{n-1}{x} p^x (1-p)^{n-1-x} \\ &= np(p + (1-p))^{n-1} = np, \end{aligned}$$

and similarly for the second moment:

$$\begin{aligned}
 \sum_{x=0}^n x^2 \binom{n}{x} p^x (1-p)^{n-x} &= \sum_{x=1}^n \frac{xn!}{(x-1)!(n-x)!} p^x (1-p)^{n-x} = \sum_{x=1}^n \frac{(x-1+1)n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x} \\
 &= \sum_{x=2}^n \frac{n!}{(x-2)!(n-x)!} p^x (1-p)^{n-x} + \sum_{x=1}^n \frac{n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x} \\
 &= n(n-1) \sum_{x=2}^n \frac{(n-2)!}{(x-2)!(n-x)!} p^x (1-p)^{n-x} + np \\
 &= n(n-1) \sum_{x=0}^{n-2} \frac{(n-2)!}{x!(n-x-2)!} p^{x+2} (1-p)^{n-2-x} + np \\
 &= p^2 n(n-1) \sum_{x=0}^{n-2} \binom{n-2}{x} p^x (1-p)^{n-2-x} + np \\
 &= p^2 n(n-1) (p + (1-p))^{n-2} + np = p^2 n^2 - np^2 + np.
 \end{aligned}$$

Finally:

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = n^2 p^2 - np^2 + np - n^2 p^2 = np(1-p).$$

Second approach (differentiating the MGF). We compute the MGF first via the binomial theorem:

$$M_X(t) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=0}^n \binom{n}{x} (e^t p)^x (1-p)^{n-x} = (pe^t + 1 - p)^n.$$

We then extract the first and second moments by differentiating $M_X(t)$ at $t = 0$:

$$\begin{aligned}
 \mathbb{E}[X] &= M'_X(0) = n(pe^0 + 1 - p)^{n-1} \cdot pe^0 = np, \\
 \mathbb{E}[X^2] &= M''_X(0) = npe^0(pe^0 + 1 - p)^{n-1} + n(n-1)p^2(e^0)^2(pe^0 + 1 - p)^{n-2} = np + n^2 p^2 - np^2, \\
 \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = np + n^2 p^2 - np^2 - n^2 p^2 = np(1-p).
 \end{aligned}$$

Third approach (representation as a sum). By Lemma 8.3, $X \sim \text{Binom}(n, p)$ is equal in distribution to the sum of n i.i.d. Bernoulli(p) random variables $X_1, \dots, X_n$. By linearity of expectation and independence for the variance:

$$\begin{aligned}
 \mathbb{E}[X] &= \sum_{i=1}^n \mathbb{E}[X_i] = np, \\
 \text{Var}(X) &= \sum_{i=1}^n \text{Var}(X_i) = np(1-p), \\
 M_X(t) &= M_{X_1+\dots+X_n}(t) = \prod_{i=1}^n M_{X_i}(t) = (pe^t + 1 - p)^n.
 \end{aligned}$$

Fourth approach (combinatorial). Consider the following problem. Among n people, we want to create a committee by choosing a subset of the n people and, among the committee, one captain member. Suppose the probability of choosing any person is p , independently. There are two distinct ways to compute the expected number of people chosen. The first is the trivial binomial choice of x people among n , multiplied by x ways to choose a captain:

$$\sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x}.$$

The second way is to first choose a captain among n people, and then compute the expected value of choosing a group of people among the remaining $n-1$. This yields:

$$np \sum_{x=0}^{n-1} \binom{n-1}{x} p^x (1-p)^{n-1-x} = np.$$

Since we are counting the same quantity, we conclude that $\mathbb{E}[X] = np$. To compute the second factorial moment, we approach the same problem, but selecting a president and a vice president. Choosing the group of x people and then electing the two leaders gives:

$$\sum_{x=0}^n x(x-1) \binom{n}{x} p^x (1-p)^{n-x} = \mathbb{E}[X^2] - \mathbb{E}[X].$$

Alternatively, we first choose the president and vice president, and then select the rest of the committee:

$$n(n-1)p^2 \sum_{x=0}^{n-2} \binom{n-2}{x} p^x (1-p)^{n-2-x} = n(n-1)p^2.$$

Therefore, $\mathbb{E}[X^2] = n^2p^2 - np^2 + np$, and

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = np(1-p).$$

□

Proposition 8.5 (Additivity of the binomial distribution). *Let $X \sim \text{Binom}(n_1, p)$ and $Y \sim \text{Binom}(n_2, p)$ be independent. Then*

$$X + Y \sim \text{Binom}(n_1 + n_2, p).$$

Proof.

First approach (MGF). By Proposition 5.9, the MGF of $X + Y$ is given by

$$M_{X+Y}(t) = M_X(t)M_Y(t) = (pe^t + 1 - p)^{n_1}(pe^t + 1 - p)^{n_2} = (pe^t + 1 - p)^{n_1+n_2},$$

which coincides with the MGF of $Z \sim \text{Binom}(n_1 + n_2, p)$. By uniqueness of the MGF (Corollary 5.23), we conclude that $X + Y \stackrel{d}{=} Z \sim \text{Binom}(n_1 + n_2, p)$.

Second approach (probabilistic). By definition, X counts the number of successes in n_1 independent Bernoulli trials with probability of success p . Similarly, Y counts the number of successes in n_2 independent Bernoulli trials with the same p . Since X and Y are independent, $X + Y$ simply counts the total number of successes in $n_1 + n_2$ independent Bernoulli trials. Hence, $X + Y \sim \text{Binom}(n_1 + n_2, p)$.

Third approach (convolution). By Corollary 3.81, the PMF p_{X+Y} is the discrete convolution of p_X and p_Y . For every $x \in \{0, 1, \dots, n_1 + n_2\}$, we have:

$$\begin{aligned} p_{X+Y}(x) &= \sum_{y=0}^x \Pr(X = y) \Pr(Y = x - y) \\ &= \sum_{y=0}^x \binom{n_1}{y} p^y (1-p)^{n_1-y} \binom{n_2}{x-y} p^{x-y} (1-p)^{n_2-(x-y)} \\ &= p^x (1-p)^{n_1+n_2-x} \sum_{y=0}^x \binom{n_1}{y} \binom{n_2}{x-y} \\ &= \binom{n_1 + n_2}{x} p^x (1-p)^{n_1+n_2-x}. \end{aligned}$$

The last equality follows from Vandermonde's identity

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}.$$

The best proof I know for Vandermonde's identity is combinatorial (and morally equivalent to the second approach presented here). Alternatively, there is a direct algebraic proof using the binomial

theorem:

$$\begin{aligned} \sum_{r=0}^{m+n} \binom{m+n}{r} x^r &= (1+x)^{m+n} = (1+x)^m (1+x)^n \\ &= \left(\sum_{i=0}^m \binom{m}{i} x^i \right) \left(\sum_{j=0}^n \binom{n}{j} x^j \right) = \sum_{r=0}^{m+n} \left(\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} \right) x^r. \end{aligned}$$

Matching coefficients of x^r on both sides yields the identity. The conclusion follows by observing that the PMF of $X + Y$ coincides with that of $Z \sim \text{Binom}(n_1 + n_2, p)$. $\square$

8.1.4 Geometric distribution

Consider an infinite sequence of independent Bernoulli trials. The geometric law counts the number of trials before the first success. If $p \in (0, 1)$ is the probability of success for a Bernoulli trial, the probability that the first success occurs exactly at the x -th trial is

$$p_X(x) = p \times \Pr(x-1 \text{ independent failures}) = p(1-p)^{x-1}, \quad x = 1, 2, \dots$$

We expect by construction that $\Pr(X \in \mathbb{N}) = 1$. This is proven analytically via the geometric series:

$$\sum_{x=1}^{\infty} p_X(x) = \sum_{x=1}^{\infty} p(1-p)^{x-1} = p \sum_{y=0}^{\infty} (1-p)^y = p \frac{1}{1-(1-p)} = \frac{p}{p} = 1.$$

Theorem 8.6 (Properties of the geometric distribution). *Let $X \sim \text{Geom}(p)$. Then X has the following properties:*

$$\begin{aligned} p_X(x) &= p(1-p)^{x-1} \mathbf{1}_{\mathbb{N}}(x), & F_X(x) &= (1 - (1-p)^{\lfloor x \rfloor}) \mathbf{1}_{[1, \infty)}(x), \\ \mathbb{E}[X] &= \frac{1}{p}, & \text{Var}(X) &= \frac{1-p}{p^2}, & M_X(t) &= \frac{pe^t}{1 - (1-p)e^t} \quad (t < -\log(1-p)). \end{aligned}$$

Proof.

The PMF p_X is given by definition. We compute the CDF at its points of discontinuity, which are the positive integers. For every $x \in \mathbb{N}$,

$$F_X(x) = \sum_{i=1}^x p(1-p)^{i-1} = p \sum_{i=0}^{x-1} (1-p)^i = p \frac{1 - (1-p)^x}{1 - (1-p)} = 1 - (1-p)^x.$$

The MGF is given by

$$\begin{aligned} M_X(t) &= \sum_{x=1}^{\infty} e^{tx} p(1-p)^{x-1} = \frac{p}{1-p} \sum_{x=1}^{\infty} (e^t(1-p))^x \\ &= \frac{p}{1-p} \left(\frac{1}{1 - e^t(1-p)} - 1 \right) = \frac{pe^t}{1 - (1-p)e^t}. \end{aligned}$$

Observe that this function exists as long as $1 - (1-p)e^t > 0$, that is, $t < -\log(1-p)$, which is strictly positive. Therefore, the MGF is well-defined in a neighborhood of zero. By Theorem 5.8, all moments are finite. This justifies the following two approaches to compute the moments:

First approach (algebraic). Let us directly compute $\mathbb{E}[X]$ via sequence shifts:

$$\begin{aligned} p \mathbb{E}[X] &= \mathbb{E}[X] - (1-p) \mathbb{E}[X] = \sum_{x=1}^{\infty} xp(1-p)^{x-1} - \sum_{x=1}^{\infty} xp(1-p)^x \\ &= \sum_{x=1}^{\infty} xp(1-p)^{x-1} - \sum_{x=2}^{\infty} (x-1)p(1-p)^{x-1} \\ &= p(1-p)^0 + \sum_{x=2}^{\infty} p(1-p)^{x-1} = p \sum_{y=0}^{\infty} (1-p)^y = p \frac{1}{p} = 1. \end{aligned}$$

Hence $\mathbb{E}[X] = 1/p$. Similarly, we directly compute $\mathbb{E}[X^2]$:

$$\begin{aligned}
 p \mathbb{E}[X^2] &= \sum_{x=1}^{\infty} x^2 p (1-p)^{x-1} - \sum_{x=1}^{\infty} x^2 p (1-p)^x \\
 &= \sum_{x=1}^{\infty} x^2 p (1-p)^{x-1} - \sum_{x=2}^{\infty} (x-1)^2 p (1-p)^{x-1} \\
 &= p + \sum_{x=2}^{\infty} (2x-1) p (1-p)^{x-1} \\
 &= p + 2 \sum_{x=2}^{\infty} x p (1-p)^{x-1} - \sum_{x=2}^{\infty} p (1-p)^{x-1} \\
 &= p + 2(\mathbb{E}[X] - p) - (1-p) = 2\mathbb{E}[X] - 1 = \frac{2-p}{p}.
 \end{aligned}$$

Therefore, $\mathbb{E}[X^2] = (2-p)/p^2$, and

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}.$$

Second approach (differentiating the MGF). We differentiate the MGF at zero:

$$\begin{aligned}
 M'_X(t) &= \frac{pe^t(1 - (1-p)e^t) - pe^t(-(1-p)e^t)}{(1 - (1-p)e^t)^2} = \frac{pe^t}{(1 - (1-p)e^t)^2}, \\
 \mathbb{E}[X] &= M'_X(0) = \frac{p}{(1 - (1-p))^2} = \frac{1}{p}, \\
 M''_X(t) &= \frac{pe^t(1 - (1-p)e^t)^2 - pe^t \cdot 2(1 - (1-p)e^t)(-(1-p)e^t)}{(1 - (1-p)e^t)^4} = \frac{pe^t(1 + (1-p)e^t)}{(1 - (1-p)e^t)^3}, \\
 \mathbb{E}[X^2] &= M''_X(0) = \frac{p(1 + (1-p))}{p^3} = \frac{2-p}{p^2}, \\
 \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}.
 \end{aligned}$$

□

Theorem 8.7 (Lack of memory of the geometric distribution). Let $X \sim \text{Geom}(p)$. Then for all $m, n \in \mathbb{N}$,

$$\Pr(X > m+n \mid X > m) = \Pr(X > n).$$

Proof.

We compute directly using the CDF evaluated at integers:

$$\begin{aligned}
 \Pr(X > m+n \mid X > m) &= \frac{\Pr(X > m+n)}{\Pr(X > m)} = \frac{1 - F_X(m+n)}{1 - F_X(m)} \\
 &= \frac{(1-p)^{m+n}}{(1-p)^m} = (1-p)^n = 1 - F_X(n) = \Pr(X > n).
 \end{aligned}$$

□

8.1.5 Negative binomial distribution

Consider an infinite sequence of independent Bernoulli trials. The negative binomial law generalizes the geometric distribution by counting the number of trials before exactly r successes are observed. If $p \in (0, 1)$ is the probability of success for a Bernoulli trial, the probability that the r -th success occurs precisely at the x -th trial, with $x \geq r \geq 1$, is

$$p_X(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \dots$$

In this PMF, the term $p^r(1-p)^{x-r}$ represents the probability of r independent successes and $x-r$ independent failures in a fixed sequence. We multiply this quantity by the number of admissible reorderings. Since the x -th trial is strictly required to be a success, we can permute only the roles of the first $r-1$ successes within the preceding $x-1$ trials. Therefore, the combinatorial factor is $\binom{x-1}{r-1}$.

We expect by construction that $\Pr(X \in \{r, r+1, \dots\}) = 1$. This is proven analytically:

$$\begin{aligned} \sum_{x=r}^{\infty} p_X(x) &= \sum_{x=r}^{\infty} \binom{x-1}{r-1} p^r (1-p)^{x-r} = p^r \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} (1-p)^k \\ &\stackrel{+}{=} p^r (1 - (1-p))^{-r} = p^r p^{-r} = 1, \end{aligned}$$

where $(+)$ corresponds to the negative binomial identity (5.1):

$$(1-x)^{-r} = \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} x^k. \quad (8.1)$$

This construction immediately suggests the following representation, which we record for later use.

Lemma 8.8 (Negative binomial distribution as a sum of i.i.d. geometric). *Let $X_1, \dots, X_r$ be i.i.d. random variables with $X_i \sim \text{Geom}(p)$ and set $S_r = \sum_{i=1}^r X_i$. Then $S_r \sim \text{NegBin}(r, p)$.*

Proof.

First approach (probabilistic). By definition, a $\text{NegBin}(r, p)$ random variable models the total number of trials required to obtain exactly r successes. If we define Y_i as the number of trials following the $(i-1)$ -th success up to and including the i -th success (with Y_1 being the number of trials up to the first success), then $Y_1, \dots, Y_r$ are inherently i.i.d. $\text{Geom}(p)$ random variables. Their sum $Y = \sum_{i=1}^r Y_i$ counts the total number of trials required to reach r successes, meaning $Y \sim \text{NegBin}(r, p)$. Thus, $S_r \sim \text{NegBin}(r, p)$.

Second approach (analytical). We proceed via MGFs. By Proposition 5.9, the MGF of the sum of independent random variables is:

$$M_{S_r}(t) = \prod_{i=1}^r M_{X_i}(t) = \prod_{i=1}^r \frac{pe^t}{1 - (1-p)e^t} = \left(\frac{pe^t}{1 - (1-p)e^t} \right)^r.$$

On the other hand, the MGF of a standard random variable $X \sim \text{NegBin}(r, p)$ is computed directly using the negative binomial identity (8.1):

$$\begin{aligned} M_X(t) &= \sum_{x=r}^{\infty} e^{tx} \binom{x-1}{r-1} p^r (1-p)^{x-r} = p^r e^{tr} \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} (e^t(1-p))^k \\ &= p^r e^{tr} [1 - (1-p)e^t]^{-r} = \left(\frac{pe^t}{1 - (1-p)e^t} \right)^r. \end{aligned}$$

Since $M_{S_r}(t) = M_X(t)$ for all $t < -\log(1-p)$, Corollary 5.23 guarantees that $S_r \sim \text{NegBin}(r, p)$. $\square$

Theorem 8.9 (Properties of the negative binomial distribution). *Let $X \sim \text{NegBin}(r, p)$. Then X has the following properties:*

$$\begin{aligned} p_X(x) &= \binom{x-1}{r-1} p^r (1-p)^{x-r} \mathbf{1}_{\{r, r+1, \dots\}}(x), \\ \mathbb{E}[X] &= \frac{r}{p}, \quad \text{Var}(X) = \frac{r(1-p)}{p^2}, \quad M_X(t) = \left(\frac{pe^t}{1 - (1-p)e^t} \right)^r \quad (t < -\log(1-p)). \end{aligned}$$

The CDF F_X does not admit a simple closed form.

Proof.

The PMF p_X is given by definition. The MGF was explicitly derived in the analytical proof of Lemma 8.8. The expectation and variance could, in principle, be computed directly, just as in the case of the geometric distribution, or by differentiating $M_X(t)$ at $t = 0$. However, the most elegant method exploits Lemma 8.8 directly. By writing $X = X_1 + \dots + X_r$, where $X_1, \dots, X_r$ are i.i.d. random variables with

$X_i \sim \text{Geom}(p)$, we obtain via linearity and independence:

$$\begin{aligned}\mathbb{E}[X] &= \sum_{i=1}^r \mathbb{E}[X_i] = r \frac{1}{p} = \frac{r}{p}, \\ \text{Var}(X) &= \sum_{i=1}^r \text{Var}(X_i) = r \frac{1-p}{p^2} = \frac{r(1-p)}{p^2}.\end{aligned}$$

□

Proposition 8.10 (Additivity of the negative binomial distribution). *Let $X \sim \text{NegBin}(r_1, p)$ and $Y \sim \text{NegBin}(r_2, p)$ be independent. Then*

$$X + Y \sim \text{NegBin}(r_1 + r_2, p).$$

Proof.

First approach (MGF). By Proposition 5.9, the MGF of $X + Y$ is given by

$$M_{X+Y}(t) = M_X(t)M_Y(t) = \left(\frac{pe^t}{1 - (1-p)e^t} \right)^{r_1} \left(\frac{pe^t}{1 - (1-p)e^t} \right)^{r_2} = \left(\frac{pe^t}{1 - (1-p)e^t} \right)^{r_1+r_2},$$

which coincides with the MGF of $Z \sim \text{NegBin}(r_1 + r_2, p)$. By uniqueness of the MGF (Corollary 5.23), we conclude that $X + Y \stackrel{d}{=} Z$.

Second approach (probabilistic). By definition, X counts the number of trials before exactly r_1 successes are observed. Similarly, Y counts the number of trials before exactly r_2 successes are observed. By independence, $X + Y$ represents the total number of trials required to reach r_1 successes, and then a further r_2 successes. Together, this is the time required for $r_1 + r_2$ successes, hence $X + Y \sim \text{NegBin}(r_1 + r_2, p)$.

Third approach (convolution). By Corollary 3.81, the PMF p_{X+Y} is the discrete convolution of p_X and p_Y . For every $x \in \{r_1 + r_2, r_1 + r_2 + 1, \dots\}$,

$$\begin{aligned}p_{X+Y}(x) &= \sum_{y=r_1}^{x-r_2} \Pr(X = y) \Pr(Y = x - y) \\ &= \sum_{y=r_1}^{x-r_2} \left(\binom{y-1}{r_1-1} p^{r_1} (1-p)^{y-r_1} \right) \left(\binom{x-y-1}{r_2-1} p^{r_2} (1-p)^{x-y-r_2} \right) \\ &= p^{r_1+r_2} (1-p)^{x-(r_1+r_2)} \sum_{y=r_1}^{x-r_2} \binom{y-1}{r_1-1} \binom{x-y-1}{r_2-1} \\ &= \binom{x-1}{r_1+r_2-1} p^{r_1+r_2} (1-p)^{x-(r_1+r_2)}.\end{aligned}$$

The last equality follows from the following shifted form of Vandermonde's identity:^a

$$\binom{n}{a+b+1} = \sum_{k=a}^{n-b-1} \binom{k}{a} \binom{n-k-1}{b}.$$

The conclusion follows by observing that the PMF of $X + Y$ coincides exactly with that of $Z \sim \text{NegBin}(r_1 + r_2, p)$. □

^aThe best proof I am aware of this identity is combinatorial. Indeed, $\binom{n}{a+b+1}$ counts the number of subsets of $\{1, \dots, n\}$ with cardinality $a + b + 1$. Given such a subset, let $k + 1$ be its $(a + 1)$ -st smallest element. Then necessarily $k \in \{a, \dots, n - b - 1\}$. For a fixed k , we must choose a elements from $\{1, \dots, k\}$ and b elements from $\{k + 2, \dots, n\}$, which gives $\binom{k}{a} \binom{n-k-1}{b}$. Summing over all admissible k yields the identity. Alternatively, one can prove it with generating functions, in a flavour similar to Example 5.3. However, this method is structurally equivalent to the first approach (MGF) presented here.

8.1.6 Poisson distribution

Recall that the binomial random variable $\text{Binom}(n, p)$ counts the number of successes in n independent Bernoulli trials. Assuming that trials are performed at discrete times $t = 1, \dots, n$, the binomial random variable counts the number of successes in the time window $[0, n]$. Imagine now performing $2n$ trials at times $t = \frac{1}{2}, 1, \frac{3}{2}, 2, \dots, n$. The distribution is given by $\text{Binom}(2n, p/2)$. Similarly, performing $3n$ trials corresponds to $\text{Binom}(3n, p/3)$. Continuing this refinement process, we create a limiting regime in which an unlikely Bernoulli event can occur at any given continuous time $t \in (0, n]$.

This construction immediately suggests the following limiting distribution, parametrized by a rate parameter $\lambda > 0$ representing the expected number of occurrences:

Definition 8.11 (Poisson distribution). Let $(X_k)_{k \in \mathbb{N}}$ be a sequence of discrete random variables such that, for every $k \in \mathbb{N}$, $X_k \sim \text{Binom}(kn, p/k)$. Let $\lambda := np$. The random variable X defined by the convergence in distribution

$$X_k \xrightarrow{d} X$$

is said to have a $\text{Pois}(\lambda)$ distribution.

By definition, the Poisson distribution models the number of successes in a given time window, under the assumption that the probability of an event occurring at any given instant is independent of time and memoryless.

Theorem 8.12 (Poisson distribution). $X \sim \text{Pois}(\lambda)$ has the following pmf:

$$p_X(x) = \frac{\lambda^x}{x!} e^{-\lambda} \mathbf{1}_{\mathbb{N}}(x).$$

Proof.

First Approach (limit of PMFs). Let $(X_k)_{k \in \mathbb{N}}$ be a sequence of random variables such that, for every $k \in \mathbb{N}$, $X_k \sim \text{Binom}(kn, p/k)$. Set $kn := N$ and $pn := \lambda$. Then $(X_k)_{k \in \mathbb{N}}$ can be reparametrized as $(X_N)_{N \in \mathbb{N}}$ with distribution $X_N \sim \text{Binom}(N, \lambda/N)$. Again, $X_N \xrightarrow{d} X$. Then

$$\begin{aligned} p_{X_N}(x) &= \binom{N}{x} (\lambda/N)^x (1 - \lambda/N)^{N-x} = \frac{N(N-1) \cdots (N-x+1)}{x!} \frac{\lambda^x}{N^x} \left(1 - \frac{\lambda}{N}\right)^N \left(1 - \frac{\lambda}{N}\right)^{-x} \\ &= \frac{\lambda^x}{x!} \frac{N(N-1) \cdots (N-x+1)}{N^x} \left(1 - \frac{\lambda}{N}\right)^N \left(1 - \frac{\lambda}{N}\right)^{-x}. \end{aligned}$$

By Theorem 6.32 with support $S = \mathbb{N}$, $p_X(x)$ is the pointwise limit of p_{X_N} . Hence

$$p_X(x) = \lim_{N \rightarrow \infty} p_{X_N}(x) = \frac{\lambda^x}{x!} \lim_{N \rightarrow \infty} \underbrace{\frac{N(N-1) \cdots (N-x+1)}{N^x}}_{\rightarrow 1} \underbrace{\left(1 - \frac{\lambda}{N}\right)^N}_{\rightarrow e^{-\lambda}} \underbrace{\left(1 - \frac{\lambda}{N}\right)^{-x}}_{\rightarrow 1} = \frac{\lambda^x}{x!} e^{-\lambda}.$$

Moreover, the support naturally becomes $\mathbb{N}$.

Second Approach (characteristic functions). If the previous computations look suspicious (since, in many courses, they are not formalized through Theorem 6.32), we can use the following method. The characteristic function φ_n of the random variable with distribution $\text{Binom}(n, \lambda/n)$ is given by

$$\varphi_n(t) = \mathbb{E}[e^{-itX}] = \sum_{x=0}^n e^{-itx} \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} = \left(\frac{\lambda \cdot e^{-it}}{n} + 1 - \frac{\lambda}{n}\right)^n = \left(1 + \frac{\lambda(e^{-it} - 1)}{n}\right)^n.$$

Letting $n \rightarrow \infty$, we deduce

$$\varphi(t) = e^{\lambda(e^{-it} - 1)}.$$

Since φ_X is continuous at 0, by Lévy's continuity theorem 6.29 it corresponds to a valid random variable X . Since X is supported in $\mathbb{N}$, Example 5.18 applies, and we recover the PMF via the inverse Fourier series:

$$\begin{aligned} p_X(k) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \varphi_X(t) e^{itk} dt = \frac{e^{-\lambda}}{2\pi} \int_{-\pi}^{\pi} e^{itk} e^{\lambda e^{-it}} dt \\ &= \frac{e^{-\lambda}}{2\pi} \sum_{m=0}^{\infty} \frac{\lambda^m}{m!} \int_{-\pi}^{\pi} e^{it(k-m)} dt = \frac{e^{-\lambda}}{2\pi} \cdot \frac{\lambda^k}{k!} \cdot 2\pi = \frac{\lambda^k}{k!} e^{-\lambda}, \end{aligned}$$

where the swap of integral and series is formalized through the Dominated Convergence Theorem 1.80. We expect by construction that $\Pr(X \in \mathbb{N}) = 1$. This is easily verified via the exponential series:

$$\sum_{x=0}^{\infty} p_X(x) = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} e^{-\lambda} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1.$$

□

Theorem 8.13 (Properties of the Poisson distribution). *Let $X \sim \text{Pois}(\lambda)$. Then X has the following properties:*

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda, \quad M_X(t) = e^{\lambda(e^t - 1)} \quad (t \in \mathbb{R}).$$

The CDF F_X does not admit a simple closed form.

Proof.

First approach (algebraic moments). Let us directly compute:

$$\begin{aligned} \mathbb{E}[X] &= \sum_{x=0}^{\infty} x \frac{\lambda^x}{x!} e^{-\lambda} = \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} e^{-\lambda} = \frac{\lambda}{e^{\lambda}} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = \cancel{\frac{\lambda}{e^{\lambda}}} = \lambda \\ \mathbb{E}[X^2] &= \sum_{x=0}^{\infty} x^2 \frac{\lambda^x}{x!} e^{-\lambda} = \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} e^{-\lambda} (x-1+1) = \sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!} e^{-\lambda} + \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} e^{-\lambda} \\ &= \frac{\lambda^2}{e^{\lambda}} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} + \frac{\lambda}{e^{\lambda}} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = \lambda^2 + \lambda. \end{aligned}$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \lambda + \lambda^2 - \lambda^2 = \lambda.$$

The MGF is computed directly as:

$$M_X(t) = \sum_{x=0}^{\infty} e^{tx} \frac{\lambda^x}{x!} e^{-\lambda} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}.$$

Second approach (differentiating the MGF). We compute the moments by differentiating the MGF at zero:

$$\begin{aligned} \mathbb{E}[X] &= M'_X(0) = \lambda e^0 e^{\lambda(e^0 - 1)} = \lambda, \\ \mathbb{E}[X^2] &= M''_X(0) = \lambda e^0 e^{\lambda(e^0 - 1)} + \lambda^2 (e^0)^2 e^{\lambda(e^0 - 1)} = \lambda + \lambda^2, \\ \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \lambda + \lambda^2 - \lambda^2 = \lambda. \end{aligned}$$

Third Approach (limit of moments). We employ the definition of the Poisson distribution as the limit of the distribution $X_n \sim \text{Binom}(n, \lambda/n)$. Since the first and second moments of a binomial distribution with parameters (n, p) are $\mathbb{E}[X] = np$ and $\text{Var}(X) = np(1-p)$, then

$$\mathbb{E}[X_n] = n \cdot \frac{\lambda}{n} = \lambda \quad \text{and} \quad \text{Var}(X_n) = n \cdot \frac{\lambda}{n} \left(1 - \frac{\lambda}{n}\right) = \lambda - \frac{\lambda^2}{n}.$$

We are tempted to take the limit as $n \rightarrow \infty$ of $\mathbb{E}[X_n]$ and $\text{Var}(X_n)$, which would yield the correct results $\mathbb{E}[X] = \lim_{n \rightarrow \infty} \lambda = \lambda$ and $\text{Var}(X) = \lim_{n \rightarrow \infty} \lambda - \frac{\lambda^2}{n} = \lambda$. To formally justify this limit procedure, we can use Theorem 6.39. We only need to show that $(|X_n|)_{n \in \mathbb{N}}$ and $(|X_n|^2)_{n \in \mathbb{N}}$ are uniformly integrable. By Lemma 6.35, it suffices to show that $\mathbb{E}[X_n^3] < \infty$ for every $n \in \mathbb{N}$. By Theorem 8.4, the MGF of X_n writes as

$$M_{X_n}(t) = \left(1 - \frac{\lambda}{n} + \frac{\lambda}{n} e^t\right)^n = \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \leq e^{\lambda(e^t - 1)} < \infty \quad \forall t \in \mathbb{R}. \quad (8.2)$$

Since M_{X_n} is finite in a neighbourhood of 0, by Theorem 5.8, $\mathbb{E}[X_n^p] < \infty$ for every $p \in \mathbb{N}$ and every $n \in \mathbb{N}$. It follows that Theorem 6.39 can be applied.

Observe that (8.2) gives us the stronger result that $M_{X_n}(t) < \infty$ for every $t \in \mathbb{R}$ and $M_{X_n}(t) \rightarrow e^{\lambda(1-e^t)}$ for every $t \in \mathbb{R}$. By Curtiss Theorem 6.30, we conclude that $M_X(t) = e^{\lambda(1-e^t)}$. □

Proposition 8.14 (Additivity of the Poisson distribution). *Let $X \sim \text{Pois}(\lambda_1)$ and $Y \sim \text{Pois}(\lambda_2)$ be independent. Then*

$$X + Y \sim \text{Pois}(\lambda_1 + \lambda_2).$$

Proof.

First approach (MGF). By Proposition 5.9, the MGF of $X + Y$ is given by

$$M_{X+Y}(t) = M_X(t)M_Y(t) = e^{\lambda_1(e^t-1)}e^{\lambda_2(e^t-1)} = e^{(\lambda_1+\lambda_2)(e^t-1)},$$

which coincides with the MGF of $Z \sim \text{Pois}(\lambda_1 + \lambda_2)$. By uniqueness of the MGF (Corollary 5.23), we conclude that $X + Y \stackrel{d}{=} Z$.

Second approach (probabilistic). By definition, X counts the number of occurrences of a phenomenon assuming the average rate is λ_1 . Similarly, Y counts occurrences with an average rate λ_2 . By independence, $X + Y$ counts the total number of occurrences of both independent phenomena. Hence, the combined average rate is $\lambda_1 + \lambda_2$, leading to $X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$.

Third approach (convolution). By Corollary 3.81, the PMF p_{X+Y} is the discrete convolution of p_X and p_Y . For every $x \in \mathbb{N}$,

$$\begin{aligned} p_{X+Y}(x) &= \sum_{y=0}^x \Pr(X = y) \Pr(Y = x - y) = \sum_{y=0}^x \left(e^{-\lambda_1} \frac{\lambda_1^y}{y!} \right) \left(e^{-\lambda_2} \frac{\lambda_2^{x-y}}{(x-y)!} \right) \\ &= e^{-(\lambda_1+\lambda_2)} \sum_{y=0}^x \frac{\lambda_1^y \lambda_2^{x-y}}{y!(x-y)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{x!} \sum_{y=0}^x \binom{x}{y} \lambda_1^y \lambda_2^{x-y} \\ &= \frac{(\lambda_1 + \lambda_2)^x}{x!} e^{-(\lambda_1+\lambda_2)}. \end{aligned}$$

The conclusion follows by observing that the PMF of $X + Y$ coincides with that of $Z \sim \text{Pois}(\lambda_1 + \lambda_2)$. $\square$

8.2 Continuous Distributions

8.2.1 Uniform distribution

If the discrete uniform distribution is the simplest discrete model, the continuous uniform distribution is its continuous counterpart. Given a domain $\mathcal{D} \subseteq \mathbb{R}^d$, a uniform distribution is characterized by a constant density function. Normalization then implies

$$f(x) = \frac{1}{\text{Vol}(\mathcal{D})} \mathbf{1}_{\mathcal{D}}(x),$$

so that $\int_{\mathbb{R}^d} f d\lambda^d = 1$. For most scalar applications, $\mathcal{D} = [a, b]$. Therefore,

$$f(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x).$$

It models complete ignorance over an interval and serves as the fundamental building block for numerically generating any other continuous distribution (through Lemma 3.64). We denote by $\mathcal{U}([a, b])$ the uniform distribution over $[a, b]$.

Theorem 8.15 (Properties of the uniform distribution). *Let $X \sim \mathcal{U}([a, b])$. Then X has the following properties:*

$$\begin{aligned} f_X(x) &= \frac{1}{b-a} \mathbf{1}_{[a,b]}(x), & F_X(x) &= \frac{x-a}{b-a} \mathbf{1}_{[a,b]}(x) + \mathbf{1}_{(b,\infty)}(x), \\ \mathbb{E}[X] &= \frac{a+b}{2}, & \text{Var}(X) &= \frac{(b-a)^2}{12}, & M_X(t) &= \frac{e^{tb} - e^{ta}}{t(b-a)} \quad (t \neq 0). \end{aligned}$$

Proof.

The PDF f_X is given by definition. From there, we compute the CDF, the moments and the MGF:

$$\begin{aligned}
 F_X(x) &= \int_a^x \frac{1}{b-a} du = \frac{x-a}{b-a} \quad \text{for } x \in [a, b], \\
 \mathbb{E}[X] &= \int_a^b \frac{x}{b-a} dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}, \\
 \mathbb{E}[X^2] &= \int_a^b \frac{x^2}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}, \\
 \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{a^2 + ab + b^2}{3} - \frac{a^2 + 2ab + b^2}{4} = \frac{(b-a)^2}{12}, \\
 M_X(t) &= \mathbb{E}[e^{tX}] = \int_a^b \frac{e^{tx}}{b-a} dx = \frac{e^{tb} - e^{ta}}{t(b-a)}.
 \end{aligned}$$

□

Proposition 8.16 (Continuous uniform distribution as a limit of discrete uniform distributions).

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of discrete random variables such that $X_n \sim \text{Unif}(n)$. Then

$$\frac{X_n}{n} \xrightarrow{d} U \sim \mathcal{U}([0, 1]).$$

Proof.

First approach (CDFs). By definition if $U \sim \mathcal{U}([0, 1])$ then

$$F_U(x) = x \mathbf{1}_{[0,1]}(x) + \mathbf{1}_{(1,\infty)}(x),$$

which is continuous everywhere. We compute the CDF of the scaled variable X_n/n . For any $x \in [0, 1]$, we have:

$$\lim_{n \rightarrow \infty} F_{X_n/n}(x) = \lim_{n \rightarrow \infty} \Pr\left(\frac{X_n}{n} \leq x\right) = \lim_{n \rightarrow \infty} \Pr(X_n \leq nx) = \lim_{n \rightarrow \infty} \frac{\lfloor nx \rfloor}{n} \stackrel{+}{=} x,$$

where $(+)$ follows since

$$nx - 1 < \lfloor nx \rfloor \leq nx \iff x - \frac{1}{n} < F_{X_n/n}(x) \leq x \implies \lim_{n \rightarrow \infty} x - \frac{1}{n} \leq F_{X_n/n}(x) \leq x.$$

For $x < 0$, $F_{X_n/n}(x) = 0 \rightarrow 0$, and for $x \geq 1$, $F_{X_n/n}(x) = 1 \rightarrow 1$. Thus, for every $x \in \mathbb{R}$:

$$\lim_{n \rightarrow \infty} F_{X_n/n}(x) = x \mathbf{1}_{[0,1]}(x) + \mathbf{1}_{(1,\infty)}(x),$$

which is exactly the CDF of $\mathcal{U}([0, 1])$. Since the limiting CDF is continuous everywhere, the sequence converges in distribution to $U \sim \mathcal{U}([0, 1])$.

Second approach (characteristic functions). We compute the characteristic functions of U and of the scaled variable X_n/n :

$$\begin{aligned}
 \varphi_U(t) &= \int_0^1 e^{-itx} dx = \left[\frac{e^{-itx}}{-it} \right]_0^1 = \frac{e^{-it} - 1}{-it} = \frac{1 - e^{-it}}{it}. \\
 \varphi_{X_n/n}(t) &= \mathbb{E}\left[e^{-it \frac{X_n}{n}}\right] = \sum_{k=1}^n e^{-it \frac{k}{n}} \Pr(X_n = k) = \sum_{k=1}^n e^{-it \frac{k}{n}} \frac{1}{n}.
 \end{aligned}$$

We immediately recognize this last expression as the right Riemann sum for the function $g(x) = e^{-itx}$ over the interval $[0, 1]$ with uniform partition size $\Delta x = 1/n$. Since g is continuous, the Riemann sum converges to the Riemann integral as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \varphi_{X_n/n}(t) = \lim_{n \rightarrow \infty} \sum_{k=1}^n e^{-it \frac{k}{n}} \frac{1}{n} = \int_0^1 e^{-itx} dx = \varphi_U(t).$$

By Lévy's continuity theorem (6.29), the sequence converges in distribution to U .

Third approach (Portmanteau). By Theorem 6.21, to prove $X_n/n \xrightarrow{d} U$, it suffices to show that $\mathbb{E}[f(X_n/n)] \rightarrow \mathbb{E}[f(U)]$ for every bounded continuous function $f \in C_b(\mathbb{R})$. Given $f \in C_b(\mathbb{R})$, we compute:

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[f \left(\frac{X_n}{n} \right) \right] = \lim_{n \rightarrow \infty} \sum_{k=1}^n f \left(\frac{k}{n} \right) \Pr(X_n = k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n f \left(\frac{k}{n} \right) \frac{1}{n} \stackrel{\dagger}{=} \int_0^1 f(x) dx = \mathbb{E}[f(U)],$$

where $\dagger$ follows by continuity of f on the compact interval $[0, 1]$, recognizing the limit of the sum as a Riemann integral of a continuous function on a compact subset of $\mathbb{R}$. $\square$

8.2.2 Exponential distribution

Recall that the Poisson distribution was conceptualized as a continuous-time limit of binomial processes. The exponential distribution is similar in spirit, emerging as the continuous limit of the geometric distribution. While the geometric distribution counts the discrete number of trials before the first success, the exponential distribution models the continuous waiting time before the first success occurs.

Consider a time interval $[0, +\infty)$ and imagine discretizing it into small "bins" of length Δt . If an event has a small probability of falling into any given bin, the waiting time for the first event is described discretely by a geometric distribution (counting the number of empty bins before the first success). By letting the bin size $\Delta t \rightarrow 0$ while keeping the average rate of events constant, the discrete "steps" of the geometric distribution smooth out into the continuous curve of the exponential distribution. Here, $\lambda > 0$ is a rate parameter (larger λ means shorter waiting times).

Definition 8.17 (Exponential distribution). Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of discrete random variables such that $X_n \sim \text{Geom}(\lambda/n)$ for n such that $\lambda/n < 1$. The continuous random variable X defined by the convergence in distribution

$$\frac{X_n}{n} \xrightarrow{d} X$$

is said to have an $\text{Exp}(\lambda)$ distribution.

The random variable X_n/n models the number of trials needed when n trials are performed in a unit of time and the probability that a trial is positive is λ/n . At the limit, this yields a continuous memoryless waiting time.

Theorem 8.18 (Density of the exponential distribution). Let $X \sim \text{Exp}(\lambda)$. Then X has the following PDF:

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{(0, \infty)}(x).$$

Proof.

By definition, $F_{X_n/n}(t) \rightarrow F_X(t)$ for every t which is a point of continuity of F_X . We compute:

$$\lim_{n \rightarrow \infty} F_{X_n/n}(t) = \lim_{n \rightarrow \infty} \Pr \left(\frac{X_n}{n} \leq t \right) = \lim_{n \rightarrow \infty} F_{X_n}(nt) = \lim_{n \rightarrow \infty} 1 - \left(1 - \frac{\lambda}{n} \right)^{\lfloor nt \rfloor},$$

where the last passage holds whenever $nt \geq 1$ in the large n limit, which is true as long as $t > 0$. Hence, we deduce that $\text{Exp}(\lambda)$ has support $(0, \infty)$.

To handle the floor function, recall that $nt - 1 < \lfloor nt \rfloor \leq nt$. Therefore, we can write the exponent as $\lfloor nt \rfloor = nt - \varepsilon_n$ for some $0 \leq \varepsilon_n < 1$. Substituting this into the limit yields:

$$\begin{aligned} F_X(t) &= \lim_{n \rightarrow \infty} 1 - \left(1 - \frac{\lambda}{n} \right)^{\lfloor nt \rfloor} = \lim_{n \rightarrow \infty} 1 - \left(1 - \frac{\lambda}{n} \right)^{nt - \varepsilon_n} \\ &= \lim_{n \rightarrow \infty} 1 - \left(1 - \frac{\lambda t}{nt} \right)^{nt} \cdot \left(1 - \frac{\lambda}{n} \right)^{-\varepsilon_n} = \lim_{n \rightarrow \infty} 1 - e^{-\lambda t} \cdot 1, \end{aligned}$$

where $(1 - \lambda/n)^{-\varepsilon_n} \rightarrow 1$ since $0 \leq \varepsilon < 1$. We observe that F_X has no points of discontinuity. Therefore, there is no $t \in \mathbb{R}$ such that $\Pr(X = t) > 0$. By Proposition 3.58, the PDF of the Exponential distribution is given by

$$f_X(t) = F'_X(t) = \frac{d}{dt}(1 - e^{-\lambda t}) = \lambda e^{-\lambda t}, \quad t > 0.$$

$\square$

We expect by construction that $\Pr(X > 0) = 1$. This is easily verified:

$$\int_0^\infty \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_0^\infty = 1.$$

Theorem 8.19 (Properties of the exponential distribution). *Let $X \sim \text{Exp}(\lambda)$. Then X has the following properties:*

$$\begin{aligned} f_X(x) &= \lambda e^{-\lambda x} \mathbf{1}_{(0,\infty)}(x), & F_X(x) &= (1 - e^{-\lambda x}) \mathbf{1}_{(0,\infty)}(x), \\ \mathbb{E}[X] &= \frac{1}{\lambda}, & \text{Var}(X) &= \frac{1}{\lambda^2}, & M_X(t) &= \frac{\lambda}{\lambda - t} \quad (t < \lambda). \end{aligned}$$

Proof.

The PDF was established previously. From there, we compute:

$$\begin{aligned} F(x) &= \int_0^x \lambda e^{-\lambda t} dt = -[e^{-\lambda t}]_0^x = 1 - e^{-\lambda x}. \\ \mathbb{E}[X] &= \int_0^{+\infty} x \lambda e^{-\lambda x} dx = [-x e^{-\lambda x}]_0^{+\infty} + \int_0^{+\infty} e^{-\lambda x} dx = [-\frac{1}{\lambda} e^{-\lambda x}]_0^{+\infty} = \frac{1}{\lambda}. \\ \mathbb{E}[X^2] &= \int_0^{+\infty} x^2 \lambda e^{-\lambda x} dx = [-x^2 e^{-\lambda x}]_0^{+\infty} + \int_0^{+\infty} 2x e^{-\lambda x} dx = \frac{2}{\lambda} \mathbb{E}[X] = \frac{2}{\lambda^2} \\ \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}^2[X] = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \\ M_X(t) &= \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx = \lambda \int_0^\infty e^{(t-\lambda)x} dx = \frac{\lambda}{t-\lambda} [e^{(t-\lambda)x}]_0^\infty = \frac{\lambda}{\lambda - t}, \end{aligned}$$

which is valid provided $t - \lambda < 0$. □

Theorem 8.20 (Moments of the exponential distribution). *Let $X \sim \text{Exp}(\lambda)$. Then for every $n \in \mathbb{N}$,*

$$\mathbb{E}[X^n] = \frac{n!}{\lambda^n}.$$

Proof.

First approach (direct integration). Let $I_n := \mathbb{E}[X^n] = \int_0^\infty x^n \lambda e^{-\lambda x} dx$. We evaluate the integral using integration by parts:

$$I_n = [-x^n e^{-\lambda x}]_0^\infty + \int_0^\infty n x^{n-1} e^{-\lambda x} dx = 0 + \frac{n}{\lambda} \int_0^\infty x^{n-1} \lambda e^{-\lambda x} dx = \frac{n}{\lambda} I_{n-1}.$$

Since the base case is $I_0 = \mathbb{E}[X^0] = \mathbb{E}[1] = 1$, recursively applying this relation yields exactly:

$$I_n = \frac{n}{\lambda} \cdot \frac{n-1}{\lambda} \cdots \frac{1}{\lambda} I_0 = \frac{n!}{\lambda^n}.$$

Second approach (MGF expansion). For an exponential distribution, the MGF is $M_X(t) = \frac{\lambda}{\lambda - t} = \frac{1}{1 - t/\lambda}$ for $t < \lambda$. Expanding this via the geometric series:

$$M_X(t) = \sum_{n=0}^\infty \frac{\mathbb{E}[X^n]}{n!} t^n = \frac{1}{1 - t/\lambda} = \sum_{n=0}^\infty \left(\frac{t}{\lambda}\right)^n = \sum_{n=0}^\infty \frac{n!}{\lambda^n} \frac{t^n}{n!}.$$

Matching the coefficients of t^n on both sides, we immediately obtain:

$$\frac{\mathbb{E}[X^n]}{n!} = \frac{1}{\lambda^n} \implies \mathbb{E}[X^n] = \frac{n!}{\lambda^n}.$$

□

Proposition 8.21 (Scaling of the exponential distribution). *Let $X \sim \text{Exp}(\lambda)$ and let $c > 0$. Then*

$$cX \sim \text{Exp}\left(\frac{\lambda}{c}\right).$$

Proof.

For $x \geq 0$, $F_{cX}(x) = \Pr(cX \leq x) = \Pr(X \leq x/c) = 1 - e^{-\lambda(x/c)} = 1 - e^{-(\lambda/c)x}$, which is the CDF of $\text{Exp}(\lambda/c)$. $\square$

Theorem 8.22 (Lack of memory of the exponential distribution). *Let $X \sim \text{Exp}(\lambda)$. Then for all $s, t \geq 0$,*

$$\Pr(X > s + t \mid X > s) = \Pr(X > t).$$

Proof.

We compute directly:

$$\begin{aligned} \Pr(X > s + t \mid X > s) &= \frac{\Pr(X > s + t)}{\Pr(X > s)} = \frac{1 - F_X(s + t)}{1 - F_X(s)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = 1 - F_X(t) = \Pr(X > t). \end{aligned}$$

$\square$

Proposition 8.23 (Exponential distribution via quantile transform). *Let $U \sim \mathcal{U}(0, 1)$. For any $\lambda > 0$*

$$X = -\frac{1}{\lambda} \log(U) \sim \text{Exp}(\lambda).$$

Proof.

Since the CDF of an exponential distribution is given by $F(x) = (1 - e^{-\lambda x})\mathbf{1}_{[0, \infty]}$, then $F^{-1}(u) = -\frac{1}{\lambda} \log(1 - u)$. By Lemma 3.64, it follows that

$$-\frac{1}{\lambda} \log(1 - U) \sim \text{Exp}(\lambda).$$

Since $1 - U \sim \mathcal{U}([0, 1])$, we conclude that

$$-\frac{1}{\lambda} \log(U) \sim \text{Exp}(\lambda).$$

$\square$

8.2.3 Normal distribution

The normal law (or Gaussian law) is the canonical model for aggregated noise and errors and it is arguably the most stable and important distribution (cf. introduction to Section 6.9). It is defined through the central limit theorem:

Definition 8.24 (Normal distribution). Let $(X_n)_{n \in \mathbb{N}} \subseteq L^2$ be a sequence of i.i.d. square-integrable random variables, with expectation μ and variance $\sigma^2 > 0$. Let $\bar{X}_n := (X_1 + \dots + X_n)/n$. The continuous random variable X defined by the convergence in distribution

$$\sqrt{n}(\bar{X}_n - \mu) + \mu \xrightarrow{d} X$$

is said to have a $\mathcal{N}(\mu, \sigma^2)$ distribution.

In other words, the normal law is defined as a universal fixed-point attractor for sums of random variables, regardless of the underlying distribution of $(X_n)_{n \in \mathbb{N}} \subseteq L^2$.

Theorem 8.25 (Density of the normal distribution). *Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then X has the following PDF:*

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Proof.

We know by the Central Limit Theorem 6.43 that

$$\lim_{n \rightarrow \infty} \varphi_{\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma}}(t) = e^{-t^2/2}.$$

By the affine properties of the characteristic function (5.21),

$$\lim_{n \rightarrow \infty} \varphi_{\sqrt{n}(\bar{X}_n - \mu) + \mu}(t) = \lim_{n \rightarrow \infty} e^{-it\mu} \varphi_{\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma}}(t\sigma) = e^{-it\mu} e^{-\frac{\sigma^2 t^2}{2}}. \quad (8.3)$$

Combining Lemma 2.58 and Example 5.19, we obtain the same result as in Example 5.20, that is, if Z is an absolutely continuous random variable with PDF

$$f_Z(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

then its CF is precisely:

$$\varphi_Z(t) = e^{-it\mu} e^{-\frac{\sigma^2 t^2}{2}}. \quad (8.4)$$

Since (8.3) and (8.4) match, the uniqueness of the characteristic function (Theorem 5.22) concludes that $X \stackrel{d}{=} Z$. $\square$

Theorem 8.26 (Properties of the normal distribution). *Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then X has the following properties:*

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, & F_X(x) &= \Phi\left(\frac{x-\mu}{\sigma}\right), \\ \mathbb{E}[X] &= \mu, & \text{Var}(X) &= \sigma^2, & M_X(t) &= e^{\mu t + \frac{\sigma^2 t^2}{2}} \quad (t \in \mathbb{R}), \end{aligned}$$

where $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$ is the standard normal CDF.

Proof.

The PDF formula was established previously. From there, we compute the CDF:

$$\begin{aligned} F(x) &= \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{(t-\mu)^2}{\sigma^2}} dt \stackrel{u=t-\mu}{=} \int_{-\infty}^{x-\mu} \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{u^2}{\sigma^2}} du \\ &\stackrel{v=u/\sigma}{=} \int_{-\infty}^{\frac{x-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv = \Phi\left(\frac{x-\mu}{\sigma}\right) \end{aligned}$$

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} \frac{x}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx = \int_{-\infty}^{+\infty} \frac{x-\mu}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx + \underbrace{\mu \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx}_{=1}$$

$$= \mu - \sqrt{\frac{\sigma}{2\pi}} \int_{-\infty}^{+\infty} -\frac{x-\mu}{\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx = \mu - \sqrt{\frac{\sigma}{2\pi}} \left[\exp\left[-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}\right] \right]_{-\infty}^{+\infty} = \mu$$

$$\begin{aligned} \text{Var}(X) &= \int_{-\infty}^{+\infty} (x-\mu)^2 \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx \stackrel{t=\frac{x-\mu}{\sigma}}{=} \sigma^2 \int_{-\infty}^{+\infty} t \cdot \frac{t}{\sqrt{2\pi}} e^{-t^2/2} dt \\ &= \sigma^2 \left(\left[t \int \frac{t}{\sqrt{2\pi}} e^{-t^2/2} dt \right]_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \left[\int \frac{t}{\sqrt{2\pi}} e^{-t^2/2} dt \right] dt \right) \\ &= \sigma^2 \left(\underbrace{\left[-\frac{t}{\sqrt{2\pi}} e^{-t^2/2} \right]_{-\infty}^{+\infty}}_{=0} + \underbrace{\int_{-\infty}^{+\infty} \frac{e^{-t^2/2}}{\sqrt{2\pi}} dt}_{=1} \right) = \sigma^2. \end{aligned}$$

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{+\infty} e^{tx} \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{x^2}{\sigma^2} + \frac{x}{\sigma^2} (2\mu + 2\sigma^2 t) - \frac{\mu^2}{2\sigma^2}} dx = \\ &= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2\sigma^2} (x - (\mu + \sigma^2 t))^2 + \mu t + \frac{\sigma^2 t^2}{2}} dx = e^{\mu t + \frac{\sigma^2 t^2}{2}}. \end{aligned}$$

Alternatively, we could have computed the MGF from the characteristic function in Equation (8.4) as

$$M_X(t) = \varphi(it) = e^{-i\mu(it)} e^{-\frac{\sigma^2(it)^2}{2}} = e^{\mu t + \frac{\sigma^2 t^2}{2}},$$

which is legitimate because both functions are holomorphic on the complex plane. Then moments are given by differentiating the MGF:

$$\begin{aligned} M'_X(t) &= (\mu + \sigma^2 t) e^{\mu t + \frac{\sigma^2 t^2}{2}} \\ \mathbb{E}[X] &= M'_X(0) = \mu \\ M''_X(t) &= (\mu + \sigma^2 t)^2 e^{\mu t + \frac{\sigma^2 t^2}{2}} + \sigma^2 e^{\mu t + \frac{\sigma^2 t^2}{2}} \\ \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = M''_X(0) - \mu^2 = \mu^2 + \sigma^2 - \mu^2 = \sigma^2. \end{aligned}$$

□

The normal PDF is symmetric around $x = \mu$. The parameter μ is a location parameter, translating the distribution. The parameter σ^2 is a scale parameter: smaller values of σ^2 correspond to more peaked densities.

Theorem 8.27 (Moments of the standard normal distribution). *Let $Z \sim \mathcal{N}(0, 1)$. Then for every $n \in \mathbb{N}$,*

$$\mathbb{E}[Z^{2n}] = (2n - 1)!!, \quad \mathbb{E}[Z^{2n+1}] = 0,$$

where $(2n - 1)!! := 1 \cdot 3 \cdot 5 \cdots (2n - 1)$.

Proof.

For a standard Gaussian, $M_Z(t) = e^{t^2/2}$. Expanding via Taylor series:

$$M_Z(t) = \sum_{n=0}^{\infty} \frac{\mathbb{E}[Z^n]}{n!} t^n = e^{t^2/2} = \sum_{m=0}^{\infty} \frac{1}{m!} \left(\frac{t^2}{2}\right)^m = \sum_{m=0}^{\infty} \frac{1}{m! 2^m} t^{2m}.$$

Matching coefficients, it trivially follows that $\mathbb{E}[Z^n] = 0$ for every odd n . For $n = 2m$ even, we have:

$$\frac{\mathbb{E}[Z^{2m}]}{(2m)!} = \frac{1}{m! 2^m} \implies \mathbb{E}[Z^{2m}] = \frac{(2m)!}{m! 2^m} = (2m - 1)!!.$$

□

Proposition 8.28 (Affine transformation of the normal distribution). *Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and let $a \neq 0$, $b \in \mathbb{R}$. Then*

$$aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

Proof.

First approach (PDF). We compute the PDF directly via the change of variables formula 3.76 in the affine case:

$$f_{aX+b}(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}} \frac{1}{|a|} = \frac{1}{\sqrt{2\pi a^2\sigma^2}} e^{-\frac{(y-b-a\mu)^2}{2a^2\sigma^2}}.$$

This matches the PDF of $\mathcal{N}(a\mu + b, a^2\sigma^2)$.

Second approach (MGF). Using the MGF properties (Proposition ??):

$$M_{aX+b}(t) = e^{bt} M_X(at) = \exp\left((a\mu + b)t + \frac{a^2\sigma^2 t^2}{2}\right),$$

which is the MGF of $\mathcal{N}(a\mu + b, a^2\sigma^2)$.

□

8.28 justifies an important procedure known as *standardization*. Essentially, any Gaussian distribution is an affine version of the standard normal. Hence, it suffices to know numerical properties for the standard normal $Z \sim \mathcal{N}(0, 1)$ and then apply a proper affine transformation

Corollary 8.29 (Standardization). If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0, 1)$.

Example 8.30. In a biomedical research study, a new therapeutic protocol is being tested for its efficacy. Preliminary data suggest that each patient enrolled in the study has a probability $p = 3/4$ of showing a "complete positive response" (success). The research team needs to ensure that at least 3000 patients respond positively for the study to achieve the required statistical power for the next phase. We are interested in determining the number of patients n that must be enrolled daily so that the probability of obtaining at least 3000 successes is 50%.

Let n be the number of enrolled patients. The total number of successful outcomes is $S_n = \sum_{i=1}^n X_i$, where $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(3/4)$. By Lemma 8.3, it follows that $S_n \sim \text{Binom}(n, 3/4)$. The exact probability of reaching the target of at least 3000 successes is given by

$$\Pr(S_n \geq 3000) = \sum_{k=3000}^n \binom{n}{k} \left(\frac{3}{4}\right)^k \left(\frac{1}{4}\right)^{n-k}.$$

This sum cannot be computed in closed form, as confirmed by Theorem 8.4 stating that the CDF of a binomial distribution has no relevant closed form.

However, since X_i are i.i.d. square integrable random variables with positive variance, and n is large, it follows that, by definition 8.24, $\sqrt{n}(\bar{X}_n - \mu) + \mu \stackrel{d}{\approx} X \sim \mathcal{N}(\mu, \sigma^2)$. Since $S_n = n\bar{X}_n$

$$\begin{aligned} \Pr(S_n \geq 3000) &= \Pr\left(\sqrt{n}(n\bar{X}_n - \mu) + \mu \geq \sqrt{n}\left(\frac{3000}{n} - \mu\right) + \mu\right) \\ &\approx \Pr\left(X \geq \sqrt{n}\left(\frac{3000}{n} - \frac{3}{4}\right) + \frac{3}{4}\right) = 1 - \Phi\left(\sqrt{n}\frac{3000 - 3n/4}{\sqrt{3n/16}}\right). \end{aligned}$$

For this probability to be at least 0.5, we need

$$\sqrt{n}\frac{3000 - 3n/4}{\sqrt{3n/16}} \leq 0,$$

that is $n \geq \lceil 3000 \cdot \frac{4}{3} \rceil = 4000$.

Remark 8.31 (Standardization and computation of normal probabilities). As Example 8.30 demonstrates, computing probabilities for the Gaussian distribution inherently relies on the CDF Φ of the standard normal distribution. Furthermore, since the Central Limit Theorem (6.43) establishes convergence specifically to the standard normal, it is computationally advantageous to reduce any normal random variable to $\mathcal{N}(0, 1)$ via standardization (cf. Corollary 8.29).

The standard normal distribution $Z \sim \mathcal{N}(0, 1)$ has several convenient properties. Because its PDF f_Z is symmetric around 0, the tail probabilities satisfy:

$$\Pr(Z \leq -x) = \Phi(-x) = 1 - \Phi(x) = 1 - \Pr(Z \leq x) = \Pr(Z > x).$$

Consequently, for any general normal variable $X \sim \mathcal{N}(\mu, \sigma^2)$, we can compute its probabilities as:

$$\Pr(X \leq x) = \Pr\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) = \Pr\left(Z \leq \frac{x - \mu}{\sigma}\right) = \begin{cases} \Phi\left(\frac{x - \mu}{\sigma}\right) & \text{if } x \geq \mu, \\ 1 - \Phi\left(-\frac{x - \mu}{\sigma}\right) & \text{if } x < \mu. \end{cases}$$

Historically, values of Φ were read from specific statistical tables. In modern computing, $\Phi(x)$ is evaluated numerically by relating it to the *error function* $\text{erf}(x)$:

$$\Phi(x) = \frac{1}{2} \left[1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right) \right], \quad \text{where} \quad \text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Proposition 8.32 (Additivity of the normal distribution). Let $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ be independent. Then

$$X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Proof.

First approach (MGF). By Proposition 5.9, the MGF of $X + Y$ is given by:

$$M_{X+Y}(t) = M_X(t)M_Y(t) = e^{\mu_1 t + \frac{\sigma_1^2 t^2}{2}} e^{\mu_2 t + \frac{\sigma_2^2 t^2}{2}} = e^{(\mu_1 + \mu_2)t + \frac{(\sigma_1^2 + \sigma_2^2)t^2}{2}},$$

which coincides with the MGF of $Z \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. By uniqueness of the MGF (Corollary 5.23), $X + Y \stackrel{d}{=} Z$.

Second approach (convolution). We compute f_{X+Y} as the continuous convolution of f_X and f_Y :

$$f_{X+Y}(x) = \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{+\infty} e^{-\frac{(y-\mu_1)^2}{2\sigma_1^2}} e^{-\frac{(x-y-\mu_2)^2}{2\sigma_2^2}} dy.$$

Let us focus on the exponent. Completing the square,

$$\begin{aligned} -\frac{1}{2} \left[\frac{(y-\mu_1)^2}{\sigma_1^2} + \frac{(x-y-\mu_2)^2}{\sigma_2^2} \right] &= -\frac{1}{2\sigma_1^2\sigma_2^2} \left[\sigma_2^2(y-\mu_1)^2 + \sigma_1^2(y-(x-\mu_2))^2 \right] \\ &= -\frac{1}{2\sigma_1^2\sigma_2^2} \left[\sigma_2^2(y^2 - 2y\mu_1 + \mu_1^2) + \sigma_1^2(y^2 - 2y(x-\mu_2) + (x-\mu_2)^2) \right] \\ &= -\frac{1}{2\sigma_1^2\sigma_2^2} \left[(\sigma_1^2 + \sigma_2^2)y^2 - 2y(\sigma_2^2\mu_1 + \sigma_1^2(x-\mu_2)) + \sigma_2^2\mu_1^2 + \sigma_1^2(x-\mu_2)^2 \right] \\ &= -\frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1^2\sigma_2^2} \left[y^2 - 2y \left(\frac{\sigma_2^2\mu_1 + \sigma_1^2(x-\mu_2)}{\sigma_1^2 + \sigma_2^2} \right) + \frac{\sigma_2^2\mu_1^2 + \sigma_1^2(x-\mu_2)^2}{\sigma_1^2 + \sigma_2^2} \right] \\ &= -\frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1^2\sigma_2^2} \left[\left(y - \frac{\sigma_2^2\mu_1 + \sigma_1^2(x-\mu_2)}{\sigma_1^2 + \sigma_2^2} \right)^2 - C \right], \end{aligned}$$

where C is explicitly computed as:

$$\begin{aligned} C &:= \left(\frac{\sigma_2^2\mu_1 + \sigma_1^2(x-\mu_2)}{\sigma_1^2 + \sigma_2^2} \right)^2 - \frac{\sigma_2^2\mu_1^2 + \sigma_1^2(x-\mu_2)^2}{\sigma_1^2 + \sigma_2^2} \\ &= \frac{\sigma_2^4\mu_1^2 + \sigma_1^4(x-\mu_2)^2 + 2\sigma_1^2\sigma_2^2\mu_1(x-\mu_2)}{(\sigma_1^2 + \sigma_2^2)^2} - \frac{\sigma_2^4\mu_1^2 + \sigma_1^4(x-\mu_2)^2 + \sigma_1^2\sigma_2^2(x-\mu_2)^2}{(\sigma_1^2 + \sigma_2^2)^2} \\ &= \frac{2\sigma_1^2\sigma_2^2\mu_1(x-\mu_2) - \sigma_1^2\sigma_2^2\mu_1^2 - \sigma_1^2\sigma_2^2(x-\mu_2)^2}{(\sigma_1^2 + \sigma_2^2)^2} \\ &= -\frac{\sigma_1^2\sigma_2^2}{(\sigma_1^2 + \sigma_2^2)^2} (\mu_1 - (x-\mu_2))^2 = -\frac{\sigma_1^2\sigma_2^2}{(\sigma_1^2 + \sigma_2^2)^2} (x - (\mu_1 + \mu_2))^2. \end{aligned}$$

Substituting the exponent back into the integral, the x -dependent term moves outside:

$$\begin{aligned} f_{X+Y}(x) &= \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{+\infty} e^{-\frac{(y-\mu_1)^2}{2\sigma_1^2}} e^{-\frac{(x-y-\mu_2)^2}{2(\sigma_1^2 + \sigma_2^2)}} dy \\ &= \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_2^2)}} e^{-\frac{(x-(\mu_1 + \mu_2))^2}{2(\sigma_1^2 + \sigma_2^2)}} \underbrace{\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{-\frac{(y-\mu_y)^2}{2\sigma_y^2}} dy}_{=1} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \end{aligned}$$

where $\mu := \mu_1 + \mu_2$, $\sigma^2 := \sigma_1^2 + \sigma_2^2$, and

$$\sigma_y^2 := \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2} \quad \text{and} \quad \mu_y := \frac{\sigma_2^2\mu_1 + \sigma_1^2(x-\mu_2)}{\sigma_1^2 + \sigma_2^2}.$$

□

8.2.4 Lognormal distribution

Definition 8.33 (Lognormal distribution). Let $Y \sim \mathcal{N}(\mu, \sigma^2)$. The *lognormal distribution* is defined as $X = e^Y \sim \text{LogNormal}(\mu, \sigma^2)$.

The lognormal distribution is not *something new*, but rather a transformed normal distribution (which is recovered by taking the natural logarithm). It is the canonical model for positive quantities whose aggregated phenomenon is driven by the multiplication of independent effects. It is remarkable that moments of the lognormal distribution are easy to compute:

Theorem 8.34 (Properties of the lognormal distribution). Let $X \sim \text{LogNormal}(\mu, \sigma^2)$. Then X has the following properties:

$$\begin{aligned} f_X(x) &= \frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} \mathbf{1}_{(0, \infty)}(x), & F_X(x) &= \Phi\left(\frac{\log x - \mu}{\sigma}\right) \mathbf{1}_{(0, \infty)}(x), \\ \mathbb{E}[X] &= e^{\mu + \frac{\sigma^2}{2}}, & \text{Var}(X) &= (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}, & \mathbb{E}[X^k] &= e^{k\mu + \frac{k^2\sigma^2}{2}} \quad (k \in [1, \infty)), \end{aligned}$$

where Φ is the standard Normal CDF. The MGF does not exist.

Proof.

We compute the PDF directly via the change of variables formula (3.76). Since the map $g(y) = e^y$ is a C^∞ -diffeomorphism from $\mathbb{R}$ to $(0, \infty)$ and $Y = \log(X) \sim \mathcal{N}(\mu, \sigma^2)$, for $x > 0$ we have:

$$f_X(x) = f_Y(\log x) \cdot \left| \frac{d}{dx} \log x \right| = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} \cdot \frac{1}{x} \mathbf{1}_{(0, \infty)}(x).$$

Similarly, for $x > 0$,

$$F_X(x) = \Pr(X \leq x) = \Pr(e^Y \leq x) = \Pr(Y \leq \log x) = F_Y(\log x) = \Phi\left(\frac{\log x - \mu}{\sigma}\right).$$

For moments, note that $X^k = e^{kY}$, hence

$$\mathbb{E}[X^k] = \mathbb{E}[e^{kY}] = M_Y(k) = e^{k\mu + \frac{k^2\sigma^2}{2}}.$$

The formulas for mean and variance are obtained by plugging $k = 1$ and $k = 2$ and simplifying. We show that $M_X(t)$ is not defined for any $t \in \mathbb{R} \setminus \{0\}$. We compute

$$\mathbb{E}[e^{tX}] = \mathbb{E}[e^{te^Y}] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} e^{te^y} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} \exp\left(\frac{2t\sigma^2 e^y - (y-\mu)^2}{2\sigma^2}\right) dy.$$

If $t > 0$, $\lim_{y \rightarrow +\infty} 2t\sigma^2 e^y - (y-\mu)^2 = \infty$. Therefore, the integral diverges. By definition, the MGF is defined to exist if it is finite in an open interval containing 0; since it diverges for all $t > 0$, the MGF does not exist. $\square$

Proposition 8.35 (Scaling of the lognormal distribution). Let $X \sim \text{LogNormal}(\mu, \sigma^2)$ and let $c > 0$. Then

$$cX \sim \text{LogNormal}(\mu + \log c, \sigma^2).$$

Proof.

If $X = e^Y$ with $Y \sim \mathcal{N}(\mu, \sigma^2)$, then $cX = ce^Y = e^{Y + \log c}$. Since $Y + \log c \sim \mathcal{N}(\mu + \log c, \sigma^2)$ by Proposition 8.28, the result follows. $\square$

Proposition 8.36 (Product of independent lognormal distributions). Let $X \sim \text{LogNormal}(\mu_1, \sigma_1^2)$ and $Y \sim \text{LogNormal}(\mu_2, \sigma_2^2)$ be independent. Then

$$XY \sim \text{LogNormal}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Proof.

Since X and Y are independent, by Proposition 3.74 $\log X$ and $\log Y$ are independent normal variables.

By Proposition 8.32, it follows that $\log X + \log Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Hence:

$$XY = e^{\log X + \log Y} \sim \text{LogNormal}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

□

Proposition 8.37 (Power of the lognormal distribution). *If $X \sim \text{LogNormal}(\mu, \sigma^2)$ and $\alpha \neq 0$, then $X^\alpha \sim \text{LogNormal}(\alpha\mu, \alpha^2\sigma^2)$. In particular, $1/X \sim \text{LogNormal}(-\mu, \sigma^2)$.*

Proof.

We compute the PDF directly via the change of variables formula 3.76. Since $t \mapsto t^\alpha$ is a C^∞ -diffeomorphism on $(0, \infty)$, for $x > 0$ we have:

$$\begin{aligned} f_{X^\alpha}(x) &= f_X(x^{1/\alpha}) \cdot \left| \frac{d}{dx} x^{1/\alpha} \right| \\ &= \frac{1}{x^{1/\alpha}} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\log(x^{1/\alpha}) - \mu)^2}{2\sigma^2}} \cdot \frac{1}{|\alpha|} x^{1/\alpha - 1} \\ &= \frac{x^{1/\alpha - 1}}{x^{1/\alpha}} \frac{1}{\sqrt{2\pi\alpha^2\sigma^2}} e^{-\frac{(\frac{1}{\alpha} \log x - \mu)^2}{2\sigma^2}} = \frac{1}{x\sqrt{2\pi\alpha^2\sigma^2}} e^{-\frac{(\log x - \alpha\mu)^2}{2\alpha^2\sigma^2}}. \end{aligned}$$

This matches exactly the PDF of $\text{LogNormal}(\alpha\mu, \alpha^2\sigma^2)$. The second result follows trivially by setting $\alpha = -1$. □

8.2.5 Gamma distribution

The Gamma law generalizes the exponential distribution by modeling the continuous waiting time before the n -th success occurs. Let us start with the following lemma:

Lemma 8.38 (Gamma distribution as a sum of i.i.d. exponential). *Let $X_1, \dots, X_n$ be a sample of i.i.d. random variables with distribution $X_i \sim \text{Exp}(\lambda)$, and let $X := X_1 + \dots + X_n$. Then X is a continuous random variable and its PDF is given by*

$$f_X(x) = \frac{\lambda^n}{(n-1)!} x^{n-1} e^{-\lambda x} \mathbf{1}_{(0,+\infty)}(x). \quad (8.5)$$

Proof.

First approach (convolution). By Corollary 3.81, the PDF f_X is the n -fold convolution of the exponential PDFs. Specifically, we have

$$f_X(x) = (f_{X_1}^{*n})(x) = (f_{X_1}^{*(n-1)} * f_{X_1})(x).$$

Let us prove (8.5) by induction on n . For $n = 1$ we obtain

$$f_X(x) = \frac{\lambda^1}{0!} x^0 e^{-\lambda x} \mathbf{1}_{(0,+\infty)}(x) = \lambda e^{-\lambda x} \mathbf{1}_{(0,+\infty)}(x) = f_{X_1}(x).$$

Assuming (8.5) is true for $n = N - 1$, we prove it for N . We have

$$\begin{aligned} f_X(x) &= (f_{X_1}^{*(N-1)} * f_{X_1})(x) = \int_0^\infty \frac{\lambda^{N-1}}{(N-2)!} y^{N-2} e^{-\lambda y} \cdot \lambda e^{-\lambda(x-y)} \mathbf{1}_{(0,\infty)}(y) \mathbf{1}_{(0,\infty)}(x-y) dy \\ &= \frac{\lambda^N}{(N-2)!} e^{-\lambda x} \cdot \int_0^\infty y^{N-2} \mathbf{1}_{(0,x)}(y) dy = \frac{\lambda^N}{(N-2)!} e^{-\lambda x} \int_0^x y^{N-2} dy \\ &= \frac{\lambda^N}{(N-2)!} e^{-\lambda x} \left[\frac{1}{N-1} y^{N-1} \right]_0^x = \frac{\lambda^N}{(N-1)!} x^{N-1} e^{-\lambda x}. \end{aligned}$$

Second approach (characteristic functions). Since $X_1, \dots, X_n$ is an i.i.d. sample, by Proposition 5.21, item (v) the characteristic function of X is given by

$$\begin{aligned}\varphi_X(t) &= \varphi_{X_1}(t) \cdots \varphi_{X_n}(t) = \left(\mathbb{E}[e^{-itX_1}] \right)^n = \left(\int_0^\infty e^{-itx} \lambda e^{-\lambda x} dx \right)^n = \left(\lambda \int_0^\infty e^{-(\lambda+it)x} dx \right)^n \\ &= \left(-\frac{\lambda}{\lambda+it} \left[e^{-(\lambda+it)x} \right]_0^\infty \right)^n = \left(\frac{\lambda}{\lambda+it} \right)^n \left(1 - \lim_{x \rightarrow \infty} e^{-(\lambda+it)x} \right) = \left(\frac{\lambda}{\lambda+it} \right)^n,\end{aligned}$$

where $\lim_{x \rightarrow \infty} e^{-(\lambda+it)x} = 0$ since

$$|e^{-(\lambda+it)x}| = |e^{-\lambda x}| |e^{-itx}| = |e^{-\lambda x}| \rightarrow 0 \quad x \rightarrow \infty.$$

Since the sum of continuous random variables is continuous, f_X can be recovered with the inverse Fourier transform as in Example 5.19 (or, equivalently, by Theorem 2.61):

$$f_X(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi_X(t) e^{ixt} dt = \frac{1}{2\pi} \int_{\mathbb{R}} \left(\frac{\lambda}{\lambda+it} \right)^n e^{ixt} dt = \frac{\lambda^n}{2\pi} \int_{\mathbb{R}} \frac{e^{ixt}}{(\lambda+it)^n} dt.$$

Let $\psi(z) := \frac{e^{ixz}}{(\lambda+iz)^n}$. We compute $\int_{\mathbb{R}} \psi(z) dz$. Consider a counterclockwise contour integration on the complex plane through the real axis (from $-R$ to R) which closes with a semicircle C_R . We distinguish whether closing the semicircle in the upper or lower semiplane. This depends on the sign of x . We write $e^{itz} = e^{i(\operatorname{Re}(z)+i\operatorname{Im}(z))x} = e^{i\operatorname{Re}(z)x} \cdot e^{-\operatorname{Im}(z)x}$.

- i) If $x > 0$ and $\operatorname{Im}(z) > 0$, then $|e^{itz}| = |e^{-\operatorname{Im}(z)x}| \rightarrow 0$ as $R \rightarrow \infty$. It can be proved (Jordan's lemma) that in this case, $\int_{C_R} \psi(z) dz \rightarrow 0$. Therefore, it is convenient to choose the semicircle corresponding to $\operatorname{Im}(z) > 0$, that is, the upper one.
- ii) Similarly, if $x < 0$ and $\operatorname{Im}(z) < 0$, then $|e^{itz}| = |e^{-\operatorname{Im}(z)x}| \rightarrow 0$ as $R \rightarrow \infty$. Again we can prove that $\int_{C_R} \psi(z) dz \rightarrow 0$, which simplifies computations. Therefore, it is convenient to choose the semicircle corresponding to $\operatorname{Im}(z) < 0$, that is, the bottom one.

By Theorem A.7,

$$\lim_{R \rightarrow \infty} \left(\int_{-R}^R \psi(z) dz + \int_{C_R} \psi(z) dz \right) = 2\pi i \sum_w \operatorname{Res}(\psi, w),$$

where w are the poles enclosed in the contour of integration. The only pole of ψ is $z = i\lambda$. If $x < 0$, selecting the bottom semicircle we enclose no poles, hence

$$\lim_{R \rightarrow \infty} \left(\int_{-R}^R \psi(z) dz + \int_{C_R} \psi(z) dz \right) = 0 \implies \int_{\mathbb{R}} \psi(z) dz = - \lim_{R \rightarrow \infty} \int_{C_R} \psi(z) dz = 0.$$

Since $\int_{\mathbb{R}} \psi(z) dz = 0$, we conclude that $f_X(x) = 0$ for $x < 0$. If $x > 0$, selecting the upper semicircle we enclose the pole $i\lambda$. Hence, we shall compute $\operatorname{Res}(\psi, i\lambda)$. Writing the Laurent series for ψ at $z_0 = i\lambda$ (setting $z := w + i\lambda$):

$$\begin{aligned}\psi(w + i\lambda) &= \frac{e^{ix(w+i\lambda)}}{(\lambda + i(w+i\lambda))^n} = \frac{e^{ixw} e^{-\lambda x}}{(\lambda + iw - \lambda)^n} \\ &= \frac{e^{-\lambda x}}{i^n w^n} e^{ixw} = \frac{e^{-\lambda x}}{i^n w^n} \sum_{k=0}^{\infty} \frac{(ixw)^k}{k!}.\end{aligned}$$

$\operatorname{Res}(\psi, i\lambda)$ corresponds to taking $k = n-1$ in the Taylor expansion, hence

$$\operatorname{Res}(\psi, i\lambda) = w \cdot \frac{e^{-\lambda x} (ixw)^{n-1}}{i^n w^n (n-1)!} = \frac{x^{n-1} e^{-\lambda x}}{i(n-1)!} = -i \frac{x^{n-1} e^{-\lambda x}}{(n-1)!}.$$

In conclusion, by Theorem A.7 we have

$$\lim_{R \rightarrow \infty} \left(\int_{-R}^R \psi(z) dz + \int_{C_R} \psi(z) dz \right) = 2\pi i \operatorname{Res}(\psi, i\lambda) = 2\pi i \left(-i \frac{x^{n-1} e^{-\lambda x}}{(n-1)!} \right) = 2\pi \frac{x^{n-1} e^{-\lambda x}}{(n-1)!}.$$

Hence, for $x > 0$,

$$f_X(x) = \frac{\lambda^n}{2\pi} \int_{\mathbb{R}} \psi(z) dz = \frac{1}{2\pi} \left(2\pi \frac{x^{n-1} e^{-\lambda x}}{(n-1)!} - \lim_{R \rightarrow \infty} \int_{C_R} \psi(z) dz \right) = \frac{\lambda^n}{(n-1)!} x^{n-1} e^{-\lambda x}.$$

It remains to compute $f_X(x)$ at $x = 0$. However, since PDFs are defined up to null sets (cf. Example 3.46), we can assign it any value. The most reasonable choice is $f_X(0) = 0$. $\square$

The PDF derived in (8.5) legitimately defines the $\text{Gamma}(n, \lambda)$ distribution for any positive integer n . However, we can significantly generalize this family by allowing the shape parameter to be any strictly positive real number $k > 0$. The real issue is to extend the term $(n-1)!$ to $n \in \mathbb{R}$. Specifically, we seek a smooth function $f : (0, \infty) \rightarrow (0, \infty)$ that satisfies the recursive property $f(x+1) = xf(x)$ along with the initial condition $f(1) = 1$. There are infinitely many functions that satisfy this extension. However, mathematicians agreed on considering one in particular due its properties: this canonical extension is called the *Gamma function*.

Lemma 8.39 (Gamma function). Define for $k > 0$,

$$\Gamma(k) := \int_0^\infty x^{k-1} e^{-x} dx.$$

Then $\Gamma(k+1) = k\Gamma(k)$ and $\Gamma(1) = 1$. In particular, for $n \in \mathbb{N}$, $\Gamma(n) = (n-1)!$.

Proof.

We compute

$$\Gamma(1) = \int_0^\infty e^{-x} dx = -[e^{-x}]_0^\infty = 1.$$

Moreover, integration by parts yields

$$\Gamma(k+1) = \int_0^\infty x^k e^{-x} dx = \left[-x^k e^{-x} \right]_0^\infty + k \int_0^\infty x^{k-1} e^{-x} dx = k\Gamma(k).$$

In particular, for $n \in \mathbb{N}$, $\Gamma(n) = (n-1)!$. □

The Gamma function allows us to extend the distribution found in Lemma 8.38 to real-valued parameters, and justifies the name of the Gamma law:

Theorem 8.40 (Properties of the Gamma distribution). Let $X \sim \text{Gamma}(k, \lambda)$, with $k, \lambda > 0$. Then X has the following properties:

$$f_X(x) = \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x} \mathbf{1}_{(0, +\infty)}(x),$$

$$\mathbb{E}[X] = \frac{k}{\lambda}, \quad \text{Var}(X) = \frac{k}{\lambda^2}, \quad M_X(t) = \left(\frac{\lambda}{\lambda - t} \right)^k \quad (t < \lambda).$$

The CDF F_X does not admit a simple closed form.

Proof.

Since f_X extends smoothly (8.5), we take this as a *bona fide* density function. Indeed, it is clear that $f_X \geq 0$ and, moreover,

$$\begin{aligned} \int_{\mathbb{R}} f_X(x) dx &= \int_0^\infty \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x} dx = \frac{1}{\Gamma(k)} \int_0^\infty (\lambda x)^{k-1} e^{-(\lambda x)} \lambda dx \\ &\stackrel{u:=\lambda x}{=} \frac{1}{\Gamma(k)} \int_0^\infty u^{k-1} e^{-u} du = \frac{\Gamma(k)}{\Gamma(k)} = 1. \end{aligned}$$

To compute the expectation and the variance, we present two instructive approaches.

First approach (direct integration). We compute the moments directly:

$$\begin{aligned} \mathbb{E}[X] &= \int_0^\infty \frac{\lambda^k}{\Gamma(k)} x^k e^{-\lambda x} dx = \frac{\Gamma(k+1)}{\lambda \Gamma(k)} \underbrace{\int_0^\infty \frac{\lambda^{k+1}}{\Gamma(k+1)} x^k e^{-\lambda x} dx}_{=1} = \frac{\Gamma(k+1)}{\lambda \Gamma(k)} = \frac{k}{\lambda}, \\ \mathbb{E}[X^2] &= \int_0^\infty \frac{\lambda^k}{\Gamma(k)} x^{k+1} e^{-\lambda x} dx = \frac{\Gamma(k+2)}{\lambda^2 \Gamma(k)} \underbrace{\int_0^\infty \frac{\lambda^{k+2}}{\Gamma(k+2)} x^{k+1} e^{-\lambda x} dx}_{=1} = \frac{\Gamma(k+2)}{\lambda^2 \Gamma(k)} = \frac{k(k+1)}{\lambda^2}, \\ \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{k(k+1) - k^2}{\lambda^2} = \frac{k}{\lambda^2}. \end{aligned}$$

Second approach (differentiating the MGF). First, we compute the MGF via integration:

$$\begin{aligned} M_X(t) &= \int_0^\infty e^{tx} \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x} dx = \int_0^\infty \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-(\lambda-t)x} dx \\ &= \left(\frac{\lambda}{\lambda-t} \right)^k \int_0^\infty \frac{(\lambda-t)^k}{\Gamma(k)} x^{k-1} e^{-(\lambda-t)x} dx = \left(\frac{\lambda}{\lambda-t} \right)^k. \end{aligned}$$

We then compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ by differentiating $M_X(t)$ evaluated at $t = 0$:

$$\begin{aligned} M'_X(t) &= \frac{d}{dt} (1-t/\lambda)^{-k} = \frac{k}{\lambda} (1-t/\lambda)^{-k-1}, \\ \mathbb{E}[X] &= M'_X(0) = \frac{k}{\lambda} (1-0)^{-k-1} = \frac{k}{\lambda}, \\ M''_X(t) &= \frac{k(k+1)}{\lambda^2} (1-t/\lambda)^{-k-2}, \\ \mathbb{E}[X^2] &= M''_X(0) = \frac{k(k+1)}{\lambda^2} (1-0)^{-k-2} = \frac{k(k+1)}{\lambda^2}, \\ \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{k(k+1)}{\lambda^2} - \frac{k^2}{\lambda^2} = \frac{k}{\lambda^2}. \end{aligned}$$

□

Proposition 8.41 (Moments of the Gamma distribution). Let $X \sim \text{Gamma}(k, \lambda)$. Then, for every $n \in \mathbb{N}$,

$$\mathbb{E}[X^n] = \frac{\Gamma(k+n)}{\Gamma(k)} \frac{1}{\lambda^n}.$$

Proof.

We compute the n -th moment directly:

$$\mathbb{E}[X^n] = \int_0^\infty x^n \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x} dx = \frac{\lambda^k}{\Gamma(k)} \frac{\Gamma(k+n)}{\lambda^{k+n}} \underbrace{\int_0^\infty \frac{\lambda^{k+n}}{\Gamma(k+n)} x^{k+n-1} e^{-\lambda x} dx}_{=1} = \frac{\Gamma(k+n)}{\Gamma(k)} \frac{1}{\lambda^n}.$$

□

Remark 8.42 (Moments of the exponential distribution). Recall that the exponential distribution is a special case of the Gamma distribution where the shape parameter is $k = 1$. By substituting $k = 1$ into the formula above, and using the properties $\Gamma(1) = 1$ and $\Gamma(1+n) = n!$, we recover the moments of the exponential distribution as in Theorem 8.20:

$$\mathbb{E}[X^n] = \frac{\Gamma(1+n)}{\Gamma(1)} \frac{1}{\lambda^n} = \frac{n!}{\lambda^n}.$$

Proposition 8.43 (Additivity of the Gamma distribution). Let $X \sim \text{Gamma}(k_1, \lambda)$ and $Y \sim \text{Gamma}(k_2, \lambda)$ be independent. Then

$$X + Y \sim \text{Gamma}(k_1 + k_2, \lambda).$$

Proof.

First approach (MGF). By Proposition 5.9, the MGF of $X + Y$ is given by

$$M_{X+Y}(t) = M_X(t)M_Y(t) = \left(\frac{\lambda}{\lambda-t} \right)^{k_1} \left(\frac{\lambda}{\lambda-t} \right)^{k_2} = \left(\frac{\lambda}{\lambda-t} \right)^{k_1+k_2},$$

which coincides with the MGF of $Z \sim \text{Gamma}(k_1 + k_2, \lambda)$. By uniqueness of the MGF (Corollary 5.23), we conclude that $X + Y \stackrel{d}{=} Z$.

Second approach (convolution). Alternatively, one can directly compute f_{X+Y} as the continuous convolution of f_X and f_Y thanks to Corollary 3.81, showing that

$$f_{X+Y}(x) = \frac{\lambda^{k_1+k_2}}{\Gamma(k_1+k_2)} x^{k_1+k_2-1} e^{-\lambda x}.$$

The conclusion follows by observing that the PDF of $X+Y$ coincides with that of $Z \sim \text{Gamma}(k_1+k_2, \lambda)$. Computing the integral requires proper rescaling so as to apply Lemma 8.50, which will be proved later. $\square$

We have already determined the relationship between the Gamma distribution and the exponential distribution. Let us establish similar relationships with other distributions.

Proposition 8.44 (Relationship between Gamma distribution and negative binomial). *Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of discrete random variables such that, for every $n \in \mathbb{N}$, $X_n \sim \text{NegBin}(r, \frac{\lambda}{n})$. Then*

$$\frac{X_n}{n} \xrightarrow{d} X \sim \text{Gamma}(r, \lambda).$$

Proof.

Intuitively, by Lemma 8.8, the negative binomial is a sum of i.i.d. geometric distributions. Since each of them converges to an exponential distribution by Definition 8.17, it follows that the negative binomial converges to a sum of i.i.d. exponential distributions ($\dagger$). By Lemma 8.38, this distribution is precisely the $\text{Gamma}(r, \lambda)$. This proof, however, is not rigorous, since Step ($\dagger$) holds by Slutsky's theorem (Theorem 6.31) only when all but one of the sequences converge to a constant.

We proceed via MGFs to provide a rigorous proof. By Lemma 8.9 and 5.7 item (ii), we compute the limit for $t \in \bigcap_{n \in \mathbb{N}} (-\infty, -n \log(1 - \frac{\lambda}{n})) = (-\infty, \lambda)$. Changing the variable to $x = 1/n$ so that $x \rightarrow 0^+$, we have:

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{X_n/n}(t) &= \lim_{n \rightarrow \infty} \left(\frac{\frac{\lambda}{n} e^{t/n}}{1 - (1 - \frac{\lambda}{n}) e^{t/n}} \right)^r = \left(\lim_{x \rightarrow 0^+} \frac{\lambda x e^{tx}}{1 - (1 - \lambda x) e^{tx}} \right)^r \\ &= \left(\lim_{x \rightarrow 0^+} \frac{\lambda x (1 + tx)}{1 - (1 - \lambda x)(1 + tx)} \right)^r = \left(\lim_{x \rightarrow 0^+} \frac{\lambda x + \lambda t x^2}{1 - (1 + tx - \lambda x - \lambda t x^2)} \right)^r \\ &= \left(\lim_{x \rightarrow 0^+} \frac{\lambda x}{(\lambda - t)x} \right)^r = \left(\frac{\lambda}{\lambda - t} \right)^r = M_X(t), \end{aligned}$$

where $X \sim \text{Gamma}(r, \lambda)$. Since the pointwise limit holds for every $t \in (-\infty, \lambda)$, by Curtiss's continuity theorem (6.30) we conclude that

$$\frac{X_n}{n} \xrightarrow{d} X \sim \text{Gamma}(r, \lambda).$$

$\square$

Proposition 8.45 (Relationship between Gamma distribution and Poisson distribution). *Let $(T_n)_{n \in \mathbb{N}}$ be a sequence of random variables where $T_n \sim \text{Gamma}(n, \lambda)$ represents the exact arrival time of the n -th event. For a fixed $t > 0$, let N_t be the discrete random variable representing the number of events occurred up to time t , defined as:*

$$N_t := \max\{n \geq 0 \mid T_n \leq t\} \quad \text{with } T_0 := 0.$$

Then $N_t \sim \text{Pois}(\lambda t)$.

Proof.

Integrating by parts:

$$\begin{aligned} \Pr(T_{k+1} \leq t) &= \int_0^t \frac{\lambda^{k+1}}{k!} x^k e^{-\lambda x} dx = \left[-e^{-\lambda x} \frac{(\lambda x)^k}{k!} \right]_0^t - \int_0^t (-e^{-\lambda x}) \frac{\lambda^k}{(k-1)!} x^{k-1} dx \\ &= -e^{-\lambda t} \frac{(\lambda t)^k}{k!} + 0 + \int_0^t \frac{\lambda^k}{(k-1)!} x^{k-1} e^{-\lambda x} dx = -e^{-\lambda t} \frac{(\lambda t)^k}{k!} + \Pr(T_k \leq t). \end{aligned}$$

By the definition of N_t , the event $\{N_t \geq k\}$ is equivalent to $\{T_k \leq t\}$. Thus:

$$\begin{aligned} \Pr(N_t = k) &= \Pr(N_t \geq k) - \Pr(N_t \geq k+1) \\ &= \Pr(T_k \leq t) - \Pr(T_{k+1} \leq t) \\ &= \Pr(T_k \leq t) - \left(\Pr(T_k \leq t) - e^{-\lambda t} \frac{(\lambda t)^k}{k!} \right) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}. \end{aligned}$$

$\square$

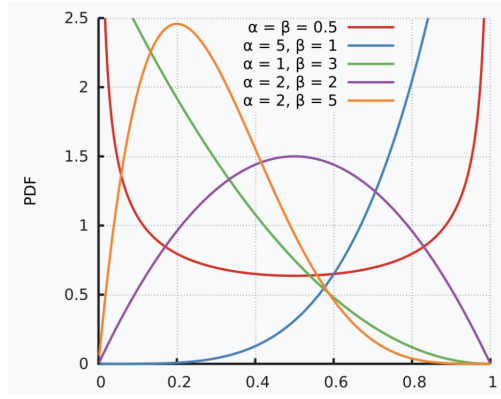


Figure 2: Beta distributions as parameters change. $\alpha = \beta = 1/2$ corresponds to the *arcsine Beta* distribution. $\alpha = \beta = 2$ corresponds to the *parabolic* distribution. $(\alpha, \beta) = (1, 2)$ corresponds to the *triangular* distribution. Image by [Wikimedia Commons](#).

Lemma 8.46 (Relationship between Gamma distribution and standard normal). Let $Z \sim \mathcal{N}(0, 1)$. Then $Z^2 \sim \text{Gamma}(1/2, 1/2)$.

Proof.

Clearly $F_{Z^2}(x) = 0$ for every $x \leq 0$. For $x > 0$, we compute

$$\begin{aligned} F_{Z^2}(x) &= \Pr(Z^2 \leq x) = \Pr(-\sqrt{x} \leq Z \leq \sqrt{x}) = \Phi(\sqrt{x}) - \Phi(-\sqrt{x}) = 2(\Phi(\sqrt{x}) - \Phi(0)) \\ &= 2 \int_0^{\sqrt{x}} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 2 \int_0^x \frac{1}{\sqrt{2\pi}} e^{-v/2} \frac{1}{2\sqrt{v}} dv. \end{aligned}$$

Thus the PDF of $X = Z^2$ is zero for $x < 0$ and

$$f_X(x) = \frac{2}{\sqrt{2\pi}} e^{-x/2} \frac{1}{2\sqrt{x}} = \frac{1}{\Gamma(1/2)} \frac{1}{\sqrt{2}} x^{-1/2} e^{-x/2},$$

using the fact that $\Gamma(1/2) = \sqrt{\pi}$. This matches the PDF of $\text{Gamma}(1/2, 1/2)$. $\square$

Corollary 8.47. If $Z_1, \dots, Z_k$ are i.i.d. random variables with standard normal distribution $\mathcal{N}(0, 1)$, then $X = Z_1^2 + \dots + Z_k^2 \sim \text{Gamma}(k/2, 1/2)$.

Proof. Combine Lemma 8.46 with Proposition 8.43. $\square$

Proposition 8.48 (Scaling of the Gamma distribution). Let $X \sim \text{Gamma}(k, \lambda)$ and let $c > 0$. Then

$$cX \sim \text{Gamma}\left(k, \frac{\lambda}{c}\right).$$

Proof.

Since $M_{cX}(t) = M_X(ct) = \left(\frac{\lambda}{\lambda - ct}\right)^k = \left(\frac{\lambda/c}{\lambda/c - t}\right)^k$, then $cX \sim \text{Gamma}\left(k, \frac{\lambda}{c}\right)$. $\square$

8.2.6 Beta distribution

The Beta distribution models proportions and probabilities, generalizing the uniform distribution on $[0, 1]$. It was introduced to provide a family of continuous distributions capable of representing a wide range of density profiles over a bounded interval. Indeed, by tuning just two parameters, a and b , the resulting PDFs can take drastically different shapes, as shown in Figure 2.

To properly capture the behavior at the boundaries while treating both endpoints symmetrically, we construct the PDF to be proportional to $x^a(1-x)^b$. To allow for asymptotic behaviors at the endpoints (such as $f(x) \propto 1/x$ as $x \rightarrow 0$) while strictly keeping the parameters positive ($a, b > 0$), it is mathematically convenient to shift the exponents, imposing $f(x) \propto x^{a-1}(1-x)^{b-1}$. The required normalizing constant is the *Beta function*, which gives this family of distributions its name:

Definition 8.49 (Beta function). The Beta function $B : (0, \infty)^2 \rightarrow \mathbb{R}$ is defined as

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Lemma 8.50. For every $a, b > 0$, it holds that

$$\int_0^1 x^{a-1}(1-x)^{b-1} dx = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Proof.

Let $\mathcal{I}(a, b) := \int_0^1 x^{a-1}(1-x)^{b-1} dx$. We first check integrability. Since the integrand is nonnegative, it suffices to check integrability near 0 and 1. We use the facts that $(1-x)^{b-1} \leq \max(1, 2^{1-b}) =: C_b$ on $[0, 1/2]$ and $x^{a-1} \leq \max(1, 2^{1-a}) =: C_a$ on $[1/2, 1]$. Hence

$$\begin{aligned} \int_0^{1/2} x^{a-1}(1-x)^{b-1} dx &\leq \int_0^{1/2} x^{a-1} C_b dx < \infty \quad (a > 0), \\ \int_{1/2}^1 x^{a-1}(1-x)^{b-1} dx &\leq \int_{1/2}^1 (1-x)^{b-1} C_a dx < \infty \quad (b > 0). \end{aligned}$$

Thus $\mathcal{I}(a, b) \in (0, \infty)$ for every $a, b > 0$. By Tonelli's theorem,

$$\Gamma(a)\Gamma(b) = \left(\int_0^\infty u^{a-1} e^{-u} du \right) \left(\int_0^\infty v^{b-1} e^{-v} dv \right) = \int_0^\infty \int_0^\infty u^{a-1} v^{b-1} e^{-(u+v)} du dv.$$

We apply the change of variables $(u, v) = (rt, r(1-t))$ with $r \in (0, \infty)$ and $t \in (0, 1)$. The Jacobian matrix is

$$J(r, t) = \begin{pmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial t} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial t} \end{pmatrix} = \begin{pmatrix} t & r \\ 1-t & -r \end{pmatrix}.$$

Thus the Jacobian factor is $|\det(J)| = |-rt - r(1-t)| = r$. Substituting this into the integral yields:

$$\begin{aligned} \Gamma(a)\Gamma(b) &= \int_0^1 \int_0^\infty (rt)^{a-1} (r(1-t))^{b-1} e^{-r} r dr dt \\ &= \left(\int_0^1 t^{a-1} (1-t)^{b-1} dt \right) \left(\int_0^\infty r^{a+b-1} e^{-r} dr \right) = \mathcal{I}(a, b) \cdot \Gamma(a+b). \end{aligned}$$

Since $\Gamma(a+b) > 0$, we conclude that $\mathcal{I}(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$. □

Theorem 8.51 (Properties of the Beta distribution). Let $X \sim \text{Beta}(a, b)$. Then X has the following properties:

$$\begin{aligned} f_X(x) &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1} \mathbf{1}_{(0,1)}(x), \\ \mathbb{E}[X] &= \frac{a}{a+b}, \quad \text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}. \end{aligned}$$

The CDF F_X does not admit a simple closed form. The MGF does not yield a clean closed-form expression.

Proof.

By hypothesis, $f_X \propto x^{a-1}(1-x)^{b-1}$ for every $x \in [0, 1]$. By Lemma 8.50, we conclude that $\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} > 0$ is the proper multiplicative constant such that f_X is a PDF:

$$\int_0^1 \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1} dx = 1.$$

From here, one computes:

$$\begin{aligned}\mathbb{E}[X] &= \int_0^1 x \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1} dx \\ &= \frac{\Gamma(a+b)}{\Gamma(a+1+b)} \frac{\Gamma(a+1)}{\Gamma(a)} \underbrace{\int_0^1 \frac{\Gamma(a+b+1)}{\Gamma(a+1)\Gamma(b)} x^a (1-x)^{b-1} dx}_{=1} \\ &= \frac{\Gamma(a+b)}{\Gamma(a+1+b)} \frac{\Gamma(a+1)}{\Gamma(a)} = \frac{a}{a+b}.\end{aligned}$$

$$\begin{aligned}\mathbb{E}[X^2] &= \int_0^1 x^2 \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1} dx \\ &= \frac{\Gamma(a+b)}{\Gamma(a+2+b)} \frac{\Gamma(a+2)}{\Gamma(a)} \underbrace{\int_0^1 \frac{\Gamma(a+b+2)}{\Gamma(a+2)\Gamma(b)} x^{a+1} (1-x)^{b-1} dx}_{=1} \\ &= \frac{\Gamma(a+b)}{\Gamma(a+2+b)} \frac{\Gamma(a+2)}{\Gamma(a)} = \frac{a(a+1)}{(a+b)(a+b+1)}.\end{aligned}$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{a(a+1)}{(a+b)(a+b+1)} - \frac{a^2}{(a+b)^2} = \frac{ab}{(a+b)^2(a+b+1)}.$$

□

Proposition 8.52 (Moments of the Beta distribution). *Let $X \sim \text{Beta}(a, b)$. Then, for every $n \in \mathbb{N}$,*

$$\mathbb{E}[X^n] = \frac{\Gamma(a+n)\Gamma(a+b)}{\Gamma(a+b+n)\Gamma(a)}$$

Proof.

We compute the n -th moment directly:

$$\begin{aligned}\mathbb{E}[X^n] &= \int_0^1 x^n \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1} dx \\ &= \frac{\Gamma(a+n)\Gamma(a+b)}{\Gamma(a+b+n)\Gamma(a)} \int_0^1 \frac{\Gamma(a+b+n)}{\Gamma(a+n)\Gamma(b)} x^{a+n-1} (1-x)^{b-1} dx \\ &= \frac{\Gamma(a+n)\Gamma(a+b)}{\Gamma(a+b+n)\Gamma(a)}\end{aligned}$$

□

The following proposition gives an intuition of the influence of the parameter a on the Beta distribution, when b is fixed to be 1:

Proposition 8.53 (From uniform to Beta distribution). *Let $U \sim \mathcal{U}([0, 1])$ and let $a > 0$. Define $X := U^{1/a}$. Then*

$$X \sim \text{Beta}(a, 1).$$

In particular, the $\text{Beta}(1, 1)$ distribution coincides with $\mathcal{U}([0, 1])$.

Proof.

For $x \in (0, 1)$, $F_X(x) = \Pr(X \leq x) = \Pr(U \leq x^a) = x^a$. Differentiating yields $f_X(x) = ax^{a-1}$ on $(0, 1)$, i.e. $\text{Beta}(a, 1)$. □

Theorem 8.54 (Beta-Gamma algebra). *Let $a, b, \lambda > 0$. Let $X \sim \text{Gamma}(a, \lambda)$ and $Y \sim \text{Gamma}(b, \lambda)$ be independent. Define*

$$Z := \frac{X}{X+Y}, \quad W := X+Y.$$

Then

$$Z \sim \text{Beta}(a, b), \quad W \sim \text{Gamma}(a + b, \lambda),$$

and Z and W are mutually independent. Moreover, the transformation is invertible:

$$X = ZW, \quad Y = (1 - Z)W.$$

Proof.

The proof is an application of the change of variables formula (Theorem 3.76). Observe that since X and Y are positive almost surely, $\Pr((X, Y) \in (0, +\infty)^2) = 1$. Define the transformation $T : (0, +\infty) \times (0, +\infty) \rightarrow (0, 1) \times (0, +\infty)$ as

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{x}{x+y} \\ \frac{y}{x+y} \end{bmatrix} = \begin{bmatrix} z \\ w \end{bmatrix} \implies T^{-1} \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} zw \\ w(1-z) \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Hence T is a C^1 -diffeomorphism between its domain and image. This shows that the transformation is invertible, and

$$X = ZW, \quad Y = (1 - Z)W.$$

. Computing the Jacobian factor,

$$|\det J_{T^{-1}}(z, w)| = \begin{vmatrix} w & z \\ -w & 1-z \end{vmatrix} = |w(1-z) - (-wz)| = |w - wz + wz| = |w| = w,$$

where the last equality holds because $w \in (0, +\infty)$. Hence,

$$\begin{aligned} f_{(Z,W)}(z, w) &= f_{(X,Y)}(T^{-1}(z, w)) \cdot |\det J_{T^{-1}}(z, w)| \\ &= f_X(zw) f_Y(w(1-z)) \cdot w \\ &= \frac{\lambda^a}{\Gamma(a)} (zw)^{a-1} e^{-\lambda zw} \mathbf{1}_{(0,+\infty)}(zw) \cdot \frac{\lambda^b}{\Gamma(b)} (w(1-z))^{b-1} e^{-\lambda w(1-z)} \mathbf{1}_{(0,\infty)}(w(1-z)) \cdot w \\ &= \frac{\lambda^{a+b}}{\Gamma(a)\Gamma(b)} e^{-\lambda(zw+w-wz)} z^{a-1} w^{a-1} w^{b-1} (1-z)^{b-1} \mathbf{1}_{(0,+\infty)}(zw) \mathbf{1}_{(0,\infty)}(w-wz) \cdot w \\ &= \frac{\lambda^{a+b}}{\Gamma(a+b)} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} e^{-\lambda w} w^{a+b-1} z^{a-1} (1-z)^{b-1} \mathbf{1}_{(0,\infty)}(w) \mathbf{1}_{(0,1)}(z) \\ &= \left[\frac{\lambda^{a+b}}{\Gamma(a+b)} w^{a+b-1} e^{-\lambda w} \mathbf{1}_{(0,\infty)}(w) \right] \cdot \left[\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} z^{a-1} (1-z)^{b-1} \mathbf{1}_{(0,1)}(z) \right] = f_Z(z) f_W(w). \end{aligned}$$

Observe that $\mathbf{1}_{(0,\infty)}(zw) \mathbf{1}_{(0,\infty)}(w-wz) = \mathbf{1}_{(0,\infty)}(w) \mathbf{1}_{(0,1)}(z)$, since $zw > 0$ and $w(1-z) > 0$ imply that $w > 0$, $z > 0$ and $1-z > 0$. Since the joint PDF $f_{(Z,W)}(z, w)$ factorizes, then Z and W are independent by Proposition 3.73. Moreover, by inspection $W \sim \text{Gamma}(a + b, \lambda)$ and $Z \sim \text{Beta}(a, b)$ distribution. $\square$

Remark 8.55 (Special cases: uniform and exponential distributions). To understand why Theorem 8.54 is remarkable, let us see some special cases:

- *Case $a = b = 1$.* If $X, Y \sim \text{Exp}(\lambda)$ are independent, then $W = X + Y \sim \text{Gamma}(2, \lambda)$ and $\frac{X}{X+Y} \sim \mathcal{U}([0, 1])$ are independent. Since the uniform distribution is preserved under the map $t \mapsto 1 - t$, then also $\frac{Y}{X+Y} = 1 - \frac{X}{X+Y} \sim \mathcal{U}([0, 1])$. This means that the proportion of the first waiting time over the total waiting time for the two events is uniformly distributed and strictly independent of the total time waited. In other words, the exponential distribution encodes *temporal homogeneity*: there is no "preferred" moment to occur.
- *Case $a = b = \lambda = \frac{1}{2}$.* Let $X, Y \sim \Gamma(\frac{1}{2}, \frac{1}{2})$ be. Then

$$W := X + Y \sim \Gamma(1, \frac{1}{2}) = \text{Exp}(\frac{1}{2}) = \chi_2^2, \quad Z := \frac{X}{X+Y} \sim \text{Beta}(\frac{1}{2}, \frac{1}{2}),$$

and W and Z are independent. The $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ law is the *arcsine distribution* on $(0, 1)$; equivalently, if $\Theta \sim \mathcal{U}(0, \pi)$ then $\sin^2 \Theta \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$. By symmetry of the density around $1/2$, the same holds for $\cos^2 \Theta$. A useful geometric interpretation comes from the representation $X = Z_1^2$ and $Y = Z_2^2$ with

$Z_1, Z_2 \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. Then

$$W = Z_1^2 + Z_2^2 = R^2 \quad \text{and} \quad Z = \frac{Z_1^2}{Z_1^2 + Z_2^2} = \cos^2 \Theta,$$

where $(Z_1, Z_2) = (R \cos \Theta, R \sin \Theta)$ in polar coordinates. Since a standard Gaussian vector in $\mathbb{R}^2$ is rotationally invariant, the radius R and the angle Θ are independent; consequently, we expect $W = R^2$ and $Z = \cos^2(\Theta)$ to be independent.

8.2.7 Chi-square distribution

The chi-square distribution arises naturally as the squared Euclidean distance from the origin of a random vector in $\mathbb{R}^k$ whose coordinates are independent standard normal variables.

Definition 8.56 (Chi-square distribution). Let $k \in \mathbb{N}$ with $k \geq 1$. A random variable X is said to have a chi-square distribution with k degrees of freedom, denoted by $X \sim \chi^2(k)$ or $X \sim \chi_k^2$, if

$$X = \sum_{i=1}^k Z_i^2,$$

where $Z_1, \dots, Z_k$ are independent identically distributed $\mathcal{N}(0, 1)$ random variables.

Theorem 8.57 (Properties of the chi-square distribution). Let $X \sim \chi^2(k)$. Then X has the following properties:

$$f_X(x) = \frac{1}{2^{k/2} \Gamma(k/2)} x^{\frac{k}{2}-1} e^{-\frac{x}{2}} \mathbf{1}_{(0, \infty)}(x),$$

$$\mathbb{E}[X] = k, \quad \text{Var}(X) = 2k, \quad M_X(t) = (1 - 2t)^{-k/2} \quad (t < 1/2).$$

Its CDF F_X does not admit a simple closed form.

Proof.

By Corollary 8.47, $\chi^2(k) \stackrel{d}{=} \text{Gamma}(k/2, 1/2)$. Therefore, all the properties follow directly from Theorem 8.40 by substituting $(k', \lambda) \mapsto (k/2, 1/2)$. $\square$

8.2.8 Cauchy distribution

We conclude the list of relevant continuous distributions with two pathological families that are useful as counterexamples. The first of them is the Cauchy family, which is the prototypical heavy-tailed distribution. It is stable under convolution but admits neither a mean nor a variance. It is defined via the PDF:

$$f_X(x) = \frac{1}{\pi} \frac{\gamma}{\gamma^2 + (x - x_0)^2} = \frac{1}{\pi\gamma} \frac{1}{1 + \left(\frac{x - x_0}{\gamma}\right)^2}, \quad x \in \mathbb{R},$$

where $x_0 \in \mathbb{R}$ is a location parameter and $\gamma > 0$ is a scale parameter.

Theorem 8.58 (Properties of the Cauchy distribution). Let $X \sim \text{Cauchy}(x_0, \gamma)$. Then X has the following properties:

$$f_X(x) = \frac{1}{\pi\gamma} \frac{1}{1 + \left(\frac{x - x_0}{\gamma}\right)^2}, \quad F_X(x) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x - x_0}{\gamma}\right),$$

$$\varphi_X(t) = e^{-ix_0 t - \gamma|t|} \quad (t \in \mathbb{R}).$$

Moreover, it is not integrable, has no finite moments, and no MGF. Furthermore, if $Y = \frac{X - x_0}{\gamma}$, then $Y \sim \text{Cauchy}(0, 1)$.

Proof.

Observe that

$$\int_{-\infty}^{+\infty} \frac{1}{\pi\gamma} \frac{1}{1 + \left(\frac{x-x_0}{\gamma}\right)^2} dx \stackrel{y=\frac{x-x_0}{\gamma}}{=} \int_{-\infty}^{+\infty} \frac{1}{\pi} \frac{1}{1+y^2} dy = \frac{1}{\pi} [\arctan y]_{-\infty}^{+\infty} = \frac{1}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = 1.$$

Hence, f_X is *bona fide* a PDF. Similarly, we compute the CDF:

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x f_X(t) dt \stackrel{y=\frac{t-x_0}{\gamma}}{=} \int_{-\infty}^{\frac{x-x_0}{\gamma}} \frac{1}{\pi} \frac{1}{1+y^2} dy = \frac{1}{\pi} [\arctan y]_{-\infty}^{\frac{x-x_0}{\gamma}} \\ &= \frac{1}{\pi} \left(\arctan \left(\frac{x-x_0}{\gamma} \right) + \frac{\pi}{2} \right) = \frac{1}{2} + \frac{1}{\pi} \arctan \left(\frac{x-x_0}{\gamma} \right). \end{aligned}$$

Since

$$F_Y(x) = \Pr(Y \leq x) = \Pr\left(\frac{X-x_0}{\gamma} \leq x\right) = \Pr(X \leq \gamma x + x_0) = F_X(\gamma x + x_0) = \frac{1}{2} + \frac{1}{\pi} \arctan(x),$$

we conclude that $Y \sim \text{Cauchy}(0, 1)$.

Furthermore, observe that for the standard Cauchy variable Y :

$$\mathbb{E}[|Y|] = \int_{-\infty}^{+\infty} |y| f_Y(y) dy = \frac{2}{\pi} \int_0^{+\infty} \frac{y}{1+y^2} dy = \frac{1}{\pi} [\log(1+y^2)]_0^{+\infty} = +\infty.$$

Since $X = \gamma Y + x_0$, it follows trivially that $\mathbb{E}[|X|] = +\infty$. By Proposition 4.24, $\mathbb{E}[|X|^p] = +\infty$ for every $p \in [1, \infty)$. As a byproduct, the Cauchy distribution does not admit an MGF.

However, it admits a characteristic function. Let us compute the CF of $Y \sim \text{Cauchy}(0, 1)$ first:

$$\varphi_Y(t) = \int_{\mathbb{R}} f_Y(z) e^{-itz} dz = \frac{1}{\pi} \int_{\mathbb{R}} \frac{e^{-itz}}{1+z^2} dz = \frac{1}{\pi} \int_{\mathbb{R}} \frac{e^{-itz}}{(z+i)(z-i)} dz.$$

Let $\psi(z) := \frac{e^{-itz}}{(z+i)(z-i)}$. We compute $\int_{\mathbb{R}} \psi(x) dx$ via contour integration. We argue verbatim as in the second approach of the proof of Lemma 8.38, considering a counterclockwise contour integration on the complex plane along the real axis (from $-R$ to R) which closes with a semicircular arc C_R . We distinguish whether to close the contour in the upper or lower half-plane depending on the sign of t . By the same arguments as in the proof of 8.38, the decay along C_R as $R \rightarrow \infty$ is achieved when $t > 0$ in the lower half-plane, and when $t < 0$ in the upper half-plane.

The function $\psi(z)$ has two simple poles of order 1 at $\pm i$. Hence,

$$\text{Res}(\psi, i) = \lim_{z \rightarrow i} (z-i) \psi(z) = \frac{e^{-it(i)}}{2i} = \frac{e^t}{2i}, \quad \text{Res}(\psi, -i) = \lim_{z \rightarrow -i} (z+i) \psi(z) = \frac{e^{-it(-i)}}{-2i} = -\frac{e^{-t}}{2i}.$$

By Cauchy's Residue Theorem A.7, if $t > 0$ (closing the contour in the lower half-plane, traversed clockwise):

$$\lim_{R \rightarrow \infty} \left(\int_{-R}^R \psi(x) dx + \int_{C_R} \psi(z) dz \right) = -2\pi i \text{Res}(\psi, -i) = \frac{e^{-t}}{2i} \cdot 2\pi i = \pi e^{-t}.$$

Thus, $\varphi_Y(t) = \frac{1}{\pi} \pi e^{-t} = e^{-t}$ for $t > 0$. If $t < 0$ (closing the contour in the upper half-plane, traversed counterclockwise):

$$\lim_{R \rightarrow \infty} \left(\int_{-R}^R \psi(x) dx + \int_{C_R} \psi(z) dz \right) = 2\pi i \text{Res}(\psi, i) = \frac{e^t}{2i} \cdot 2\pi i = \pi e^t.$$

Thus, $\varphi_Y(t) = \frac{1}{\pi} \pi e^t = e^t$ for $t < 0$. Since $\varphi_Y(0) = 1$ regardless of the sign, it follows that

$$\varphi_Y(t) = e^{-|t|}, \quad \forall t \in \mathbb{R}.$$

Finally, $\varphi_X(t)$ is computed via the affine property (Proposition 5.21, item (v)). Since $\gamma > 0$:

$$\varphi_X(t) = \varphi_{\gamma Y + x_0}(t) = e^{-itx_0} \varphi_Y(\gamma t) = e^{-itx_0} e^{-|\gamma t|} = e^{-ix_0 t - \gamma |t|}, \quad \forall t \in \mathbb{R}..$$

□

Proposition 8.59 (Affine transformation of the Cauchy distribution). *If $Y \sim \text{Cauchy}(0, 1)$ and $b \in \mathbb{R}$, $a \in \mathbb{R} \setminus \{0\}$, then $aY + b \sim \text{Cauchy}(b, |a|)$.*

Proof.

To prove this result, we present three distinct approaches. Let $X := aY + b$.

First approach (CDF evaluation). We compute the CDF of X directly, recalling the observation from the previous theorem that $F_Y(y) = \frac{1}{2} + \frac{1}{\pi} \arctan(y)$. We must distinguish the sign of a . If $a > 0$, then

$$F_X(x) = \Pr(aY + b \leq x) = \Pr\left(Y \leq \frac{x-b}{a}\right) = F_Y\left(\frac{x-b}{a}\right) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x-b}{a}\right).$$

If $a < 0$, then

$$\begin{aligned} F_X(x) &= \Pr(aY + b \leq x) = \Pr\left(Y \geq \frac{x-b}{a}\right) = 1 - F_Y\left(\frac{x-b}{a}\right) \\ &= 1 - \left(\frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x-b}{a}\right)\right) = \frac{1}{2} - \frac{1}{\pi} \arctan\left(\frac{x-b}{a}\right) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x-b}{|a|}\right). \end{aligned}$$

In both cases, $F_X(x)$ exactly matches the CDF of a $\text{Cauchy}(b, |a|)$ distribution.

Second approach (change of variables). We compute the PDF directly via the change of variables formula 3.76 in the affine case:

$$f_X(x) = \frac{1}{|a|} f_Y\left(\frac{x-b}{a}\right) = \frac{1}{\pi \left(1 + \left(\frac{x-b}{a}\right)^2\right)} \cdot \frac{1}{|a|} = \frac{1}{\pi |a|} \frac{1}{1 + \left(\frac{x-b}{|a|}\right)^2}.$$

Third approach (characteristic functions). From the previous theorem, we know that the characteristic function of the standard Cauchy distribution is $\varphi_Y(t) = e^{-|t|}$. By the affine transformation property of characteristic functions (Proposition 5.21, item (v)), we evaluate the CF of $X = aY + b$:

$$\varphi_X(t) = \varphi_{aY+b}(t) = e^{-itb} \varphi_Y(at) = e^{-itb} e^{-|at|} = e^{-ibt - |a||t|}.$$

We recognize this expression as the characteristic function of a Cauchy distribution with location parameter b and scale parameter $|a|$. By the uniqueness of the characteristic function (Theorem 5.22), we conclude that $X \sim \text{Cauchy}(b, |a|)$. $\square$

Corollary 8.59 is not the only stability property of the Cauchy distribution. The next two properties are remarkable:

Proposition 8.60 (Additivity of the Cauchy distribution). *Let $X \sim \text{Cauchy}(x_1, \gamma_1)$ and $Y \sim \text{Cauchy}(x_2, \gamma_2)$ be independent. Then*

$$X + Y \sim \text{Cauchy}(x_1 + x_2, \gamma_1 + \gamma_2).$$

Proof.

By Proposition 5.21, item (v), the characteristic function of $X + Y$ is given by

$$\varphi_{X+Y}(t) = \varphi_X(t) \cdot \varphi_Y(t) = e^{-ix_1 t - \gamma_1 |t|} e^{-ix_2 t - \gamma_2 |t|} = e^{-i(x_1 + x_2)t - (\gamma_1 + \gamma_2)|t|},$$

which coincides with the CF of $Z \sim \text{Cauchy}(x_1 + x_2, \gamma_1 + \gamma_2)$. By uniqueness of the characteristic function (Theorem 5.22), we conclude that $X + Y \stackrel{d}{=} Z \sim \text{Cauchy}(x_1 + x_2, \gamma_1 + \gamma_2)$. $\square$

Proposition 8.61 (Reciprocal of the standard Cauchy distribution). *Let $X \sim \text{Cauchy}(0, 1)$. Then*

$$\frac{1}{X} \sim \text{Cauchy}(0, 1).$$

Proof.

This follows from the property of the arctangent function:

$$\arctan(1/x) = \begin{cases} \frac{\pi}{2} - \arctan(x) & x > 0 \\ -\frac{\pi}{2} - \arctan(x) & x < 0 \end{cases}.$$

To see why it holds, let $\psi(x) = \arctan(1/x) + \arctan(x)$ for every $x \neq 0$. Differentiating yields

$$\psi'(x) = -\frac{1}{x^2} \frac{1}{1+1/x^2} + \frac{1}{1+x^2} = -\frac{1}{1+x^2} + \frac{1}{1+x^2} = 0.$$

We conclude that ψ is constant on connected intervals. Since ψ is defined on the open intervals $(-\infty, 0)$ and $(0, \infty)$, we evaluate it at $x = \pm 1$:

$$\begin{aligned} \arctan(1/x) + \arctan(x) &= \psi(1) = \arctan(1) + \arctan(1) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \quad (x > 0), \\ \arctan(1/x) + \arctan(x) &= \psi(-1) = \arctan(-1) + \arctan(-1) = -\frac{\pi}{4} - \frac{\pi}{4} = -\frac{\pi}{2} \quad (x < 0). \end{aligned}$$

Hence, for every $y > 0$,

$$\begin{aligned} F_{1/X}(y) &= \Pr(1/X \leq y) = \Pr(X \leq 0) + \Pr(X \geq 1/y) = F_X(0) + 1 - F_X(1/y) \\ &= \frac{1}{2} + 1 - \left(\frac{1}{2} + \frac{1}{\pi} \arctan(1/y) \right) = -\frac{1}{\pi} \arctan(1/y) + 1 \\ &= -\frac{1}{\pi} \left(\frac{\pi}{2} - \arctan(y) \right) + 1 = \frac{1}{2} + \frac{1}{\pi} \arctan(y) = F_X(y). \end{aligned}$$

Similarly, for every $y < 0$,

$$\begin{aligned} F_{1/X}(y) &= \Pr(1/X \leq y) = \Pr(1/y \leq X \leq 0) = F_X(0) - F_X(1/y) \\ &= \frac{1}{2} - \left(\frac{1}{2} + \frac{1}{\pi} \arctan(1/y) \right) = -\frac{1}{\pi} \arctan(1/y) \\ &= -\frac{1}{\pi} \left(-\frac{\pi}{2} - \arctan(y) \right) = \frac{1}{2} + \frac{1}{\pi} \arctan(y) = F_X(y). \end{aligned}$$

Finally, $F_{1/X}(0) = \Pr(1/X \leq 0) = \Pr(X \leq 0) = F_X(0)$, concluding the proof. $\square$

Proposition 8.62 (Ratio of standard normal distributions). *Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 1)$ be independent. Then*

$$\frac{X}{Y} \sim \text{Cauchy}(0, 1).$$

Proof.

The proof is a direct application of the change of variables formula (Theorem 3.76). Since Y is an absolutely continuous random variable, $\Pr(Y = 0) = 0$. Thus, the random vector (X, Y) takes values in $\mathbb{R} \times (\mathbb{R} \setminus \{0\})$ almost surely, meaning the ratio is well-defined.

Define the transformation $T : \mathbb{R} \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R} \times (\mathbb{R} \setminus \{0\})$ as

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{x}{y} \\ y \end{bmatrix} = \begin{bmatrix} z \\ w \end{bmatrix} \implies T^{-1} \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} zw \\ w \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

This transformation T is a C^1 -diffeomorphism on its domain. Let us compute the Jacobian factor:

$$|J_{T^{-1}}(z, w)| = \begin{vmatrix} w & z \\ 0 & 1 \end{vmatrix} = |w \cdot 1 - z \cdot 0| = |w|.$$

Hence

$$\begin{aligned} f_{(Z, W)}(z, w) &= f_{(X, Y)}(T^{-1}(z, w)) \cdot |J_{T^{-1}}(z, w)| = f_X(zw) f_Y(w) \cdot |w| \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{(zw)^2}{2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}} |w| = \frac{1}{2\pi} |w| e^{-\frac{w^2}{2}(z^2+1)}. \end{aligned}$$

Marginalizing with respect to Z :

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{+\infty} f_{(Z,W)}(z,w)dw = 2 \int_0^{+\infty} \frac{1}{2\pi} w e^{-\frac{w^2}{2}(z^2+1)} dw = \frac{1}{\pi} \int_0^{+\infty} w e^{-\frac{w^2}{2}(z^2+1)} dw. \\ &= \frac{1}{\pi} \left[-\frac{1}{z^2+1} e^{-\frac{w^2}{2}(z^2+1)} \right]_0^{+\infty} = \frac{1}{\pi} \left(0 - \left(-\frac{1}{z^2+1} \right) \right) = \frac{1}{\pi(1+z^2)}. \end{aligned}$$

This is exactly the PDF of a Cauchy(0, 1) random variable, concluding the proof. $\square$

8.2.9 Pareto distribution

The Pareto family is another heavy-tailed distribution frequently used in counterexamples. If the Cauchy family does not admit moments of any order, the Pareto distribution controls which moments exist through its parameters.

Definition 8.63 (Pareto distribution). Let $x_m > 0$ and $\alpha > 0$. We say that $X \sim \text{Pareto}(x_m, \alpha)$ if $X \geq x_m$ and

$$\Pr(X > x) = \left(\frac{x_m}{x} \right)^\alpha, \quad x \geq x_m.$$

Theorem 8.64 (Properties of the Pareto distribution). Let $X \sim \text{Pareto}(x_m, \alpha)$. Then X has the following properties:

$$\begin{aligned} f_X(x) &= \frac{\alpha}{x} \left(\frac{x_m}{x} \right)^\alpha \mathbf{1}_{[x_m, \infty)}(x), & F_X(x) &= \left(1 - \left(\frac{x_m}{x} \right)^\alpha \right) \mathbf{1}_{[x_m, \infty)}(x), \\ \mathbb{E}[X^r] &= \begin{cases} \frac{\alpha}{\alpha - r} x_m^r & \text{if } r < \alpha \\ +\infty & \text{if } r \geq \alpha \end{cases}. \end{aligned}$$

The MGF does not exist. Furthermore, its characteristic function has no simple closed form.

Proof.

The formula for F_X trivially follows from Definition 8.63. Differentiating with respect to x , we obtain, for every $x > x_m$,

$$f_X(x) = \frac{dF_X(x)}{dx} = \frac{\alpha}{x} \left(\frac{x_m}{x} \right)^\alpha.$$

Let $r \geq 1$. Then

$$\mathbb{E}[X^r] = \int_{x_m}^{\infty} x^r \alpha x_m^\alpha x^{-(\alpha+1)} dx = \alpha x_m^\alpha \int_{x_m}^{\infty} x^{r-\alpha-1} dx.$$

If $r - \alpha - 1 \geq -1$; that is, $r \geq \alpha$, then $\mathbb{E}[X^r] = +\infty$. Otherwise, we have

$$\mathbb{E}[X^r] = \alpha x_m^\alpha \int_{x_m}^{\infty} x^{r-\alpha-1} dx = \alpha x_m^\alpha \left[\frac{x^{r-\alpha}}{r-\alpha} \right]_{x_m}^{\infty} = \frac{\alpha x_m^r}{\alpha - r},$$

concluding the proof. $\square$

A Complex Analysis

In this section we recall some fundamental results that will be used throughout the notes.

Definition A.1 (Complex differentiability). Given an open set $U \subset \mathbb{C}$, a function $f : U \rightarrow \mathbb{C}$ and a point $z_0 \in U$, we say that f is complex differentiable at z_0 if there exists a complex number $\alpha \in \mathbb{C}$ such that

$$f(z) = f(z_0) + \alpha(z - z_0) + o(|z - z_0|) \iff \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \in \mathbb{C}.$$

In such case, we call $f'(z_0) \in \mathbb{C}$ the limit $\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$. We say that f is **holomorphic** if it is complex differentiable at all points $z_0 \in U$.

Definition A.2 (Power series). Let $(\alpha_k)_{k \in \mathbb{N}} \subseteq \mathbb{C}$ be a sequence of complex numbers. A *power series* is a function $f : U \rightarrow \mathbb{C}$ defined by

$$f(z) := \sum_{k=0}^{\infty} \alpha_k (z - z_0)^k,$$

where $U \subseteq \mathbb{C}$ is the set of points z for which the series converges (equivalently, converges absolutely). One can show that $U = B_R(z_0) := \{z \in \mathbb{C} : |z - z_0| < R\}$, where R is called *radius of convergence* and is given by

$$R := \frac{1}{\limsup_{k \rightarrow \infty} \sqrt[k]{|\alpha_k|}}.$$

Furthermore, it can be shown that f is holomorphic on $B_R(z_0)$, and its complex derivative is obtained by differentiating the series term by term.

Theorem A.3 (Holomorphic functions are analytic). Assume that $f : U \rightarrow \mathbb{C}$ is holomorphic and $z_0 \in U$. Then on $B_R(z_0)$ there exists a sequence $(\alpha_k)_{k \in \mathbb{N}} \subseteq \mathbb{C}$ such that

$$f(z) = \sum_{k=0}^{\infty} \alpha_k (z - z_0)^k, \quad \forall z \in B_R(z_0), \quad \text{where} \quad R = \begin{cases} \inf_{z \in \mathbb{C} \setminus U} |z - z_0| & \text{if } U \neq \mathbb{C} \\ +\infty & \text{if } U = \mathbb{C}. \end{cases}$$

As a consequence, holomorphic functions are of class C^∞ and their complex derivatives of any order are holomorphic. In that case, the sequence $(\alpha_k)_{k \in \mathbb{N}}$ is unique and given by

$$\alpha_k = \frac{f^{(k)}(z_0)}{k!}.$$

The following important theorem shows that two holomorphic functions agreeing on a set with an accumulation point must coincide. This is not true for C^∞ functions in $\mathbb{R}$:

Theorem A.4 (Rigidity of Holomorphic Functions). If $f_1, f_2 : U \rightarrow \mathbb{C}$ are holomorphic functions defined on a connected open set $U \subseteq \mathbb{C}$ and there exists $z_0 \in U$ and a sequence of points $(z_i)_{i \in \mathbb{N}} \subseteq U \setminus \{z_0\}$ converging to z_0 such that $f_1(z_i) = f_2(z_i)$ for every $i \in \mathbb{N}$, then $f_1 \equiv f_2$.

Definition A.5 (Meromorphic functions). Assume $U \subseteq U' \subseteq \mathbb{C}$ are two open sets. We say that $f : U \rightarrow \mathbb{C}$ is a meromorphic function on U' if $U' \setminus U$ consists of isolated points, i.e., for any $z_0 \in U' \setminus U$ there exists $\delta > 0$ such that

$$B_\delta(z_0) \cap ((U' \setminus U) \setminus \{z_0\}) = \emptyset.$$

Moreover, for each $z_0 \in U' \setminus U$, we can write

$$f(z) = \frac{g(z)}{(z - z_0)^k},$$

with g holomorphic defined on an open set containing z_0 , $k \in \mathbb{N}$, and $z \neq z_0$ sufficiently close to z_0 . Since g is holomorphic, we can equivalently write a meromorphic function around a *pole* z_0 as

$$f(z) = \sum_{k=k_0}^{\infty} \alpha_k (z - z_0)^k,$$

with $k_0 \leq 0$. We call *residue* of f at z_0 , denoted $\text{Res}(f, z_0)$, the coefficient α_{-1} in the previous expansion.

Definition A.6 (Loop, simple loop, and null-homotopy). Let $V \subseteq \mathbb{C}$ be open. A *loop* in V is a continuous map $\gamma : [0, 1] \rightarrow V$ such that $\gamma(0) = \gamma(1)$. It is *injective* (or *simple*) if γ is injective on $[0, 1]$. We say that γ is *homotopic to a constant within* V if there exist $z_* \in V$ and a continuous map

$$H : [0, 1] \times [0, 1] \rightarrow V$$

such that $H(s, 0) = \gamma(s)$, $H(s, 1) = z_*$, and $H(0, t) = H(1, t) = z_*$ for all $s, t \in [0, 1]$.

Theorem A.7 (Cauchy's residue theorem). Given a function $f : U \rightarrow \mathbb{C}$ holomorphic and meromorphic on $U' \supseteq U$, consider an injective loop γ in U traveled counterclockwise. If γ is homotopic to a constant within the set where f is holomorphic (equivalently, within U' avoiding the poles), then

$$\frac{1}{2\pi i} \int_{\gamma} f(z) dz = \sum_w \text{Res}(f, w),$$

where w denote the poles of f enclosed by γ .

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